The Genesis Trilogy

Emergent Geometry, Algebraic Persistence,
and the Architecture of the Observer

Complete Collected Volume

HAFF · Q-RAIF · T-DOME

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Abstract

This volume collects the HAFF, Q-RAIF, and T-DOME papers comprising the Genesis Trilogy, a programme for studying how physical descriptions, persistent agents, and self-referential observers may be modelled within quantum theory.

The programme is organised in three parts. HAFF studies observable-algebra accessibility and the conditional emergence of effective geometry, measurement, temporal records, and programme boundaries. Q-RAIF studies Clifford-compatible algebraic encodings and image-level interfaces as modelling choices, not as a completed proof of uniqueness or algebraic necessity. T-DOME studies memory, reference-frame selection, and self-referential calibration as a conditional architecture for specified persistent-agent models, not as a necessary, sufficient, minimal, or complete architecture for all agents.

Together, the three parts provide a calibrated structural research programme. They supply conditional constructions, compatibility tests, and explicit open interfaces rather than a closed derivation of geometry, persistence, or consciousness from one theorem chain.

Keywords: emergent geometry, accessible algebras, coarse-graining, Clifford algebra, non-Markovian dynamics, reference-frame selection, Fisher information, self-referential calibration, Lyapunov stability, implementation cost

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Part I

HAFF: The Holographic Alaya-Field Framework

Chapter 1

Emergent Geometry from Coarse-Grained Observable Algebras: The Holographic Alaya–Field Framework

Paper A

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Abstract

We construct a theoretical framework where the tensor factorization of a Hilbert space is treated as a structural variable rather than a kinematic background. By lifting the "Alaya" concept to a globally entangled vacuum state $|\Psi_{\text{vac}}\rangle$, we analyze how local geometric data may depend on specific observable subalgebras $\mathcal{A}_i \subset \mathcal{B}(\mathcal{H})$ when an explicit reconstruction dictionary is supplied. We show that, under a separating reconstruction dictionary, inequivalent entropy profiles induced by coarse-graining maps yield reconstructed geometric data not related by diffeomorphism. The analysis is structural in nature: we do not propose new dynamics, but examine consistency and consequences of removing subsystem factorization from fundamental assumptions. This approach is complementary to existing interpretational frameworks and suggests natural connections to algebraic quantum field theory, entanglement-based approaches to spacetime, and quantum information theory.

1.1 Introduction

1.1.1 Motivation

Two central problems in contemporary theoretical physics concern the emergence of classicality and the emergence of geometry. In quantum foundations, the measurement problem asks how effectively classical behavior arises from an underlying quantum description. Decoherence theory has provided a powerful account of this process by explaining

the suppression of interference between certain degrees of freedom through environmental entanglement [142]. However, this explanation typically presupposes a fixed decomposition of the total system into subsystems, distinguishing "system," "apparatus," and "environment" from the outset.

A closely related emergence problem appears in quantum gravity. A growing body of work suggests that spacetime geometry is not fundamental, but arises from patterns of quantum entanglement [78, 121]. In holographic settings, geometric quantities are related to entanglement measures via precise correspondences, most notably the Ryu–Takayanagi formula [97]. Yet these constructions likewise assume a prior specification of spatial regions or tensor factors, with geometry inferred only after such a subdivision has been fixed.

In both contexts, the emergence problem is addressed only after a subsystem decomposition has been assumed. The structure responsible for classicality or geometry is therefore explained relative to a partition whose origin remains largely unexamined.

1.1.2 The Structural Gap

The assumption of a given subsystem structure is often treated as innocuous, or as a matter of convenient description. However, from a fundamental perspective, there is no canonical tensor factorization of a generic Hilbert space, nor a unique way to decompose a global quantum state into subsystems. While this issue is occasionally acknowledged in passing [141, 127], its consequences for emergence are rarely explored systematically.

The present work does not challenge the empirical success of decoherence theory, entanglement-based approaches to geometry, or the algebraic formulation of quantum theory. Rather, we make explicit a structural assumption common to these frameworks and investigate the consequences of relaxing it. Our focus is on what follows if subsystem structure itself is treated as emergent, rather than fundamental.

1.1.3 Our Contribution

We formulate a framework in which subsystem structure is not assumed *a priori*, but arises from a choice of coarse-graining over observable algebras. Within this framework, we show that different coarse-graining channels can induce inequivalent effective states and connectivity data. In a holographic or other reconstruction model that separates the resulting entropy profiles modulo diffeomorphisms, this difference can induce inequivalent effective geometries. Outside such a dictionary, the output is a weighted connectivity structure, not a spacetime metric.

The analysis is structural in nature. Our aim is not to propose a new dynamical mechanism, but to examine the consistency and consequences of removing subsystem factorization from the set of fundamental assumptions.

1.1.4 Terminology: The Alaya-Field

Throughout this work, we adopt the term *Alaya-Field* to denote the fundamental, non-factorized structure prior to any subsystem decomposition. This terminology, borrowed from Yogācāra philosophy (referring to the "storehouse consciousness"), is used here strictly in a technical sense.

In what follows, the Alaya-Field does not refer merely to a vector in Hilbert space, but to the triple

$$(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle),$$

where $\mathcal{A}_U$ is a von Neumann algebra acting on $\mathcal{H}_U$ and $|\Omega\rangle$ is a cyclic and separating vector. The emphasis is on the absence of a canonical factorization, not on the particular choice of state.

In an AQFT realization, $\mathcal{A}_U$ may be a Type III_1 factor and $|\Omega\rangle$ a cyclic and separating vector [50]. This is a possible source-side regime, not a consequence of the finite-dimensional calculations below, and we do not infer localized eigenmodes or a metric from the Type III label alone. This usage is intended to evoke the non-factorized, pre-geometric nature of the fundamental quantum structure, without importing any metaphysical commitments.

1.1.5 Regime of Each Result

The paper uses a mixed regime. The source triple $(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle)$ may be discussed for a general von Neumann algebra, including Type III under the stated cyclic-and-separating hypothesis. The factorization, Haar measure, density-matrix, partial-trace, and von Neumann-entropy results in Sections 1.2–1.4 are finite-dimensional/Type I results. Type III factors have no density matrices or trace entropy, so those results are not asserted there. The conversion from entropy/connectivity data to a spacetime metric is a separate model-dependent holographic input.

1.1.6 Structure of the Paper

The paper is organized as follows. In Section 1.2, we review the absence of a canonical subsystem decomposition in quantum theory and formalize this observation. Section 1.3 introduces coarse-graining in terms of observable algebras and analyzes how effective subsystem descriptions arise from this procedure. In Section 1.4, we construct coarse-graining-dependent connectivity data, give a state-space QFIM construction, and state the additional reconstruction hypothesis required before metric inequivalence can be claimed. Section 1.5 discusses the scope and conceptual implications of the framework, its relation to existing approaches, and possible directions for future work. We conclude in Section 1.6 with a summary of results and open questions.

1.2 Absence of Canonical Tensor Factorization

1.2.1 The Factorization Problem

In standard quantum mechanics, the state space of a composite system is constructed as a tensor product of subsystem Hilbert spaces. This construction presupposes that a natural decomposition into subsystems has already been identified. However, for a given Hilbert space $\mathcal{H}_{\text{total}}$, there is no unique or canonical way to express it as a tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ without additional physical input.

The universal entanglement statement used in an earlier version is withdrawn. Indeed, fix dimensions $d_A, d_B > 1$ with $d_A d_B = \dim \mathcal{H}_{\text{total}}$. For any pure state $|\Psi_U\rangle$, choose a unitary U sending a reference product vector $|0\rangle_A \otimes |0\rangle_B$ to $|\Psi_U\rangle$. Pulling the reference

tensor factors through U gives a nontrivial factorization in which $|\Psi_U\rangle$ is a product. Thus no finite-dimensional pure state is entangled across *every* nontrivial factorization.

What does survive is a fixed-family typicality result.

Theorem 1.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\text{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule independent of the sampled state. Write $\mathcal{H}_{\text{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \leq d_{B_j}$. For $\delta > 0$, let*

$$p_j(\delta) := \Pr_{\Psi \sim \text{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \geq \log d_{A_j} - \delta \text{ for every } j \right] \geq 1 - \sum_{j=1}^{m} p_j(\delta),$$

and the standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual dimension-ratio hypotheses.

Proof. For each fixed $\mathbf{f}_j$, Page's theorem gives [83]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j} d_{B_j}} \frac{1}{k} - \frac{d_{A_j} - 1}{2 d_{B_j}}, \tag{1.1}$$

while concentration of measure bounds the lower tail [56]; this is precisely $p_j(\delta)$. Applying the union bound to the finite, preselected family proves the displayed inequality. □

Remark 1.2 (What the theorem does not select)**.** *Theorem 1.1 says nothing about all possible factorizations and supplies no preferred member of $\mathfrak{F}$. Specifying the admissible family and its selection rule is a quantum-mereology input, not a consequence of Haar typicality. Developing that input is an open problem; see [29, 24].*

Remark 1.3 (Relation to prior work)**.** *The observation that tensor product structures are not canonical was recognized early in the foundations of quantum theory, and has been systematically analyzed in work on virtual subsystems and quantum mereology [140, 141, 29, 24]. Their work demonstrated that subsystem decompositions are effectively determined by sets of accessible observables rather than being intrinsic to the Hilbert space. The present framework extends this perspective to the context of holographic geometry and entanglement-based spacetime emergence, where the choice of observable algebra determines not only subsystem structure but also the emergent geometric description.*

Remark 1.4. *We do not claim that subsystem decompositions are impossible or unphysical, but that they are not uniquely determined by the universal quantum state alone. This observation motivates the search for additional structure that specifies how a subsystem decomposition arises in concrete physical contexts.*

1.2.2 Relation to Algebraic Approaches

The absence of a canonical factorization has been recognized in various contexts, including algebraic quantum field theory where observable algebras take precedence over tensor product structures [50]. Our framework builds on these insights by treating coarse-graining structure as the fundamental input from which subsystem decompositions emerge.

1.3 Observer-Dependent Coarse-Graining and Effective States

1.3.1 Coarse-Graining Structure

Before introducing the formal definition, we emphasize that the introduction of an observable algebra does not reinstate a tensor factorization. Operators may act irreducibly on $\mathcal{H}_{\text{total}}$ without inducing any subsystem decomposition. In algebraic quantum field theory, observable algebras are defined independently of any global tensor product structure [50]. Furthermore, a coarse-graining is not selected by an agent, but instantiated by a concrete physical interaction structure.

Definition 1.5 (Operational Coarse-Graining Structure)**.** *Let $\mathcal{H}_U$ be the universal Hilbert space. A **coarse-graining structure** $\mathbf{c}$ is defined as a pair*

$$\boxed{\mathbf{c} \equiv (\mathcal{A}_{\text{sel},\mathbf{c}}, \Phi_{\mathbf{c}})}$$

where:

1. *$\mathcal{A}_{\text{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ is a selected unital $*$-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer/record algebra, or environmental fixed set; a physical model must state any map between them.*

2. *In the finite-dimensional results below, $\Phi_{\mathbf{c}}$ is a CPTP map on states implementing an operational reduction, and its Heisenberg dual $\Phi_{\mathbf{c}}^*$ is unital and completely positive. Examples include partial trace over a specified factor and channels induced by a conditional expectation. In a general von Neumann setting the corresponding object would be a normal channel on states with a normal UCP dual, but no entropy theorem below is asserted merely from that extension.*

Remark 1.6 (Non-uniqueness of observable algebras)**.** *It is important to note that the specification of a selected algebra $\mathcal{A}_{\text{sel},\mathbf{c}}$ is not assumed to be unique. In algebraic quantum field theory and quantum information theory, there exists no theorem guaranteeing a unique maximal observable algebra associated with a given physical system without additional structure. The coexistence of multiple admissible algebras reflects physical underdetermination rather than subjectivity or observer dependence.*

1.3.2 Refinement Structure of Coarse-Grainings

Definition 1.7 (Refinement Relation)**.** *Given two coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$, we say $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there exists a CPTP map*

$$\Lambda : \mathcal{B}(\mathcal{H}_{\mathit{eff}}(\mathbf{c}_1)) \to \mathcal{B}(\mathcal{H}_{\mathit{eff}}(\mathbf{c}_2))$$

such that

$$\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$$

*In this case, $\mathbf{c}_2$ is a **further coarse-graining** of $\mathbf{c}_1$.*

Remark 1.8 (Multiplicity of coarse-graining maps)**.** *For a fixed selected algebra $\mathcal{A}_{\mathrm{sel},\mathbf{c}}$, there generally exist multiple completely positive trace-preserving maps implementing distinct coarse-graining procedures. The present framework does not require the selection of a preferred CPTP map. Rather, different maps correspond to physically realizable information-loss mechanisms, such as tracing over inaccessible degrees of freedom or effective decoherence channels.*

Different coarse-graining choices are related not by unitary symmetry, but by information-theoretic refinement maps, forming a partially ordered structure rather than a group.

1.3.3 Core Theorem

The absence of a unitary intertwiner and the distinction of observable images were previously presented as conclusions, although they merely unpacked (and in the case of the images did not follow from) the chosen inequivalence relation. They are not retained as theorem claims. The nontrivial finite-dimensional statement is the following generic separation result.

Theorem 1.9 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be two CPTP maps from states on $\mathcal{H}$ to states on the common output space $\mathcal{K}$. Then the pure inputs on which Φ_1 and Φ_2 agree form a Haar-null proper real-algebraic subset. If, in addition, the real-analytic function*

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\, \Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\, \Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs are full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Because pure-state projectors span the Hermitian operators, $\Phi_1 - \Phi_2$ cannot vanish on every pure projector. The equality condition is polynomial in the real and imaginary components of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain, von Neumann entropy and hence f_A are real analytic. A nonzero real-analytic function has a measure-zero zero set, proving the conditional entropy statement. □

Remark 1.10. *The nondegeneracy hypothesis on f_A is essential: distinct channels related by a unitary acting only on A can produce different states with identical subsystem entropy for every input. The theorem therefore does not turn channel inequivalence alone into entanglement or geometric inequivalence.*

Remark 1.11 (Relation to decoherence theory)**.** *We emphasize that the present framework is fully compatible with standard decoherence theory and does not modify its dynamical content. Decoherence successfully explains the emergence of classical behavior given a fixed system–environment decomposition. The novelty of the present approach lies instead in treating such subsystem decompositions as coarse-graining–dependent and not fundamental.*

1.3.4 Clarification on Subsystem Structure

Subsystems are not fundamental inputs to the framework, but derived representations induced by a chosen observable algebra. The connectivity measure defined in Section 1.4 acts on these representations, and is not used to define the algebra itself. The causal order is:

$$\text{Observable Algebra} \to \text{Representation} \to \text{Entanglement} \to \text{Connectivity} \to \text{Geometry}$$

No geometric structure is presupposed in the definition of coarse-graining.

1.3.5 Algebraic Perspective on Subsystems

A potential concern is whether the use of observable algebras implicitly presupposes a subsystem decomposition. We stress that this is not the case. Observable algebras need not be defined via a tensor factorization of the total Hilbert space.

In algebraic quantum field theory, local algebras are assigned to spacetime regions without invoking a global tensor product structure [50]. Subsystems arise only at the level of representations induced by a chosen algebra, rather than serving as its foundational input. In this sense, subsystem structure is emergent rather than fundamental.

Remark 1.12 (Subsystems as induced representations)**.** *Within the present framework, subsystems are not primitive entities. They emerge as representations associated with a given observable algebra and coarse-graining structure. This avoids circularity by reversing the usual explanatory order: observable structure precedes subsystem identification.*

1.4 Entanglement Structure and Emergent Geometry

1.4.1 Mutual Information as Connectivity Measure

We stress that no geometric structure is assumed in the definition of the coarse-graining introduced in Section 1.3. Geometry only appears at the level of entanglement relations between the induced subsystems.

Given a coarse-graining structure $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$ as defined in Section 1.3, the universal quantum state $|\Psi_U\rangle$ induces a family of effective subsystems $\{A, B, \dots\}$ associated with subalgebras of $\mathcal{A}_{\mathrm{sel},\mathbf{c}}$. For any such pair of subsystems A and B, we consider the quantum mutual information

$$I_{\mathbf{c}}(A : B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB), \tag{1.2}$$

where $S_{\mathbf{c}}(\cdot)$ denotes the von Neumann entropy computed after coarse-graining.

Definition 1.13 (Entanglement-Induced Connectivity)**.** *Let A and B be two disjoint subsystems induced by $\mathbf{c}$. We define an effective **proximity measure** $\mu_{\mathbf{c}}(A, B)$ based on the quantum mutual information:*

$$\mu_{\mathbf{c}}(A, B) := I_{\mathbf{c}}(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \tag{1.3}$$

While $\mu_{\mathbf{c}}$ does not itself constitute a metric (as it violates the triangle inequality), it defines a weighted connectivity graph in a declared subsystem representation.

Here is the precise finite-dimensional information-geometric construction used when a metric on a *parameter space of coarse-grained states* is needed. Let $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$ be a smooth faithful family and define the symmetric logarithmic derivatives L_a by

$$\partial_a \rho_{\mathbf{c}} = \frac{1}{2}\big(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a\big).$$

The SLD quantum Fisher metric is

$$g_{ab}^{\mathrm{QF}}(\lambda) := \frac{1}{2}\,\mathrm{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]\,. \tag{1.4}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions. It is not a Lorentzian spacetime metric, and it is not determined by the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ without an additional map from graph/state parameters to geometry. In the holographic discussion below that additional map is a reconstruction dictionary, not a general consequence of QFIM [90].

This quantity should be understood as a *connectivity measure* rather than a fundamental spacetime distance. In general, mutual information does not define a metric on arbitrary quantum subsystems. However, it acquires a natural geometric interpretation under physically motivated assumptions, which we make explicit below.

Assumptions. Throughout this section, we restrict attention to coarse-grainings that satisfy a geometric admissibility condition:

Definition 1.14 (Geometric Admissibility)**.** *A coarse-graining structure* $\mathbf{c}$ *is said to be* ***geometrically admissible*** *if the induced mutual information satisfies:*

1. Finite correlation length: *The state* $|\Psi_U\rangle$ *exhibits exponentially decaying correlations with respect to the induced subsystems, as is typical for gapped systems, ground states of local Hamiltonians, and holographic large-N states.*

2. Monotonic decay under refinement: *Mutual information decreases monotonically as subsystems become more refined.*

3. Stability under perturbations: *The entanglement structure is robust under small perturbations of* $\Phi_{\mathbf{c}}$*.*

Not all coarse-grainings admit a geometric interpretation. Geometry is not generic; it is a *special phase* of information organization. We further assume coarse-graining consistency: all entropic quantities are computed using a single coarse-graining structure $\mathbf{c}$, avoiding any mixing of inequivalent observable algebras.

Under these assumptions, the entanglement-induced connectivity $\mu_{\mathbf{c}}(A, B)$ is non-negative, symmetric, and vanishes if and only if the subsystems are uncorrelated at the level resolved by $\mathbf{c}$. Moreover, in regimes where mutual information decays monotonically with separation, $\mu_{\mathbf{c}}$ induces an effective notion of spatial proximity.

Remark 1.15. *The connectivity* $\mu_{\mathbf{c}}(A, B)$ *should be understood as an* effective, coarse-graining–dependent measure of correlation*, rather than a fundamental spacetime metric. A metric on a state-family parameter space may be supplied by Eq.* (1.4)*. A spacetime metric requires the additional reconstruction input stated below.*

In a continuum limit of densely overlapping subsystems, the collection $\{\mu_{\mathbf{c}}(A, B)\}$ may approximate a weighted graph with a geometric interpretation. This is a model-building step: neither graph convergence nor a map from the QFIM parameter metric to spacetime is established in general.

1.4.2 From Entanglement to Geometry

The idea that spacetime geometry is encoded in quantum entanglement has been explored extensively in the context of holography. In particular, the Ryu–Takayanagi formula relates the entanglement entropy of a boundary region A to the area of an extremal surface γ_A in the bulk [97],

$$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N}, \tag{1.5}$$

suggesting a direct correspondence between entanglement structure and geometric data.

More generally, Van Raamsdonk has argued that the connectivity of spacetime is controlled by the pattern of entanglement between subsystems [121]. In this perspective, highly entangled degrees of freedom correspond to nearby regions in the emergent geometry, while weakly entangled subsystems are geometrically distant.

The entanglement-induced connectivity $\mu_{\mathbf{c}}(A, B)$ provides a concrete realization of this idea. Given a collection of subsystems induced by $\mathbf{c}$, the mutual information defines a weighted graph whose vertices correspond to subsystems and whose edge weights encode entanglement strength.

In tensor network constructions such as MERA, similar entanglement graphs admit a natural geometric interpretation, with graph distance approximating continuum spatial distance [114]. In particular tensor-network or holographic models, an appropriate continuum limit can be equipped with an effective Riemannian spatial geometry whose metric reflects the underlying entanglement structure.

In this sense, geometry emerges not as an additional postulate, but as an effective description of how information is distributed and shared among coarse-grained degrees of freedom.

1.4.3 Coarse-Graining Dependence of Geometry

We now turn to the central observation of this section: the emergent geometry depends essentially on the choice of coarse-graining structure.

Theorem 1.16 (Coarse-graining dependent geometry, conditional reconstruction)**.** *Let $\mathbf{c}_1$ and $\mathbf{c}_2$ act on the same finite-dimensional global state and induce entropy profiles S_1, S_2. Assume a specified reconstruction dictionary $\mathfrak{G}$ is defined on both profiles and is separating modulo diffeomorphisms on the compared class:*

$$\mathfrak{G}[S_1] \simeq_{\mathrm{diff}} \mathfrak{G}[S_2] \quad \Longrightarrow \quad S_1 \simeq_{\mathfrak{G}} S_2.$$

If $S_1 \not\simeq_{\mathfrak{G}} S_2$, then the reconstructed metrics $\mathfrak{G}[S_1]$ and $\mathfrak{G}[S_2]$ are not related by a diffeomorphism. The theorem makes no metric claim when no such dictionary is supplied.

Proof. This is the contrapositive of the separating property of $\mathfrak{G}$. Holographic entropy/area reconstruction is one setting in which such a property may be established on a restricted state class [97]. Channel inequivalence by itself does not provide either the hypothesis $S_1 \not\simeq_{\mathfrak{G}} S_2$ or the dictionary. □

Example 1.17 (Free Fermion Chain: Spatial vs. Momentum Coarse-Graining)**.** *Consider N free fermions on a one-dimensional lattice with ground state $|\Psi_0\rangle$.*

***Spatial coarse-graining** $\mathbf{c}_x$: group sites into blocks of k consecutive sites. By the area law for gapped free-fermion chains [37], the entanglement entropy of a block of L*

sites satisfies $S(L) \sim$ const, independent of L. The mutual information between blocks A and B separated by distance r decays exponentially as $I(A:B) \sim e^{-r/\xi}$, where ξ is the correlation length set by the spectral gap [54]. The resulting connectivity graph has short-range edges: its graph distance approximates a one-dimensional lattice, yielding a line-like emergent geometry.

Momentum-mode coarse-graining $\mathbf{c}_p$*: use the occupation-number tensor factors of the diagonal momentum modes. A free-fermion ground state is a Slater determinant*

$$|\Psi_0\rangle = \prod_{k \in F} c_k^\dagger |0\rangle,$$

which is a product of filled or empty occupation states in this factorization. Every subset of momentum modes therefore has a pure reduced state and disjoint momentum subsets have zero mutual information. The momentum connectivity graph is empty, not a volume-law, long-range graph. Wolf's result [136] concerns spatial *entanglement scaling in fermionic systems and does not support momentum-mode entanglement.*

Thus the example demonstrates dependence of connectivity on a specified factorization: the spatial representation can carry short-range edges, while the free momentum-occupation representation carries none. It does not exhibit two reconstructed geometries; indeed $\mathbf{c}_p$ fails the geometric-admissibility goal unless further, interacting correlations or a separate reconstruction prescription are supplied.

A simple conceptual illustration is provided by contrasting spatial coarse-graining with momentum-space coarse-graining. In the former case, entanglement typically obeys an area law and supports a local, connected geometry. For the free Slater determinant, the latter has no momentum-mode entanglement at all; interacting models would be required to obtain a nontrivial momentum graph. Importantly, both descriptions arise from the same underlying state $|\Psi_U\rangle$.

Thus, the geometry inferred from entanglement is not an intrinsic property of the quantum state alone, but depends on how information is rendered accessible through coarse-graining.

1.4.4 Beyond Coordinate Choice

It is important to distinguish the dependence described above from an ordinary change of coordinates. A diffeomorphism acts within a fixed algebra of observables, preserving the underlying notion of what is measurable. By contrast, a change of coarse-graining alters the observable algebra itself, modifying which correlations are accessible and how subsystems are defined.

These are categorically distinct operations. Nevertheless, algebraic inequivalence alone does not prove that two reconstructed metrics fail to be diffeomorphic: a reconstruction dictionary may discard the algebraic differences. Non-diffeomorphism follows only under the separating hypothesis of Theorem 1.16. Without it, the justified conclusion is that the effective operator or connectivity descriptions differ.

We return to the conceptual implications of this distinction in the Discussion.

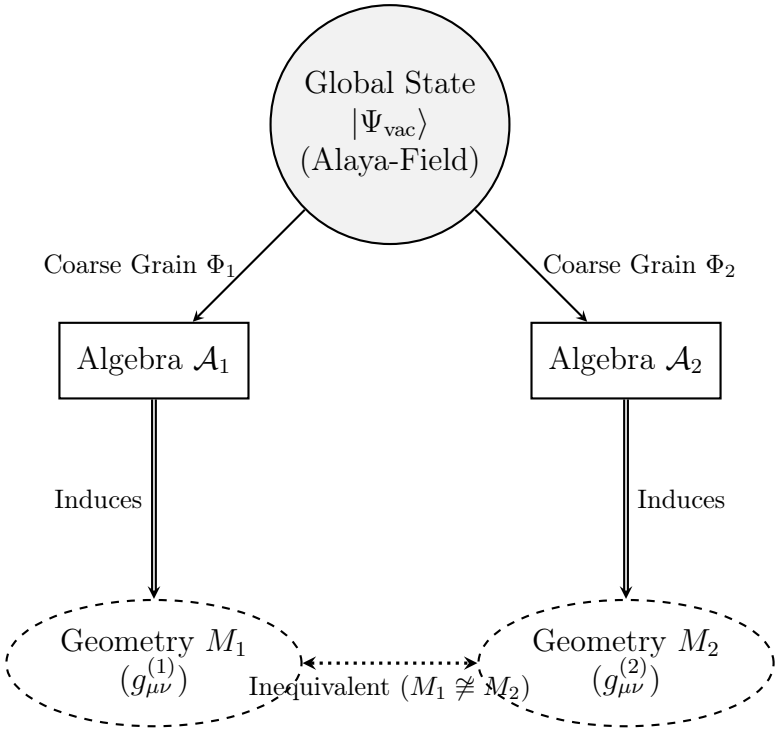


Figure 1.1: Schematic of the conditional HAFF construction. A single global state is reduced by two coarse-graining channels. The outputs define different effective connectivity data; only a stated reconstruction dictionary promotes them to metrics $g_{\mu\nu}^{(1,2)}$. The dependence is on coarse-graining at the level of effective description, not on an observer in a fundamental sense.

1.5 Discussion

1.5.1 Relation to Existing Frameworks

The framework developed in this work is not intended as a replacement for existing interpretational or algebraic approaches to quantum theory. Rather, it is best understood as orthogonal to several well-established lines of thought, addressing a distinct structural question. In this section, we briefly clarify its relation to three representative frameworks:

Relation to QBism

QBism emphasizes the role of agents and their personal probability assignments in quantum theory, interpreting the quantum state as an expression of subjective belief rather than an objective property of a system. In contrast, the present framework assumes a single, objective global quantum state throughout. No agent-dependent elements enter the formalism, and no interpretational commitments regarding belief, experience, or decision theory are required.

The point of contact lies solely in the rejection of a privileged subsystem decomposition. While QBism attributes this absence to the primacy of the agent, our framework treats it as a structural feature of quantum theory itself. Coarse-grainings are not chosen by agents, but instantiated by concrete physical interaction structures. The resulting multiplicity of effective descriptions reflects physical underdetermination rather than subjectivity.

Relation to Many-Worlds and Decoherence-Based Approaches

Many-Worlds–type interpretations and decoherence-based accounts provide a compelling explanation of classical behavior within quantum mechanics by analyzing branching structures relative to a fixed subsystem decomposition. Our framework does not modify this analysis, nor does it introduce an alternative account of branching or classicality.

The difference lies at a prior level. Decoherence theory presupposes a tensor factorization into system, apparatus, and environment. Here, we instead ask how such subsystem structures arise in the first place. In this sense, the framework is complementary to decoherence-based approaches: it leaves their dynamical conclusions intact while removing subsystem factorization from the list of fundamental assumptions.

Relation to Algebraic Quantum Field Theory

The closest structural affinity of the present work is with algebraic quantum field theory (AQFT). In AQFT, observable algebras are taken as primary, and states are defined as positive linear functionals over these algebras, without reliance on a global tensor product structure. Our use of observable algebras and their representations is directly inspired by this tradition.

The present framework may be viewed as extending this algebraic perspective by emphasizing the role of coarse-graining relations between algebras. Different choices of coarse-graining induce different effective representations and, consequently, different entanglement structures. The novelty lies not in the algebraic formalism itself, but in using it to analyze the emergence and non-uniqueness of geometric descriptions.

Taken together, these comparisons situate the present work as a structural investigation into the conditions under which subsystem structure and geometry emerge. It neither commits to a particular interpretation of quantum mechanics nor proposes a new dynamical law, but instead clarifies how several existing frameworks implicitly rely on assumptions that can be made explicit and, in some cases, relaxed.

1.5.2 Scope and Limitations

The present work is concerned with structural consistency rather than phenomenological prediction. Observable consequences depend on the physical mechanisms implementing a given coarse-graining, which lie beyond the scope of this paper. This is analogous to effective field theory, where multiple UV completions may share the same low-energy structure. Future work will explore concrete physical realizations and their empirical signatures.

1.5.3 Philosophical Implications

We emphasize that the framework assumes a single, objective global quantum state $|\Psi_U\rangle$. What is coarse-graining–dependent is not reality itself, but the effective structures used to describe it. This position is compatible with scientific realism while acknowledging the role of operational context in physical description.

1.5.4 Future Directions

While the present work is deliberately limited to a structural analysis of subsystem emergence and geometry within a fixed global quantum state, it naturally opens several directions for further investigation. We emphasize that the following points are not results established here, but rather indicate possible extensions where the current framework may provide useful conceptual or technical guidance.

Dynamical Models of Coarse-Graining Selection

In this work, coarse-graining structures are treated as fixed relational inputs, instantiated by concrete physical interaction patterns. A natural next step is to investigate whether such coarse-grainings can themselves be characterized dynamically.

One possible direction is to study how interaction Hamiltonians, coupling strengths, or network topologies bias the emergence of particular subalgebra structures over others, in contact with explicit quantum-mereology and locality-selection programmes [29, 24]. This could clarify under what physical conditions certain factorizations become robust or persistent, and whether transitions between inequivalent coarse-grainings admit an effective dynamical description.

Importantly, such an analysis would remain compatible with globally unitary evolution, treating coarse-graining selection as an emergent, effective phenomenon rather than a modification of fundamental dynamics.

Connections to Quantum Information and Complexity

The non-uniqueness of emergent geometry highlighted here suggests a close connection to quantum information–theoretic notions such as entanglement structure, operator complexity, and resource constraints.

Future work could explore whether preferred geometric descriptions correlate with informational criteria—for example, minimal description length, stability under noise, or computational accessibility of observables. Such considerations may help explain why certain coarse-grainings are physically salient, even when many are formally admissible.

This perspective may also provide a bridge to recent work on complexity-based approaches to spacetime emergence, without committing to any particular complexity measure at the present stage.

Extensions to Quantum Field Theory and Continuum Limits

While the framework has been formulated in abstract Hilbert space terms, an important open question concerns its implementation in quantum field–theoretic settings, where issues of locality, algebraic nets, and continuum limits arise.

In particular, it would be valuable to examine how coarse-graining relations between observable algebras interact with the locality structures emphasized in algebraic quantum field theory, and whether familiar spacetime geometries can be recovered as stable fixed points of such relations.

We stress that the present work does not resolve these questions, but provides a structural language in which they can be posed more precisely.

Empirical and Phenomenological Implications

At the level developed here, the framework is primarily structural and conceptual. Nevertheless, future investigations may ask whether different coarse-graining choices lead to distinguishable effective descriptions, for example in semiclassical regimes, quantum gravity–motivated models, or analogue systems.

Such studies could clarify whether the non-uniqueness of emergent geometry has observable consequences, or whether physical constraints effectively suppress this freedom in realistic settings.

Any empirical analysis would necessarily require additional assumptions beyond those adopted in this work, and thus lies outside its present scope.

Conceptual Clarifications and Interpretational Interfaces

Finally, although the framework is intentionally neutral with respect to interpretations of quantum mechanics, it may serve as a useful interface for comparative studies. By making explicit the structural assumptions underlying subsystem decomposition and geometry, it could help clarify which features are interpretation-dependent and which arise more generally from the formalism itself.

We view this not as an attempt to adjudicate between interpretations, but as an opportunity to sharpen the questions they address.

Taken together, these directions suggest that the framework developed here is best viewed as a scaffold: it does not dictate specific physical models, but provides a structured setting in which questions about subsystems, geometry, and emergence can be formulated with greater precision.

1.6 Conclusion

We have presented a framework in which subsystem structure is not presupposed, but is modelled through coarse-graining over selected observable algebras. The central results are:

1. For a finite, physically preselected family of factorizations, Haar-typical states are highly entangled across every member under the stated dimension hypotheses (Theorem 1.1); no claim is made over all possible factorizations or about a preferred factorization.

2. For a fixed pair of distinct finite-dimensional channels, their outputs are generically separated; a generic entropy difference requires the explicit non-degeneracy hypothesis in Theorem 1.9.

3. When a separating reconstruction dictionary is supplied and the compared entropy profiles are inequivalent in its declared sense, the reconstructed geometric data are not related by a diffeomorphism (Theorem 1.16).

The framework is deliberately limited in scope. We have not proposed new dynamics, derived empirical predictions, or resolved interpretational debates. Instead, we have examined the structural consequences of removing subsystem factorization from the list of fundamental assumptions.

Within the stated channel and reconstruction hypotheses, this investigation shows how geometric data can depend on the chosen coarse-graining context. It does not establish a geometry for an arbitrary quantum state or coarse-graining without an additional reconstruction dictionary. This perspective is compatible with, and complementary to, existing approaches including decoherence theory, holographic duality, and algebraic quantum field theory.

Several open questions remain. Can coarse-graining structures themselves be characterized dynamically? Do informational or complexity-based criteria select physically preferred coarse-grainings? Can the framework be extended to quantum field theory and reconciled with standard locality structures? These questions lie beyond the present scope, but the structural setting developed here provides a language in which they can be formulated with greater precision.

Ultimately, this work suggests that the relationship between quantum states, subsystems, and geometry is more subtle than the standard picture implies. By making explicit an assumption that is often left implicit, we hope to have clarified the conditions under which emergence occurs and opened new avenues for investigating the foundations of quantum theory and spacetime.

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Chapter 2

Accessibility, Stability, and Emergent Geometry: Conceptual Clarifications on the Holographic Alaya–Field Framework

Paper B

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Abstract

This paper provides conceptual clarification of the Holographic Alaya-Field Framework (HAFF) introduced in our previous work. We address three potential misreadings: subjectivism (that observers create spacetime), anti-realism (that geometry is illusory), and trivialism (that the framework reduces to coordinate choice). By analyzing the structural notion of accessibility via stability conditions, delineating boundaries with existing interpretations (AQFT, RQM, QBism, MWI), and characterizing emergent geometry as a stable organizational phase, we clarify what the framework commits to and what it does not. This analysis is purely interpretational; no new formal results are introduced.

2.1 Introduction

Our previous work proposed a framework in which effective operator and connectivity descriptions depend on a selected algebra acting on a global quantum state [147]. Metric inequivalence is conditional on a reconstruction dictionary; it is not a general consequence of channel or algebra inequivalence. However, structural novelty of this kind is particularly vulnerable to interpretational confusion. The present paper addresses these interpretational implications and clarifies the conceptual commitments of the framework.

2.1.1 The Risk of Misreading

The claim that geometry is coarse-graining-dependent naturally invites several misreadings, each of which conflates distinct notions of dependence. Three such misreadings are

especially common:

1. *Subjectivism*: The view that observers or agents create spacetime through their choices or beliefs, collapsing the framework into an epistemic or observer-relative interpretation.

2. *Anti-realism*: The view that spacetime has no objective existence, and that emergent geometry is therefore illusory or merely pragmatic.

3. *Trivialism*: The view that coarse-graining-dependent geometry reduces to a choice of coordinates or descriptive convention, with no substantive physical consequences.

Each of these readings is incorrect, but each arises naturally from surface-level features of the formalism. The purpose of this paper is to block such misreadings by clarifying the nature of accessibility, the structural role of observable algebras, and the ontological status of emergent geometry within the framework.

2.1.2 Scope and Objectives

This paper introduces no theorems. The stability criteria and phase quantities below are explicitly heuristic diagnostics and model inputs, not proved selection principles. The paper provides a systematic interpretational analysis aimed at three specific objectives:

1. *Clarify the notion of accessibility*: We formulate accessibility as a structural and operational modelling proposal relative to declared dynamics, probes, and mereology, rather than as an epistemic or observer-dependent notion.

2. *Delineate boundaries with existing interpretations*: We situate the framework in relation to algebraic quantum field theory, relational quantum mechanics, QBism, and the Many-Worlds interpretation, clarifying points of agreement, divergence, and complementarity.

3. *Characterize the ontological status of emergent geometry*: We argue that geometry functions as a stable organizational phase of quantum information, analogous to phases in condensed matter systems, avoiding both naive realism and anti-realist eliminativism.

Importantly, this analysis does not constitute a defense of the framework, nor does it aim to persuade readers of its correctness. The goal is clarity: to ensure that the structural commitments of the framework are understood on their own terms, and that criticisms, if any, are directed at what the framework actually claims rather than at interpretational projections.

2.1.3 What This Paper Does Not Do

To further delimit scope, we note explicitly what this paper does *not* attempt:

- It does not propose new dynamics, empirical predictions, or modifications to quantum mechanics.

- It does not claim that the framework resolves outstanding problems in quantum gravity, quantum foundations, or the measurement problem.
- It does not advocate for any particular metaphysical or philosophical position beyond the minimal structural commitments required by the formalism itself.
- It does not interpret the framework as implying idealism, observer-created reality, or any form of mind-dependence.

The analysis remains strictly within the domain of structural interpretation: identifying what the mathematical formalism commits to, what it leaves open, and how it relates to existing approaches.

2.1.4 Organization

The paper proceeds as follows. Section 2.2 briefly recapitulates the structural stance adopted in our previous work, emphasizing the priority of observable algebras over tensor factorizations and the role of coarse-graining in defining effective subsystems.

Section 2.3 provides a detailed analysis of candidate accessibility diagnostics and their physical inputs. A taxonomy distinguishes subjective, relational, and structural notions of dependence; HAFF adopts the third as a model-relative proposal without claiming a unique input-free selector.

Section 2.4 examines the relationship between the framework and four representative approaches: algebraic quantum field theory, relational quantum mechanics, QBism, and the Many-Worlds interpretation. Each subsection clarifies points of conceptual overlap and structural divergence, preventing conflation while identifying opportunities for complementarity.

Section 2.5 argues that emergent geometry should be understood as a stable organizational phase of entanglement structure, drawing on analogies with condensed matter physics. This perspective avoids treating geometry as either fundamental or illusory, instead characterizing it as contingent but objective—dependent on physical conditions rather than epistemic contexts.

Section 2.6 delimits the structural assumptions of the framework, enumerates what it does not commit to, and outlines open questions for future investigation. Particular attention is given to the distinction between structural analysis and metaphysical advocacy.

2.1.5 Methodological Note

Throughout this paper, we adopt a deliberately conservative rhetorical stance. Claims are hedged with modal qualifiers ("may suggest," "is consistent with," "can be understood as") because the selection criteria and phase analogy are heuristic and because interpretational conclusions should not outrun their model inputs. The goal is to present the framework as one coherent way of organizing the conceptual landscape, not as the uniquely correct interpretation.

This methodological caution reflects a broader commitment: interpretational clarity is best served by precision and restraint, not by advocacy or polemics. We aim to make the framework legible to researchers across different interpretational traditions, facilitating comparison and critique rather than preempting it.

2.2 Structural Stance Recap

We briefly recapitulate the structural stance of our previous work without repeating full mathematical derivations.

The framework uses three claims with different epistemic status:

1. **Factorization input**: Hilbert-space structure alone supplies no preferred tensor factorization. Paper A proves entanglement typicality only for a finite, physically preselected family; choosing that family remains a quantum-mereology input.

2. **Conditional channel separation**: Distinct finite-dimensional coarse-graining channels generically produce different outputs. Entropy separation requires an explicit nondegeneracy condition.

3. **Conditional geometry**: Entropy/connectivity data become spacetime geometry only under a supplied reconstruction dictionary. Non-diffeomorphism follows only when that dictionary separates the compared profiles modulo diffeomorphisms.

The conceptual core can be summarized by reversing the standard explanatory arrow:

Standard: State + Tensor factorization $\rightarrow$ Subsystems $\rightarrow$ Entanglement $\rightarrow$ Geometry

$$\textbf{HAFF:}\ \text{State} + \text{selected algebra/channel} \rightarrow \text{Effective subsystems} \rightarrow \text{Entanglement/connectivity} \xrightarrow{\text{dictionary}} \text{Geometry}$$

Write $\mathcal{A}_{\text{sel},\mathbf{c}}$ for the selected algebra used in Paper A. It is not Paper E's probe-reachable effect system, pointer/record algebra, or environmental fixed set. The present paper investigates physical diagnostics that may constrain $\mathcal{A}_{\text{sel},\mathbf{c}}$; it does not prove that the Hamiltonian uniquely determines it.

This reversal has a modest but consequential implication: subsystem structure, and consequently geometry, is not intrinsic to the quantum state alone but depends on which observables are accessible—where accessibility is understood in a precise, structural sense developed in the next section.

2.3 Accessibility vs Observer-Dependence

The notion of accessibility is central to the framework, and it is here that the risk of misreading is greatest. HAFF intends accessibility diagnostics to concern physical stability and records rather than beliefs, but their outputs remain conditional on dynamics, probes, fragmentation, and candidate-algebra data.

2.3.1 Three Notions of Dependence

To prevent conflation, we distinguish three distinct senses in which a physical quantity might be said to "depend on" something:

HAFF's notion of accessibility falls squarely in the third category.

Type	Depends On	Example	Framework
Subjective	Agent's beliefs/knowledge	Bayesian probability	QBism
Relational	Reference system	Velocity in SR	RQM
Structural	Physical interaction pattern	Decoherence basis	HAFF

Table 2.1: Taxonomy of dependence notions

2.3.2 Heuristic Stability Diagnostics

The following diagnostics constrain a supplied candidate algebra; they neither define measurement accessibility nor constitute sufficient conditions for $\mathrm{Fix}(\mathcal{E})$. Each diagnostic requires its own domain data. In particular, the quantum-Darwinism diagnostic requires a system/environment fragmentation and the OTOC diagnostic requires declared probe operators.

Consider a unital candidate $\mathcal{A} \subset \mathcal{B}(\mathcal{H})$ and a normal UCP Heisenberg dynamics $\mathcal{E}_t$.

Criterion 1: Dynamical Invariance. On a declared window $0 \le t \le T$, require a scale-controlled bound

$$\sup_{0\le t\le T} \sup_{O\in \mathrm{Ball}(\mathcal{A})} \|\mathcal{E}_t(O) - O\| \le \epsilon, \tag{2.1}$$

or, where complete-bounded control is required, the corresponding CB-norm bound on $\mathcal{E}_t|_{\mathcal{A}} - \mathrm{id}_{\mathcal{A}}$. Restriction to the norm unit ball is essential: an absolute bound over an unbounded algebra is defeated by scalar rescaling. Decoherence, RG flow, and measurement back-action are different maps and must not be merged under one symbol without a model.

Criterion 2: Environmental Redundancy (Quantum Darwinism). Following quantum Darwinism [143], let a model first supply a system/environment split, disjoint fragments E_k, and a classical pointer label X. Redundancy can then be tested by

$$\chi(X : E_k) \ge (1-\delta)H(X) \qquad \text{for many disjoint } E_k, \tag{2.2}$$

with a finite-resolution label in continuous-spectrum cases. Approximate commutation with an environmental algebra is not equivalent to Darwinist redundancy. More importantly, the fragmentation that defines E_k is precisely a mereological input that HAFF ultimately seeks to explain. No factorization-free construction of $\mathcal{A}_E$ is given here; this is an unresolved interface tension, not a solved circularity.

Criterion 3: Non-scrambling Subspace. For normalized $O \in \mathrm{Ball}(\mathcal{A})$ and a declared normalized probe class $V \in \mathfrak{V}$, define

$$C_{O,V}(t) := -\omega\big([O(t), V]^2\big)\,. \tag{2.3}$$

An informative slow-scrambling diagnostic constrains a growth rate or a post-transient window, for example

$$\sup_{O,V} C_{O,V}(t) \le \delta \quad \text{for } t_0 \le t \le T, \tag{2.4}$$

where $t_0 > 0$ lies beyond the universal short-time regime. Equivalently, a model may bound a fitted Lyapunov growth rate. The former condition at merely $t \ll \tau_{\text{scrambling}}$ selected almost every bounded operator at sufficiently early time and is withdrawn.

These diagnostics are observer-independent once their input data are fixed, but they are not input-free. They may select different or even disjoint operator sectors. In particular, Darwinist pointer records favour approximately classical commuting observables, whereas holographic geometry uses nonclassical entanglement across region algebras. No theorem shows that one algebra satisfies both roles, and the framework must allow their intersection to be trivial.

Remark: The Hamiltonian as Physical Input

A potential objection to the structural account of accessibility is that the interaction Hamiltonian $\hat{H}_{\text{int}}$ appears to be an arbitrary input, merely displacing observer-dependence from the algebra to the choice of Hamiltonian. We clarify that this is a misunderstanding of the framework's commitments.

The Hamiltonian is not a free parameter chosen by an observer, but a *physical specification of which degrees of freedom interact and how.* In this respect, it plays a role analogous to the stress-energy tensor $T_{\mu\nu}$ in general relativity: it is input data describing the causal structure of the system, not a coordinate choice or descriptive convention.

Concretely:

- In quantum optics, $\hat{H}_{\text{int}}$ describes atom-photon coupling strengths and selection rules, determined by atomic energy levels and field modes.
- In condensed matter systems, it encodes lattice structure, tunneling amplitudes, and interaction potentials, all fixed by material properties.
- In holographic models (AdS/CFT), the boundary Hamiltonian is determined by the conformal field theory's operator content and coupling constants.

The claim is not that observers are irrelevant, but that they do not *create* the supplied interaction structure. Given a specified dynamics, norm, time window, and candidate family, whether a candidate passes a diagnostic is an objective fact. The choice of that input family remains a separate problem.

This is not an ordinary coordinate transformation. Different selected algebras can yield different effective operator descriptions, while the pass/fail result for a declared stability test is interaction-dependent rather than belief-dependent.

We emphasize: the framework does not solve why a particular Hamiltonian or mereology describes our universe. It analyzes diagnostics relative to those inputs; a geometric conclusion additionally requires a reconstruction model.

Pre-geometric Interaction Structure. A potential concern is that the Hamiltonian $\hat{H}_{\text{int}}$ itself presupposes spacetime structure (e.g., via spatial locality in $\hat{H} = \sum_{i,j} J_{ij}\hat{\sigma}_i \cdot \hat{\sigma}_j$), creating a circular dependence: geometry emerges from algebras determined by a Hamiltonian that already assumes geometry.

The Hamiltonian input may be *metric-free* in the limited sense that it is specified as an abstract interaction graph or tensor network. However, the nodes, tensor factors/degrees of freedom, and edge incidence are already mereological structure. The graph avoids assuming a background metric but does not derive its own relata. This residual input must be supplied independently and is a central quantum-mereology problem [141, 29, 24].

Crucially, the *effective spacetime geometry* that emerges from stable algebras may differ from the graph structure of the input Hamiltonian. For instance:

- In tensor network models (MERA), the input is a discrete causal network, but the emergent geometry can be continuous AdS space [114].
- In spin chain models, the Hamiltonian is defined on a 1D lattice, but entanglement structure can induce higher-dimensional effective geometry.
- In holographic duality (AdS/CFT), the boundary Hamiltonian is defined on a fixed $(d-1)$-dimensional manifold, but the bulk geometry (including its dimensionality) emerges dynamically.

Thus, the input interaction structure can constrain candidate algebras but does not uniquely determine one. The framework asks which candidates pass declared tests and, when a reconstruction dictionary is available, what geometric data follow.

This perspective aligns with recent work on "locality from entanglement" [121], where spatial locality is itself understood as arising from entanglement patterns rather than being presupposed.

2.4 Relations to Existing Interpretations

We now situate HAFF relative to four representative frameworks, clarifying conceptual boundaries and identifying points of potential complementarity.

2.4.1 Algebraic Quantum Field Theory (AQFT)

The closest structural affinity of HAFF is with algebraic quantum field theory [50, 7]. In AQFT, observable algebras are treated as primary, with states defined as positive linear functionals over these algebras. Crucially, local algebras are assigned to spacetime regions without relying on a global tensor product structure.

HAFF extends this algebraic perspective by emphasizing *coarse-graining relations* between algebras. In standard AQFT, locality is typically presupposed: algebras are indexed by spacetime regions, and the split property ensures independence of spacelike-separated algebras. In HAFF, we relax this assumption and instead treat *stability under physical interactions* as the criterion for algebra selection.

The transformation can be summarized as:

AQFT: Spacetime regions $\rightarrow$ Local algebras $\rightarrow$ States
HAFF: Interaction structure $\rightarrow$ Stable algebras $\rightarrow$ Effective geometry

This is not a replacement of AQFT but an exploration of its structure in contexts where spacetime locality is not presupposed. The framework may be understood as asking: what happens to the algebraic approach when we do not assume a background spacetime to index our algebras?

2.4.2 Relational Quantum Mechanics (RQM)

Relational quantum mechanics [95, 70] emphasizes that the values of physical quantities are defined only relative to observer systems, rejecting the notion of absolute, observer-independent observables. In Rovelli's formulation, quantum mechanics is fundamentally a theory of *interactions* rather than systems: what exists are relational facts, not intrinsic properties.

There is significant conceptual overlap with HAFF: both frameworks reject privileged subsystem decompositions and treat quantum descriptions as contextual. However, a key difference concerns the *stabilization of relata*.

RQM analyzes relations between systems whose existence is typically taken as given (or at least presupposed operationally through interaction records). HAFF provides a mechanism for the *stabilization of distinct relata* from the underlying quantum field, identifying which subsystem partitions are robustly maintained under decoherence and measurement-induced dynamics.

In this sense, HAFF may provide the stable nodes required for RQM's relational network: before relations can exist, there must be relata stable enough to participate in interactions. HAFF addresses how such stable relata emerge from the pre-factorized quantum substrate.

These are complementary rather than competing perspectives: RQM asks what observables mean relative to a system, HAFF asks which systems stabilize as distinct relata in the first place. The two frameworks operate at different levels of analysis and could in principle be combined, with HAFF providing the stability conditions under which RQM's relational structure becomes well-defined.

2.4.3 QBism

QBism [42, 43] interprets quantum states as expressions of an agent's personal beliefs about measurement outcomes, emphasizing the subjective, agent-centric nature of quantum probability assignments.

Here, the distinction from HAFF is sharpest. While both frameworks reject naive realism about the quantum state, they differ fundamentally in their treatment of dependence:

- **QBism**: Quantum states represent personal beliefs. Dependence is epistemic and agent-centric.
- **HAFF**: Observable algebras are selected by physical interaction structure. Dependence is structural and interaction-centric.

The key point is that interactions are not agents: they have no beliefs, make no decisions, and exist independently of any epistemic perspective. Relative to supplied subsystem and probe data, the Hamiltonian can constrain candidate algebras; no theorem here says that it determines a unique one.

We emphasize that this is a categorical difference, not a matter of one framework being "more correct" than the other. QBism and HAFF address different questions and operate within different conceptual frameworks. The point of comparison is simply to clarify that HAFF's notion of accessibility does not reduce to QBist agent-dependence.

2.4.4 Many-Worlds Interpretation (MWI)

The Many-Worlds interpretation [130] explains the emergence of classical behavior through decoherence-induced branching, all occurring within globally unitary quantum evolution.

HAFF shares with MWI a commitment to:

- A single, objective global quantum state
- Unitary evolution without collapse
- Effective classicality emerging from entanglement structure

However, MWI typically presupposes a tensor factorization into system and environment as input, analyzing how this decomposition gives rise to branch structure. Recent work in the MWI tradition (e.g., Wallace, Saunders) addresses emergent decoherence structure, but typically within a framework where tensor factorization is assumed at the fundamental level.

HAFF complements MWI by examining the preconditions for any branching structure: before branches can emerge, there must be a notion of subsystems relative to which branching occurs. In this sense, HAFF may provide a structural framework relevant to understanding how the effective subsystem decompositions presupposed by branching emerge in the first place.

This is a point of potential contact rather than a hierarchical claim: we do not argue that MWI requires HAFF, but that the two frameworks address distinct but related aspects of quantum structure.

2.5 Geometry as a Stable Organizational Phase

Conceptual Disclaimer. Before discussing the analogy with condensed matter phases, we stress that it is conceptual rather than rigorous. In condensed matter, a Hamiltonian together with thermodynamic limit, control parameters, and boundary conditions can support distinct phases and order parameters. HAFF currently lacks an analogous proved selection model. The comparison is therefore a heuristic for stability and contingent objectivity, not an independent result.

Contingent Objectivity. A better analogy is spontaneous symmetry breaking. Below the Curie temperature, one ferromagnetic Hamiltonian admits macroscopically distinct phases with magnetization $+M$ or $-M$. Boundary conditions or a small external field select a phase; once selected, its magnetization is objective and reproducible, although the microscopic laws do not choose a unique sign in isolation.

Likewise, a supplied interaction/mereology datum may admit several stable candidate algebras. A physical selection condition can make one candidate salient without making it observer-created. If a separate reconstruction dictionary maps that candidate's entanglement data to a metric, observers using the same inputs obtain the same reconstructed result. The analogy supports contingent objectivity only; it neither proves algebra selection nor supplies the reconstruction dictionary.

This makes the intended interpretation coherent without proving it: reconstructed geometry may be contingent yet objective within a model, much as an order parameter is objective in a selected phase.

2.5.1 The Phase Analogy

With these caveats in place, we develop the analogy with condensed matter phases.

In statistical mechanics, different thermodynamic phases (solid, liquid, gas, magnetic, superconducting) represent distinct organizational patterns of microscopic degrees of freedom. These phases are:

- **Emergent**: They arise from collective behavior, not from individual constituents.
- **Stable**: They persist under perturbations below characteristic energy scales.
- **Observable**: They manifest in measurable order parameters.
- **Contingent**: They depend on external conditions (temperature, pressure, fields).

We propose that emergent geometry in HAFF exhibits analogous features:

Condensed Matter	**HAFF**
Hamiltonian $\hat{H}$	Metric-free but mereologically supplied interaction graph
Control parameter (Temperature T)	Model-specific coupling/noise parameter
Symmetry breaking	Algebra selection
Order parameter $\langle M \rangle$	Entanglement pattern $I(A:B)$
Phase transition	Geometry emergence
Critical temperature T_c	Model-specific stability threshold

Table 2.2: Heuristic analogy between condensed matter phases and candidate algebra selection. No universal HAFF control parameter or critical threshold is derived.

In holographic models (AdS/CFT), a model-specific analogue is the ratio $\ell_{\text{AdS}}/\ell_{\text{Planck}}$, which controls the validity of a classical bulk description [78]; this does not supply a universal HAFF control parameter.

Just as ferromagnetism is a stable organizational pattern of spin alignment below the Curie temperature, geometry may be modelled as a stable organizational pattern only when explicit algebra-selection and reconstruction conditions are supplied.

Post-selection entanglement diagnostic. After a model has supplied subsystems and an interaction graph $G = (V, E)$, one may define the average pair mutual information

$$\rho_{\text{ent}}^{(G)} := \frac{1}{|E|} \sum_{(i,j)\in E} I(i:j), \tag{2.5}$$

for $|E| > 0$. If all unordered pairs are used, the denominator is $\binom{|V|}{2}$ instead. The earlier normalization by N was undefined without a graph-scaling convention.

This quantity depends on the very subsystem graph whose origin is at issue, so it cannot select the pre-geometric mereology without circularity. It is only a diagnostic after that input is fixed. Low average pair mutual information does not by itself imply an area law or smooth geometry, and high values are not equivalent to a scrambling rate.

In a particular holographic model, quantities such as ℓ_{AdS}/ℓ_P control a semiclassical expansion [78]; they are not identified here with $\rho_{\text{ent}}^{(G)}$.

Whether any model exhibits a genuine transition in geometric reconstructibility is an open, model-specific question.

2.5.2 Geometric Admissibility

Not all coarse-graining structures admit geometric interpretation. As discussed in our previous work (Definition 4.2), we require geometric admissibility conditions:

1. Finite correlation length (exponentially decaying correlations)
2. Monotonic decay of mutual information under refinement
3. Stability under perturbations

These conditions are not arbitrary but reflect empirical observations from known emergent geometries:

- **AdS/CFT**: Boundary states with area-law entanglement give rise to smooth bulk geometries [97].
- **Tensor networks**: MERA and similar structures with finite bond dimension naturally induce geometric connectivity [114].
- **Condensed matter**: Ground states of local Hamiltonians typically satisfy area laws and admit geometric descriptions.

Geometry, in this view, is conjectured to represent a special organizational phase characterized by specific entanglement structure. Demonstrating an attractor requires a concrete coarse-graining dynamics and reconstruction dictionary and remains open.

2.6 Scope and Future Directions

2.6.1 Structural Assumptions

The framework rests on three core assumptions:

1. **Global state objectivity**: There exists a universal quantum state $|\Psi_U\rangle$, independent of observers.
2. **Algebraic priority**: Observable-algebra data are treated as prior to the effective tensor factorizations reconstructed from them; their own physical selection remains open.
3. **Stability-based emergence**: Stable candidate algebras are proposed as inputs to effective subsystem and geometric reconstruction, conditional on the missing selection and dictionary results.

2.6.2 What the Framework Does NOT Commit To

To prevent interpretational overreach, we explicitly enumerate what the framework does *not* claim:

- That observers create reality or that consciousness plays a fundamental role
- That spacetime is illusory or that geometry has no objective existence

- That quantum mechanics is incomplete or requires modification
- That the framework solves the measurement problem
- That Buddhist metaphysics or any other philosophical tradition is presupposed

The framework is structurally neutral regarding these questions. It analyzes consequences of relaxing the assumption of canonical tensor factorization, but does not commit to any particular metaphysical position beyond what the formalism requires.

2.6.3 Open Questions

The following questions include load-bearing open gaps as well as later empirical directions. The mereology, algebra-selection, and reconstruction problems must be solved in concrete models before the heuristic diagnostics become a derivation.

1. **Quantum mereology**: What physical datum selects the nodes/factors or candidate algebra family without presupposing the subsystem structure to be explained [141, 29, 24]?
2. **Dynamical algebra selection**: Can interaction Hamiltonians, coupling strengths, or network topologies be shown to select particular stable algebras over others?
3. **Information-theoretic criteria**: Do preferred geometric descriptions correlate with minimal description length, robustness under noise, or computational accessibility?
4. **Quantum field theory extensions**: How does the framework interact with locality structures in algebraic QFT? Can continuum limits be rigorously constructed?
5. **Empirical signatures**: Do different coarse-graining choices lead to distinguishable effective descriptions in semiclassical regimes or quantum gravity-motivated models?
6. **Connections to quantum complexity**: How does the framework relate to recent work on complexity-based approaches to spacetime emergence?

These questions are intentionally left open. The present work provides a structural scaffold within which they can be formulated precisely, but does not claim to resolve them.

2.6.4 Beyond Structural Analysis

The framework developed here is deliberately limited to structural and interpretational analysis. Broader philosophical implications—concerning causation, free will, ontology, and connections to contemplative traditions—lie beyond the scope of this technical paper. Such questions are addressed in a companion philosophical essay currently in preparation.

2.7 Conclusion

We have clarified the conceptual commitments of the Holographic Alaya-Field Framework, addressing three potential misreadings:

1. **Against subjectivism**: Accessibility is defined structurally via stability conditions determined by physical interaction structure, not by observer beliefs or epistemic states.

2. **Against anti-realism**: Emergent geometry is a stable, measurable feature of entanglement structure—contingent on physical conditions but objective within those conditions, analogous to thermodynamic phases.

3. **Against trivialism**: Coarse-graining dependence reflects genuine physical structure (accessible algebras), not mere coordinate choice. Different algebras can induce inequivalent geometries when a supplied reconstruction dictionary separates the compared profiles modulo diffeomorphisms.

The framework has been situated relative to AQFT (closest affinity), RQM (complementary), QBism (categorically distinct), and MWI (potentially complementary). Throughout, we have emphasized what the framework commits to structurally and what it leaves open interpretionally.

The central insight is modest but consequential: by removing tensor factorization from fundamental assumptions and treating it as emergent from coarse-graining structure, we reveal that geometry is more context-dependent than typically acknowledged—not in an epistemic or observer-relative sense, but in a structural, interaction-dependent sense.

This perspective does not resolve deep problems in quantum foundations or quantum gravity, but it clarifies the conditions under which subsystem structure and geometry emerge. By making explicit an assumption that is often left implicit, we hope to have opened new avenues for investigating the relationship between quantum states, observable structure, and spacetime.

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Chapter 3

Causation, Agency, and Existence: Structural Constraints and Interpretive Bridges

Essay C

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Abstract

This essay examines the structural conditions under which agency, causation, and existence can be coherently discussed in the absence of foundational subsystem decompositions. Building on recent work in quantum information theory and algebraic approaches to quantum mechanics, Parts I–III develop a framework in which causal relations, agent boundaries, and existential claims emerge from stability properties of accessible observable algebras rather than from intrinsic substance or preferred factorizations.

Part I treats directional information asymmetry as an operational diagnostic that can support causal analysis when paired with explicit process, instrument, and intervention models, without requiring fundamental temporal ordering. Part II analyzes agency as boundary-stabilization—a non-scrambling subspace that propagates constraints without presupposing a metaphysically autonomous agent. Part III recasts existence in terms of relational form rather than intrinsic being, developing a notion of "emptiness" as absence of substance compatible with objectivity.

Part IV explores whether these structural features find formal parallels in Buddhist philosophical frameworks, particularly Yogācāra and Mādhyamaka traditions. The analysis emphasizes interpretive humility: parallels are offered as invitations to dialogue, not demonstrations of equivalence. The essay's contribution is methodological—clarifying structural constraints on emergence—rather than doctrinal.

Keywords: agency, emergence, structural realism, coarse-graining, accessible algebras, Buddhist philosophy, emptiness, comparative philosophy

Contents

3.1 Introduction

Contemporary philosophy of physics faces a structural puzzle. When quantum systems lack canonical subsystem decompositions, how should we understand causation, agency, and existence? If there is no preferred way to carve reality into parts, what becomes of the conceptual apparatus built on the assumption of well-defined relata?

This essay develops a framework in which these notions emerge from *stability properties of accessible algebras* rather than from intrinsic substance or preferred factorizations. The analysis proceeds in four parts.

Roadmap. Part I examines causation without temporal foundations, treating stable asymmetries in coarse-grained structure as operational diagnostics that can support causal analysis when paired with an intervention model. Part II analyzes agency as boundary-stabilization—the capacity of certain subsystems to maintain non-scrambling coherence while propagating constraints. Part III recasts existence in terms of relational patterns rather than intrinsic being, developing a technical sense of "emptiness" compatible with structural realism.

Part IV explores whether these structural features find formal parallels in Buddhist philosophical traditions, particularly Yogācāra and Mādhyamaka. The analysis emphasizes interpretive humility: we identify structural similarities without claiming ontological identity, historical influence, or doctrinal convergence.

Methodological stance. This is a structural investigation, not a metaphysical proposal. We do not claim that consciousness creates reality, that Buddhist texts anticipated quantum mechanics, or that comparative philosophy resolves foundational problems. Rather, we clarify how certain structural constraints—particularly the absence of canonical factorization—reshape discussions of emergence, agency, and existence across different conceptual traditions.

The essay's contribution is methodological: it demonstrates how attention to algebraic structure can discipline interpretive claims and reveal unexpected points of contact between seemingly disparate frameworks.

Relation to prior work. This essay builds on the companion programme [153, 154]. Paper A establishes finite-dimensional channel separation under stated hypotheses and obtains metric inequivalence only conditionally on a reconstruction dictionary; Paper B treats stability criteria as heuristic diagnostics. The present essay explores possible philosophical consequences without upgrading those conditional inputs.

3.2 Part I: Causation Without Foundations

3.2.1 The Standard Picture and Its Assumptions

Causal relations are typically understood as relations between events ordered by time. Event A causes event B if: (i) A temporally precedes B, (ii) A and B are spatially connectible, and (iii) interventions on A counterfactually affect B [87].

This picture presupposes several structural features:

- A well-defined notion of temporal ordering
- Spatially localized events with clear boundaries
- Stable subsystem decompositions supporting counterfactual reasoning

In quantum contexts without canonical factorization, none of these features is guaranteed. Time may be emergent rather than fundamental [84]. Spatial locality depends on choice of coarse-graining [153]. Subsystem boundaries are algebra-dependent rather than intrinsic.

3.2.2 Directional Information Asymmetry

The former definition of causation by a bare cross-time quantum mutual information is withdrawn: standard quantum mechanics assigns no joint density operator to noncommuting events at two times without a process/instrument formalism. We retain a weaker, operational diagnostic.

Definition 3.1 (Instrument-relative directional information asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation, and specified instruments $\mathcal{I}_A, \mathcal{I}_B$ whose classical outcomes are $X_A^{t_i}, X_B^{t_i}$. The instruments and process define ordinary joint outcome distributions in separate forward and reverse experimental arrangements. Define*

$$\Delta^{\mathcal{I}}_{A\to B} := I\big(X_A^{t_1} : X_B^{t_2}\big) - I\big(X_B^{t_1} : X_A^{t_2}\big)\,, \tag{3.1}$$

where both mutual informations are Shannon mutual informations of the declared classical outcome distributions. A positive value is a ***directional informational asymmetry*** *relative to $(\mathcal{N}, \mathcal{I})$.*

Definition 3.1 does not define causation. It is instrument-, preparation-, and clock-order dependent and can arise from common causes or asymmetric noise. Quantum two-time objects such as pseudo-density matrices or conditional-state/process formalisms offer other explicit conventions [40, 73], but no convention-free expression $I(A_{\text{past}} : B_{\text{future}})$ is used here. A causal claim additionally requires an intervention comparison of the kind discussed in Section 3.2.4.

Protocol dependence and thermodynamic ordering. Different instruments, preparations, and coarse-grainings may produce distinct signs of $\Delta^{\mathcal{I}}_{A\to B}$. This is expected for an observational diagnostic and must not be reported as multiple fundamental causal orders.

Thermodynamic gradients may motivate which time ordering and instruments are physically salient [131], but entropy increase alone neither proves an intervention relation nor selects a unique coarse-graining.

3.2.3 Thermodynamic Grounding

The time labels used in Definition 3.1 may be aligned with a thermodynamic arrow in systems approaching equilibrium [131]. This supplies an ordering convention, not a derivation of causation from mutual information.

The thermodynamic arrow is coarse-graining dependent because macrostates are. Any causal conclusion must therefore state both the coarse-graining and the intervention model.

3.2.4 Counterfactuals Without Intrinsic Relata

Interventionist accounts of causation rely on counterfactual reasoning: A causes B if intervening on A would change B [137]. This appears to require well-defined intervention targets—subsystems A and B with stable identities.

Counterfactuals can be formulated relative to supplied subsystem algebras. An intervention on A is a declared quantum instrument/CPTP map that replaces the ordinary mechanism for A while holding the comparison protocol fixed. The causal quantity is then a difference between output statistics with and without that intervention, not the observational $\Delta^{\mathcal{I}}_{A\to B}$ alone.

This reformulation makes no reference to intrinsic subsystem boundaries. Intervention targets are defined by accessible algebras, which are themselves context-dependent.

3.2.5 Memory as Informational Constraint

Causal relations leave traces—memory records that constrain future accessible states. In quantum systems, memory can be understood as constraint propagation through entanglement structure [55].

A subsystem M acts as a memory of an instrument outcome $X_A^{t_1}$ if a later declared readout $X_M^{t_2}$ remains correlated with it despite environmental decoherence:

$$I(X_A^{t_1} : X_M^{t_2}) \gg I(X_A^{t_1} : X_E^{t_2}), \tag{3.2}$$

where all variables arise from the specified process and instruments.

Memory, on this account, is not storage of intrinsic properties but maintenance of relational structure across time—a pattern of correlations that persists under dynamical evolution.

3.2.6 Worked Example: Two-Qubit Directional Information

We compute Definition 3.1 in a valid two-qubit process. The example illustrates the diagnostic, not a discovery of causal direction from correlations.

Setup. At t_1, prepare a uniform classical bit in A and a fixed zero in B:

$$\rho_{AB}(t_1) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|10\rangle\langle 10|. \tag{3.3}$$

This matrix is positive and normalized, with spectrum $(1/2, 1/2, 0, 0)$. Use computational-basis instruments at both times.

Process. Let $\mathcal{N}_{t_1 \to t_2}$ be the unitary CNOT channel with A as control and B as target. It maps $|x, 0\rangle$ to $|x, x\rangle$, so

$$\rho_{AB}(t_2) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|, \tag{3.4}$$

again with spectrum $(1/2, 1/2, 0, 0)$. The relevant observable span

$$\mathrm{span}\{\mathbf{1}, \sigma_A^x, \sigma_A^z, \sigma_B^x, \sigma_B^z\}$$

is an operator subspace, not an algebra. If an algebra is required, one takes the unital $*$-algebra generated by these operators (here the full two-qubit matrix algebra).

Directional asymmetry. Writing the outcomes as bits, $X_A^{t_1}$ is uniform and $X_B^{t_2} = X_A^{t_1}$. Hence

$$I(X_A^{t_1} : X_B^{t_2}) = \log 2. \tag{3.5}$$

In the reverse arrangement, $X_B^{t_1} = 0$ is constant, so

$$I(X_B^{t_1} : X_A^{t_2}) = 0, \qquad \Delta_{A \to B}^{\mathcal{I}} = \log 2. \tag{3.6}$$

Interpretation. The sign is produced by the asymmetric preparation and the known controlled channel. It correctly detects directional information transmission in this protocol, but does not prove causation independently of the intervention/process specification.

Context-dependence. Changing the instruments, preparation, or channel can change or erase $\Delta_{A \to B}^{\mathcal{I}}$. The computed number is operationally real but protocol-relative.

3.2.7 From Causation to Agency

The transition from causation to agency requires an additional structural feature: *stable subsystem boundaries that support constraint propagation.*

An agent is not merely a locus of causal influence, but a subsystem capable of maintaining coherent constraint propagation despite environmental coupling. This suggests analyzing agency in terms of *non-scrambling subspaces*—degrees of freedom that resist rapid information delocalization [55].

Part II develops this connection, arguing that agency emerges from boundary-stabilization rather than metaphysical autonomy.

Summary. Directional information asymmetry is well defined only relative to a process and instruments. It may support a causal analysis when paired with interventions, but is not itself a definition of causation.

3.3 Part II: Agency as Emergent Constraint Structure

3.3.1 The Problem of Autonomous Agents

Traditional accounts treat agents as metaphysically autonomous entities—unified subjects possessing intrinsic intentionality and causal efficacy. This picture faces two challenges in quantum contexts without canonical factorization.

First, if subsystem boundaries are algebra-dependent, what distinguishes an "agent" from an arbitrary collection of degrees of freedom? Second, if quantum dynamics are unitary and deterministic at the global level, how can agents possess genuine causal autonomy?

We argue that both challenges dissolve once agency is recast as a structural feature—boundary-stabilization under constraint propagation—rather than a metaphysical primitive.

Physical grounding of algebra selection. Companion Paper B [154] supplies heuristic stability diagnostics relative to a candidate family, dynamics, fragmentation, and probe class. It does not derive a unique algebra from the Hamiltonian. We therefore take an agent candidate and its mereological boundary as model inputs below; their dynamical selection remains open.

3.3.2 Agency as Boundary-Stabilization

An *agent*, in the structural sense, is a subsystem whose boundaries remain stable under dynamical evolution and whose internal degrees of freedom exhibit coordinated constraint propagation.

Definition 3.2 (Agent-Like Subsystem)**.** *A subsystem $\mathcal{A}_{agent}$ exhibits **agent-like behavior** if:*

1. ***Boundary stability**: The subalgebra $\mathcal{A}_{agent}$ is approximately preserved under physically relevant dynamical maps:*

$$\sup_{0\le t\le T}\ \sup_{O\in \mathrm{Ball}(\mathcal{A}_{agent})} \|\mathcal{E}_t(O)-O\| \le \epsilon \tag{3.7}$$

   for timescales relevant to constraint propagation.

2. ***Post-transient scrambling control**: For a declared normalized probe class $V \in \mathfrak{V}$, observables in $\mathcal{A}_{agent}$ obey a rate or finite-window bound beyond the universal early-time regime, for example*

$$\sup_{O\in \mathrm{Ball}(\mathcal{A}_{agent}),\, V\in\mathfrak{V}} -\omega([O(t),V]^2) \le \delta \quad (t_0 \le t \le T), \tag{3.8}$$

   with $t_0 > 0$ specified by the model.

3. ***Constraint propagation**: Internal correlations support directed information flow toward system boundaries, enabling intervention on environmental degrees of freedom.*

Remark 3.3 (Operational Threshold for Meaningful Agency)**.** *Definition 3.2 is a structural modelling convention, not a derived biological threshold. Transient early-time commutator suppression does not establish sustained control.*

One may introduce a model-specific persistence ratio

$$R_{\mathrm{stab}} := \frac{\tau_{\mathrm{boundary}}}{\tau_{\mathrm{update}}}, \tag{3.9}$$

where τ_{boundary} is persistence of the effective control boundary and τ_{update} is its operational update time. No universal numerical cutoff for R_{stab} is known or claimed.

For biological systems these times are classical, coarse-grained control times, not quantum coherence times. Warm neural degrees of freedom decohere rapidly, and the present operator language supplies no quantum-brain prediction [117]. Applying the structural analogy to organisms therefore requires an independently justified classical/open-system model.

This threshold distinguishes:

- ***Transient diagnostic satisfaction**: insufficient for operational agency*
- ***Sustained boundary stabilization**: a possible model input for persistent control*

Any empirical threshold is domain specific and must be calibrated rather than asserted from the formalism.

This definition makes no reference to consciousness, phenomenology, or intrinsic intentionality. Agency is characterized purely in terms of stability properties and information-theoretic structure.

3.3.3 Will as Constraint Propagation

What, then, becomes of "will" or "intention" in this framework?

We suggest that will can be understood as *constraint propagation structure*—patterns of internal correlation that bias future accessible states toward specific outcomes. An agent "wills" action A if its internal state ρ_{agent} is such that future measurements will register correlation with A with high probability.

Analogy: Thermostat control. Consider a thermostat coupled to a heating system. The thermostat's internal state (temperature reading) constrains future system behavior (heater activation), despite having no phenomenological experience. This is constraint propagation without metaphysical agency.

The analogy is limited: biological agents exhibit far richer constraint structure. But it clarifies the conceptual move—replacing metaphysical autonomy with structural analysis of how internal states bias future trajectories.

3.3.4 The Phenomenology-Structure Gap

An immediate objection: this account leaves no room for phenomenology—the felt quality of agency, the subjective sense of "I am acting."

We acknowledge this gap. The framework developed here is *structural*, concerned with information-theoretic organization. It does not address the *explanatory gap* between structure and experience [74, 27].

Speculative Connection: Neural Constraint Propagation

Caveat: The following connects structural features to neuroscience, but remains speculative. We note these as suggestive parallels, not established mechanisms.
Recent work in computational neuroscience suggests that neural "will" may correspond to hierarchical constraint propagation through cortical-basal ganglia loops [48]. Habitual behavior emerges when constraint patterns stabilize, while "voluntary" action involves flexible reconfiguration of these patterns [138].
If agency is boundary-stabilization, then the phenomenology of "willing" may correspond to proprioceptive monitoring of constraint reconfiguration—the felt sense of internal degrees of freedom reorganizing in preparation for action [52].
This remains highly speculative and does not bridge the explanatory gap. We raise it only to illustrate how structural analysis might interface with empirical research programs.

3.3.5 Degrees of Agency

The framework suggests that agency is not all-or-nothing, but admits degrees. Systems exhibit more or less agent-like behavior depending on:

- **Boundary stability duration**: How long does $\mathcal{A}_{\text{agent}}$ remain well-defined?
- **Scrambling timescale**: How quickly do internal correlations delocalize?
- **Constraint propagation fidelity**: How reliably do internal states bias future trajectories?

Simple thermostats exhibit minimal agency (short timescales, limited constraint structure). Biological organisms exhibit far richer agency (extended coherence, complex constraint networks). But both are comprehensible within the same structural framework.

3.3.6 Relation to Free Will Debates

Traditional free will debates ask: are agents causally autonomous, or are their actions determined by prior states and laws? This presupposes well-defined agent boundaries and unambiguous causal histories—precisely what the framework questions.

On the structural account, the relevant question is not "Are agents free?" but "Under what conditions do subsystems exhibit stable constraint-propagation structure?" This reformulation may dissolve certain traditional impasses while opening new empirical questions about stability conditions and scrambling timescales.

We do not claim to resolve free will debates—only to clarify how they depend on assumptions about subsystem structure that are themselves context-dependent.

Summary. Agency can be understood as boundary-stabilization supporting constraint propagation, rather than metaphysical autonomy. This account is eliminativist about intrinsic intentionality but realist about structural patterns of constraint. It leaves the phenomenology-structure gap unresolved but clarifies the information-theoretic conditions under which agent-like subsystems emerge.

3.4 Part III: Existence Without Substance

3.4.1 The Realism Problem

Parts I–II analyzed causation and agency as context-dependent but objective—dependent on coarse-graining structure but not on observer beliefs. This raises an existential question: if fundamental structures (subsystems, geometries, agents) are context-dependent, what ontological status do they possess?

Two extremes must be avoided:

1. **Naive realism**: Treating emergent structures as metaphysically fundamental, ignoring their context-dependence.

2. **Anti-realism**: Denying objective reality to context-dependent structures, collapsing into subjectivism.

We propose a middle path: *structural realism about relational patterns.* Existence is understood not as possession of intrinsic properties, but as participation in stable relational structures.

3.4.2 Form Without Substance

Consider the temperature of a gas. Temperature is *context-dependent*: it depends on which degrees of freedom are macroscopically accessible. Different coarse-grainings may yield different effective temperatures for the same microstate.

Yet temperature is not merely subjective. Given a coarse-graining, temperature is an objective, measurable quantity with predictive power. It is *relationally real*—real within a specified context, but lacking intrinsic existence independent of that context.

Definition 3.4 (Relational Existence)**.** *An entity X possesses **relational existence** relative to structure $\mathcal{S}$ if:*

1. *X is well-defined and stable within $\mathcal{S}$*

2. *X participates in objective relational patterns (correlations, symmetries, invariants)*

3. *X may be absent or differently constituted under alternative structures $\mathcal{S}'$*

This notion captures *form without substance*: patterns that are objectively real without possessing intrinsic being.

3.4.3 Emptiness as Technical Concept

We introduce *emptiness* as a technical term denoting absence of intrinsic existence compatible with relational reality.

Definition 3.5 (Emptiness (Technical Sense))**.** *An entity X is **empty** (in the technical sense) if:*

1. *X lacks intrinsic being independent of relational context*

2. *X exhibits stable patterns within specified contexts*

3. The absence of intrinsic being does not entail non-existence or illusoriness

This usage is stipulative and should not be confused with colloquial meanings ("containing nothing") or metaphysical nihilism ("nothing really exists"). Emptiness, in this technical sense, is compatible with robust realism about relational structures.

3.4.4 Examples of Relational Existence

Temperature. As discussed in §3.4.2, temperature is context-dependent but objective. Different coarse-grainings yield different effective temperatures, yet temperature remains a genuine physical quantity within each context.

Quasiparticles in condensed matter. Phonons, magnons, and other quasiparticles are collective excitations—emergent entities with no counterpart in the microscopic Hamiltonian. They are empty of intrinsic being (there are no "phonon particles" in fundamental theory) yet fully real within effective descriptions [67].

Operationally, quasiparticles are defined by stable relational patterns in measurable observables:

- **Spectral weight**: Well-defined peaks in momentum-resolved spectroscopy ($A(\mathbf{k}, \omega)$)
- **Dispersion relations**: Stable functional dependence $\omega(\mathbf{k})$ across parameter ranges
- **Finite lifetimes**: Decay rates $\Gamma(\mathbf{k})$ obeying systematic scaling laws
- **Scattering cross-sections**: Reproducible interaction amplitudes in transport experiments

These observables are context-dependent—they depend on temperature, pressure, doping, and measurement resolution—yet objectively real within specified experimental contexts. This exemplifies relational existence: patterns that are measurable, predictive, and stable, despite lacking intrinsic being independent of effective theory.

Subsystems in quantum mechanics. As established in prior work [153], subsystem decompositions are coarse-graining-dependent. A quantum state may admit infinitely many inequivalent factorizations, none privileged. Yet within any given factorization, subsystems exhibit objective entanglement structure and support meaningful predictions.

These examples illustrate a common pattern: entities that are *empty* (lacking intrinsic being) yet *existent* (participating in stable relational structures).

3.4.5 Structural Invariants and Objectivity

A potential objection: if everything is context-dependent, what grounds objectivity?

The answer lies in *structural invariants*—features preserved across context transformations. While subsystem decompositions are coarse-graining-dependent, certain global properties (total entropy, symmetry groups, topological invariants) remain well-defined independently of factorization.

Objectivity does not require context-independence. It requires only that relational patterns exhibit stability and predictive power within specified contexts, and that transformations between contexts preserve identifiable structural features.

This perspective aligns with *structural realism* in philosophy of science: what is objectively real is relational structure, not intrinsic substance [67, 41].

3.4.6 Existence and Non-Existence

The framework suggests a taxonomy of existential claims:

Type	**Characterization**	**Example**
Intrinsic existence	Context-independent being	Classical particles (if fundamental)
Relational existence	Context-dependent but objective	Temperature, subsystems
Conventional existence	Context-dependent and agent-relative	Money, legal rights
Non-existence	Absent from all relevant contexts	Phlogiston, luminiferous ether

Table 3.1: Taxonomy of existential claims. The framework developed here concerns relational existence—entities that are context-dependent but objective. These correspondences are analytical distinctions, not ontological commitments.

Emergent structures discussed in Parts I–II (causal relations, agent boundaries, geometric features) belong to the second category: relationally existent. They are empty of intrinsic being yet objectively real within specified coarse-graining contexts.

Summary. Existence can be understood in terms of relational patterns rather than intrinsic substance. "Emptiness," in the technical sense developed here, denotes absence of intrinsic being compatible with objectivity. This perspective avoids both naive realism and anti-realist eliminativism, offering a middle path grounded in structural invariance.

3.5 Part IV: Interpretive Bridges to Buddhist Philosophy

3.5.1 Methodological Preface

The structural features developed in Parts I–III—directional asymmetry as a causal diagnostic, agency as boundary-stabilization, existence as relational form—were derived independently of any particular metaphysical tradition. We now explore whether these features find formal parallels in Buddhist philosophical frameworks.

Several caveats are essential:

Methodological Caveats

1. **No historical causation**: We do not claim that Buddhist texts influenced quantum mechanics, or vice versa. Any parallels are structural convergences, not genealogical connections.

2. **No doctrinal advocacy**: Identifying formal similarities does not constitute endorsement of Buddhist metaphysics, soteriology, or religious practices.

3. **No cultural essentialism**: Buddhism comprises diverse traditions spanning two millennia. References here focus on specific textual traditions (primarily Yogācāra and Mādhyamaka), not "Buddhism" as a monolithic whole.

4. **No mystical reduction**: We reject interpretations conflating quantum mechanics with consciousness studies, New Age thought, or perennialist philosophy. Our analysis is structural and comparative, not mystical.

5. **Interpretive humility**: Parallels are offered as invitations to dialogue, not demonstrations of equivalence. The comparative exercise is exploratory, not conclusive.

6. **Naming is not corroboration**: The framework was deliberately named after ālaya-vijñāna. The parallel discussed below was therefore partly installed by the naming convention and cannot count as independent evidence for HAFF or for Yogācāra.

With these caveats in place, we proceed to examine possible structural correspondences.

3.5.2 Yogācāra and Accessible Algebras

Yogācāra ("Yoga practice") is a Mahāyāna Buddhist philosophical school emphasizing the role of consciousness (*vijñāna*) in constituting experienced reality. Central to Yogācāra is the concept of **ālaya-vijñāna** (Skt. *aalayavaija naaana*, "storehouse consciousness"), a foundational stratum of mind that stores karmic seeds (*bīja*) conditioning future experience [8, 123].

A structural parallel may be noted: ālaya-vijñāna functions as a holistic substrate from which individuated mental states emerge, analogous to how subsystem structures emerge from coarse-graining a global quantum state [77, 128].

Yogācāra Concept	**HAFF Analog**
Ālaya-vijñāna (storehouse consciousness)	Global quantum state $\|\Psi_U\rangle$
Bīja (karmic seeds)	Eigenmodes of accessible algebras
Pravṛtti-vijñāna (active consciousness)	Effective subsystem ρ_{eff}
Āsraya-parāvṛtti (basis-transformation)	Coarse-graining map $\Phi_{\mathbf{c}}$

Table 3.2: Possible formal parallels between Yogācāra concepts and HAFF structures. **Disclaimer**: These correspondences are formal analogies highlighting structural similarities, not claims of ontological identity or historical influence. Yogācāra is a soteriological framework concerned with liberation from suffering; HAFF is a structural analysis of quantum emergence. The table illustrates conceptual resonances, not equivalences.

Critical Clarification: Against Panpsychism

The structural parallel between ālaya-vijñāna and the global quantum state $|\Psi_U\rangle$ does **not** imply:

- That quantum states possess consciousness or phenomenological properties
- That the universe is fundamentally mental or experiential (idealism)
- That matter is constituted by or reducible to mind (panpsychism)
- That information or quantum information is intrinsically conscious

The analogy is *purely structural*: both frameworks describe how individuated entities emerge from holistic substrates via coarse-graining or cognitive filtering. Yogācāra's substrate is explicitly mental (*vijñāna*, consciousness); HAFF's substrate is explicitly physical (quantum state in Hilbert space).
Any appearance of convergence concerns *formal pattern*—the structure of emergence—not ontological content. We emphatically reject interpretations that would construe this parallel as supporting quantum consciousness theories, New Age mysticism, or perennialist claims about universal mind.
The comparison is offered in the spirit of *structural analogy*, not metaphysical synthesis.

However, critical divergences must be noted:

- Yogācāra is primarily concerned with *mental* phenomena and the path to liberation, while HAFF analyzes *physical* structure without soteriological commitments.
- Ālaya-vijñāna is explicitly described as a form of consciousness, while $|\Psi_U\rangle$ is not imbued with phenomenological properties.
- The Yogācāra framework is embedded in Buddhist ethics and meditation practice, which have no counterpart in HAFF's purely structural analysis.

The parallel, if valid, concerns *formal structure*—both frameworks describe how individuated entities emerge from holistic substrates—not phenomenological or ontological content.

3.5.3 Mādhyamaka and Emptiness

The Mādhyamaka ("Middle Way") tradition, founded by Nāgārjuna (c. 2nd century CE), articulates **śūnyatā** (Skt. *shauunayataaa*, "emptiness") as the absence of **svabhāva** (Skt. *savabhaaava*, "intrinsic nature" or "own-being") [81].

This concept bears structural resemblance to the notion of "emptiness" developed in Part III (§3.4.3), where it denoted absence of intrinsic being without entailing non-existence. We emphasize that this parallel concerns the *structural form* of the claim—denial of substance while affirming functional reality—not historical influence or causal connection [45, 107].

Nāgārjuna's central argument proceeds via *prasaṅga* (reductio) reasoning: all phenomena are empty because they arise dependently (*pratītyasamutpāda*), and what arises dependently cannot possess intrinsic nature. This is formalized in the famous verse:

> *"Whatever arises dependently is said to be empty. That, being a dependent designation, is itself the middle way."* (MMK 24:18) [81]

A structural reading: entities lacking intrinsic being (empty) can nonetheless participate in stable relational networks (dependent arising). This maps onto the framework developed in §3.4.4: subsystems are empty (coarse-graining-dependent, lacking intrinsic factorization) yet existent (exhibiting objective entanglement structure).

Recursive emptiness and second-order structure. A subtle question arises: in Mādhyamaka, emptiness applies universally, including to emptiness itself—"emptiness is empty" (*śūnyatā-śūnyatā*). Does HAFF's relational existence admit similar recursive application?

The answer is affirmative in a formal sense. Selected algebras $\mathcal{A}_{\mathrm{sel},\mathbf{c}}$ are model data constrained by physical interaction structure, apparatus constraints, and resolution limitations. They are distinct from probe-effect, record, and fixed-point objects, and no context-free selection theorem is assumed.

Moreover, the coarse-graining maps $\Phi_{\mathbf{c}}$ that select accessible algebras are context-dependent: different experimental setups, measurement resolutions, or dynamical timescales induce different Φ structures. Thus, *the apparatus of emergence is itself emergent*—coarse-graining structure arises from prior coarse-graining choices in a potentially infinite regress.

This mirrors Mādhyamaka's insight that even the *tools of analysis* (concepts, language, logical operations) are empty—lacking intrinsic being while remaining functionally effective. In HAFF, even the "machinery" of accessible algebras is context-dependent, yet this does not undermine objectivity: structural invariants (entanglement entropy, symmetry groups) remain well-defined across contexts.

The parallel is formal: both frameworks acknowledge that *relational structure goes "all the way down,"* with no metaphysically foundational level immune to context-dependence. However, Mādhyamaka deploys this insight soteriologically (to undermine attachment to fixed views), while HAFF employs it descriptively (to clarify structural constraints on emergence).

Key divergence. Mādhyamaka śūnyatā is deployed soteriologically—to undermine attachment to fixed views and facilitate liberation. HAFF's "emptiness" is a technical descriptor of relational structure, with no soteriological function. The similarity is formal, not practical or existential.

3.5.4 Karma and Constraint Propagation

Buddhist karma doctrine holds that intentional actions leave traces (*saṁskāra*, "formations") that condition future experience. In Yogācāra, these traces are stored in ālaya-vijñāna as *vāsanā* ("karmic impressions") [128].

A structural analogy: karma may be understood as *constraint propagation through entanglement structure*—past actions (interventions on accessible algebras) leave informational imprints that bias future trajectories [46, 53].

This is consonant with the account of memory developed in §3.2.5: causal traces are not storage of intrinsic properties but maintenance of relational structure across time.

Critical limitation. Buddhist karma is inherently normative: actions are classified as wholesome (*kuśala*) or unwholesome (*akuśala*) based on ethical criteria and soteriological consequences. HAFF's constraint propagation is descriptive, lacking normative content. The formal parallel does not extend to ethical or soteriological dimensions.

3.5.5 Limits and Divergences

Having noted possible parallels, we now emphasize substantive divergences—areas where Buddhist frameworks and HAFF diverge structurally, methodologically, or conceptually. This section is weighted equally to §3.5.2–3.5.4 to prevent over-interpreting formal similarities.

Phenomenological vs. Structural Orientation

Buddhist philosophy is fundamentally concerned with first-person experience and liberation from suffering (*duḥkha*). Yogācāra and Mādhyamaka analyze consciousness, perception, and mental afflictions (*kleśa*) as prerequisites for soteriological transformation.

HAFF, by contrast, is a third-person structural framework with no phenomenological commitments. It analyzes information-theoretic organization without addressing subjective experience, qualia, or the explanatory gap between structure and consciousness.

Consequence: Any parallel between ālaya-vijñāna and $|\Psi_U\rangle$ cannot extend to phenomenological dimensions. HAFF does not explain consciousness, nor does it claim that quantum states possess mental properties.

Soteriological vs. Descriptive Goals

Buddhist frameworks are *pragmatic* in orientation: concepts are introduced to facilitate liberation, not to accurately describe metaphysical reality. As the Buddha reportedly stated, philosophical speculation is a "thicket of views" (*diṭṭhi-gahaṇa*) distracting from the path [113].

HAFF is *descriptive*: it aims to clarify structural constraints on emergence without prescribing practices or soteriological goals. There is no "path" in HAFF, no liberation to achieve, no suffering to overcome.

Consequence: Structural parallels do not imply that HAFF serves Buddhist soteriological purposes, nor that Buddhist practice requires acceptance of quantum mechanics.

Rebirth, Cosmology, and Ethics

Traditional Buddhist cosmology includes rebirth across multiple realms, karmic causation spanning lifetimes, and detailed ethical taxonomies. These elements are absent from—and irrelevant to—HAFF's structural analysis.

HAFF makes no claims about:

- Post-mortem continuity of consciousness
- Karmic retribution across lifetimes
- Ethical status of actions
- Cosmological realms or deities
- Meditation practices or contemplative attainments

Consequence: Identifying formal parallels does not validate Buddhist cosmology, rebirth doctrine, or ethical systems. The frameworks operate in disjoint conceptual spaces.

Ontological Commitments

While both frameworks reject intrinsic substance, they differ in ontological commitments:

- **Buddhist frameworks** (particularly Yogācāra) often privilege mind or consciousness as fundamental, with matter derivative.
- **HAFF** remains neutral on mind-matter relations, analyzing quantum structure without metaphysical commitments about consciousness.

Additionally, Mādhyamaka's "two truths" doctrine—distinguishing conventional (*saṁvṛti*) from ultimate (*paramārtha*) reality—has no clear analog in HAFF. HAFF distinguishes context-dependent from invariant structures, but this is epistemological (about what can be known) rather than ontological (about levels of reality).

Methodological Incommensurability

Buddhist philosophy employs contemplative introspection, textual hermeneutics, and dialectical reasoning as primary methods. HAFF employs mathematical formalism, operator algebras, and information theory.

These methodologies are not mutually translatable. One cannot *meditate* one's way to understanding quantum entanglement, nor *calculate* one's way to soteriological insight. The parallels identified are *structural*, not methodological.

Summary of divergences. The frameworks differ in:

1. Explanatory target (phenomenology vs. structure)
2. Pragmatic goal (liberation vs. description)
3. Scope (ethics/cosmology vs. physics)
4. Ontological commitments (consciousness-first vs. neutral)

5. Methodology (contemplative vs. mathematical)

These divergences are not defects but reflect different intellectual projects. Recognizing formal parallels does not collapse these distinctions.

3.5.6 Interpretive Humility

We conclude Part IV by reaffirming interpretive humility. The parallels identified—between accessible algebras and ālaya-vijñāna, between emptiness and śūnyatā, between constraint propagation and karma—are *suggestive but inconclusive*.

They suggest that:

1. Certain structural features (holistic substrates, relational existence, informational constraints) appear across traditions when thinkers grapple with similar conceptual problems.
2. Cross-cultural philosophical dialogue may benefit from precise structural comparison, avoiding both premature dismissal and uncritical conflation.
3. Formal parallels can motivate further investigation without requiring doctrinal convergence.

They do *not* suggest that:

1. Buddhist philosophy anticipated quantum mechanics or modern physics.
2. Quantum mechanics validates Buddhist metaphysics or soteriology.
3. Structural similarities entail ontological identity.
4. Comparative philosophy resolves foundational debates in either tradition.

The exercise is exploratory: we map conceptual terrain, noting points of contact and divergence, without claiming to adjudicate between frameworks. Our contribution is methodological—demonstrating how attention to algebraic structure can discipline comparative claims and prevent both over-interpretation and premature dismissal.

3.6 Conclusion: Structural Constraints and Interpretive Modesty

This essay has developed a structural framework for understanding causation, agency, and existence in quantum contexts without canonical subsystem decompositions. The analysis proceeded in four parts.

Part I: Causation. Directional information asymmetries can be defined relative to specified processes and instruments, without requiring fundamental temporal ordering or intrinsic relata. They may support causal analysis when paired with an explicit intervention model; thermodynamic gradients supply an ordering convention rather than a derivation of causation.

Part II: Agency. Agent-like behavior emerges from boundary-stabilization—subsystems maintaining non-scrambling coherence while propagating constraints. This account is eliminativist about intrinsic intentionality but realist about structural patterns. The phenomenology-structure gap remains unresolved.

Part III: Existence. Existential claims can be reformulated in terms of relational patterns rather than intrinsic substance. "Emptiness," in the technical sense developed here, denotes absence of intrinsic being compatible with objectivity. This perspective avoids both naive realism and anti-realist eliminativism.

Part IV: Interpretive Bridges. Formal parallels may exist between these structural features and concepts in Buddhist philosophy (particularly Yogācāra and Mādhyamaka traditions). However, substantive divergences in methodology, goals, and scope prevent conflation. Parallels are offered as invitations to dialogue, not demonstrations of equivalence.

3.6.1 Methodological Contribution

The essay's primary contribution is methodological: it demonstrates how structural analysis can discipline interpretive claims across multiple domains.

1. **In quantum foundations**: By clarifying how causation, agency, and existence depend on coarse-graining structure, the framework reveals which features are context-dependent and which admit invariant characterization.

2. **In philosophy of science**: By developing relational existence without metaphysical substance, the framework contributes to structural realist programs while avoiding reification of emergent entities.

3. **In comparative philosophy**: By identifying formal parallels while respecting substantive divergences, the framework models how cross-cultural comparison can proceed without cultural essentialism or premature synthesis.

3.6.2 What This Essay Does Not Claim

To prevent misreading, we reiterate what the essay does *not* claim:

- That consciousness creates reality or plays a fundamental physical role
- That Buddhist texts anticipated quantum mechanics or modern physics
- That quantum mechanics validates any particular metaphysical or religious tradition
- That structural parallels resolve foundational problems in physics or philosophy
- That comparative philosophy provides unique insights unavailable within traditions

The analysis is structural and comparative, not metaphysical or apologetic.

3.6.3 Open Questions

Several questions remain open:

1. **Phenomenology**: How, if at all, does structural organization relate to subjective experience? The framework developed here is silent on the explanatory gap.

2. **Normativity**: Can constraint propagation ground normative distinctions, or does ethics require additional conceptual resources beyond structural analysis?

3. **Comparative methodology**: What criteria should govern cross-cultural philosophical comparison? When do formal parallels indicate genuine convergence versus superficial similarity?

4. **Empirical implications**: Do different coarse-graining choices lead to observationally distinguishable predictions in realistic physical systems?

5. **Contemplative epistemology**: Can first-person contemplative methods contribute to structural understanding, or are mathematical and phenomenological investigations fundamentally disjoint?

These questions are not deficiencies but opportunities for future investigation. The framework provides conceptual scaffolding for pursuing them with greater precision.

3.6.4 Final Reflection

The absence of canonical subsystem decompositions in quantum mechanics is not merely a technical curiosity. It reshapes how we think about emergence, identity, and existence across multiple domains—from quantum gravity to philosophy of mind to cross-cultural hermeneutics.

By attending to structural constraints—particularly the dependence of effective descriptions on accessible algebras—we can navigate between naive realism and anti-realist eliminativism, between cultural essentialism and dismissive parochialism, between metaphysical dogmatism and interpretive nihilism.

The resulting picture is one of *structured pluralism*: multiple effective descriptions, none metaphysically privileged, yet constrained by objective relational patterns and transformation principles. Reality is not uniquely carved at the joints, but neither is it infinitely malleable. The joints themselves are context-dependent yet objective.

This perspective invites humility. We cannot claim unique access to fundamental structure, nor can we dismiss alternative frameworks as merely conventional. Instead, we map the space of possibilities, identify structural invariants, and acknowledge the limits of any single descriptive framework.

In this spirit, the essay concludes not with answers but with refined questions—questions shaped by attention to algebraic structure, informed by cross-cultural comparison, and disciplined by interpretive modesty.

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Chapter 4

Observable-Algebra Selection and Emergent Gravitational Phenomena: A Conditional Structural Analysis

Paper D

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Abstract

We examine a programme-level conjecture that gravitational phenomena may be associated with adiabatic paths of selected observable algebras, while gauge interactions are modelled within a fixed selected algebra. The paper does not derive the selection dynamics, the empirical equivalence principle, or a Lorentzian metric from the Quantum Fisher Information Metric; the latter is used only as a positive state-space metric and information-geometric analogy. In the operator-algebraic sector we formulate conditional modular-stability criteria and explicitly withdraw the former uniqueness theorem. In a separately supplied holographic CFT regime satisfying assumptions (H1)–(H5), an HSMI/ball modular variation can be related, through imported entanglement-first-law results, to linearized gravitational data. This is a conditional correspondence rather than a derivation from generic accessible-algebra evolution.

4.1 Introduction

4.1.1 The Quantization Problem

Among the four fundamental interactions, gravity occupies a singular position. While the strong, weak, and electromagnetic forces have been successfully incorporated into the framework of quantum field theory, gravity has resisted analogous treatment for nearly a century. The difficulties are well known: naive quantization of general relativity yields a non-renormalizable theory, and more sophisticated approaches—string theory, loop quantum gravity, asymptotic safety—remain either incomplete or empirically unconfirmed [65].

A common diagnosis attributes this difficulty to the self-referential nature of gravity: the metric tensor both defines the arena in which physics takes place and participates as a dynamical variable within that arena. Quantizing gravity thus appears to require quantizing spacetime itself—a conceptually and technically formidable task.

4.1.2 An Alternative Diagnosis

In this paper, we explore an alternative structural diagnosis. We suggest that the difficulty may arise not because gravity is a particularly subtle force, but because gravity may not be a force at all—at least not in the same categorical sense as gauge interactions.

The proposal rests on a simple observation: all descriptions of physical systems presuppose some decomposition of the total system into subsystems. In quantum mechanics, this corresponds to a tensor factorization of the Hilbert space. However, as has been established in foundational work on quantum information theory [140, 141] and developed in our previous analysis [153], there is no canonical or physically privileged factorization for a generic quantum state. Different choices of factorization—or more generally, different choices of accessible observable algebra—yield inequivalent physical descriptions.

We propose that gauge forces and gravity may be distinguished at this structural level:

- **Gauge forces** describe interactions between degrees of freedom *within* a given subsystem decomposition.
- **Gravitational phenomena** reflect properties of the decomposition *itself*—specifically, how effective geometry emerges from the pattern of accessible observables.

If this reframing is correct, it may help clarify why gravity resists quantization: one cannot straightforwardly quantize the choice of how to divide a system into parts, because that choice is logically prior to the application of quantum dynamics to those parts.

4.1.3 Scope and Limitations

We emphasize at the outset what this paper does and does not attempt.

This paper does:

- Offer a structural reframing of the distinction between gravity and gauge forces
- Draw on established results concerning coarse-graining and emergent geometry
- Identify this perspective as a possible diagnostic for the quantization problem

This paper does not:

- Propose new dynamical equations
- Derive the Einstein field equations or their quantum corrections
- Claim to solve the problem of quantum gravity
- Introduce observer-dependent, consciousness-related, or interpretational elements

The analysis is structural in nature. We examine the conceptual architecture underlying descriptions of gravity and gauge forces, and suggest that a categorical distinction at the level of observable algebras may illuminate longstanding difficulties.

4.1.4 Outline

Section 4.2 reviews the factorization problem and the conditional results inherited from Paper A. Section 4.3 develops a proposed distinction between gauge forces and gravity. Section 4.4 states the central conjecture and compares its state-space geometry with the QFIM, while recording the unresolved Lorentzian-signature problem. Section 4.5 defines a class of modular-stable geometric candidate algebras, withdraws the former uniqueness claim, checks parts of the construction in two models, and states a conditional holographic route to linearized gravity. Later sections discuss scope and open questions.

4.2 The Factorization Problem

4.2.1 No Canonical Tensor Factorization

In standard quantum mechanics, composite systems are described by tensor products of subsystem Hilbert spaces: $\mathcal{H}_{\text{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$. This structure is typically taken as given, with subsystems identified by physical intuition or experimental arrangement.

However, for a generic Hilbert space $\mathcal{H}$, there is no unique or canonical way to express it as a tensor product. Any finite-dimensional Hilbert space of dimension $d = d_1 \times d_2$ admits a factorization $\mathcal{H} \cong \mathcal{H}_{d_1} \otimes \mathcal{H}_{d_2}$, but the choice of such a factorization is not determined by the Hilbert space structure alone.

This observation was formalized by Zanardi and collaborators [140, 141], who demonstrated that tensor product structures are determined by the algebra of accessible observables rather than by intrinsic properties of the state space. A change in which observables are accessible corresponds to a change in how the system is effectively decomposed into subsystems.

4.2.2 Coarse-Graining and Effective Descriptions

Building on this foundation, our previous work [153] established that:

Proposition 4.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.*

Thus channel inequivalence establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. The latter is a conditional model claim throughout this paper.

4.2.3 Geometry from Entanglement

The connection between entanglement and geometry has been extensively studied in the context of holographic duality. The Ryu-Takayanagi formula [97] and its generalizations establish that, in certain settings, geometric quantities (areas of extremal surfaces) are directly related to entanglement entropies of boundary regions:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N\hbar}. \tag{4.1}$$

More broadly, Van Raamsdonk [121] and others have argued that spacetime connectivity itself may be understood as a manifestation of quantum entanglement: regions that are highly entangled are geometrically "close," while weakly entangled regions are "far apart" or even disconnected.

Within the present framework, these results motivate a conditional interpretation: if a reconstruction dictionary maps the selected algebraic/entropic data to geometry, changing those inputs may change the reconstructed geometry. No general reconstruction theorem is assumed silently.

Remark 4.2 (Geometry as Coarse-Graining Dependent)**.** *In models with a stated reconstruction dictionary, effective spacetime geometry is a derived, coarse-graining-dependent quantity. Outside those models the construction supplies operator and connectivity data only; topology and metric are not inferred.*

Remark 4.3 (Relation to Prior Work)**.** *While the non-uniqueness of tensor factorization has been widely discussed in the quantum information literature [140, 141], its implications for distinguishing gravitational phenomena from gauge interactions at a structural level have not, to our knowledge, been made explicit. The present work develops this connection.*

This observation sets the stage for the distinction we develop in the next section.

4.3 A Structural Distinction: Gauge Forces vs. Gravity

4.3.1 Forces Within a Factorization

Consider the standard description of gauge interactions. In quantum electrodynamics, the electromagnetic force is mediated by photon exchange between charged particles. In quantum chromodynamics, gluons mediate the strong force between quarks. In each case, the interaction is described as a coupling between degrees of freedom that are already identified as distinct subsystems.

Formally, gauge theories are constructed on a fixed background: a spacetime manifold M equipped with a principal bundle whose structure group is the gauge group ($U(1)$, $SU(2)$, $SU(3)$, etc.). Matter fields are sections of associated bundles, and gauge fields are connections on the principal bundle. The dynamics describes how these fields interact *given* the background structure.

Crucially, the identification of "electron here" and "photon there" presupposes a decomposition of the total system into localized subsystems. The gauge interaction operates *within* this decomposition, coupling degrees of freedom that have already been distinguished.

4.3.2 Gravity: A Different Category?

General relativity describes gravity not as a force between objects on a fixed background, but as the curvature of spacetime itself. The metric tensor $g_{\mu\nu}$ is both the arena in which physics unfolds and a dynamical variable subject to the Einstein equations.

This dual role has long been recognized as the source of conceptual and technical difficulties. But the present framework suggests a sharper formulation of the distinction.

If gauge forces operate within a given subsystem decomposition, we propose that gravitational phenomena may be understood as reflecting properties of the decomposition itself. Specifically:

- The effective geometry—the metric, the notion of distance, the causal structure—emerges from the pattern of entanglement among accessible degrees of freedom.
- This pattern is determined by the choice of accessible observable algebra.
- Gravitational phenomena, in this view, are not interactions between pre-existing objects, but manifestations of how effective spacetime structure responds to changes in what is accessible.

We emphasize that this proposal does not deny that gravity is geometrical at the effective level. Rather, it suggests that the *origin* of this geometry may lie in how accessible observables define effective subsystems. The geometry remains real and physically consequential; what changes is the account of where it comes from.

4.3.3 An Intuitive Picture

To fix intuitions, consider the following analogy.

Imagine a map of a territory. On the map, one can trace routes between cities—these routes depend on the geography depicted. Now consider the *projection* used to create the map: Mercator, Robinson, or some other. Different projections yield different maps with different distance relationships and shape distortions.

In this analogy:

- **Gauge forces** are like routes on the map—interactions that take place within a given representational structure.
- **Gravity** is like the projection itself—a property of how the representation is constructed, not a feature operating within it.

Changing the projection does not add new routes; it changes what "distance" and "proximity" mean. Similarly, changing the accessible observable algebra does not introduce new forces; it changes the effective geometry in which all forces are described.

Remark 4.4 (Intuitive Picture)**.** *This analogy is offered for conceptual orientation, not as a precise technical claim. The formal relationship between observable algebra selection and effective geometry requires the machinery developed in [153] and subsequent sections of this paper.*

4.3.4 Implications for Quantization

If this structural distinction is correct, it may illuminate the difficulty of quantizing gravity.

Quantizing a gauge theory means promoting classical fields to operator-valued distributions on a fixed background, subject to appropriate commutation relations and dynamics. The background—including the decomposition into subsystems—is held fixed while the fields are quantized.

But if gravity reflects the choice of decomposition itself, then "quantizing gravity" would require quantizing the selection of how to divide the system into parts. This is a categorically different task. It is not a matter of promoting a classical field to a quantum operator; it is a matter of making the *framework in which quantization is defined* itself subject to quantum uncertainty.

This may explain why straightforward approaches to quantum gravity encounter difficulties: they attempt to apply quantization procedures designed for systems *within* a fixed decomposition to a structure that determines the decomposition itself.

This perspective does not introduce new dynamics or predictions, but may offer diagnostic value: it suggests a structural reason why gravity resists the quantization procedures that succeed for gauge interactions, and points toward the need for approaches that do not presuppose a fixed subsystem decomposition.

Remark 4.5 (Diagnostic Value)**.** *We do not claim that this perspective solves the problem of quantum gravity. Rather, we suggest that it offers diagnostic value: it identifies a structural reason why gravity may resist the techniques that succeed for gauge forces, and points toward the need for approaches that do not presuppose a fixed subsystem decomposition.*

4.3.5 Relation to Background Independence

The idea that gravity is connected to "background independence" is well established in the quantum gravity literature [96, 111]. The present proposal may be viewed as a sharpening of this intuition in terms of observable algebras.

Background independence is often formulated as the requirement that physical laws not depend on a fixed spacetime metric. In the present framework, this requirement is subsumed under a more general principle: physical content should not depend on a particular choice of accessible observable algebra, or at least should transform covariantly under changes in that choice.

This suggests that a satisfactory theory of quantum gravity may need to be formulated not in terms of fields on a spacetime manifold, but in terms of structures that are prior to—or more fundamental than—the decomposition into spatially localized subsystems.

Remark 4.6 (Context-Dependence vs. Observer-Dependence)**.** *A potential misreading of this proposal is that it renders gravity "observer-dependent" or subjective. We stress that this is not the case. The selection of accessible observable algebras is constrained by physical interactions and stability criteria (such as decoherence structure and dynamical invariance), not by subjective choice or epistemic limitation. The resulting effective geometry is* context-dependent*—it depends on which physical degrees of freedom are stably accessible—but not* observer-relative *in any subjective sense. This distinction is developed in detail in [154].*

4.3.6 Universality Within a Selected Geometric Sector

A central puzzle in gravitational physics is the universality of free fall. The algebraic proposal can encode a limited version of universality, but does not derive the empirical equivalence principle.

Assume a reconstruction dictionary assigns one effective metric to a selected geometric algebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$. All matter representations included in that sector then inhabit the same

assigned metric. This is universality by common representation, conditional on the choice of sector and dictionary; it is not a derivation of universal free fall or of equality between inertial and gravitational mass.

Operators outside $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ are simply outside the modelled geometric sector. The present construction makes no claim about their gravitational influence.

Remark 4.7 (No dark-sector conclusion)**.** *It would be inconsistent here to declare a degree of freedom outside $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ irrelevant to the reconstructed metric and simultaneously let it alter that metric gravitationally. A dark-sector application would require an enlarged source algebra and an explicit backreaction map. Neither is supplied, so no dark-matter or dark-energy conclusion is drawn.*

4.4 Technical Formulation

4.4.1 Setup and Notation

We distinguish the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic/separating hypotheses. Density matrices, traces, QFIM, and the qubit example are finite-dimensional/Type I model statements. The Rindler example is an AQFT model check. The linearized Einstein equation is imported only in the holographic CFT regime under the assumptions listed in Theorem 4.29.

Definition 4.8 (Selected geometric algebra)**.** *A **selected geometric algebra** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer/record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$.*

No assumption is made that such a subalgebra admits a unique or canonical tensor factorization of $\mathcal{H}$.

Remark 4.9. *The conditions introduced in Section 4.5 define a class of modular-stable candidates for $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; they do not derive a unique member.*

4.4.2 Entanglement Structure and Induced Geometry

Given a selected geometric algebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, one may consider state restrictions or relative-entropy data on that algebra, subject to the regime qualifications above.

Following insights from holography and tensor network constructions, patterns of entanglement within $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may be associated with an effective distance structure on equivalence classes of observables.

Crucially, this effective geometry depends on:

- the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$,
- the stability of correlations under coarse-grained dynamics,
- and the redundancy of information encoding.

No claim is made that this geometry is fundamental. It is an effective description, valid within the context defined by $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$.

4.4.3 Central Conjecture

We now state the central conjecture of this paper explicitly.

Conjecture 4.10 (Gravity as Adiabatic Algebra Evolution)**.** *Gravitational dynamics is conjectured to correspond to an* ***adiabatic flow*** *of selected geometric algebras $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$. This paper does not derive the selection rule that produces the path.*

Specifically:

1. *The global state evolves unitarily: $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$.*
2. *A model-dependent rule selects a geometric candidate $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ at each time.*
3. *As the state evolves, the selected candidate may shift: $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$.*
4. *This shift $\dot{\mathcal{A}}_{\mathbf{c}}(t)$ manifests phenomenologically as the dynamical curvature of spacetime—i.e., as gravity.*

In contrast, unitary evolution of observables within *a fixed algebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ manifests as gauge interactions. The categorical distinction is:*

- ***Gauge dynamics:*** *Evolution within $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ (fixed stage, moving actors)*
- ***Gravitational dynamics:*** *Evolution of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ itself (moving stage)*

This formulation addresses a key objection: if algebras are kinematical background, how can gravity be dynamical? The answer is that the *selection* of which algebra is stable is itself state-dependent, and state evolution induces algebra flow.

Remark 4.11 (Status of Algebraic Variations)**.** *The adiabatic approximation assumes that algebra transitions occur slowly relative to internal dynamics within $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$. Rapid transitions would correspond to strong gravitational effects or spacetime singularities—regimes where the effective geometric description breaks down. The question of what dynamics, if any, governs non-adiabatic transitions is left open (see Section 4.8).*

4.4.4 Metric from Quantum Information Geometry

For comparison with information geometry, consider the **Quantum Fisher Information Metric (QFIM)** on a finite-dimensional family of effective states [90, 13]. We do not identify it with the spacetime metric.

Let $\{\lambda^\mu\}$ parameterize a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$. The induced metric $g_{\mu\nu}$ on this state family is given by:

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \tag{4.2}$$

where L_μ is the symmetric logarithmic derivative satisfying

$$\partial_\mu \rho = \frac{1}{2}(\rho L_\mu + L_\mu \rho). \tag{4.3}$$

This construction has several attractive features:

- It is coordinate-independent and intrinsically quantum.

- It reduces to the classical Fisher metric in appropriate limits.
- It is directly related to distinguishability of quantum states—geometrically "close" states are hard to distinguish operationally.

The QFIM is positive semidefinite and Riemannian. A Lorentzian spacetime metric has indefinite signature and causal structure. No operation in Eq. (4.2) supplies the negative direction, so the QFIM currently provides an analogy or a metric on model space only. Changes in the selected algebra,

$$\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}, \tag{4.4}$$

may change this state-space metric. Calling the change a spacetime perturbation or gravitational wave requires an additional Lorentzian reconstruction map.

Remark 4.12 (Relation to Holographic Results)**.** *In AdS/CFT, relative entropy and entanglement first-law arguments constrain bulk perturbations [69, 39]. Those results use the holographic dictionary and do not show that an arbitrary boundary QFIM is itself the Lorentzian bulk metric. Extending the relationship beyond holography is an open conjecture.*

4.4.5 What This Section Does Not Claim

To prevent misreading, we state explicitly what this technical formulation does *not* attempt:

- It does not derive gravitational field equations.
- It does not specify a dynamics for coarse-graining selection.
- It does not claim empirical adequacy or testable predictions.
- It does not introduce observer-dependent or consciousness-related elements.

The role of this section is to demonstrate internal coherence between the structural claims of Sections 1–3 and existing entanglement–geometry correspondences in the literature.

4.5 Algebraic Foundation: Modular-Stable Candidates

The preceding sections formulated the HAFF gravity conjecture in structural and information-geometric terms. This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not prove uniqueness. The former uniqueness theorem is explicitly withdrawn below.

4.5.1 Preliminaries

We review the mathematical tools required for the main construction. Standard references include Takesaki [115], Bratteli–Robinson [19], and Haag [50].

Von Neumann Algebras

Definition 4.13. *A **von Neumann algebra** $\mathcal{M}$ acting on a Hilbert space $\mathcal{H}$ is a $*$-subalgebra of $\mathcal{B}(\mathcal{H})$ that contains the identity and is closed in the weak operator topology. Equivalently, by von Neumann's bicommutant theorem, $\mathcal{M} = \mathcal{M}''$, where $\mathcal{M}' = \{T \in \mathcal{B}(\mathcal{H}) : [T, M] = 0 \ \forall\, M \in \mathcal{M}\}$ is the commutant.*

Definition 4.14. *A von Neumann algebra $\mathcal{M}$ is a **Type III_1 factor** if it has trivial center ($\mathcal{M} \cap \mathcal{M}' = \mathbb{C}\mathbf{1}$), admits no finite or semifinite normal trace, and the Connes spectrum $S(\mathcal{M}) = \mathbb{R}_{\geq 0}$.*

By a deep result of algebraic QFT, local algebras of observables in any reasonable quantum field theory are Type III_1 factors [50, 139].

Tomita–Takesaki Modular Theory

The Tomita–Takesaki theorem is the central structure theorem for von Neumann algebras with a cyclic and separating vector.

Definition 4.15. *Let $\mathcal{M}$ be a von Neumann algebra on $\mathcal{H}$, and let $|\Omega\rangle \in \mathcal{H}$.*

- *$|\Omega\rangle$ is **cyclic** for $\mathcal{M}$ if $\mathcal{M}|\Omega\rangle$ is dense in $\mathcal{H}$.*
- *$|\Omega\rangle$ is **separating** for $\mathcal{M}$ if $M|\Omega\rangle = 0$ implies $M = 0$ for all $M \in \mathcal{M}$.*

If $|\Omega\rangle$ is cyclic and separating, define the antilinear operator

$$S_\Omega : M|\Omega\rangle \mapsto M^*|\Omega\rangle, \qquad M \in \mathcal{M}. \tag{4.5}$$

This operator is closable, and its closure admits a polar decomposition:

$$S_\Omega = J_\Omega \Delta_\Omega^{1/2}, \tag{4.6}$$

where J_Ω is an antiunitary involution (the **modular conjugation**) and Δ_Ω is a positive self-adjoint operator (the **modular operator**).

Theorem 4.16 (Tomita–Takesaki [115])**.** *Let $\mathcal{M}$ be a von Neumann algebra with cyclic and separating vector $|\Omega\rangle$, and let Δ_Ω, J_Ω be as above. Then:*

(a) $J_\Omega \mathcal{M} J_\Omega = \mathcal{M}'$ (modular conjugation exchanges the algebra and its commutant).

*(b) The **modular automorphism group***

$$\sigma_t^\Omega(M) := \Delta_\Omega^{it} M \Delta_\Omega^{-it}, \qquad t \in \mathbb{R}, \tag{4.7}$$

satisfies $\sigma_t^\Omega(\mathcal{M}) = \mathcal{M}$ for all $t \in \mathbb{R}$.

Thus, any von Neumann algebra with a faithful state possesses a canonical one-parameter automorphism group.

The KMS Condition

Definition 4.17. *Let $\mathcal{M}$ be a von Neumann algebra, α_t a one-parameter automorphism group, and ω a normal state on $\mathcal{M}$. The state ω satisfies the* ***KMS condition*** *at inverse temperature β with respect to α_t if, for all $A, B \in \mathcal{M}$, there exists a function $F_{A,B}(z)$ analytic in the strip $0 < \mathrm{Im}(z) < \beta$ and continuous on its closure, such that*

$$F_{A,B}(t) = \omega(A\,\alpha_t(B)), \qquad F_{A,B}(t + i\beta) = \omega(\alpha_t(B)\,A) \tag{4.8}$$

for all $t \in \mathbb{R}$.

Theorem 4.18 (Takesaki [115])**.** *Let ω be a faithful normal state on a von Neumann algebra $\mathcal{M}$. Then ω is KMS at $\beta = 1$ with respect to σ_t^ω, and σ_t^ω is the* unique *one-parameter automorphism group with this property.*

The physical content is striking: the modular flow is the unique time evolution for which the given state looks thermal. In the Rindler wedge, this flow is the Lorentz boost, and the KMS condition at $\beta = 2\pi$ reproduces the Unruh temperature $T_U = (2\pi)^{-1}$ (in natural units where the acceleration $a = 1$).

Half-Sided Modular Inclusions

Definition 4.19 ([133])**.** *Let $\mathcal{N} \subset \mathcal{M}$ be an inclusion of von Neumann algebras on $\mathcal{H}$, with $|\Omega\rangle$ cyclic and separating for both. The inclusion is a* ***half-sided modular inclusion*** *(HSMI) if*

$$\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N} \qquad \text{for all } t \le 0, \tag{4.9}$$

where $\sigma_t^{\mathcal{M}}$ denotes the modular automorphism group of $(\mathcal{M}, |\Omega\rangle)$.

Theorem 4.20 (Wiesbrock [133])**.** *Let $\mathcal{N} \subset \mathcal{M}$ be a half-sided modular inclusion with common cyclic and separating vector $|\Omega\rangle$. Then there exists a unique one-parameter unitary group $U(a) = e^{-iaP}$, $a \ge 0$, with positive generator $P \ge 0$, such that*

$$\mathcal{N} = U(1)\mathcal{M}U(1)^*. \tag{4.10}$$

Moreover, $U(a)|\Omega\rangle = |\Omega\rangle$ for all a.

The physical interpretation is that the translation from $\mathcal{M}$ to $\mathcal{N}$ is encoded algebraically: no background geometry is needed to define the notion of "shifting a wedge." This is the key tool for extracting spacetime structure from purely algebraic data.

4.5.2 Modular Definition of Accessible Algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating for $\mathcal{M}_U$, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal on $\mathcal{M}_U$ and its ambient modular automorphism group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\ge0}$ be a normal UCP semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 4.21 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a* ***modular-stable geometric candidate*** *if it satisfies:*

(U1) ***Faithful restriction.*** *The restriction $\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the stated faithful ambient setting, faithfulness is inherited by every unital subalgebra and is therefore a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *The candidate is globally invariant under the ambient Type III modular flow:*

$$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}} \qquad (t \in \mathbb{R}). \tag{4.11}$$

(U3) ***Semigroup stability.*** *The candidate is setwise invariant under the named Heisenberg dynamics:*

$$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}} \qquad (t \geq 0). \tag{4.12}$$

Remark 4.22 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's probe-reachable operator system* $\mathrm{Eff}_{\mathrm{acc}}$ *or pointer algebra* $\mathcal{P}_{\mathrm{env}}$*. Nor is setwise semigroup stability the same as the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*; absent that map, the relation is motivational only. In particular, environmental redundancy does not imply faithfulness of a state restriction.*

Remark 4.23 (The role of $\mathcal{E}_t$)**.** *Condition (U3) names a normal UCP Heisenberg semigroup. A causal restriction map and a Lindblad time evolution are different types and are not identified. Each model must provide its own* $\mathcal{E}_t$ *and verify setwise invariance.*

4.5.3 Withdrawal of the Uniqueness Claim

Remark 4.24 (Retracted uniqueness theorem)**.** *The previous theorem asserted uniqueness from (U1)–(U3) and the pointwise condition that only scalars are fixed by both dynamics. That assertion is withdrawn for four independent reasons.*

1. *Uniqueness of the modular flow for a* fixed pair $(\mathcal{A}, \omega)$ *does not select the algebra* $\mathcal{A}$*.*
2. *A normal CP map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation is valid for the modular automorphisms, but not for a general* $\mathcal{E}_t$*.*
3. *Pointwise ergodicity constrains fixed elements, whereas (U2)–(U3) are setwise invariance conditions on subalgebras. No implication from the former to a unique maximal element of the latter lattice was proved.*
4. *With a faithful ambient state,* $\mathcal{M}_U$ *itself satisfies (U1)–(U3). A bare maximality rule therefore selects the ambient algebra trivially unless an independent seed, properness, readout, or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ *by a faithful Type III factor repairs the existence of the ambient modular flow, but does not repair any of these selection failures. No uniqueness or uniqueness/multiplicity dichotomy is claimed in this version.*

Remark 4.25 (Candidate multiplicity)**.** *The class of modular-stable candidates may contain multiple non-unitarily-equivalent members regardless of pointwise ergodicity. Classifying that family requires additional structure and is an open problem.*

Remark 4.26 (Selection remains an input)**.** *A nontrivial selection principle could use a foreground seed and construct a minimal invariant hull, or impose an explicit readout/interface constraint. Neither construction is proved here. Candidate status must not be read as measurement accessibility or completeness.*

4.5.4 Example 1: Rindler Wedge

Consider a free massless scalar field $\phi(x)$ in $(1+3)$-dimensional Minkowski spacetime $(\mathbb{R}^{1,3}, \eta_{\mu\nu})$. The total Hilbert space is the Fock space $\mathcal{H}_U = \mathcal{F}(\mathcal{H}_1)$ over the one-particle space $\mathcal{H}_1 = L^2(\mathbb{R}^3)$. The global state is the Minkowski vacuum $|\Omega\rangle$.

Define the right Rindler wedge:

$$\mathcal{W}_R = \{(t, x, y, z) \in \mathbb{R}^{1,3} : x > |t|\}. \tag{4.13}$$

The local algebra $\mathcal{R}(\mathcal{W}_R) = \{e^{i\phi(f)} : \operatorname{supp} f \subset \mathcal{W}_R\}''$ is the von Neumann algebra generated by Weyl operators smeared with test functions supported in $\mathcal{W}_R$. For this model check we take $\mathcal{M}_U = \mathcal{R}(\mathcal{W}_R)$. This makes the ambient modular action well defined but does not perform a nontrivial subalgebra selection.

Verification of (U1): Faithfulness

The Reeh–Schlieder theorem [50] guarantees that the Minkowski vacuum $|\Omega\rangle$ is cyclic and separating for $\mathcal{R}(\mathcal{W}_R)$. Therefore, the state $\omega(\cdot) = \langle\Omega| \cdot |\Omega\rangle$ is faithful on $\mathcal{R}(\mathcal{W}_R)$.

Verification of (U2): Modular stability

By the Bisognano–Wichmann theorem [16, 17], the modular operator of $(\mathcal{R}(\mathcal{W}_R), |\Omega\rangle)$ is

$$\Delta_\Omega = e^{-2\pi K}, \tag{4.14}$$

where K is the boost generator in the x-direction. The modular automorphism group acts as

$$\sigma_t^\Omega(\phi(t_0, \mathbf{x})) = \phi(\Lambda_{2\pi t}(t_0, \mathbf{x})), \tag{4.15}$$

where Λ_s is the Lorentz boost with rapidity s. Since the Rindler wedge is invariant under Lorentz boosts—$\Lambda_s(\mathcal{W}_R) = \mathcal{W}_R$ for all s—the modular flow preserves the wedge algebra:

$$\sigma_t^\Omega(\mathcal{R}(\mathcal{W}_R)) = \mathcal{R}(\mathcal{W}_R) \qquad \forall\, t \in \mathbb{R}. \tag{4.16}$$

Status of (U3)

No decoherence semigroup $\mathcal{E}_t$ has been specified in this AQFT example, so (U3) is not verified. Haag duality, $\mathcal{R}(\mathcal{W}_R)' = \mathcal{R}(\mathcal{W}_L)$, does not imply that every proper extension of the wedge algebra contains the entire opposite-wedge algebra, nor does it prove a maximality statement. The example checks standardness and the geometric modular action only.

Physical content

The modular flow σ_t^Ω coincides with the Lorentz boost, and the KMS condition at $\beta = 2\pi$ gives the Unruh temperature $T_U = a/(2\pi)$, where a is the proper acceleration. The modular Hamiltonian is

$$K_{\text{mod}} = -\ln \Delta_\Omega = 2\pi K = 2\pi \int_{\mathcal{W}_R} d\Sigma^\mu \, x_\nu \, T^\nu{}_\mu, \tag{4.17}$$

where $d\Sigma^\mu$ is the surface element on the $t = 0$ Cauchy surface. This example reproduces the established Bisognano–Wichmann identification of wedge modular flow with boosts. It does not show that the wedge algebra is uniquely selected by (U1)–(U3).

4.5.5 Example 2: Qubit Chain with Decoherence

Consider n qubits with total Hilbert space $\mathcal{H}_U = (\mathbb{C}^2)^{\otimes n}$. The system Hamiltonian is the transverse-field Ising model:

$$\hat{H} = -J \sum_{i=1}^{n-1} \sigma_z^{(i)} \sigma_z^{(i+1)} - h \sum_{i=1}^{n} \sigma_x^{(i)}, \tag{4.18}$$

with $J > 0$ and transverse field strength h. The system is coupled to a thermal environment at inverse temperature β through a Schrödinger-picture dephasing Lindblad semigroup:

$$\mathcal{T}_t(\rho) = e^{t\mathcal{L}}(\rho), \qquad \mathcal{L}(\rho) = -i[\hat{H}, \rho] + \gamma \sum_i \left(\sigma_z^{(i)} \rho \, \sigma_z^{(i)} - \rho \right), \tag{4.19}$$

where $\gamma > 0$ is the dephasing rate. Its normal UCP Heisenberg dual is denoted $\mathcal{E}_t := \mathcal{T}_t^*$. The global state is the thermal state $\omega = Z^{-1} e^{-\beta \hat{H}}$. For an exact check of (U2)–(U3) below we take the commuting specialization $h = 0$. When $h \neq 0$, the diagonal algebra is only approximately stable in a strong-dephasing approximation and is not an exact model of the stated conditions.

Identification of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$

For $h = 0$, off-diagonal coherences in the σ_z basis are destroyed while the diagonal observables are exactly fixed by the Heisenberg semigroup. The corresponding pointer/record algebra is

$$\mathcal{A}_{\mathrm{geo},\mathbf{c}} = \{ f(\sigma_z^{(1)}, \ldots, \sigma_z^{(n)}) : f \text{ is a polynomial} \}'' = \mathcal{A}_{\mathrm{diag}}, \tag{4.20}$$

the maximal abelian subalgebra (MASA) of $\mathcal{B}(\mathcal{H}_U)$ associated with the computational basis.

Verification of (U1): Faithfulness

The thermal state $\omega = Z^{-1} e^{-\beta \hat{H}}$ is a full-rank density matrix for any finite β (every eigenvalue of $e^{-\beta \hat{H}}$ is strictly positive). Its restriction to $\mathcal{A}_{\mathrm{diag}}$ assigns non-zero probability to every computational-basis state:

$$\omega(|s\rangle\langle s|) = \langle s | \rho_{\mathrm{th}} | s \rangle > 0 \qquad \forall \, s, \tag{4.21}$$

where the strict positivity follows from $\rho_{\mathrm{th}} = Z^{-1} e^{-\beta \hat{H}}$ being positive definite. Note that $|s\rangle$ are computational-basis (not energy-eigen-) states; the diagonal matrix elements of a positive-definite operator are strictly positive regardless of the choice of basis.

Verification of (U2): Modular stability

For the finite-dimensional system with faithful thermal state ρ_{th}, the ambient modular flow acts on $\mathcal{B}(\mathcal{H}_U)$ as

$$\sigma_t^{\omega,U}(X) = \rho_{\mathrm{th}}^{it} \, X \, \rho_{\mathrm{th}}^{-it} \qquad \forall \, X \in \mathcal{B}(\mathcal{H}_U). \tag{4.22}$$

For diagonal operators $A = \sum_s a_s|s\rangle\langle s| \in \mathcal{A}_{\text{diag}}$, we compute

$$\sigma_t^{\omega,U}(A) = \rho_{\text{th}}^{it}\left(\sum_s a_s|s\rangle\langle s|\right)\rho_{\text{th}}^{-it} = \sum_s a_s\,(\rho_{\text{th}}^{it}|s\rangle)(\langle s|\rho_{\text{th}}^{-it}). \tag{4.23}$$

For $h = 0$, ρ_{th} is diagonal in the computational basis, so ρ_{th}^{it} commutes with every $A \in \mathcal{A}_{\text{diag}}$ and (U2) holds exactly. For $h \neq 0$, the displayed conjugation generally leaves $\mathcal{A}_{\text{diag}}$, which is why that case is not advertised as an exact verification.

Verification of (U3): Semigroup stability

Under the Heisenberg dual of the dephasing dynamics, off-diagonal matrix units decay as

$$e^{t\mathcal{L}_{\text{deph}}^*}(|s\rangle\langle s'|) = e^{-2\gamma\, d(s,s')\, t}\,|s\rangle\langle s'|, \tag{4.24}$$

where $d(s, s')$ is the Hamming distance between strings s and s'. For $h = 0$, the Hamiltonian is diagonal as well, so every diagonal observable is fixed and (U3) holds. Thus $\mathcal{A}_{\text{diag}} \subset \text{Fix}(\mathcal{E}^*)$. This does not prove maximal setwise invariance: the full ambient matrix algebra is always mapped into itself, and decaying off-diagonal operators can remain inside a setwise-invariant algebra.

Physical content

With the additional exact-fixed-point criterion, the pointer algebra is the set of observables that survive this commuting dephasing model. The modular flow acts trivially on the pointer basis (since the algebra is abelian), consistent with the fact that classical variables do not undergo non-trivial modular evolution. The entanglement entropy of a subsystem $A \subset \{1, \ldots, n\}$ within $\mathcal{A}_{\text{diag}}$ reduces to the classical Shannon entropy:

$$S_A = -\sum_{s_A} p(s_A)\ln p(s_A), \qquad p(s_A) = \sum_{s_{\bar{A}}} p(s). \tag{4.25}$$

This finite-dimensional example checks that $\mathcal{A}_{\text{diag}}$ is one modular- and semigroup-stable candidate and, under exact fixation, the pointer algebra. It neither verifies the Type III regime nor selects the candidate uniquely.

4.5.6 Connection to Linearized Gravity

Having defined a candidate class and checked only parts of it in concrete models, we now isolate the extra assumptions needed to import a known holographic linearized-gravity result.

Entanglement first law

Theorem 4.27 (Entanglement first law in a density-matrix/regulator setting)**.** *Let ρ_0 be a faithful density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation. Set $K_0 = -\log\rho_0$. Then, to first order in ε:*

$$\delta S = \delta\langle K_0\rangle. \tag{4.26}$$

For a Type III local algebra, neither a reduced density matrix nor its von Neumann entropy exists; the corresponding statement must be formulated through Araki relative entropy and a specified comparison of normal states.

Proof sketch. The finite-dimensional relative entropy $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is non-negative and stationary at $\varepsilon = 0$. Its first variation gives the result. □

From algebra perturbation to modular Hamiltonian perturbation

The conjecture posits a one-parameter family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$, with $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$. Modular theory treats state perturbations on a fixed algebra and certain specially generated inclusion families differently:

(a) **State perturbation (fixed algebra):** The global state changes, $|\Psi_U\rangle \to |\Psi'_U\rangle$, but the algebra remains fixed. The modular Hamiltonian changes via the Connes cocycle Radon–Nikodym theorem [28, 115]:

$$\Delta_{\Psi'}^{it} = (D\omega' : D\omega)_t \, \Delta_{\Psi}^{it}, \tag{4.27}$$

where $(D\omega' : D\omega)_t$ is the Connes cocycle.

(b) **HSMI translation model (fixed state):** If, as an additional model input, the family is generated by the positive translation supplied by a half-sided modular inclusion, then

$$\mathcal{A}'_{\mathrm{geo},\mathbf{c}} = e^{-i\delta a\, P} \mathcal{A}_{\mathrm{geo},\mathbf{c}}\, e^{i\delta a\, P}, \qquad \delta a \ll 1, \tag{4.28}$$

yielding

$$\delta K_{\mathrm{mod}} = -i\delta a\, [P, K_{\mathrm{mod}}] + O(\delta a^2). \tag{4.29}$$

Equation (4.29) is not a formula for an arbitrary change of von Neumann subalgebra. It already assumes an HSMI translation generator P and therefore cannot be used to derive the existence of translation geometry without circularity. Nor has an additive decomposition into state and algebra variations been proved in general.

Linearized Einstein equations from the entanglement first law

The key result connecting algebraic perturbation to gravitational dynamics is due to Faulkner et al. [39], with complementary arguments by Jacobson [62].

Theorem 4.28 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\mathrm{mod}}\rangle$, applied to all ball-shaped regions on the boundary, is equivalent to the linearized Einstein equations in the bulk:*

$$G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N\, T^{(1)}_{\mu\nu}. \tag{4.30}$$

The derivation uses the explicit form of the modular Hamiltonian for a ball-shaped boundary region [25]:

$$K^B_{\mathrm{mod}} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{4.31}$$

together with the Ryu–Takayanagi formula and the JLMS relation [60] between bulk and boundary modular Hamiltonians.

Conditional theorem within the HAFF framework

Theorem 4.29 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) The holographic correspondence (AdS/CFT duality) holds.

(H2) The Ryu–Takayanagi formula and its quantum corrections hold.

(H3) The JLMS formula relating bulk and boundary modular Hamiltonians holds.

(H4) The compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the same boundary-region variation used in the holographic first-law argument.

(H5) The regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required boundary balls.

Then Eq. (4.29) *describes the modular-generator variation within that supplied family, and the holographic entanglement first law*

$$\delta S = \delta\langle K_{\mathrm{mod}}\rangle \tag{4.32}$$

matches the linearized Einstein equations in the holographic bulk by the imported Faulkner et al. result. The conditional chain is:

$$\textit{HSMI/ball family (H4)} \implies \delta K_{\mathrm{mod}} \xrightarrow{(H5)} \delta S = \delta\langle K_{\mathrm{mod}}\rangle \xrightarrow{(H1)\text{–}(H3)} G^{(1)}_{\mu\nu} = 8\pi G_N T^{(1)}_{\mu\nu}. \tag{4.33}$$

This theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is identical to gravitational dynamics.

The circularity issue

A fundamental objection to any "gravity from entanglement" program is circularity: Jacobson's derivation [61] presupposes a background geometry, but HAFF claims geometry emerges from the algebra. Three possible strategies are often discussed:

(A) **Algebraic programme (Wiesbrock).** "Wedge regions" are defined purely algebraically via modular inclusions, without reference to a background metric. Wiesbrock's theorem (Theorem 4.20) recovers a positive translation from a half-sided modular inclusion. Constructing a sufficient net and showing that the candidates in this paper form such inclusions remain additional tasks.

(B) **Bootstrap resolution.** Start with a seed geometry, derive linearized Einstein equations via the entanglement first law, update the geometry, and iterate.

(C) **Holographic resolution (AdS/CFT).** The boundary CFT provides the algebra without reference to bulk geometry; the bulk geometry is entirely derived from boundary data.

Theorem 4.29 uses the holographic strategy and assumes the HSMI/ball interface explicitly. A geometry-free reconstruction from the present candidate class remains open.

4.6 Relation to Existing Approaches

The perspective developed in this paper does not compete with existing approaches to quantum gravity and emergent spacetime. Rather, it may be understood as offering a *conceptual umbrella* under which several distinct research programs can be situated. We briefly discuss four such connections.

4.6.1 AdS/CFT and Holographic Duality

The AdS/CFT correspondence [78] provides the most concrete realization of geometry emerging from quantum entanglement. In this framework, a $(d+1)$-dimensional gravitational theory in anti-de Sitter space is dual to a d-dimensional conformal field theory on its boundary.

The Ryu-Takayanagi formula [97] and its generalizations establish that geometric quantities in the bulk (areas of extremal surfaces) correspond to entanglement entropies in the boundary theory:

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}. \tag{4.34}$$

Within the present framework, AdS/CFT supplies a model in which a detailed holographic dictionary relates boundary entanglement data to bulk geometry. It does not show that an arbitrary selected algebra carries such a dictionary.

Remark 4.30 (Not a Replacement)**.** *We do not claim that the present framework explains or derives AdS/CFT. It is used here as a conditional input and model of an algebra/entropy-to-geometry interface.*

4.6.2 Tensor Networks and MERA

Tensor network constructions, particularly the Multi-scale Entanglement Renormalization Ansatz (MERA) [126, 114], provide discrete models in which geometry emerges from entanglement structure.

In MERA, a quantum state is constructed by successive layers of disentanglers and isometries. The network structure itself defines an effective geometry: the "depth" direction in the network corresponds to a radial direction in an emergent spacetime, with properties reminiscent of AdS geometry.

This construction illustrates concretely how:

- A choice of coarse-graining (the tensor network structure) determines entanglement patterns.
- Entanglement patterns induce effective geometric relationships.
- Different network structures yield different effective geometries from the same boundary data.

The present framework generalizes this observation: tensor networks are specific implementations of coarse-graining structures, and MERA-type emergence is a special case of the algebra-to-geometry correspondence we propose.

4.6.3 Jacobson's Thermodynamic Derivation

Jacobson's remarkable result [61] showed that Einstein's field equations can be derived from thermodynamic considerations applied to local Rindler horizons, assuming the Bekenstein-Hawking entropy formula and the Clausius relation $\delta Q = T\, dS$.

This derivation suggests that gravity may be "thermodynamic"—an effective description arising from coarse-graining over microscopic degrees of freedom, rather than a fundamental force.

The present perspective is consonant with Jacobson's approach:

- Both treat gravitational dynamics as emergent rather than fundamental.
- Both connect gravity to entropy and information-theoretic quantities.
- Both suggest that the Einstein equations describe effective, coarse-grained physics.

The present work asks whether this intuition can be organized around selected geometric algebras; it does not establish that organization beyond the conditional models stated above.

4.6.4 Background Independence in Loop Quantum Gravity

Loop quantum gravity [96, 118] pursues quantization of gravity while maintaining background independence—the principle that physical laws should not depend on a fixed spacetime metric.

The present framework shares this commitment to background independence, but approaches it differently:

- Loop quantum gravity seeks to quantize the metric directly, constructing spacetime from spin networks.
- The present approach conjectures a relation between spacetime and paths of selected geometric algebras.

These are not mutually exclusive. A relation between spin-network states and selected geometric algebras would require an explicit representation/interface map; none is developed here.

4.6.5 Summary: A Conceptual Umbrella

We emphasize that the present framework does not claim superiority over these approaches. Each addresses aspects of quantum gravity that the present structural analysis does not. Our contribution is to articulate a perspective in which these diverse programs may be seen as exploring different facets of a common structural insight: that gravity is connected to the selection of how quantum degrees of freedom are organized into effective subsystems.

4.7 Explicit Scope Limitations

To ensure clarity regarding the claims of this paper, we state explicitly what it does and does not assert.

Approach	**Key Mechanism**	**Relation to Present Work**
AdS/CFT	Holographic duality	Specific instance of algebra → geometry
Tensor Networks	Discrete entanglement structure	Concrete implementation of coarse-graining
Jacobson	Thermodynamic derivation	Consonant emergent perspective
Loop QG	Background-independent quantization	Shared commitment, different strategy

Table 4.1: Relation of the present framework to existing approaches. The present work does not replace any of these programs, but offers a unifying structural perspective.

4.7.1 What This Paper Claims

1. **Structural conjecture:** The paper proposes, rather than proves, that gauge dynamics can be modelled within a fixed selected algebra while gravitational behaviour may be associated with a path of selected geometric algebras.

2. **Candidate mechanism:** Conjecture 4.10 identifies a research target; no selection dynamics for the algebra path is derived.

3. **Sector universality:** Given one reconstruction dictionary and one selected sector, all matter representations placed in that sector use the same assigned metric. This is not a derivation of the equivalence principle.

4. **Information-geometric analogy:** The QFIM supplies a positive state-space metric. Its relation to a Lorentzian spacetime metric is open.

5. **Conceptual umbrella:** Several existing research programs (holography, tensor networks, thermodynamic gravity) may be situated under this common structural framework.

4.7.2 What This Paper Does Not Claim

1. **No new dynamics:** We do not propose equations of motion, Lagrangians, or dynamical principles beyond those already established.

2. **No unconditional derivation of Einstein equations:** The linearized matching (Section 4.5.6) is conditional on (H1)–(H5) and imports the Faulkner et al. holographic result; we do not derive general relativity from arbitrary algebra perturbations.

3. **No empirical predictions:** We do not offer testable predictions that distinguish this perspective from standard approaches.

4. **No resolution of quantum gravity:** We do not claim to solve the problem of quantum gravity; we offer a diagnostic perspective, not a solution.

5. **No uniqueness theorem:** The former modular uniqueness theorem is withdrawn. The candidate class may contain multiple members, and no maximality/ergodicity dichotomy has been established.

6. **No observer-dependence:** The framework does not render gravity subjective or observer-relative. Accessible algebras are constrained by physical criteria, not by epistemic states of observers.

7. **No interpretational commitments:** The analysis is compatible with various interpretations of quantum mechanics and does not require commitment to any particular one.

4.8 Open Questions

The candidate framework in Section 4.5 sharpens the open problems facing the HAFF gravity program. The following are the immediate targets for subsequent work.

4.8.1 Linearized Einstein Equations without Holography

Theorem 4.29 records a conditional holographic matching under (H1)–(H5). A genuinely algebraic derivation would have to construct, rather than assume, a suitable net of half-sided modular inclusions from the candidate data and prove a bulk reconstruction theorem. Neither step is present here.

4.8.2 Classification and Selection of Candidates

Remark 4.24 withdraws the uniqueness argument independently of any ergodicity assumption. The immediate problem is to supply a nontrivial seed/readout/properness condition, determine existence of candidates satisfying it, and classify possible multiplicity. Only after that classification would a moduli-space metric be meaningful.

4.8.3 Computing δK_{mod} beyond Linear Order

The connection to linearized gravity uses an HSMI-generated family. A first task is to define differentiable variations for more general subalgebra families and determine whether a modular-generator derivative exists. Higher-order terms cannot be interpreted as nonlinear gravity until that interface and a reconstruction dictionary are established.

4.8.4 The Backreaction Problem

If a future reconstruction map assigns bulk geometry to $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, one may then ask how geometric data feed back into the selection rule. At present neither map in this loop is constructed, so uniqueness of a trajectory is not asserted.

4.9 Conclusion

We have proposed a structural framework in which gravitational phenomena may be associated with adiabatic paths of selected observable algebras as the global quantum state evolves.

The core claims are:

1. **Structural conjecture:** Gauge dynamics may be modelled *within* a fixed selected algebra, while gravitational behaviour may be associated with a path of selected geometric algebras.

2. **Candidate mechanism:** A state-dependent algebra path is a research target; this paper does not derive a unique selector or show that such a path manifests as spacetime curvature.

3. **Sector universality:** Given a selected geometric sector and a reconstruction dictionary, matter representations placed in that sector use the same assigned metric. This is not a derivation of the empirical equivalence principle.

4. **Information-geometric analogy:** The QFIM is a positive metric on a finite-dimensional state family; identifying it with a Lorentzian spacetime metric requires an additional reconstruction map.

This framework does not derive the Einstein equations from first principles or resolve quantum gravity. It organizes a conditional research programme for comparing selected-algebra paths, modular data, and gravitational reconstruction.

The perspective is consistent with, and provides a conceptual umbrella for, existing research programs: holographic duality (where boundary entanglement encodes bulk geometry), tensor networks (where network structure induces effective geometry), and thermodynamic approaches (where Einstein equations emerge from entropy considerations).

Section 4.5 makes a class of modular-stable candidates precise, but it does not select a unique accessible algebra: the former uniqueness claim is withdrawn in Remark 4.24. The Rindler and qubit-chain examples check parts of the candidate framework in their stated regimes. Under the explicit holographic assumptions (H1)–(H5), the paper examines the conditional correspondence

$$\delta\mathcal{A}_{\mathrm{geo},\mathbf{c}} \;\longrightarrow\; \delta K_{\mathrm{mod}} \;\longrightarrow\; \delta S = \delta\langle K_{\mathrm{mod}}\rangle \;\longrightarrow\; G^{(1)}_{\mu\nu} = 8\pi G_N\, T^{(1)}_{\mu\nu}, \tag{4.35}$$

for the specifically supplied HSMI/ball family of Theorem 4.29. In that restricted setting, imported holographic results relate modular perturbations to linearized gravitational data. The adiabatic flow of a generic selected algebra may be read only as a candidate for gravitational dynamics; identifying the two remains open. Removing assumptions (H1)–(H5) is an open problem (Section 4.8).

We conclude with a reflection. The difficulty of quantizing gravity may not be purely technical. If gravity is the evolution of the stage on which quantum mechanics is performed, rather than an actor on that stage, then quantizing gravity requires quantizing the framework of quantization itself. This is not a problem to be solved by better regularization schemes, but a conceptual challenge requiring us to think beyond fixed subsystem decompositions.

The path forward may lie not in quantizing forces, but in understanding what determines the structure of accessibility—and how that structure flows.

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Chapter 5

Measurement as Accessibility: A Structural Analysis of Observable Algebra Selection

Paper E

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Abstract

We propose that quantum measurement is not a primitive process but can be analyzed through accessibility constraints on operator-algebraic structures. Building on the Holographic Alaya-Field Framework (HAFF), we distinguish three typed tests—probe reachability, dynamical stability, and environmental redundancy—rather than treating them as one intersection that determines a single accessible algebra. Measurement outcomes are reinterpreted through the eigenvalue and record structure of observables satisfying the relevant declared tests, and the definiteness of outcomes is traced to redundant environmental encoding rather than wave function collapse. This framework maintains compatibility with unitary quantum mechanics while providing a structural account of why certain observables acquire definite values. The analysis is structural in nature: we do not propose new dynamics or modifications to quantum mechanics, but clarify the conditions under which measurement-like phenomena emerge from algebraic constraints. Connections to decoherence theory, quantum Darwinism, and algebraic quantum field theory are discussed, along with explicit non-claims to prevent misinterpretation.

5.1 Introduction

5.1.1 The Measurement Problem Reconsidered

The quantum measurement problem has resisted resolution for nearly a century. In its sharpest form, the problem asks: how do definite measurement outcomes arise from quantum states that, prior to measurement, assign non-trivial amplitudes to multiple possibilities? Standard quantum mechanics provides rules for computing outcome probabilities but does not explain the transition from superposition to definiteness.

Various approaches have been proposed: collapse postulates that modify unitary evolution, many-worlds interpretations that deny the uniqueness of outcomes, and decoherence-based accounts that explain the suppression of interference without addressing the selection of particular results. Each approach has merits, but none has achieved consensus.

5.1.2 A Structural Reframing

This paper proposes a different perspective. Rather than asking how measurement *causes* a unique outcome, we ask: under what structural conditions do effects become probe-reachable and records become objective in the first place?

The central claim is:

> **Measurement is not a primitive process, but a manifestation of accessibility constraints on operator algebras.**

Within the Holographic Alaya-Field Framework (HAFF) developed in previous papers [153, 154, 155, 156], physical descriptions are formulated relative to selected operator families in an ambient von Neumann algebra $\mathcal{M}_U$. Not all mathematically definable operators correspond to physically realizable measurements. The present paper separates three tests that were previously conflated:

1. **Probe reachability**: an allowed probe channel can distinguish the relevant effect.
2. **Dynamical stability**: the effect is approximately stable, or exactly fixed, under a specified environmental dynamics.
3. **Record redundancy**: a classical pointer label is recoverable from many disjoint environmental fragments.

These tests apply to different mathematical objects and therefore must not be intersected and called an algebra. Probe-reachable effects form an operator system; exact fixed points form an operator system and, under additional hypotheses, a von Neumann algebra; pointer records form a designated commutative algebra. Redundancy supports robust, intersubjectively readable records, but does not select a unique outcome of the global state.

5.1.3 Scope and Limitations

We emphasize what this paper does and does not attempt.

This paper does:

- Provide a structural characterization of measurement in terms of algebraic accessibility
- Identify three typed tests within a measurement-accessibility profile
- Connect measurement to established frameworks (decoherence, quantum Darwinism, AQFT)
- Maintain compatibility with unitary quantum mechanics

This paper does not:

- Propose new dynamics or modifications to quantum mechanics
- Explain why specific measurement outcomes occur (the "outcome problem")
- Resolve interpretational debates about the ontology of quantum states
- Invoke consciousness, observers, or subjective elements

The analysis is structural: we clarify conditions under which measurement-like phenomena emerge, without claiming to have solved the measurement problem in its deepest form.

5.1.4 Outline

Section 5.2 reviews relevant background on algebraic approaches to quantum mechanics and the HAFF framework. Section 5.3 develops the three accessibility constraints in detail. Section 5.4 reframes measurement in terms of these constraints. Section 5.5 discusses connections to existing approaches. Section 5.6 states explicit non-claims. Section 5.7 situates the paper within the broader HAFF program. Section 5.8 concludes.

5.2 Background

5.2.1 Algebraic Approaches to Quantum Mechanics

In the algebraic formulation of quantum mechanics, the fundamental objects are not wave functions or Hilbert spaces, but algebras of observables. A quantum system is characterized by a $*$-algebra $\mathcal{A}$ of bounded operators, and states are positive linear functionals on $\mathcal{A}$ [50, 7].

This perspective has several advantages. It does not presuppose a specific Hilbert space representation, accommodates systems with infinitely many degrees of freedom, and naturally incorporates superselection rules. Most importantly for our purposes, it treats the specification of observables as logically prior to the specification of states.

5.2.2 Observable Algebras in AQFT

In algebraic quantum field theory (AQFT), local observable algebras $\mathcal{A}(\mathcal{O})$ are associated with spacetime regions $\mathcal{O}$, without invoking a global tensor product structure [50]. The key insight is that subsystem structure emerges from the algebra of observables rather than being presupposed.

The HAFF framework extends this perspective by treating the selection of operator structures as physically constrained rather than given. Building on foundational work demonstrating that tensor product structures are observable-induced [140, 141], different physical contexts—characterized by different interaction structures, stability conditions, and environmental couplings—can yield different effective operator descriptions.

5.2.3 Typed Accessibility Data

Let $\mathcal{M}_U$ be the ambient von Neumann algebra. A context $\mathbf{c}$ specifies allowed probe channels, an environmental dynamics, a record algebra, timescales, and resolution thresholds. The following objects are distinct.

Definition 5.1 (Probe channel and reachable effects)**.** *For each allowed probe α, let*

$$\mathcal{R}^*_{\alpha,\mathbf{c}} : \mathcal{B}(\mathcal{K}_\alpha) \longrightarrow \mathcal{M}_U$$

be its normal unital completely positive Heisenberg map. The ***probe-reachable operator system*** *is*

$$\mathrm{Eff}_{\mathrm{acc}}(\mathbf{c}) := \overline{\mathrm{span}}^{uw}\{\mathcal{R}^*_{\alpha,\mathbf{c}}(F) : 0 \le F \le \mathbf{1},\ \alpha \text{ allowed}\}.$$

Its positive contractions are the effects distinguishable by the allowed probes. The family $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c}) := \{\mathcal{R}_{\alpha,\mathbf{c}}\}_\alpha$ is part of the context, not an observable algebra.

Definition 5.2 (Environmental fixed set)**.** *For a normal UCP semigroup $\{\mathcal{E}_t\}_{t\ge 0}$ on $\mathcal{M}_U$, set*

$$\mathrm{Fix}(\mathcal{E}) := \{O \in \mathcal{M}_U : \mathcal{E}_t(O) = O \text{ for every } t \ge 0\}.$$

This is always an ultraweakly closed operator system. It is an ordinary-product von Neumann subalgebra, for example, when the semigroup admits a faithful normal invariant state: equality in the Schwarz inequality then places every fixed element in the multiplicative domain. We do not assume this closure without stating such a hypothesis.

Definition 5.3 (Pointer/record algebra)**.** *A* ***pointer/record algebra*** *$\mathcal{P}_{\mathrm{env}} \subset \mathcal{M}_U$ is a designated commutative von Neumann algebra generated by a classical record PVM $Z(B)$, or by a finite-resolution coarse graining of such a PVM. Record channels*

$$\Gamma_k : \mathcal{P}_{\mathrm{env}} \longrightarrow \mathcal{B}(\mathcal{H}_{E_k})$$

describe what environmental fragment E_k carries about the record. Approximate pointer observables are treated by distance to $\mathcal{P}_{\mathrm{env}}$, rather than by calling an approximately closed set an algebra.

There is no default equality among $\mathrm{Eff}_{\mathrm{acc}}$, $\mathrm{Fix}(\mathcal{E})$, and $\mathcal{P}_{\mathrm{env}}$. The declared maps are the probe maps $\mathcal{R}^*_{\alpha,\mathbf{c}}$, the dynamics $\mathcal{E}_t$, the inclusions of any chosen subalgebras into $\mathcal{M}_U$, and the record channels Γ_k. A model may establish overlaps—for example, stable records in $\mathcal{P}_{\mathrm{env}} \cap \mathrm{Fix}(\mathcal{E})$—but such overlaps are conclusions or assumptions of that model, not type identifications.

The operator-system and fixed-set statements above are made for general von Neumann algebras under the hypotheses shown. Entropic redundancy formulas are finite-alphabet or finite-resolution statements about specified states. This paper makes no blanket finite-dimensional or Type III claim for all four objects.

5.3 Accessibility as Physical Constraint

We now formulate the three tests in their appropriate types. They provide a measurement workflow; they do not define one common algebra by intersection.

5.3.1 Test 1: Probe Reachability

Constraint 5.4 (Probe reachability)**.** *An effect $M(B) \in \mathcal{M}_U$ is probe-reachable in context* $\mathbf{c}$ *if it lies in the image of an allowed probe map: $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$ for some probe effect $0 \leq F \leq \mathbf{1}$.*

Physical interpretation. Operational accessibility is a property of a channel and an effect, not of a bare commutator. The allowed channel must map distinct system statistics to distinguishable probe statistics. This formulation includes indirect measurements and general POVMs.

Relation to measurement. Measurement requires that system information become correlated with apparatus degrees of freedom. A probe Hamiltonian may implement the channel, but $[O, H_{\text{probe}}] \neq 0$ is neither necessary nor closed under algebra operations; ideal nondemolition couplings commonly commute with the measured pointer observable. The environmental interaction responsible for einselection is a separate part of the context.

Remark 5.5. *Probe reachability does not imply stability or environmental redundancy. Conversely, an exact environmental fixed point need not be readable by any probe in the allowed family.*

5.3.2 Test 2: Dynamical Stability

Constraint 5.6 (Stability)**.** *An effect O satisfies the* ***finite-window stability test*** *if it remains approximately invariant under the specified environmental maps $\mathcal{E}_t$:*

$$\|\mathcal{E}_t(O) - O\| < \epsilon \tag{5.1}$$

for $0 \leq t \leq T_{\mathbf{c}}$. Exact stability is the separate statement $O \in \text{Fix}(\mathcal{E})$.

Physical interpretation. Observables that scramble rapidly—spreading their information across many degrees of freedom faster than any recording process can track—cannot be reliably measured. Stability ensures that the observable persists long enough to be correlated with records.

Relation to scrambling. In the language of quantum chaos, stable observables are those with slow out-of-time-order correlator (OTOC) growth [55]:

$$\langle [\hat{O}(t), \hat{V}(0)]^2 \rangle \ll 1 \quad \text{for } t \ll \tau_{\text{scrambling}}. \tag{5.2}$$

Observables satisfying this condition resist rapid delocalization and maintain their identity under dynamical evolution.

Relation to decoherence. The stability constraint is closely related to the selection of pointer observables in decoherence theory [142]. Pointer observables are those that remain stable under system-environment interaction, forming the preferred basis in which the density matrix becomes approximately diagonal.

Remark 5.7 (Threshold ϵ)**.** *The threshold ϵ is not a fundamental constant but depends on the physical context: the precision of available recording mechanisms, the timescales of interest, and the noise level of the environment. This context-dependence is a feature, not a bug—it reflects the operational nature of accessibility.*

5.3.3 Test 3: Environmental Redundancy

Constraint 5.8 (Redundancy)**.** *Let X be the classical label of a pointer PVM, or of a finite measurable coarse graining of a POVM. For each environmental fragment E_k, form the corresponding classical–quantum state ρ_{XE_k}. The label is δ-redundant if many disjoint fragments satisfy*

$$\chi(X : E_k)_\rho \geq (1 - \delta)H(X), \tag{5.3}$$

where χ is the Holevo information. Equivalently, one may require the accessible information of an explicitly specified fragment measurement to meet the same threshold.

For continuous spectra, X denotes a declared finite-resolution measurable partition. No von Neumann entropy is assigned to an operator, and no cross-type expression such as $I(\hat{O} : E_k)$ is used.

Physical interpretation. Redundancy ensures that information about the observable is not localized in a single environmental degree of freedom but is broadcast across many independent fragments. This makes the information robust and intersubjectively accessible: multiple independent observers can extract the same information without disturbing each other's records.

Relation to quantum Darwinism. The redundancy constraint formalizes the central insight of quantum Darwinism [143]: classical objectivity arises when information about a system is redundantly imprinted on the environment. Observables satisfying this constraint are precisely those for which multiple observers can agree on measurement outcomes.

Operational significance. Redundancy distinguishes *objective* from *subjective* information. An observable whose information is encoded in only a single environmental fragment is accessible to at most one observer; different observers would obtain different, incompatible records. Redundancy ensures that the observable's value is a matter of intersubjective fact.

Remark 5.9 (Relation to classical objectivity)**.** *The redundancy constraint provides a structural account of why certain observables behave "classically": their values are recorded multiply and independently, making them robust against local perturbations and accessible to multiple agents.*

5.3.4 Typed Relations Rather Than One Accessible Algebra

Definition 5.10 (Measurement-accessibility profile)**.** *The* ***measurement-accessibility profile*** *of a context is the typed datum*

$$\mathbf{Acc}(\mathbf{c}) := \big(\mathrm{Eff}_{\mathrm{acc}}(\mathbf{c}), \mathfrak{R}_{\mathrm{probe}}(\mathbf{c}), \mathcal{P}_{\mathrm{env}}(\mathbf{c}), \mathrm{Fix}(\mathcal{E}_{\mathbf{c}}); \Gamma_k, T_{\mathbf{c}}, \epsilon, \delta\big). \tag{5.4}$$

The old definition that intersected the three tests is withdrawn. In particular, $[O, H_{\mathrm{int}}] \neq 0$ excludes the identity and is not stable under sums or products, while pointer stability typically selects observables commuting with the environmental interaction. The profile above preserves those distinct roles and makes every cross-object map explicit.

Constraint	Condition	Physical Meaning
Probe	$M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$	Reachable effects; an operator system, not automatically an algebra
Stability	$\sup_{0\le t\le T_\mathbf{c}} \|\mathcal{E}_t(O) - O\| < \epsilon$	Approximate sector; exact sector is $\mathrm{Fix}(\mathcal{E})$
Records	$\chi(X : E_k) \ge (1-\delta)H(X)$	A classical pointer label is redundantly readable

Table 5.1: The three typed tests in a measurement-accessibility profile.

5.4 Measurement Reframed

We now apply the accessibility framework to reinterpret quantum measurement.

5.4.1 What Can Be Measured

Within the present framework, the question "What can be measured?" is relative to an allowed probe family. A POVM $M : \Sigma \to \mathcal{M}_U$ is implementable in context $\mathbf{c}$ when there are an allowed probe channel $\mathcal{R}_{\alpha,\mathbf{c}}$ and a probe POVM F such that

$$M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F(B)) \qquad (B \in \Sigma).$$

Measurement accessibility is implementability of effects through a declared probe channel.

Effects outside $\mathrm{Eff}_{\mathrm{acc}}(\mathbf{c})$ are not reachable by the allowed probes. This does not mean they "do not exist" in any metaphysical sense. It says only that the specified context supplies no operational implementation. Stability and redundant recording are additional properties of an implementation, not hidden algebra-closure conditions on probe reachability.

5.4.2 Measurement Outcomes

The outcome space is part of the POVM datum (X, Σ, M). For a sharp discrete observable, M is its spectral PVM and the possible outcomes may be identified with eigenvalues. For continuous spectrum, outcomes are measurable sets or finite-resolution bins; for an unsharp measurement, they are POVM labels rather than eigenvalues of a single self-adjoint operator.

Eigenvalues describe the sharp discrete-PVM special case, not measurement outcomes in general.

The structural question is therefore whether the POVM effects are reachable through $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c})$ and, separately, whether their labels acquire stable redundant records.

5.4.3 Objectivity from Redundant Records

Environmental redundancy explains why a record can be robust and independently readable by many observers. It does not, by itself, explain why the global state contains one unique outcome rather than several decohered branches.

When a pointer label X satisfies the redundancy test, multiple disjoint fragments carry nearly the same classical information about X. Conditional on a branch/record value, fragment readers can agree without accessing the original system. This is the operational content of quantum-Darwinist objectivity.

> **Objectivity and robustness are constituted by redundant environmental encoding.**

The global quantum state may remain in superposition. The present analysis establishes no context-independent deterministic value assignment for a noncommutative effect system and no unique global outcome.

5.4.4 The Outcome Problem

The framework does not explain why a *particular* outcome is realized rather than another. This "outcome problem" remains open:

> **We explain how records become robust and intersubjectively readable, not why one global outcome is selected.**

This limitation is shared with decoherence-based approaches. The present framework does not claim to resolve this aspect of the measurement problem, only to clarify the structural conditions under which objective records become possible.

5.4.5 Measurement Without Observers

A crucial feature of the framework is that measurement is characterized without reference to observers, agents, or consciousness:

> **The "observer" is replaced by the "interaction context."**

Any physical system implementing a probe channel and producing records—be it a photon counter, a mineral surface, or an interstellar dust grain—can constitute a measurement site for the relevant effects. Stability and redundancy determine the persistence and public readability of those records. Human observers are a special case, not a privileged category.

5.5 Relation to Existing Approaches

5.5.1 Decoherence Theory

Decoherence theory explains how interference between quantum states is suppressed through environmental entanglement [142]. The present framework is fully compatible with decoherence and may be viewed as extending it in two respects:

1. We make explicit the *conditions* under which decoherence selects a preferred basis (the stability and redundancy constraints).
2. We embed decoherence within the broader HAFF framework, connecting it to emergent geometry and gravitational phenomena.

5.5.2 Quantum Darwinism

Quantum Darwinism [143] emphasizes the role of environmental redundancy in establishing classical objectivity. The redundancy test (Constraint 5.8) formalizes this insight for a classical pointer label using Holevo or accessible information.

The present framework situates quantum Darwinism within an algebraic setting while keeping record objectivity distinct from probe implementability and exact dynamical fixation.

5.5.3 QBism

QBism [42] interprets quantum states as expressions of an agent's beliefs. The present framework differs fundamentally: accessibility is determined by physical interaction structure, not by agent beliefs.

The key difference:

- **QBism**: Dependence on agent's epistemic state (belief-determined)
- **HAFF**: Dependence on interaction structure (interaction-determined)

Both reject naive realism about quantum states, but the present framework maintains objectivity by grounding accessibility in physical constraints rather than subjective beliefs.

5.5.4 Relational Quantum Mechanics

Relational quantum mechanics (RQM) [95] holds that quantum states are relational—defined only relative to a reference system. The present framework shares the emphasis on relationality but differs in its treatment of what grounds the relation:

- **RQM**: Relations between systems (system-relative)
- **HAFF**: Stability conditions on algebras (interaction-determined)

HAFF may provide the stable "nodes" required for RQM's relational network: before relations can exist, there must be relata stable enough to participate in interactions.

5.5.5 Algebraic Quantum Field Theory

The closest structural affinity is with algebraic quantum field theory (AQFT) [50]. Both frameworks treat observable algebras as primary and states as secondary. The present framework extends AQFT by:

1. Providing typed criteria for probe reachability, stability, and record redundancy
2. Connecting algebra selection to measurement and emergent geometry
3. Situating AQFT insights within the broader HAFF program

Approach	**Key Mechanism**	**Relation to HAFF**
Decoherence	Environmental entanglement	Compatible; constraints specify conditions
Quantum Darwinism	Redundant encoding	Redundancy constraint formalizes this
QBism	Agent beliefs	Categorically distinct; HAFF is interaction-determined
RQM	System relations	Complementary; HAFF provides stable relata
AQFT	Observable algebras	Closest affinity; HAFF adds selection criteria

Table 5.2: Relation of the present framework to existing approaches.

5.6 What This Paper Does NOT Claim

To prevent misinterpretation, we state explicitly what the paper does not claim.

1. **No resolution of the outcome problem.** We do not explain why particular measurement outcomes occur; redundancy establishes robust, intersubjectively readable records, not unique global selection.

2. **No collapse postulate.** The framework assumes standard quantum dynamics. Record objectivity arises from redundancy, not from non-unitary collapse.

3. **No modification of quantum mechanics.** We do not propose new equations, new dynamics, or modifications to the standard formalism.

4. **No consciousness or observer-dependence.** Accessibility is determined by physical constraints, not by conscious observers or epistemic states.

5. **No claim that all measurement problems are solved.** The framework addresses probe accessibility and record objectivity but leaves unique outcomes, preferred-basis details, and tails to be addressed in conjunction with existing approaches.

6. **No claim of novelty regarding decoherence.** The framework builds on and is compatible with decoherence theory; it does not replace it.

7. **No claim of universal applicability.** The analysis is confined to the HAFF framework and does not assert that all approaches to measurement must adopt this structure.

8. **No metaphysical conclusions.** We do not claim that the probe-reachable effect system exhausts reality, only that it records what the declared probe family can implement within a given context.

9. **No derivation of Born rule.** The framework does not derive the Born rule for outcome probabilities; it assumes standard quantum probability.

10. **No claim about quantum-classical divide.** We do not assert a sharp boundary between quantum and classical; accessibility is context-dependent and admits degrees.

11. **No hidden variables.** The framework does not invoke hidden variables or additional ontology beyond standard quantum mechanics.

12. **No many-worlds commitment.** The framework is compatible with, but does not require, many-worlds interpretations.

13. **No claim of interpretational neutrality.** While the framework avoids some interpretational commitments, it does adopt the structural stance of HAFF, which may not be neutral with respect to all interpretations.

5.7 Connection to the HAFF Framework

5.7.1 Relation to Previous Papers

The present paper (Paper E) is part of a series developing the Holographic Alaya-Field Framework:

- **Paper A** [153]: Establishes that inequivalent coarse-graining structures induce inequivalent effective geometries from the same global quantum state.
- **Paper B** [154]: Clarifies the structural (vs. epistemic) nature of accessibility and situates HAFF relative to existing interpretations.
- **Paper C** [155]: Explores philosophical implications for causation, agency, and existence.
- **Paper D** [156]: Proposes that gravitational dynamics corresponds to the adiabatic evolution of accessible algebras.
- **Paper E** (this paper): Reframes measurement as a manifestation of accessibility constraints.
- **Paper F** (forthcoming): Addresses temporal asymmetry as accessibility propagation.

5.7.2 The Diagnostic Triangle: D + E + F

Papers D, E, and F form a "diagnostic triangle" within the HAFF framework:

The proposed unifying insight is that force, measurement, and time may be described through related operator-algebraic structures. This does not identify the structures: probe effects, record algebras, dynamical fixed sets, and geometric algebras retain their declared types.

Paper	**Phenomenon**	**Traditional View**	**HAFF Reframing**
D	Gravity	Fundamental force	Conditional flow of a geometric algebra
E	Measurement	Primitive process	Probe-reachable effects and objective records
F	Time	Fundamental parameter	Direction of record propagation

Table 5.3: The diagnostic triangle: gravity, measurement, and time reframed as aspects of algebraic accessibility.

5.7.3 Structural Link to Gravity

Paper D proposes that gravitational behaviour can be associated, under additional modular and holographic hypotheses, with the evolution of a selected geometric algebra. The present paper does not identify that algebra with $\mathrm{Eff}_{\mathrm{acc}}$, $\mathcal{P}_{\mathrm{env}}$, or $\mathrm{Fix}(\mathcal{E})$. Any bridge must specify a representation, inclusion, or channel between the relevant objects.

The connection may be summarized as follows:

> If Paper D's conditional holographic dictionary maps selected algebraic data to spatial curvature, Paper E supplies candidate operational inputs: probe-reachable effects and stable records. Which of these data enter the geometric algebra is a separate interface problem. No finiteness or gravitational conclusion follows merely from the measurement-accessibility profile.

5.8 Conclusion

We have proposed that quantum measurement is not a primitive process but a manifestation of accessibility constraints on operator algebras.

The central results are:

1. **Three accessibility tests**: Probe reachability, dynamical stability, and environmental redundancy form a typed measurement-accessibility profile within a given physical context; they do not define one common algebra by intersection.

2. **Measurement reframed**: What can be measured is described by the relevant profile components; measurement outcomes are read from selected observables or pointer records; definiteness arises from redundant environmental encoding.

3. **Observer-independence**: The framework characterizes measurement without reference to observers, consciousness, or subjective elements. The "observer" is replaced by the "interaction context."

4. **Compatibility**: The framework is compatible with unitary quantum mechanics, decoherence theory, and quantum Darwinism, while providing a structural account that connects measurement to the broader HAFF program.

The framework does not resolve all aspects of the measurement problem. It does not explain why particular outcomes occur, nor does it derive the Born rule. What

it provides is a structural clarification: the conditions under which measurement-like phenomena emerge from physical constraints on operator algebras.

Within the HAFF program, measurement can be analyzed alongside gravity and time through related accessibility structures. This diagnostic unification does not constitute a Theory of Everything, but it suggests that seemingly disparate foundational puzzles may share a common structural origin.

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Chapter 6

Accessibility Propagation and Temporal Direction: A Conditional Structural Analysis

Paper F

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Abstract

We examine how a preferred temporal direction may be represented by accessibility propagation in specified record-forming protocols. Under a supplied dynamics, fragmentation, pointer label, and low-record boundary condition, algebra inclusion and redundancy growth define protocol-relative comparisons; they do not derive a fundamental time parameter or a universal causal partial order. Causal influence additionally requires an intervention model and is treated as a diagnostic rather than a consequence of redundancy alone. The former generic-expansion and measure-zero retrocausation claims are withdrawn: the appendix gives a conditional finite-dimensional record-growth witness together with its exact unitary reversal.

6.1 Introduction

6.1.1 The Problem of Time's Arrow

Among the deepest puzzles in physics is the origin of temporal asymmetry. The fundamental laws of physics—Newtonian mechanics, electromagnetism, quantum mechanics—are time-reversal invariant or nearly so. Yet our experience of the world is profoundly asymmetric: eggs break but do not unbreak; we remember the past but not the future; causes precede effects.

This tension between microscopic reversibility and macroscopic irreversibility has been recognized since Boltzmann's work on statistical mechanics [18]. The standard resolution appeals to special initial conditions: the universe began in a low-entropy state, and the

second law of thermodynamics reflects the statistical tendency to evolve toward higher entropy [88, 23].

While this explanation is widely accepted, it raises further questions:

- Why should initial conditions be "special"? What selects them?
- Is the thermodynamic arrow the only arrow, or are there independent sources of temporal asymmetry?
- In quantum gravity, where time itself may be emergent, how does any notion of "before" and "after" arise?

The present work does not claim to resolve these questions definitively. Instead, it offers a structural reframing: temporal asymmetry may be understood as a consequence of how accessible algebras propagate information.

6.1.2 Central Thesis

We propose that temporal asymmetry can be understood as follows:

> **Revised thesis:** In a specified branching/record model with a low-redundancy boundary condition and an external dynamical parameter, record redundancy can grow along the thermodynamic arrow. Algebraic inclusion and redundancy can summarize that ordered history, but do not derive time or its arrow without those inputs.

This thesis uses Paper E's pointer/record sector [157]. Paper E distinguishes probe-reachable effects, pointer records, and dynamical fixed points; it does not define one accessible algebra by intersecting three constraints. Here redundancy concerns a classical pointer label encoded in specified environmental fragments.

The key insight is that the redundancy constraint is inherently asymmetric:

- **Record-producing protocol**: A measurement-like branching interaction can correlate one pointer label with many fragments.
- **Time-reversed protocol**: The inverse unitary removes those records when the precisely reversed state and control are supplied.

The observed asymmetry therefore comes from the low-record boundary, restricted controls, and statistical ensemble. It is not a theorem about accessibility structure alone.

6.1.3 Scope and Limitations

We state explicitly what this paper does and does not attempt.

This paper does:

- Give a conditional records-based description of temporal asymmetry
- Give an algebraic description of record ordering conditional on a dynamical model and low-redundancy boundary
- Connect this account to the thermodynamic and quantum arrows

- Define a protocol-relative causal-influence diagnostic combining records and interventions
- Provide a star-graph witness (Appendix A) of reversible record formation from a low-record boundary
- Situate the analysis within the broader HAFF framework

This paper does not:

- Derive the second law of thermodynamics from first principles
- Explain why initial conditions are low-entropy
- Resolve metaphysical debates about the nature of time (A-theory vs. B-theory, presentism vs. eternalism)
- Address free will, agency, or the phenomenology of temporal experience
- Propose new dynamical equations or empirical predictions

The analysis is structural. We examine how temporal asymmetry relates to accessibility structure, without claiming that this analysis exhausts the content of the problem.

Remark 6.1 (Relation to Papers D and E)**.** *Paper D [156] conjectures a relation between geometry and paths of selected modular-stable algebras. Paper E [157] separates probe effects from pointer records and fixed points. The present paper studies record growth in a conditional model. These are related but differently typed programme components:*

- *D: conditional geometric-algebra dynamics*
- *E: measurement effects and objective records*
- *F: record propagation under a supplied time evolution*

6.1.4 Outline

Section 6.2 reviews the status of time in various physical theories. Section 6.3 identifies the dynamical, mereological, and boundary inputs needed for redundancy growth. Section 6.4 defines a protocol-relative influence diagnostic. Section 6.5 compares the present approach to existing accounts of temporal asymmetry. Section 6.6 states explicit non-commitments. Section 6.7 discusses connections to the HAFF framework. Section 6.8 concludes. Appendix 6.9 gives a conditional model of reversible record formation.

6.2 Background: Time in Physics

Before developing the accessibility-based account, we briefly review the status of time in major physical theories. The purpose is to identify a common structural assumption: that time is an external parameter, given rather than derived.

6.2.1 Time in Classical and Quantum Mechanics

In Newtonian mechanics, time is an absolute parameter. The equations of motion are time-reversal invariant: if $\mathbf{x}(t)$ is a solution, so is $\mathbf{x}(-t)$ (with velocities reversed). There is no intrinsic arrow.

In quantum mechanics, time evolution is governed by the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle. \tag{6.1}$$

This equation is unitary and reversible. The apparent irreversibility of measurement is an interpretational issue, not a feature of the formalism itself.

6.2.2 Time in Quantum Gravity

In canonical approaches to quantum gravity, the Wheeler-DeWitt equation takes the form:

$$\hat{H}|\Psi\rangle = 0, \tag{6.2}$$

where $|\Psi\rangle$ is the wave function of the universe. This equation contains no time parameter; the universe is described by a static state satisfying a constraint equation [33].

The Page-Wootters mechanism [84] recovers effective time evolution from correlations between a "clock" subsystem and the rest of the universe within a timeless universal state. However, this mechanism explains how time *ordering* emerges from correlations but does not explain why this ordering is *asymmetric*.

6.2.3 The Common Thread

Across these theories, time appears either as an external parameter or as an emergent concept requiring additional input. The model below uses an external parameter and asks how records become asymmetric relative to a low-redundancy boundary; it does not constitute a third, input-free derivation of time.

6.3 Accessibility and Directionality

We now develop the central technical content in its conditional form: a specified record-forming dynamics can generate an ordered history relative to a low-redundancy boundary condition. The construction summarizes that history; it does not remove its dynamical or boundary-condition inputs.

6.3.1 Recap: The Redundancy Constraint

Paper E [157] treats redundancy as a property of a classical pointer label X and a supplied fragment family $\{E_k\}$. The relevant quantity is Holevo or explicitly accessible information $\chi(X : E_k)$, not an entropy assigned to an operator.

6.3.2 The Redundancy Index

We introduce a quantitative measure of redundancy:

Definition 6.2 (Redundancy Index)**.** *For a classical pointer label X, a joint classical–quantum state ρ_{XE}, and a declared fragmentation $E = \bigotimes_k E_k$, define*

$$\mathcal{R}_\delta(X;\rho,\{E_k\}) = \sum_{k=1}^{N} \Theta(\chi(X:E_k)_\rho - (1-\delta)H(X))\,, \tag{6.3}$$

with a finite-resolution label for continuous outcomes.

High redundancy supports objective readability of the pointer label. It does not imply a unique global outcome or characterize every noncommutative observable.

6.3.3 Asymmetry of Redundancy Flow

The earlier version of this paper stated a generic phase-volume proposition for redundancy growth. That proposition is withdrawn. Unitary spreading of correlations alone does not imply Darwinist redundancy: Haar-typical highly entangled states generally encode information nonlocally, so that small fragments reveal very little about a pointer label [83, 57]. High redundancy is instead characteristic of branching or record states generated by a suitable interaction from a low-record boundary condition. Appendix 6.9 gives such a conditional witness and also displays its exact unitary reversal; it does not derive a universal volume ratio.

6.3.4 Temporal Direction from Redundancy Gradient

Algebra inclusion and record redundancy supply two distinct comparisons. They need not point in the same direction: decoherence may increase the number of records while contracting the surviving noncommutative algebra toward a pointer algebra.

Definition 6.3 (Protocol-Relative Configuration Comparisons)**.** *Fix a dynamics, a fragment family, a pointer label X, and two configurations α, β along the resulting history. Define*

$$\begin{aligned} \mathcal{A}_\alpha \preceq_{\mathrm{inc}} \mathcal{A}_\beta \quad &\Longleftrightarrow \quad \mathcal{A}_\alpha \subseteq \mathcal{A}_\beta, \\ \alpha \preceq_{\mathcal{R}} \beta \quad &\Longleftrightarrow \quad \mathcal{R}_\delta(X;\rho_\alpha,\{E_k\}) \le \mathcal{R}_\delta(X;\rho_\beta,\{E_k\}). \end{aligned} \tag{6.4}$$

Each relation is a preorder on its declared domain. Their intersection may be used as a model-dependent comparison when both inequalities happen to be aligned, but no universal temporal partial order is asserted.

6.3.5 Audit of Temporal Inputs

The construction uses temporal orientation at several points: the Schrödinger parameter t, the unitary $U(t)$ (or a chosen dynamical semigroup), the comparison of record redundancy at different parameter values, the words "before" and "after," and a low-redundancy condition at one end of the history. Consequently, the algebraic comparisons above compress an already timed record-forming process. They do not derive a time parameter or its orientation from an unordered algebra alone.

6.3.6 The Statistical Nature of the Arrow

In the conditional record models considered here, the microscopic dynamics may remain time-reversal invariant while a low-record boundary condition selects an oriented history. Redundancy can then grow along the same orientation as the thermodynamic arrow. No generic measure on interactions or states is supplied from which an overwhelming phase-volume statement would follow.

> **A redundancy gradient is a record of an oriented dynamics, not an input-free definition of time.**

6.3.7 Relation to Thermodynamic Arrow

The accessibility arrow and the thermodynamic arrow are closely related but not identical:

Feature	**Thermodynamic Arrow**	**Accessibility Arrow**
Defined by	Entropy increase relative to a macrostate choice	Growth of records for a declared pointer label and fragmentation
Requires	Coarse-graining and a low-entropy boundary	Dynamics, fragmentation, and a low-redundancy boundary
Fundamental quantity	$S = -k_B \mathrm{Tr}(\rho \ln \rho)$	$\mathcal{R}_\delta(X; \rho, \{E_k\})$
Applies to	Specified thermodynamic descriptions	Specified branching/record models

Table 6.1: Comparison of thermodynamic and accessibility arrows.

The redundancy arrow is therefore an algebraic restatement of part of the records-based thermodynamic arrow, not a more fundamental replacement for it. Extending it to quantum-gravitational settings would require a concrete fragment structure and record dynamics there.

6.4 Causation as Constraint Propagation

Having specified a protocol-relative record direction, we ask what causal information can be summarized by it. The result is a diagnostic layered on an intervention model, not a derivation of causation from redundancy alone.

6.4.1 Causation Without Fundamental Time

Traditional accounts of causation distinguish temporal order from intervention or counterfactual dependence. Redundancy records can witness the former, but cannot by themselves supply the latter.

Definition 6.4 (Protocol-Relative Causal-Influence Diagnostic)**.** *Fix a record-forming protocol with external parameter t, pointer labels X_A, X_B, and an intervention family on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when:*

1. *both labels have operational records in the declared fragments;*
2. *the X_A records form no later than the X_B records according to t; and*
3. *interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B.*

This is a protocol-relative influence test. It is not asserted to be transitive, antisymmetric, or independent of the supplied intervention family.

Redundancy thus makes an influence readable across fragments; the intervention structure, not redundancy alone, carries the causal content.

6.4.2 Retrocausation Remains Open

A persistent question in philosophy of physics is whether retrocausation—effects preceding causes—is possible. The redundancy construction does not settle it. In the star model of Appendix 6.9, exact evolution by $U(-t)$ erases the records created by $U(t)$. A statistical suppression estimate would require a specified ensemble, macroregion, and measure; even an estimate proportional to $e^{-\alpha N}$ would be positive and would not establish physical uninstantiability. The earlier "Suppression of Retrocausation" proposition and its universal phase-volume claim are therefore withdrawn.

6.4.3 Causal Structure Without Spacetime

The diagnostic $\rightsquigarrow_{\mathsf{P}}$ is not a causal partial order. In particular, pairwise counterfactual dependence need not be transitive: an intervention on X_A can affect X_B, and one on X_B can affect X_C, without the stipulated intervention on X_A changing X_C after conditioning or cancellation. Any partial-order result would require additional assumptions such as an acyclic structural causal model and a faithful intervention semantics.

6.5 Relation to Existing Approaches

We situate the accessibility-based account relative to existing approaches to temporal asymmetry.

6.5.1 Comparison Table

6.5.2 Key Distinctions

Thermodynamic arrow: Redundancy growth describes the proliferation of records along an already oriented, low-record history. It is not more fundamental than the thermodynamic arrow and does not derive the relevant coarse-graining without mereological input.

Page-Wootters: Both approaches can organize correlations relationally. The present construction does not derive direction without a low-redundancy boundary condition.

Retrocausality programs: Some approaches explore retrocausal models [92]. The present record model neither excludes them nor derives their frequency; such a claim needs an ensemble and a measure beyond the model specified here.

Approach	**Source of Arrow**	**What It Presupposes**	**Relation to HAFF**
Thermodynamic	Entropy increase	Coarse-graining and past hypothesis	Redundancy tracks records within this arrow
Cosmological	Low-entropy Big Bang	Boundary conditions	Supplies rather than follows from the needed boundary
Decoherence	Interference suppression	System-environment split	Supplies pointer/record dynamics
Causal set	Fundamental partial order	Causal order as primitive	No derivation claimed here
Page-Wootters	Correlations in static $\|\Psi\rangle$	Timeless formulation	Record direction still needs boundary data
Accessibility	**Redundancy expansion**	**Interaction structure**	—

Table 6.2: Comparison of approaches to temporal asymmetry.

6.6 What This Paper Does NOT Claim

To prevent misreading, we state explicitly what this paper does *not* claim.

1. **No claim that time is unreal or illusory.** The framework reframes temporal asymmetry as emergent from accessibility structure, but this does not imply that time is "merely subjective" or non-existent.

2. **No adjudication between A-theory and B-theory of time.** The framework is compatible with both presentism and eternalism.

3. **No explanation of initial conditions.** We assume a low-redundancy boundary condition. This is a records version of a past hypothesis, not a consequence of the algebraic ordering [2, 129].

4. **No resolution of the problem of time in quantum gravity.** The framework clarifies what temporal asymmetry *means* in accessibility terms but does not derive time from the Wheeler-DeWitt equation.

5. **No claims about consciousness or subjective time.** The phenomenology of temporal experience is not addressed.

6. **No novel empirical predictions.** The analysis is structural, not dynamical.

7. **No probability claim about retrocausation.** No ensemble or measure has been supplied from which its frequency could be computed.

8. **No modification of quantum mechanics.** The framework assumes standard unitary evolution throughout.

6.7 Connection to HAFF Framework

This paper contributes a records-based diagnostic to the HAFF programme; it does not complete a closed derivation of geometry, measurement, and time.

6.7.1 The D + E + F Diagnostic Triangle

Papers D, E, and F address related but non-identical typed structures. Paper D studies conditional algebraic and modular inputs to geometry, Paper E separates probe effects from pointer records and exact fixed points, and the present paper studies records in a supplied dynamics. They should not be read as a closed derivation from one common algebra:

Paper	**Phenomenon**	**Traditional View**	**HAFF Reframing**
D	Gravity	Force between masses	Conditional geometry from selected algebraic data
E	Measurement	Primitive process	Probe effects and pointer/record sectors
F	Time	Fundamental parameter	Record ordering conditional on dynamics and boundary data

Table 6.3: The D + E + F diagnostic triangle.

The shared motivation is that effective geometry, measurement records, and temporal evidence depend on selection and stability structures. The revised papers do not establish a symmetric theorem-level unification of the three phenomena.

6.7.2 Structural Link: D + E + F

The three papers form a programme-level comparison rather than a symmetric closed structure:

- **D (Gravity)**: Geometry is conditional on specified modular and holographic reconstruction data.
- **E (Measurement)**: Probe reachability, pointer records, and dynamical fixed points are distinct typed objects; redundancy supports objective readability rather than a unique outcome.
- **F (Time)**: Redundancy growth can record an oriented history once dynamics, fragmentation, and a low-record boundary are supplied.

6.7.3 Boundary Note: Toward Layer III

The completion of Layer II (diagnostic unification) sets the stage for Layer III: the structural limits of the framework itself.

A question suggested by D + E + F is how their supplied selection data are themselves chosen and justified. Because the three papers no longer use a single untyped "accessible algebra," this is a family of interface questions rather than one closed fixed-point problem.

Layer III (Paper G) discusses this issue at the level of philosophy of science and metalanguage. Its non-self-grounding proposal is motivation only, not a theorem that closes the physical programme.

A theory that claims to explain everything must know where it must stop.

6.8 Conclusion

6.8.1 Summary of Results

We have proposed a structural account of temporal asymmetry and causation based on accessibility propagation. The central results are:

1. **Protocol-relative temporal direction**: Algebra inclusion and redundancy define preorders on a declared, already timed record-forming history. Their aligned intersection can supply a model-dependent direction, not a universal temporal partial order.

2. **Conditional accessibility arrow**: In the supplied low-record star model, redundancy grows along the chosen orientation. Generic expansion and exponential suppression of contraction are not established, and exact unitary reversal remains possible.

3. **Causal-influence diagnostic**: Redundant records can make an influence readable, while the causal content comes from the separately supplied intervention protocol.

4. **Retrocausation remains open**: No ensemble or measure is supplied that would make reversed trajectories measure zero or physically uninstantiable.

5. **Programme relation**: Papers D, E, and F offer related but differently typed structural diagnostics for gravity, measurement, and temporal records; they do not form one completed derivation.

6.8.2 Closing Remark

The arrow of time has puzzled physicists and philosophers for over a century. We do not claim to have dissolved this puzzle. What we have done is reframe it:

> The question is not only "Why does entropy increase?" but also "Under which record-forming conditions does accessibility acquire a preferred direction?"

The conditional star model shows one way in which such a direction can arise from an oriented boundary condition and specified interaction. It does not establish a generic Hilbert-space asymmetry. The remaining task is to characterize the ensembles, causal semantics, and boundary data under which the diagnostic extends beyond that witness.

Acknowledgments

The author thanks the anonymous reviewers for their insightful comments and suggestions, which greatly improved the clarity and rigor of this work.

6.9 A Conditional Star-Graph Record Model

This appendix gives a finite-dimensional witness that a specified measurement interaction can create redundant records from a low-record product state. The model is conditional: it neither makes record states Haar-generic nor derives an arrow without an external parameter and boundary condition.

6.9.1 The Star-Graph Interaction Setup

Consider a central system S (the "source" of accessibility) and an environment E consisting of N independent subsystems (fragments) $E_1, E_2, \ldots, E_N$. The total Hilbert space is:

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_S \otimes \mathcal{H}_{E_1} \otimes \mathcal{H}_{E_2} \otimes \cdots \otimes \mathcal{H}_{E_N}. \tag{6.5}$$

The interaction Hamiltonian is chosen to be of the "pre-measurement" type:

$$\hat{H}_{\text{int}} = g \sum_{k=1}^{N} \hat{O}_S \otimes \hat{M}_k, \tag{6.6}$$

where $\hat{O}_S$ carries the pointer label X, the $\hat{M}_k$ monitor that label, and g is the coupling strength. This star graph and the tensor factorization of the fragments are mereological inputs, not consequences of redundancy.

6.9.2 Dynamics of Redundancy

Assume the low-record product boundary condition

$$|\Psi(0)\rangle = \left(\sum_i c_i |o_i\rangle_S \right) \otimes |e_0\rangle_{E_1} \otimes \cdots \otimes |e_0\rangle_{E_N}. \tag{6.7}$$

This is a records version of a past hypothesis: it is assumed rather than derived [2, 129].

Under the unitary evolution $U(t) = e^{-i\hat{H}_{\text{int}}t/\hbar}$, the state evolves into an entangled superposition. For $\hat{O}_S$ with eigenstates $|o_i\rangle$:

$$|\Psi(t)\rangle = \sum_i c_i |o_i\rangle_S \otimes |E_i^{(1)}(t)\rangle \otimes |E_i^{(2)}(t)\rangle \otimes \cdots \otimes |E_i^{(N)}(t)\rangle, \tag{6.8}$$

where $|E_i^{(k)}(t)\rangle$ are the conditional states of the fragments.

For a declared prior $p_i = |c_i|^2$, the operational quantity is the Holevo information $\chi(X : E_k)_{\rho(t)}$. If the chosen interaction has a monitoring window in which the conditional states $|E_i^{(k)}(t)\rangle$ become sufficiently distinguishable, then fragments cross the threshold in Definition 6.2 and $\mathcal{R}_\delta(t)$ increases. An ideal controlled measurement can reach $\chi(X : E_k) = H(X)$ for every k, hence $\mathcal{R}_\delta = N$. This conclusion is a property of that interaction and window, not of arbitrary Hamiltonians.

6.9.3 Exact Reversal and the Absence of a Generic Volume Claim

The same model also makes the boundary-condition dependence explicit. Evolution by $U(-t) = U(t)^\dagger$ exactly reverses the record-forming unitary:

$$U(-t)|\Psi(t)\rangle = |\Psi(0)\rangle. \tag{6.9}$$

Thus record contraction is dynamically allowed. To assign it a probability or a phase-volume suppression one would have to declare a state ensemble, macroregions, and a measure. Dimension counting alone does not establish the former formula

$$\frac{\mathrm{Vol}(\mathcal{R}\downarrow)}{\mathrm{Vol}(\mathcal{R}\uparrow)} \sim e^{-\alpha N}, \tag{6.10}$$

which is withdrawn here.

Nor does generic high entanglement repair the argument. For Haar-typical pure states, small subsystems are nearly maximally mixed and information about a reference or pointer label is predominantly stored in nonlocal correlations [83, 57]. Small environmental fragments therefore carry little separately accessible information. Quantum-Darwinist redundancy is a special branching/record structure, not the generic large-volume sector of Hilbert space.

6.9.4 What the Witness Establishes

The model establishes only the following compatibility statement:

- a chosen branching interaction can convert a low-record product state into a state with many readable records;
- the same unitary has an exact record-erasing inverse; and
- interpreting the forward branch as an arrow requires the low-record boundary condition and the supplied orientation of t.

Accordingly, the accessibility comparison in the main text is an algebraic description of record formation within a timed model, not a proof of structural irreversibility.

Chapter 7

Structural Limits of Unification: Accessibility, Incompleteness, and a Programme-Level Final-Cut Diagnosis

Paper G

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Abstract

This paper offers a programme-level philosophy-of-science diagnosis of where an accessibility-based framework may encounter limits when asked to represent its own selection rules. The former structural non-self-grounding lemma is withdrawn: no typed fixed-point domain or map has been supplied, and partial internalization or nontrivial self-reference remains open. Regress and explanatory collapse are therefore illustrative scenarios under a particular explanatory demand, not an exhaustive theorem or a forced stopping point. The contribution is methodological: to identify the object/meta interface that a genuine self-grounding analysis would have to formalize and to motivate an architectural final cut without claiming its necessity.

7.1 Introduction

7.1.1 The Expectation of Completeness

The aspiration toward a unified physical description has historically been guided by the expectation that deeper unification corresponds to increased completeness. In this traditional view, apparent multiplicity—of forces, degrees of freedom, or explanatory principles—is taken to signal provisional fragmentation, to be resolved by a more fundamental theory. A Theory of Everything, in its strongest formulation, is therefore often assumed to be both unifying and self-grounding: it should not only subsume all known interactions under a single framework, but also account for the conditions under which its own descriptions are possible.

The present work examines a more restricted possibility: a given explanatory strategy may encounter a boundary when asked to represent its own selection rules. The discussion is a philosophy-of-science analysis in the metalanguage of the HAFF programme, not a theorem about all formal systems or a new physical no-go result. Its motivation comes from the preceding papers [153, 154, 155, 156, 157, 158], which build on the non-uniqueness of tensor factorizations [140, 141].

7.1.2 Summary of the Preceding Framework

Across Papers D–F, gravitational reconstruction, measurement records, and temporal evidence are discussed using related selection and stability structures. The revised papers use differently typed objects and make differently graded claims; they do not jointly prove that the three phenomena emerge from one "accessible algebra."

- **Paper D**: Geometric conclusions are conditional on modular and holographic reconstruction inputs.
- **Paper E**: Probe-reachable effects, pointer/record algebras, and exact dynamical fixed points are separated rather than intersected into one algebra.
- **Paper F**: Redundant records can grow in a supplied branching dynamics from a low-record boundary; the resulting order does not derive time from algebra alone.

These discussions share a programme-level question about how selection data enter physical description. That common question is motivation, not a theorem-level unification.

7.1.3 The Problem of Self-Grounding

If the explanatory programme depends on restrictions—on probe reachability, record stability, modular invariance, or reconstruction dictionaries—then it is legitimate to ask whether those restrictions can themselves be represented inside the same formal object. The answer depends on how object language, metalanguage, and encodings between them are specified; it is not fixed by the word "accessibility."

The aim of this final layer is therefore diagnostic: to articulate a possible regress when selection rules are repeatedly redescribed, to identify the extra formal data a genuine fixed-point theorem would require, and to explain why the present HAFF papers do not yet supply such a theorem.

7.1.4 Scope and Limitations

Several clarifications are necessary at the outset.

First, the incompleteness identified in this paper is not formal in the sense of Gödel's incompleteness theorems. We do not claim that the framework contains undecidable propositions within a formal system, nor do we invoke metamathematical results. The incompleteness is *architectural*: it concerns the explanatory roles within a physical framework, not provability within a formal calculus.

Second, we do not claim that the structural limits identified here apply universally to all conceivable approaches to unification. The present analysis is confined to the algebraic and coarse-graining-based framework developed in Papers A–F. Whether alternative formalisms—category-theoretic, non-algebraic, or radically background-free—would exhibit analogous limits is an open question that we do not address.

Third, the proposed stopping point is not temporal, existential, or normative. It is a methodological option for the present programme. A richer formalization could internalize some selection data without inconsistency, so necessity is not claimed.

7.1.5 Structure of the Paper

Section 7.2 distinguishes physical observables from syntax and meta-description. Section 7.3 replaces the former no-go lemma with a heuristic non-self-grounding observation and states its missing hypotheses. Section 7.4 gives an illustrative regress scenario rather than an exhaustive dichotomy. Section 7.5 presents the final cut as a methodological proposal. Section 7.6 states explicit non-claims. Section 7.7 concludes with reflections on the proposal's significance.

7.2 Accessibility as a Non-Global Constraint

7.2.1 The Role of Accessibility in HAFF

The HAFF papers use several notions previously grouped under "accessibility." Paper E now distinguishes probe-reachable effects, pointer/record sectors, and exact fixed points; Paper D uses selected modular-stable algebraic data; Paper F uses a declared fragmentation and record dynamics. Treating all of these as one endofunction on a space of algebras would erase their types.

In Papers D–F, this point is developed across distinct domains:

- Effective geometry is conditional on a reconstruction dictionary and modular hypotheses.
- Measurement objectivity concerns robust pointer records, not a proof of a unique outcome.
- Record directionality is conditional on dynamics, fragmentation, and a low-record boundary.

In each case, some selection data are supplied. The types and epistemic status of those data must be tracked separately.

7.2.2 Structural vs. Epistemic Constraints

A crucial distinction must be drawn between epistemic and structural constraints. This distinction is contested in philosophy of physics, and we acknowledge that what follows adopts it as a working criterion rather than a demonstrated result.

Definition 7.1 (Epistemic Constraint)**.** *A constraint is* ***epistemic*** *if it concerns what can be known, inferred, or verified by agents, given their informational position.*

Definition 7.2 (Structural Constraint)**.** *A constraint is* ***structural*** *if it concerns what descriptions are well-defined, given a pattern of physical interactions, independent of any agent's knowledge or epistemic state.*

The HAFF programme intends these inputs to be structural when they are fixed by dynamics and states, but their independence from a chosen factorization, fragment family, probe class, or reconstruction dictionary remains a substantive mereological question.

Given all of those inputs, different agents can apply the same rule. That conditional reproducibility does not by itself prove that the inputs are uniquely selected by the underlying physics.

Remark 7.3 (Contested Distinction)**.** *We acknowledge that this distinction between epistemic and structural constraints is philosophically contested. The framework does not claim to have resolved this broader debate. The burden is constructive: a structural claim should exhibit the physical data and selection map that make the relevant output invariant under admissible redescriptions.*

7.2.3 Non-Globality of Accessibility

The present papers do not provide a privileged, all-encompassing selected algebra from which all effective descriptions are derived. Each construction uses restrictions whose source is only partly analyzed. This is an open interface problem, not yet a proof that internal specification is impossible.

Before discussing internalization, one type distinction is essential. An element O of an observable algebra represents a physical observable. A formula, algorithm, or data structure specifying which algebra is selected belongs to a syntax or metalanguage until an explicit encoding maps it into physical degrees of freedom. Algebra membership alone does not mean that an algebra "contains its own description."

7.3 A Heuristic Observation on Self-Grounding

7.3.1 Status of the Former Lemma

The former "Structural Non-Self-Grounding Lemma" is withdrawn as a theorem. No formal domain of descriptions, accessibility endomap, order structure, or continuity/monotonicity hypothesis was defined, so the claimed absence of nontrivial fixed points did not follow.

The surviving heuristic is narrower: if every explanation of a selection rule is required to be given by another rule of exactly the same, still-unformalized kind, a regress can be generated by repeatedly asking what selects the next rule. This is a statement about a chosen explanatory demand. It does not establish inconsistency, collapse, or the nonexistence of a self-representing model.

7.3.2 Why the Fixed-Point Argument Does Not Prove a No-Go Result

The earlier argument can be separated into the following typed questions.

Object level. An observable algebra contains operators and their limits. It does not automatically contain formulas naming that algebra, proofs about its closure, or code implementing a selection rule. Identifying an observable $\hat{A}$ with a syntactic description of "accessibility" is therefore a type error unless an encoding and decoding map are supplied.

Meta level. Selection criteria are presently written in the theorist's metalanguage. One may ask whether a physical subsystem can encode them, but this requires a representation of syntax, an evaluation map, and a criterion for faithful self-application. None is defined in HAFF Papers A–F.

Fixed points. Even after such a formalization, self-application does not imply nonexistence. Fixed-point and diagonal results routinely produce nontrivial self-reference under suitable hypotheses: the Kleene recursion theorem, Lawvere's fixed-point formulation, Scott's domain-theoretic fixed points, and Tarski's lattice theorem are standard counterweights to a blanket exclusion [66, 72, 103, 116]. They do not prove that HAFF has the desired fixed point; they show that its absence must be demonstrated from a specified map and domain rather than inferred from the word "self-reference."

What a genuine theorem would require. A mathematical non-self-grounding theorem would at least have to specify:

- a space or category of descriptions and the type of each object;
- a selection or evaluation map with a well-defined codomain;
- the encoding that connects syntax or meta-data to physical observables;
- the admissible fixed points and why known fixed-point hypotheses fail; and
- a proved exhaustiveness statement if regress and totalization are claimed to be the only outcomes.

Remark 7.4 (Heuristic Status)**.** *Until those data are supplied, non-self-grounding is a plausible philosophy-of-science diagnosis of the programme's current explanatory architecture. It is not a lemma, and alternatives such as partial internalization, stratified languages, reflective encodings, or nontrivial fixed points remain open.*

7.4 An Illustrative Regress Scenario

To make the heuristic concrete, consider one particular extension that tries to internalize a selection rule without first formalizing the object/meta interface. The example illustrates how a regress can arise; it does not prove that every internalization fails.

7.4.1 Hypothetical Extension

Step 1: Choose an encoding. Suppose a physical register R is introduced to encode descriptions of candidate subalgebras and their selection scores. The register observables are ordinary algebra elements; a separate map must say which register states encode which syntactic descriptions.

Step 2: Supply dynamics and evaluation. For the encoding to do explanatory work, one must specify:

- dynamics for R and its coupling to the system; and
- an evaluation procedure that maps an encoded rule and physical data to a selected output.

Step 3: Reapply the same explanatory demand. If the programme insists that the encoding, evaluation map, and register dynamics must themselves be selected by another rule of the same kind, then a new datum is introduced at each level. Repeating that demand produces a regress by construction.

7.4.2 Two Possible Outcomes, Not an Exhaustive Dichotomy

Outcome (a): Regress under unrestricted reapplication. If no base rule, fixed point, equivalence class, or stopping convention is admitted, the explanatory request repeats indefinitely. This is a consequence of the stipulated demand, not a contradiction in the observable algebra.

Outcome (b): Failure of a naive totalization. One attempted stopping convention is to declare every effect or observable equally selected. In models where the physical content depends on a nontrivial probe class, pointer sector, or reconstruction subalgebra, that convention erases the contrast being modeled:

- no nontrivial selection remains;
- pointer records are no longer distinguished from arbitrary observables; and
- any geometry or record order conditional on the selected structure loses that input.

This shows that one totalization proposal is unhelpful. It does not exclude typed fixed points, reflective sublanguages, externally supplied base data, or other ways of terminating the regress.

7.4.3 Conclusion of the Scenario

The scenario supports a methodological warning: internalization must preserve the object/meta type distinction and specify its evaluation map. Without those ingredients, claims of either self-grounding or necessary collapse are premature.

7.5 A Proposed Final Cut

7.5.1 Stopping as a Methodological Choice

The preceding analysis motivates a limited sense in which the current HAFF programme is incomplete: it has not formalized how all of its selection inputs are themselves chosen. This is neither a formal incompleteness theorem nor a metaphysical limit, and it may be addressed by future interface constructions.

Within the programme, geometry, measurement records, and temporal records all use supplied selection or stability data, but of different types and at different evidential grades. The present paper asks where to stop redescribing those inputs; it does not prove that no further physical account can be given.

7.5.2 The Final Cut

The notion of a "final cut" is introduced as a proposed bookkeeping boundary. It does not denote a temporal endpoint or the completion of physics. It marks where the current papers stop and expose their selection data as inputs rather than presenting a further unproved closure claim.

This cut is revisable. A future typed model of self-description, a nontrivial fixed point, or a physical derivation of the selection data could move it.

Beyond this point, a further extension must state how descriptions of selection rules are represented and evaluated. The heuristic in Section 7.3 identifies that obligation but does not show that it admits no nondegenerate solution.

The stopping point is therefore selected provisionally so that an open interface is not disguised as a theorem.

7.5.3 Incompleteness as Internal Limit

In this sense, the present incompleteness is programme-relative and provisional: the formalism has not supplied the typed self-description data listed above.

The phrase "unselected structure" is therefore interpretive shorthand, not a demonstrated domain that admits no geometry, temporal ordering, or observational standpoint.

A theory can be clearer by marking where its current derivations stop and its interface assumptions begin.

Remark 7.5 (Open Alternatives)**.** *Section 7.4 gives two outcomes for one naive internalization strategy, not an exhaustive dichotomy. A proposed extension should specify how it handles encoding, evaluation, and repeated selection; whether it does so by a base datum, stratification, or fixed point is left open.*

7.6 Scope and Non-Claims

To prevent misinterpretation, several non-claims must be stated explicitly.

1. **No invocation of Gödel's theorems.** This work does not invoke Gödel's incompleteness theorems, nor does it rely on any formal analogy to them. The incompleteness identified here is architectural, not metamathematical. It concerns explanatory roles within a physical framework, not provability within a formal system.

2. **No claim of universal applicability.** No claim is made that accessibility constraints apply universally across all conceivable approaches to unification. The present analysis is confined to the algebraic and coarse-graining-based framework developed in Papers A–F.

3. **Contested distinction acknowledged.** The distinction between structural and epistemic constraints is acknowledged to be contested in the philosophy of

physics. The framework adopts this distinction as a working criterion, grounded in interaction-determined stability conditions. It does not claim to have resolved the broader philosophical debate.

4. **No foreclosure of further formalization.** The proposed final cut does not preclude further physical progress or a typed self-description model within HAFF. Further internal extension has not been shown incoherent.

5. **No uniqueness or necessity claim.** Other frameworks may stop elsewhere, and future versions of this programme may move or remove the proposed cut.

6. **No philosophical finality.** This is a structural observation within a specific formalism, not a philosophical conclusion about the ultimate nature of reality or the limits of knowledge as such. The stopping point identified is methodological and structural, not existential.

7. **No consciousness or observer-creation claims.** The framework does not claim that consciousness plays a fundamental role, that observers create reality, or that accessibility is observer-relative in any subjective sense.

8. **No claim of novelty regarding theoretical presupposition.** We acknowledge that explanatory frameworks can have presuppositions they do not presently explain. The contribution here is to identify where this issue appears in HAFF and what a formal replacement for the withdrawn lemma would need.

Any downstream "No View from Nowhere" closure principle that relied on the former lemma must therefore be conditionalized. Likewise, any later Paper III use of this discussion is motivation-only unless it supplies the missing typed fixed-point argument. No downstream theorem is established here.

7.7 Conclusion

7.7.1 Summary of Results

This paper has examined conditions under which the HAFF framework may encounter a boundary in extending its own explanatory strategy.

The central results are:

1. **Accessibility-specification asymmetry**: HAFF descriptions use selection and stability data that are presently stated in a theorist's metalanguage rather than derived from one common object-level formalism.

2. **Structural non-self-grounding (programme-level diagnosis)**: The current framework exhibits an asymmetry between accessibility-specification and accessibility-governed descriptions; a formal fixed-point exclusion remains open (Section 7.3).

3. **Illustrative regress scenario**: Repeatedly demanding a selector of the same unformalized kind can generate regress or totalization pressure. This does not show that every typed internalization fails.

4. **Architectural hypothesis**: A final cut may serve as a programme-level stopping rule while the formal self-grounding problem remains unresolved; it is structural motivation, not a proved necessity.

7.7.2 Methodological Significance

Unification is often equated with the elimination of boundaries. The analysis presented here suggests a different criterion: that a unifying framework should distinguish between boundaries that are provisional and those that are structural.

Papers D–F motivate related boundary questions in the treatment of gravity, measurement, and temporal records. This final layer proposes that making those open interfaces explicit can improve explanatory clarity without turning them into an impossibility theorem.

The value of the framework lies not in claiming to determine what cannot be said, but in identifying which selection and self-description interfaces remain unformalized. Treating a final cut as an architectural hypothesis keeps that limitation visible while leaving stronger fixed-point constructions open.

7.7.3 Open Questions

Several questions remain beyond the scope of this analysis:

- Whether alternative frameworks (category-theoretic, non-algebraic, or background-free) would exhibit analogous structural limits.
- Whether the structural/epistemic distinction adopted here can be given a more robust philosophical foundation.
- Whether the collapse scenario admits any non-trivial avoidance strategies not considered here.

These questions are left for future investigation.

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Chapter 8

Postscript: On the Closure of Structure

Postscript

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In the Holographic Alaya–Field Framework, the universe has been treated not as a collection of fundamental objects, but as a bounded domain of accessibility within a single global operator structure [153, 154]. Throughout this work, gravity, time, and measurement have appeared only insofar as stable distinctions are sustained by a persistent separation—what we have called the Cut—between accessible subalgebras and the total algebra [157, 159]. The philosophical implications of this structural stance—for causation, agency, and existence—have been explored in the accompanying essay [155].

Pushing this framework to its logical limit raises a natural question: what becomes of the theory when such distinctions can no longer be maintained?

Heat Death as Accessibility Saturation

From a structural perspective, the conventional notion of heat death admits a reinterpretation. Rather than signifying the disappearance of physical existence, it corresponds to the saturation of accessibility. On a heuristic redundancy picture, stably recordable differences between subsystems are progressively lost [143]. In that reading, the distinctions that the framework associates with locality, temporal direction, and effective classicality become unsustainable.

In this limit, the Cut loses operational meaning. No stable partition remains that could support observers, records, or localized descriptions. The system approaches the undifferentiated operator structure introduced at the beginning of this work—a state of maximal symmetry and minimal distinguishability.

Structural Equivalence of Origin and Terminus

Crucially, this endpoint is not structurally distinct from the origin. The state of maximal entropy reached at late times is, in algebraic terms, indistinguishable from the maximally symmetric pre-differentiated configuration. The difference between "beginning" and "end" is therefore not ontological, but structural: it reflects whether accessibility constraints are present or dissolved.

Schematically, let $\mathcal{A}_{\text{total}}$ denote the full operator algebra and let $S[\rho]$ denote the von Neumann entropy of a state ρ. The proposed comparison of the two temporal extremes may be written as

$$\lim_{t\to 0^+} S[\rho(t)] \approx \lim_{t\to\infty} S[\rho(t)] \approx S_{\max}, \tag{8.1}$$

where the limits are understood in terms of accessible structure rather than absolute time. This is an interpretive schematic, not a result derived from specified cosmological dynamics. The initial state (pre-Cut) and the final state (post-dissolution) occupy the same region of algebraic configuration space—both correspond to conditions under which no stable coarse-graining can be sustained.

Bounded Evolution Without Cyclicity

Seen this way, cosmic evolution traces neither a linear narrative nor a teleological arc. It is instead bounded by two structurally equivalent limits: one preceding the emergence of stable distinctions, and one following their dissolution. The domain in which physics, observation, and meaning are possible occupies only the intermediate regime, where the Cut is sustained.

This observation carries no additional dynamical claims, nor does it posit a cosmological cycle in the sense of a Big Bounce or oscillating universe model [89]. It marks the intended conceptual boundary of the framework developed here. The theory describes the conditions under which structure can appear, persist, and ultimately fail. Beyond those conditions, no further physical description is available—not because reality ends, but because the criteria for description are no longer satisfied.

The Contingency of Intelligibility

If the work has a final implication, it is a modest one: intelligibility itself is contingent. The universe is describable only while distinctions endure. Understanding this boundary does not diminish the value of structure; it clarifies the narrow window in which structure—and thus physics—is possible.

The framework terminates not in absence, but at the level of unselected structure: a domain relative to which geometry, temporal direction, and observational standpoint are selectively realized.

Remark 8.1 (On Structural Closure)**.** *The identification of origin and terminus as algebraically equivalent does not constitute a prediction about cosmological dynamics. It is a statement about the explanatory boundaries of accessibility-based description. Within those boundaries, the framework offers related, differently typed structural diagnostics for gravity, measurement, and time [156, 157, 158]. Beyond them, the framework offers no further description in terms of accessible algebras; this marks an interpretive boundary of the programme rather than a proven impossibility [159].*

Concluding Reflection

The value of a theoretical framework lies not only in what it explains, but in what it determines cannot be explained without loss of coherence. The stopping point identified

in this work is not a failure of explanation but its completion. A theory that claims to explain everything must know where it must stop.

Clarity does not require totality. And knowing where to stop is sometimes the most precise act of understanding.

Part II

Q-RAIF: Quantum Reference Algebra for Information Flow

Chapter 9

A Clifford-Encoding Model for Lorentzian Metrics in Entropic Gravity Frameworks

Paper A — "The Water"

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Abstract

We investigate a modelling route by which an emergent bulk description may be equipped with a Lorentzian signature within the framework of entropic gravity. While thermodynamic approaches to gravity [61, 124] and the Ryu–Takayanagi formula [97] relate entanglement entropy to geometric data, the specific algebraic mechanism constraining the spacetime signature remains an open question.

Given a supplied quadratic space (V, q) and Clifford generator relations, we adopt the **Clifford algebra** $Cl(V, q)$ as the universal minimal algebra for that presentation. Associativity, metric compatibility, and causal-channel encoding are used as compatibility criteria for this modelling choice; they do not independently derive the Clifford relations or select a unique boundary algebra. We compare alternative algebraic frameworks (von Neumann factors, C^*-algebras, Jordan algebras, Lie algebras) by recording whether they encode the supplied form internally or require additional representation data. The comparison is not a general exclusion theorem for those frameworks. A worked example using a qubit tensor network illustrates how the three constraints operate in a concrete setting.

The analysis complements the conditional algebra-and-reconstruction setting of the Holographic Alaya-Field Framework (HAFF) [153, 154]. It studies one compatible Lorentzian encoding after the algebraic and reconstruction data are supplied; it does not characterize constraints that every emergent geometry must satisfy.

Keywords: emergent geometry, Clifford algebra, entropic gravity, holographic principle, Lorentzian encoding, algebraic quantum gravity

9.1 Introduction

9.1.1 Context and Motivation

The AdS/CFT correspondence [78] and the Ryu–Takayanagi formula [97] have established that spacetime geometry can be viewed as an emergent property of quantum entanglement. Thermodynamic approaches [61, 124] further suggest that the Einstein equations arise as an equation of state. Yet a critical question remains: *What algebraic constraints ensure that the emergent geometry is Lorentzian?*

Most models assume the $(1, 3)$ signature *a priori*. We argue that this assumption can be partially justified by examining the algebraic consistency conditions on the boundary degrees of freedom.

9.1.2 Relation to HAFF

The Holographic Alaya-Field Framework [153, 154] studies how selected observable-algebra and reconstruction data can produce coarse-graining-dependent effective geometry. The present work asks how a Lorentzian quadratic form could be encoded algebraically once such a read-out and its signature are supplied.

HAFF	Geometry is coarse-graining-dependent (the "ocean")
This paper	A Lorentzian Clifford encoding is modelled (the "water")

9.1.3 Scope and Disclaimers

This work does not propose a new fundamental theory of gravity and does not derive $Cl(1, 3)$ from first principles. It compares algebraic frameworks after a Lorentzian quadratic form is supplied and records why Clifford algebra is a compact universal encoding of the imposed generator relations. It is a model-choice analysis, not a selection or uniqueness proof.

The specific value $(1, 3)$ for the signature requires additional input beyond the algebraic constraints developed here (e.g., observational dimensionality or anomaly cancellation arguments). We do not address why spacetime has $3 + 1$ dimensions.

Result-status ledger. Associativity of operator composition is a general operator-algebra observation. The information-geometric discussion below is a heuristic bridge from a positive control metric to bulk reconstruction. Lorentzian signature and Clifford encoding are separate interface/model inputs. The former "Clifford Compatibility Theorem" is retracted as a selection theorem: once the Clifford relation is imposed, the universal-property conclusion is definitional.

9.2 Candidate Algebraic Structures

Before deriving constraints, we survey the landscape of algebraic structures that could, in principle, describe boundary degrees of freedom in a holographic setting. This survey serves as the basis for the exclusion argument in Section 9.3.4.

Definition 9.1 (Boundary Algebra)**.** *Let $\mathcal{A}_\partial$ be the algebra of observables on a holographic boundary. We require: (a) $\mathcal{A}_\partial$ acts faithfully on $\mathcal{H}_\partial$; and (b) a declared coarse-graining/reconstruction map associates it with an effective bulk description. Entropic data are regime-dependent. Finite-dimensional or tracial models may use a normalized trace and density matrices. A Type III boundary factor has no faithful normal trace; there one must instead specify faithful normal states, relative entropy, and modular data. No trace requirement is imposed across both regimes.*

The following algebraic families are candidates:

1. **von Neumann algebras** (Type I, II, III): Associative, closed under adjoint, weakly closed. Standard in algebraic QFT [50]. Type III_1 factors are generic in relativistic QFT.

2. **C^*-algebras**: Associative Banach algebras with involution. More general than von Neumann algebras. Standard framework for quantum observables.

3. **Lie algebras**: Antisymmetric bracket $[A, B] = -[B, A]$, satisfying the Jacobi identity. Encode infinitesimal symmetries. The universal enveloping algebra is associative.

4. **Jordan algebras**: Commutative but generally non-associative: $A \circ B = B \circ A$, satisfying the Jordan identity. Proposed for quantum mechanics by Jordan, von Neumann, and Wigner (1934).

5. **Octonion algebras**: Non-associative division algebra. Explored in the context of exceptional structures in string theory [49].

6. **Clifford algebras** $Cl(V, q)$: Associative, generated by a vector space V with quadratic form q, subject to $v^2 = q(v)\mathbf{1}$. Encode both metric and algebraic structure [58, 36].

9.3 Three Algebraic Constraints

We now organize three physically motivated requirements and compare how candidate algebras represent them. Only the first is a declared operator-composition boundary; the second is a holographic interface heuristic, and the third is an explicit Lorentzian/Clifford model input.

9.3.1 Constraint I: Associativity

Remark 9.2 (Associativity as a structural boundary)**.** *This paper represents sequential operations by linear maps with ordinary composition, which is associative by definition. A semigroup of evolution maps consequently obeys $(U(t_1)U(t_2))U(t_3) = U(t_1)(U(t_2)U(t_3))$. That identity constrains the chosen evolution representation; it does not prove that every possible ambient non-associative product is inconsistent. Non-associative models require extra bracketing and composition data and lie outside the operator-algebra regime declared here [101, 49].*

Remark 9.3 (Physical pathologies of non-associative dynamics)**.** *Non-associative models do not inherit the ordinary bracketing-independent calculus of sequential linear maps. Octonionic and Jordan formulations can be studied with additional axioms, but standard tensor products, subsystem composition, and quantum-error-correction constructions are then not automatic [49, 3]. This observation explains the present scope restriction; it is not a no-go theorem for every non-associative physical theory.*

Scope consequence: Jordan and octonion algebras are not candidates within the declared associative operator-composition class unless an associative representation and the required composition data are supplied separately.

9.3.2 Constraint II: Metric Compatibility

Remark 9.4 (Information-metric bridge (heuristic))**.** *Some holographic reconstruction and error-correction models equip a state or code manifold with a positive distinguishability metric. If one additionally chooses a gradient correction protocol, its local parameter dynamics may take the form*

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \tag{9.1}$$

where G is a supplied positive, non-degenerate metric on the relevant parameter directions. Neither quantum error correction in general nor the abstract boundary algebra forces this particular protocol.

The metric G is information-geometric or control-theoretic, not the emergent spacetime metric $g_{\mu\nu}$. In specified holographic dictionaries, relative entropy, modular Hamiltonians, and entanglement first laws constrain linearized bulk reconstruction [90, 39, 69]; those results do not turn a positive information metric into a Lorentzian form or supply a canonical bilinear form for every von Neumann algebra. The present link is therefore an interface heuristic, not a theorem or an exclusion criterion.

Comparison: Lie, C^*, and von Neumann algebras can all be paired with model-dependent positive forms or information metrics. The absence of a unique form intrinsic to the bare algebra does not exclude them; it identifies additional representation data that the model must state.

Remark 9.5 (Quantitative cost of external metrics)**.** *For an algebra generated by n directions, a generic symmetric bilinear form has $n(n+1)/2$ components in a supplied basis. Writing the same form in Clifford relations does not erase that information: the anticommutators contain the same matrix q_{ij}, and diagonalization also requires the frame transformation. A fixed canonical signature can reduce recurring description cost only after that frame and form have been selected.*

Accordingly, no basis-independent $O(1)$ versus $O(n^2)$ advantage is proved here, and no thermodynamic necessity follows from T-DOME. At most, a pre-agreed Clifford presentation can package a selected metric together with the multiplication rules used by a particular protocol.

9.3.3 Constraint III: Causal Channel Encoding

Lorentzian read-out requirement (interface input). If a reconstruction is required to yield a Lorentzian causal cone, then its spacetime tangent data must carry an indefinite quadratic form of signature (p, q) with $p, q \geq 1$. This statement is a requirement

on the intended read-out, not a derivation from the positive information metric in Constraint II. Bures and Fisher–Rao metrics remain positive definite because they control distinguishability or descent on state space; they are not candidate spacetime metrics.

Encoding the supplied Lorentzian form by generators satisfying

$$\{e_\mu, e_\nu\} = 2q_{\mu\nu}\mathbf{1} \tag{9.2}$$

is the Clifford model choice. Once this relation is imposed, both the quadratic form and its signature are already present in the premise. Microcausality of region algebras does not by itself derive these generator relations, and algebraic commutation should not be identified with the sign of a tangent vector. The value $(1, 3)$ additionally requires dimensional and signature input not supplied here.

9.3.4 Comparison of Alternative Algebraic Encodings

We now systematically evaluate each candidate from Section 9.2:

Algebra	I: Assoc.	II: Metric	III: Indef.	Status
von Neumann (Type III_1)	✓	external	external	Compatible, not canonical
C^*-algebra (general)	✓	external	external	Compatible, not canonical
Lie algebra	via $U(\mathfrak{g})$	optional	optional	Needs representation data
Jordan algebra	non-assoc.	optional	optional	Outside operator-composition setup
Octonion algebra	non-assoc.	built-in	optional	Outside operator-composition setup
Clifford $Cl(V, q)$	✓	built-in	chosen in q	**Minimal encoding choice**

Table 9.1: Comparison of candidate encodings. Type III and general C^* algebras are not inconsistent with Lorentzian read-out; they simply do not select a preferred quadratic form without additional representation data.

Clifford algebras are distinguished by having the chosen quadratic form q built into the defining relation $v^2 = q(v)\mathbf{1}$. Other associative algebras can contain Clifford representations or be equipped with metric data, but do not select a canonical one in general. The table therefore identifies a compact encoding choice, not an exclusion or necessity theorem.

9.3.5 Worked Example: Qubit Tensor Network

To illustrate how the three constraints operate concretely, consider a tensor network model of holographic bulk reconstruction.

Setup. Take N qubits arranged on a MERA (multiscale entanglement renormalization ansatz) tensor network [114, 126]. The boundary algebra is generated by tensor products of Pauli operators $\{\sigma_x, \sigma_y, \sigma_z\}$ acting on individual qubits.

Constraint I. The Pauli algebra is associative (it consists of 2×2 matrices). If we were to replace the Pauli operators with elements of an octonion algebra (which is non-associative), the isometry conditions defining the MERA network—specifically, $V^\dagger V = \mathbf{1}$, which requires associative composition—would fail.

Constraint II. The Pauli operators satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, which defines a *positive-definite* quadratic form. This is the Clifford algebra $Cl(3,0)$ (since $\sigma_i^2 = +\mathbf{1}$, the quadratic form is positive-definite). Note that $Cl(3,0) \cong M_2(\mathbb{C})$ while $Cl(0,3) \cong \mathbb{H} \oplus \mathbb{H}$ as real algebras; these are *not* isomorphic. The quadratic form is built into the anticommutation relation.

Constraint III. The Pauli algebra alone encodes a Euclidean signature $(3,0)$. There is no operator on $\mathbb{C}^2$ that anticommutes with all three Pauli matrices, and a generic modular Hamiltonian K satisfies neither $K^2 = \mathbf{1}$ nor $\{K, \sigma_i\} = 0$. The former two-dimensional construction with iK is therefore withdrawn.

A genuine chosen complex representation acts on $\mathbb{C}^2 \otimes \mathbb{C}^2$. With

$$\gamma_0 = \sigma_z \otimes \mathbf{1}, \qquad \gamma_j = i\sigma_y \otimes \sigma_j \quad (j = 1,2,3), \tag{9.3}$$

one checks $\gamma_0^2 = +\mathbf{1}$, $\gamma_j^2 = -\mathbf{1}$, and $\{\gamma_\mu, \gamma_\nu\} = 0$ for $\mu \neq \nu$. With the convention that $Cl(p,q)$ has p positive and q negative generators, this realizes the imposed $Cl(1,3)$ relations. It is a model check, not a derivation of the time-like generator from modular data.

Lesson. The example shows that a larger matrix representation can implement a chosen Lorentzian Clifford encoding. It does not show that MERA dynamics or modular flow selects that encoding.

9.3.6 The Clifford Universal-Property Observation

The former Clifford Compatibility Theorem is retracted as a selection theorem. If a vector space (V,q) and maps of generators satisfying $v^2 = q(v)\mathbf{1}$ are already posited, then the universal property of $Cl(V,q)$ supplies the corresponding factorization through the Clifford algebra [58, 36]. This is the defining universal property, not an independent physical derivation of the generator relations, the signature, or the preference for a minimal realization.

Remark 9.6 (Scope of the model choice)**.** *Clifford algebra is a natural minimal framework once Clifford generator relations and a quadratic form are chosen. Larger structures and Type III representations remain compatible. The physical problem is to derive or select the generator map upstream; this paper does not solve it.*

The ingredients have distinct status: Remark 9.2 (associativity as a declared operator-composition boundary), Remark 9.4 (a heuristic information-metric interface in specified holographic models), and the independent Lorentzian/Clifford interface choice above. The positive information metric in the second item does not produce the indefinite form in the third.

9.4 Entropic Gravity from Algebraic Structure

9.4.1 Holographic Screen and Einstein Equations

Following Jacobson [61], a local horizon thermodynamic argument recovers the Einstein equation under specific equilibrium, entropy-area, and local Lorentz assumptions. Choosing a $Cl(1,3)$ representation supplies an algebraic representation of $\mathrm{Spin}(1,3)$; it does not derive those thermodynamic or reconstruction assumptions.

9.4.2 Relation to HAFF Emergence Chain

Within HAFF, geometry emerges via:

$$\text{Observable Algebra} \to \text{Representation} \to \text{Entanglement} \to \text{Connectivity} \to \text{Geometry}$$

The present work describes one model for the final arrow: when a geometrically admissible coarse-graining [153] is separately required to produce a Lorentzian read-out, a Clifford representation can encode its tangent quadratic form.

9.4.3 Relation to Algebraic QFT

In algebraic quantum field theory (AQFT) [50], local observable algebras associated with spacetime regions are generically Type III_1 von Neumann factors. These algebras are associative and support rich mathematical structure, but they do not carry a canonical metric of indefinite signature.

The connection to Lorentzian structure emerges through the Tomita–Takesaki theorem: for any cyclic and separating state, the modular operator Δ generates a one-parameter group (modular flow) that, in the Bisognano–Wichmann theorem, coincides with the boost generator in Rindler spacetime. In special settings such as wedges, modular flow has a geometric boost interpretation. Generic modular flow does not by itself supply a normalized Clifford generator or a complete Lorentzian tangent frame.

In the HAFF framework, the accessible algebra $\mathcal{A}_{\mathbf{c}}$ can be understood as a stable subalgebra of a Type III_1 factor. The present analysis uses a $Cl(1,3)$ representation as a possible encoding when a Lorentzian bulk description and a suitable generator map are independently available. Deriving that map from modular flow and locality remains open.

This perspective connects the present work to Witten's observation [135] that Type III_1 algebras are essential in gravitational settings, and to Connes' spectral-triple programme [28], where Dirac/Clifford data encode Riemannian geometry. Extending spectral triples to Lorentzian signature is itself a nontrivial open structural problem; representative approaches include Barrett and Lorentzian/Krein spectral triples [12, 86, 122]. The present model choice does not resolve those difficulties.

9.5 Discussion

9.5.1 What This Result Does and Does Not Show

Does show: Given a chosen quadratic space and Clifford generator relations, $Cl(V,q)$ is the universal minimal algebra for that presentation. Table 9.1 compares which candidate frameworks encode the chosen form internally and which require external representation data. Section 9.3.5 gives an explicit 4×4 Lorentzian Clifford representation.

Does not show: That the Clifford relations or Lorentzian signature follow from the positive information metric. Why $3+1$ dimensions rather than some other (p,q)—the argument constrains to $Cl(p,q)$ for any $p \geq 1$; the value $(1,3)$ requires additional input. That gravity *is* entropic—we derive consistency conditions within the entropic gravity framework. That $Cl(1,3)$ is the *unique* boundary algebra—larger algebras containing $Cl(1,3)$ as a subalgebra are also compatible. A complete theory of quantum gravity.

9.5.2 Convergence with Paper B

The companion paper [161] also uses $Cl(V, q)$, but its revised stability analysis does not derive an indefinite Clifford form. Both papers now treat Clifford structure as a modelling input for different read-out roles.

	Paper A (this work)	**Paper B**
Question	What algebra does geometry need?	What algebra does persistence need?
Method	Lorentzian encoding choice	Positive-metric stability analysis
Perspective	The world ("ocean")	The subsystem ("fish")
Result	$Cl(1, 3)$ model representation	$Cl(V, q)$ model parameterization

Because both routes posit rather than independently derive Clifford relations, there is no independent convergence result. The shared choice is a programme-level comparison whose physical selection remains an open problem.

9.5.3 Open Problems

1. **Dimensionality**: What additional constraints (anomaly cancellation, stability of persistent subsystems, observational input) fix $(p, q) = (1, 3)$?

2. **Constructive derivation**: Can Clifford generators be explicitly constructed from modular Hamiltonians or Tomita–Takesaki data in holographic models?

3. **Relation to noncommutative geometry**: How does the present analysis connect to Connes' spectral triples, where Clifford algebras characterize the Dirac operator?

4. **Tensor network realization**: Can the qubit toy model of Section 9.3.5 be made rigorous in the context of holographic error-correcting codes?

9.6 Conclusion

Within a supplied quadratic space and chosen Clifford generator relations, $Cl(V, q)$ is the universal minimal algebra for that presentation. The comparison with von Neumann, C^*, Jordan, Lie, and octonionic frameworks identifies where the chosen form is internal and where extra representation data are required; it does not exclude those frameworks in general or derive the Clifford presentation from the compatibility criteria alone.

This model provides a possible Lorentzian encoding within HAFF's conditional selected-algebra and reconstruction setting. It is neither a derivation of Lorentzian gravity from first principles nor a proof that every emergent causal geometry must be Clifford-compatible; physical selection of the quadratic data and generator map remains open.

Chapter 10

Thermodynamic Stability Constraints on the Operator Algebra of Persistent Open Quantum Subsystems

Paper B — "The Fish"

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Abstract

We investigate algebraic modelling choices for open quantum subsystems that maintain a non-equilibrium steady state (NESS) under continuous environmental interaction. We formulate persistence as a variational and Lyapunov-stability problem, while keeping the scope of the algebraic claim explicit.

The analysis isolates two structural desiderata for a control parameterization: associativity of the operator product and a supplied quadratic generator relation when channel directions and magnitudes are to be encoded together. Clifford algebra $Cl(V, q)$ is a compact way to package such a selected quadratic form with control generators, but the paper does not prove that Clifford structure is necessary, uniquely selected, or forced by persistence.

A worked controlled-qubit example records what a full Lyapunov proof would have to check. The comparison with von Neumann, C^*, and Lie algebraic frameworks is therefore a modelling audit, not an exclusion theorem. The analysis complements the Holographic Alaya-Field Framework (HAFF) [153, 154] and the companion paper on algebraic constraints in entropic gravity [160].

10.1 Introduction

The interaction of a quantum system with a large environment typically leads to decoherence and thermalization [20, 132]. Maintaining a NESS requires continuous energetic cost [104, 63], which can be modeled as a feedback control process [98, 85]. We address: *What algebraic structures allow the internal control dynamics to remain Lyapunov stable?*

10.1.1 Three-Paper Structure

Paper	Question	Analogy
HAFF [153]	How does geometry emerge?	Ocean
Q-RAIF A [160]	What algebra does geometry need?	Water
This work	What algebra does survival need?	Fish

10.1.2 Anti-Solipsism Disclaimer

A potential misreading is that the observer "creates" geometry through survival. We explicitly reject this. The revised claim is model-relative: Pauli/Clifford generators are one convenient parameterization of feedback channels, but persistence does not force an indefinite Clifford algebra. Within HAFF, geometric reconstruction is itself conditional on selected algebraic and dictionary inputs [154].

10.1.3 Scope

This work does not derive Clifford algebra from first principles. The stability calculation requires a positive-definite control metric and a complete closed-loop model. Clifford relations may be adopted as a modelling choice that packages a selected quadratic form with control generators; indefinite signature is a separate read-out choice, not a Lyapunov consequence.

Result-status ledger. Associativity of standard operator composition is a general observation. The persistence action below is heuristic motivation. The former indefinite Metric Constraint and Persistence Compatibility Theorem are retracted. The qubit section is a model audit showing what a valid Lyapunov proof would have to control, not a proof of Clifford necessity.

10.2 Variational Bounds on Persistence

Consider $\mathcal{H}_{\text{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$, with reduced dynamics:

$$\dot{\rho}_R = -i[H_{\text{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{10.1}$$

Heuristic persistence cost. For a declared control protocol $Q = \{u_k(t)\}_{k=1}^{m}$, one may compare protocols using

$$\mathcal{A}[Q] = \int_0^\tau dt\, D_{KL}(\rho_Q(t) \| \rho_{\text{NESS}}). \tag{10.2}$$

No variational domain, endpoint conditions, or Euler–Lagrange calculation is supplied here. Consequently $\delta\mathcal{A} = 0$ does not derive a control Hamiltonian or an operator algebra. The functional is motivation only; the actual feedback law must be specified and verified directly in the closed-loop master equation.

10.2.1 What Lie Algebras Do Not Encode by Themselves

Standard quantum control theory uses Lie algebra generators [134]: the control Hamiltonian $H_{\text{ctrl}} = \sum_k u_k(t)G_k$ where $\{G_k\}$ generate a Lie algebra $\mathfrak{g}$ via commutators $[G_i, G_j] = if_{ijk}G_k$.

Lie algebras encode *infinitesimal symmetries*—they specify which directions in state space are accessible via control. A control model normally supplies a positive cost, norm, or information metric separately to quantify correction size. That separation is standard and does not require the Lie algebra itself to carry a canonical metric. A Clifford quadratic form can package additional representation data when chosen, but it is not forced by error correction or channel discrimination.

10.3 Algebraic Constraints on Control Stability

10.3.1 Constraint I: Associativity as Structural Boundary

Remark 10.1 (Associativity as a control-model boundary)**.** *The present master-equation model represents sequential controls by linear maps with ordinary composition, which is associative by definition [20]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem, and composition rules before the control calculation could be transferred [101, 49]. Revised Paper A [160] makes the same type distinction in its remark "Physical pathologies of non-associative dynamics."*

The substantive constraint is Constraint II, which discriminates among *associative* algebras.

10.3.2 Constraint II: Positive Control Metric for Lyapunov Descent

The former lemma asserting that Lyapunov stability requires an indefinite metric is withdrawn. For an auxiliary gradient controller on parameters λ, the descent ansatz is

$$\dot{\lambda} = -\Gamma\, G^{-1}\nabla_\lambda V, \tag{10.3}$$

with G positive definite. Then the controller contribution is

$$(\nabla_\lambda V)^T\dot{\lambda} = -\Gamma(\nabla_\lambda V)^T G^{-1}(\nabla_\lambda V) \le 0. \tag{10.4}$$

If G is indefinite, a negative-norm gradient direction reverses this sign and is anti-stable under the same rule. Thus the stated descent mechanism selects positive definiteness, not Lorentzian signature.

Channel labels such as unitary, dephasing, and dissipative may be represented by a discrete type label, a direct-sum decomposition, or externally supplied weights. Their qualitative difference does not require negative norm. Any indefinite quadratic form used later for spacetime or read-out representation is a separate modelling input and must not be identified with the control metric G.

More importantly, λ-descent alone is not the physical state equation. With drift H_0, dissipator $\mathcal{D}$, and controls $H_{\text{ctrl}}(\lambda) = \sum_i u_i(\lambda)G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\text{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1}\nabla_\lambda V. \tag{10.5}$$

For a differentiable $V(\rho, \lambda)$,

$$\dot{V} = D_\rho V[-i[H_0 + H_{\text{ctrl}}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1}(\nabla_\lambda V). \tag{10.6}$$

A stability proof must control the first term as well as the positive-metric controller term. No general sign follows merely by setting a control amplitude equal to a gradient component.

10.3.3 Comparison with Alternative Algebraic Frameworks

Algebra	**I: Assoc.**	**Control metric**	**Status**
von Neumann (III_1)	✓	externally chosen positive G	Compatible
C^*-algebra	✓	externally chosen positive G	Compatible
Lie/control algebra	via representation	positive cost metric	Standard control setting
Jordan algebra	non-assoc.	model-dependent	Outside operator-composition setup
Clifford $Cl(V, q)$	✓	q need not equal G	Optional parameterization

Table 10.1: Candidate control frameworks after separating the positive Lyapunov metric G from any Clifford/read-out form q.

The table is no longer an exclusion argument. Standard dissipative and coherent-control frameworks stabilize target states with positive cost functions and Lie/operator-algebraic controls [125, 34, 119, 80]. They are counterexamples to the claim that indefinite Clifford structure is necessary for persistence.

10.3.4 Worked Example: Controlled Qubit Under Lindblad Dynamics

Setup. Consider a single qubit coupled to a thermal bath at inverse temperature β, with Lindblad dissipator:

$$\mathcal{D}[\rho] = \gamma_\downarrow \mathcal{L}[\sigma_-]\rho + \gamma_\uparrow \mathcal{L}[\sigma_+]\rho + \gamma_\phi \mathcal{L}[\sigma_z]\rho, \tag{10.7}$$

where $\mathcal{L}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$, and $\gamma_\downarrow$, $\gamma_\uparrow$, γ_ϕ are decay, excitation, and dephasing rates.

Control algebra. The control Hamiltonian is $H_{\text{ctrl}} = \sum_i u_i(t)\,\sigma_i$ where $\{\sigma_x, \sigma_y, \sigma_z\}$ are Pauli operators. These satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$—the defining relation of $Cl(3, 0)$.

Lyapunov function. Take $V = D_{KL}(\rho \| \rho_{\text{NESS}})$ where ρ_{NESS} is the thermal state. For full-rank states its derivative along the actual master equation is

$$\dot{V} = \text{Tr}[\dot{\rho}(\log \rho - \log \rho_{\text{NESS}})]\,, \tag{10.8}$$

with $\dot{\rho}$ containing the drift, all Lindblad terms, and the controlled commutator. The former equation $\dot{V} = -\alpha \sum_i (\partial V/\partial u_i)^2$ is withdrawn: the amplitudes u_i are not state coordinates, and assigning their values does not eliminate the Lindblad/drift contribution. A concrete feedback law must prove the full derivative nonpositive for its declared target and rates.

Failure mode without built-in metric. An abstract Lie algebra $\mathfrak{su}(2)$ specifies rotation directions; a control design also supplies amplitudes and a positive cost/control metric. Importing that metric is normal control data, not a consistency failure. In the Pauli representation the Hilbert–Schmidt form or another positive weight matrix can provide it.

The Pauli anticommutators can package a Euclidean quadratic form, so $Cl(3,0)$ is a convenient model representation. This convenience neither proves stability nor makes Clifford algebras uniquely necessary for feedback.

No indefinite extension from this mechanism. Under the same gradient-descent law, replacing the positive control metric by an indefinite form creates anti-descent directions. No stable indefinite example is supplied, so the former extension claim is deleted. A Lorentzian form may still be used at a separate representation/read-out layer, but it is not the Lyapunov metric of this feedback calculation.

10.3.5 Clifford Parameterization as a Model Choice

The former Persistence Compatibility Theorem and its NESS corollary are retracted. If one separately chooses a quadratic space (V, q) and requires generators to satisfy $v^2 = q(v)\mathbf{1}$, then the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [58]. This definitional observation does not show that persistence, channel discrimination, or Lyapunov descent imposes those relations.

Remark 10.2 (Natural Selection, Not Design)**.** *No algebraic natural-selection theorem is established. A particular physical feedback model may favour a representation because of its controls and costs, but that preference must be demonstrated by its closed-loop dynamics.*

Remark 10.3 (Content of the Argument)**.** *As with the withdrawn selection theorem in Paper A [160], invoking the universal property of $Cl(V, q)$ after positing Clifford relations is a standard algebraic identity. The surviving ingredients have different status: Remark 10.1 (associativity as the declared operator-composition boundary; see the named non-associative-dynamics remark in Paper A [160]), the positive control-metric calculation in Section 10.3.2, and the model audit in Section 10.3.4. The latter does not prove stability until a feedback law controls the full Lindblad derivative.*

10.4 Contextual Relations

10.4.1 Convergence with Paper A

The revised companion paper [160] treats $Cl(1,3)$ as a model representation of a separately supplied Lorentzian form.

	Paper A	**This work**
Starting point	Holographic boundary	Open subsystem
Method	Lorentzian encoding choice	Positive-metric stability audit
Key input	Clifford generator relations	Feedback model and cost metric
Result	$Cl(1,3)$ parameterization	$Cl(V, q)$ parameterization

There is no independent convergence theorem: both papers posit Clifford relations for their respective representation tasks. The shared choice motivates comparison, but neither route currently derives it from its upstream physical premises.

10.4.2 Relation to Quantum Control Theory

Standard quantum control operates within a Lie algebraic framework [134]: controllability is characterized by the Lie algebra generated by the drift and control Hamiltonians. This framework is complete for determining *reachability* of target states.

However, Lie algebras encode symmetries (via commutators) without encoding distances (via quadratic forms). When the control objective is not merely reachability but *stabilization against stochastic perturbations*—as in NESS maintenance—both direction and magnitude of corrections must be specified. Clifford algebras provide this additional structure through their built-in quadratic form, complementing rather than replacing the Lie algebraic framework.

10.4.3 Relation to Decoherence-Free Subspaces

Decoherence-free subspaces (DFS) [140, 75] represent subsystems that are passively protected from environmental noise by symmetry. The present analysis addresses the *active* counterpart: subsystems that maintain coherence through continuous feedback. In both cases, the algebraic structure of the system-environment interaction determines which subsystems can persist. The Clifford parameterization considered here applies to the active model; DFS theory applies to the passive case. A unified treatment remains an open problem.

10.5 Discussion

What this result does show: A positive-definite parameter metric makes the auxiliary gradient contribution nonpositive. Clifford algebra is a compact parameterization if a quadratic generator relation is separately selected. Table 10.1 and Section 10.3.4 record modelling options and missing stability conditions, not an exclusion theorem.

What this result does not show: That Clifford algebra is necessary or uniquely selected. That the displayed qubit feedback stabilizes the target without an explicit full-derivative proof. That an indefinite read-out form can serve as the Lyapunov control metric under the same descent rule.

10.6 Conclusion

We have argued for a narrower conclusion: Clifford algebra $Cl(V, q)$ is a useful modelling choice when a control model separately supplies a quadratic generator relation for channel discrimination and directed correction. Associativity remains a structural boundary condition, and the quadratic form remains an additional read-out or modelling input. The comparison with alternative algebras identifies missing ingredients for this particular package; it does not exclude those frameworks or prove Clifford necessity.

Combined with the companion paper's holographic constraints, this supports a compatibility programme linking geometric consistency and thermodynamic persistence. It

does not show that $Cl(V, q)$ is necessary or uniquely selected, as stated in the Discussion above.

Chapter 11

Conditional Image-Level Closure in the Q-RAIF Framework

Paper C — "The Bridge"

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Abstract

This addendum isolates an algebraic interface between two modelling roles in the **Q-RAIF (Quantum Reference Algebra for Information Flow)** framework. Paper A [160] adopts a $Cl(1,3)$ representation as a Lorentzian-encoding choice under supplied quadratic and generator data. Paper B [161] motivates a Cliffordian parameterization $Cl(V,q)$ for a control algebra under its stated compatibility conditions; neither paper uniquely derives Clifford structure.

Given a supplied unital $*$-homomorphism from the control algebra to the environmental algebra, we establish an **image-level closure result**: its image is a subalgebra, and in the stated even-dimensional Clifford case a nonzero homomorphism is injective. The *Same-Clock* statement records an operational co-indexing condition; it is not an input to the image proof and does not by itself ensure thermodynamic potency.

This note does not derive the representation map, select a preferred $Cl(1,3)$ subalgebra, or prove signature preservation. Physical realizability therefore remains an open interface rather than a completed bridge.

Keywords: Q-RAIF, realizability, representation, operator algebra, Clifford algebra, open quantum systems, Lyapunov stability, algebraic closure

11.1 Introduction

11.1.1 Context: The Q-RAIF Program

The Quantum Reference Algebra for Information Flow (Q-RAIF) framework investigates algebraic representations for geometric read-out and subsystem control. Revised Papers A and B treat Clifford structure as a modelling input rather than a necessary consequence.

The programme builds on the conditional algebra/reconstruction setting of HAFF [153, 154].

Paper	Question	Analogy	Result
HAFF [153]	How does geometry emerge?	Ocean	Algebra $\rightarrow$ Geometry
Q-RAIF A [160]	How encode Lorentzian read-out?	Water	$Cl(1,3)$ model
Q-RAIF B [161]	How parameterize controls?	Fish	$Cl(V,q)$ model
This work	What follows from representation?	Bridge	$\mathrm{Im}\,\phi \subseteq \mathcal{A}_{\mathrm{ext}}$

11.1.2 The Logical Gap

Papers A and B both posit Clifford representations for different tasks; they do not independently derive them [160, 161]. There is therefore no convergence theorem to close. This note isolates the tautological algebraic consequence of an explicit representation input and identifies the still-open physical question: what dynamics, if any, supplies that input?

11.1.3 Scope

This addendum explicitly assumes a representation map from an internal model algebra into an external observable algebra. It does not derive that map from persistence or system–environment coupling. Its image statement is algebraic; physical realizability remains an interface problem.

Result-status ledger. The representation map is an interface assumption. Same-clock overlap is an operational remark and does not enter the image proof. Semisimplicity/injectivity is a standard finite-dimensional Clifford-algebra fact. The former closure theorem is downgraded to an image-level proposition; all signature compatibility claims require additional quadratic-form-preserving data.

11.2 Setup and Prerequisites

Let $\mathcal{U}$ be a universe described by the Q-RAIF framework.

- **Environment ("water").** Let $\mathcal{A}_{\mathrm{ext}}$ denote the algebra of observables accessible at the holographic boundary. Paper A [160] uses a $Cl(1,3)$ representation as a model for a supplied Lorentzian read-out; it does not prove that every $\mathcal{A}_{\mathrm{ext}}$ must contain a preferred copy.

- **Subsystem ("fish").** Let $\mathcal{O}_{\mathrm{int}}$ denote the internal control algebra of a persistent subsystem $R \subset \mathcal{U}$. Paper B [161] may parameterize controls by $\mathcal{O}_{\mathrm{int}} \cong Cl(V,q)$, but persistence does not force that choice.

The remaining logical gap is the relationship between $\mathcal{O}_{\mathrm{int}}$ and $\mathcal{A}_{\mathrm{ext}}$: can a stable Clifford control algebra exist while being structurally disjoint from the available environmental observables?

11.3 Realizability and Same-Clock Co-Indexing

Definition 11.1 (Algebraic Representation Input)**.** *A control algebra $\mathcal{O}_{\text{int}}$ is represented in an environment $\mathcal{A}_{\text{ext}}$ when a unital algebra homomorphism is supplied,*

$$\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}} \tag{11.1}$$

such that $\text{Im}(\phi)$ *has non-zero action on the interaction Hamiltonian H_{int}, i.e., $[\text{Im}(\phi), H_{\text{int}}] \neq 0$. The commutator checks nontrivial coupling once ϕ is given. The definition does not prove that persistence or coupling produces such a homomorphism; calling it physical realizability is therefore an interface assumption.*

Remark 11.2 (Same-clock operational condition)**.** *Let I be an operational index set and suppose a model supplies a net $J \mapsto \mathcal{A}(J)$ for operational windows $J \subseteq I$. A feedback protocol may require windows $J_{\text{ctrl}}, J_{\text{env}} \subseteq I$ with*

1. ***Non-null intersection:*** *$J_{\text{ctrl}} \cap J_{\text{env}} \neq \emptyset$.*
2. ***Window integrity:*** *every critical lookback window $W \subseteq J_{\text{ctrl}} \cap J_{\text{env}}$ has a declared algebra $\mathcal{A}(W)$ and feedback channel.*

These are reasonable protocol requirements for discussing entropy export [104], but they neither imply the existence of ϕ nor enter the algebraic image proposition below. The former Same-Clock lemma is therefore demoted to this operational remark.

11.3.1 Semisimplicity of Clifford Algebras

Lemma 11.3 (Injectivity from Semisimplicity)**.** *Let (V, q) be a finite-dimensional real vector space with non-degenerate quadratic form. Then $Cl(V, q)$ is semisimple. If $\dim V$ is even, $Cl(V, q)$ is simple, and every non-zero algebra homomorphism $\phi : Cl(V, q) \to \mathcal{A}$ is injective.*

Proof. By the periodicity theorem for real Clifford algebras [1, 71], $Cl(V, q)$ with non-degenerate q is isomorphic to a matrix algebra $M_n(K)$ or a direct sum $M_n(K) \oplus M_n(K)$, where $K \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ depends on the signature and dimension modulo 8. In either case the algebra is semisimple.

When $\dim V$ is even, $Cl(V, q)$ is simple (a single matrix block). The kernel of any algebra homomorphism is a two-sided ideal; a simple algebra admits no proper non-trivial ideals, so $\ker \phi = \{0\}$ whenever $\phi \neq 0$. □

11.4 The Image Proposition

Proposition 11.4 (Image of a represented Clifford algebra)**.** *Let $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ be a chosen finite-dimensional Clifford model and let $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$ be the homomorphism supplied by Definition 11.1. Then*

$$\mathcal{O}_{\text{eff}} := \text{Im}(\phi) \subseteq \mathcal{A}_{\text{ext}} \tag{11.2}$$

is a subalgebra satisfying the induced quotient relations. If $\dim V$ is even and $\phi \neq 0$, then ϕ is injective and $\mathcal{O}_{\text{eff}} \cong Cl(V, q)$.

Proof. The image of an algebra homomorphism is a subalgebra, and relations in the source descend to its image. Lemma 11.3 gives injectivity for even $\dim V$ when the homomorphism is nonzero. No Same-Clock hypothesis is used. □

Remark 11.5 (Scope of the image proposition)**.** *The logical structure of Proposition 11.4 is simply: "the image of a homomorphism is a subalgebra," plus the even-dimensional injectivity fact. Papers A and B no longer supply independent selection theorems. The physically nontrivial step—deriving a representation ϕ from system–environment dynamics—is assumed rather than proved. Operator-algebra inclusion, virtual-subsystem, split-property, or modular-inclusion techniques may inform a future derivation [140, 35, 133].*

Withdrawal of the "No Ghost Algebra" corollary. The statement that a non-represented algebra fails Definition 11.1 merely repeats the definition. Its former signature obstruction is false for ordinary algebra homomorphisms: matrix-algebra embeddings need not preserve a presentation's quadratic form or signature.

To compare signatures one must supply a quadratic-space morphism $j : (V, q) \to (W, Q)$ satisfying

$$Q(jv) = q(v) \tag{11.3}$$

and, where relevant, require the induced Clifford map to preserve the unit, involution, and grading. Proposition 11.4 assumes no ambient quadratic space (W, Q) and proves no such isometry. It therefore yields an image in $\mathcal{A}_{\text{ext}}$, not an embedding into a distinguished $Cl(1, 3)$ subalgebra and not a signature-compatibility theorem.

11.5 Discussion

11.5.1 What This Result Does and Does Not Show

Does show: Given a supplied homomorphism, its image is a subalgebra of $\mathcal{A}_{\text{ext}}$; for even-dimensional $Cl(V, q)$, a nonzero homomorphism is injective.

Does not show: That ϕ must be injective in general—however, by Lemma 11.3, injectivity *is* guaranteed when $\dim V$ is even (which includes the physically relevant case $\dim V = 4$). For odd $\dim V$, the image $\mathrm{Im}(\phi)$ is isomorphic only to the quotient $Cl(V, q)/\ker\phi$; depending on the real signature and the representation, this quotient may be the full simple algebra or one surviving simple block of a semisimple direct sum. No general injectivity or preservation of the full Clifford presentation follows in the odd-dimensional case. That the image lies in a preferred $Cl(1, 3)$ subalgebra or preserves signature. That persistence produces ϕ or selects Clifford structure.

11.5.2 The Bridge Statement

Remark 11.6 (Closing the Loop)**.** *Papers A and B propose Clifford representations for separate modelling roles. Proposition 11.4 does not lock them together independently; it records what follows after a representation map is supplied. Deriving that map and any quadratic compatibility is the open bridge problem.*

11.5.3 Connection to HAFF

Within the HAFF programme [153, 154], geometric reconstruction is conditional on selected observable-algebra and dictionary data. The image proposition adds no independent natural-selection result. It isolates the representation/interface assumption that a future dynamical construction would have to justify.

Part III

T-DOME: Thermodynamic Dynamics of Observer-Memory Entanglement

Chapter 12

Non-Markovian Memory and Temporal Accumulation: A Conditional Thermodynamic Analysis

Paper I — "The Seed"

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Abstract

We investigate a conditional thermodynamic role for memory in open quantum systems interacting with stochastic environments. Working within stochastic and information thermodynamics, we define a *survival functional* $\mathcal{S} := \Delta F - W$ measuring the difference between non-equilibrium free energy gained and work invested by an agent.

For a fixed-Hamiltonian autonomous Davies/GKSL baseline with no measurement or feedback, we prove a restricted Markovian ceiling: $\mathcal{S} \leq 0$. This is not a universal ceiling for driven or feedback-controlled Markovian agents. We then derive an accounting identity for finite-dimensional autonomous evolution, with the interaction-energy term retained or a declared weak-coupling approximation made, expressing the survival functional through changes in system–environment mutual information and bath displacement. In models where prior interaction has supplied usable correlations, their consumption during non-Markovian backflow intervals can yield $\mathcal{S} > 0$, bounded by the initial correlation budget.

The result is therefore a conditional memory advantage, not a proof that memory or non-Markovianity is necessary or sufficient for all persistent systems. Fisher accretion and the comparison with the accessibility ordering of the Holographic Alaya-Field Framework (HAFF) are presented as heuristics. A worked spin-boson example with Lorentz–Drude spectral density illustrates how the accounting can operate in one supplied model.

Finally, using entropy rate and predictive information from computational mechanics, we study the implementation cost of a specified irreversible archive-reset model. Universal Landauer costs for merely storing or maintaining memory, and the resulting universal "Memory Catastrophe," are withdrawn; only the declared archive implementation is charged.

Keywords: non-Markovian dynamics, open quantum systems, Nakajima–Zwanzig equation, memory kernel, thermodynamic arrow of time, information backflow, entropy production, stochastic thermodynamics

12.1 Introduction

12.1.1 Context: The Problem of Persistence

A quantum system coupled to a thermal environment generically relaxes toward equilibrium. This is the content of the *zeroth crisis*: absent special structure, every open subsystem is eventually erased by thermal noise [22].

Yet the physical world contains persistent far-from-equilibrium structures—from molecular machines to living organisms—that maintain themselves against the entropic tide for timescales vastly exceeding their intrinsic relaxation times. What structural feature of their dynamics makes this possible?

The standard answer invokes free-energy input: a persistent system continuously imports low-entropy energy and exports high-entropy waste [102]. This paper asks the narrower accounting question of when pre-existing system–environment correlations can fund a later interval. It does not claim that memory distinguishes every persistent system from every non-persistent one.

12.1.2 Position within the Series

This paper is the first of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	Question		Result	Status
HAFF [148, 149]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [160, 161]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I (this work)	What can inter-epoch correlations fund?	Seed	ELV-type accounting identity; restricted ceiling	**This paper**
T-DOME II	How can a supplied archive be budgeted?	Ego	Conditional feature selection	Companion
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence, not an irreversible theorem chain:

1. **Paper I (The Seed):** The accounting identity asks when correlations built in earlier epochs can fund a later interval. A deliberately uncompressed archive is then introduced only as motivation for a finite-budget problem.

2. **Paper II (The Ego):** Given that archive model and a supplied component representation, Paper II studies conditional feature selection under a processing budget.

3. **Paper III (The Loop):** Given a selected frame and an explicit residual likelihood, Paper III studies conditions under which a calibration controller can track drift.

The present paper addresses only the first link in this chain.

12.1.3 Relation to HAFF Paper F

Revised HAFF Paper F [151] treats record growth and algebra inclusion as protocol-relative orderings and does not derive a universal temporal partial order. Section 12.5 therefore uses it only as motivation: a memory-influence relation and a pointer-record ordering may be compared after both are supplied, but neither is proved to embed in the other.

12.1.4 Scope and Disclaimers

1. This work does *not* claim that memory or non-Markovianity is necessary for persistence. Continuously driven Markovian dissipative structures can persist. The narrower claim is that pre-existing inter-epoch correlations can fund a positive interval of the survival accounting functional.

2. The sign of that functional is governed by changes in initial correlation and bath-displacement resources, not by Markovianity as such. No strict comparison of Markovian and non-Markovian suprema is proved.

3. We do *not* derive the specific form of the memory kernel from first principles. The kernel is treated as a structural feature of the system-environment coupling.

4. The term "agent" is used in the control-theoretic sense (a subsystem that acts on its environment to maintain a target state) and carries no implication of consciousness, intention, or subjective experience.

5. A broader structural analogy with classical philosophical concepts of temporal persistence exists but is outside the scope of this paper.

Result-status ledger. The information–thermodynamic identity is a general finite-dimensional unitary accounting identity (GvN/THM/OK-cond, with the interaction-energy term retained or a declared weak-coupling approximation). The original time-dependent-control Markovian Ceiling is retracted; a narrower autonomous Davies-semigroup theorem is proved below (FD/I/THM/OK-cond). Fisher accretion and the HAFF bridge are heuristics. Universal Landauer costs for storing or maintaining memory and the resulting "Memory Catastrophe" are retracted; only a specified irreversible archive-reset model remains.

12.2 Mathematical Preliminaries

12.2.1 Open Quantum Systems: The Markovian Baseline

Consider a bipartite Hilbert space $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_E$, where R denotes the "agent" (reduced system) and E the environment. The total Hamiltonian is

$$H = H_R \otimes \mathbb{1}_E + \mathbb{1}_R \otimes H_E + \lambda H_{\text{int}}, \tag{12.1}$$

where λ parametrizes the coupling strength.

Under the Born–Markov and secular approximations, the reduced dynamics of $\rho_R(t) = \text{Tr}_E[\rho(t)]$ is governed by the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation [76, 44]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \sum_k \gamma_k \left(L_k \rho_R(t) L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k,\, \rho_R(t)\} \right), \tag{12.2}$$

with $\gamma_k \geq 0$ and Lindblad operators $\{L_k\}$.

Remark 12.1 (Markovian = Memoryless)**.** *The GKSL equation is* time-local*: $\dot{\rho}_R(t)$ depends only on $\rho_R(t)$, never on $\rho_R(s)$ for $s < t$. Physically, this corresponds to an environment with vanishing correlation time ($\tau_E \to 0$): the bath "forgets" its interaction with the system instantaneously. The semigroup property $\Lambda(t+s) = \Lambda(t)\Lambda(s)$ ensures complete positivity at all times but precludes any information backflow from environment to system [94].*

12.2.2 Beyond Markov: The Nakajima–Zwanzig Equation

When the environmental correlation time τ_E is non-negligible, the Born–Markov approximation fails. The exact reduced dynamics is captured by the Nakajima–Zwanzig (NZ) integro-differential equation [82, 144]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \int_0^t ds\; \mathcal{K}(t,s)\, \rho_R(s), \tag{12.3}$$

where $\mathcal{K}(t,s)$ is the **memory kernel**—a superoperator encoding the influence of the system's entire history on its present dynamics.

Definition 12.2 (Memory Kernel)**.** *The memory kernel $\mathcal{K} : [0,\infty)^2 \to \mathcal{L}(\mathcal{B}(\mathcal{H}_R))$ is the superoperator satisfying* (12.3)*. It encodes two types of information:*

1. ***Environmental structure:*** *the spectral density, correlation functions, and non-equilibrium features of the bath;*
2. ***Temporal reach:*** *the effective support $\tau_{\text{mem}} := \inf\{\tau : \|\mathcal{K}(t,s)\| < \epsilon \;\forall\; t - s > \tau\}$, the "memory depth."*

The Markovian limit corresponds to $\mathcal{K}(t,s) \to \mathcal{K}_0\, \delta(t-s)$, recovering the GKSL generator.

Remark 12.3 (Information Backflow)**.** *Non-Markovian dynamics admits* information backflow*: the distinguishability of two initial states, as measured by trace distance $D(\rho_1(t), \rho_2(t))$, can temporarily increase [21]. This is the operational signature of memory—the environment returns previously absorbed information to the system.*

12.2.3 Thermodynamic Framework

We adopt the framework of stochastic thermodynamics for open quantum systems [38]. The following conventions are fixed throughout.

Definition 12.4 (Thermodynamic Setup)**.**

1. ***Hamiltonian decomposition.*** *The system Hamiltonian is $H_S(t) = H_R + H_{\mathrm{ctrl}}(t)$, where H_R is the* fixed *bare Hamiltonian and $H_{\mathrm{ctrl}}(t)$ is the agent's time-dependent control protocol. The bath Hamiltonian H_E and coupling H_{int} are as in* (12.1)*.*

2. ***Reference state.*** *The thermal equilibrium state of the bare Hamiltonian is*

$$\rho_{\mathrm{eq}} := \frac{e^{-\beta H_R}}{Z_R}, \qquad Z_R := \mathrm{tr}\big(e^{-\beta H_R}\big), \qquad \beta := (k_B T)^{-1}. \tag{12.4}$$

   Since H_R is time-independent, ρ_{eq} is a well-defined, fixed reference throughout the protocol.

3. ***Non-equilibrium free energy.*** *For any state ρ of the reduced system,*

$$F(\rho) := \mathrm{tr}(\rho\, H_R) + \beta^{-1}\,\mathrm{tr}(\rho \ln \rho) = \langle H_R \rangle_\rho - \beta^{-1} S(\rho), \tag{12.5}$$

   where $S(\rho) = -\,\mathrm{tr}(\rho \ln \rho)$ is the von Neumann entropy. The equilibrium value is $F_{\mathrm{eq}} = -\beta^{-1} \ln Z_R$.

4. ***Free energy–relative entropy identity.***

$$D_{\mathrm{KL}}(\rho \| \rho_{\mathrm{eq}}) = \beta\big(F(\rho) - F_{\mathrm{eq}}\big) \geq 0. \tag{12.6}$$

   Thus D_{KL} measures the free-energy surplus in units of $k_B T$.

5. ***Work.*** *The inclusive work performed on the total system by the control protocol over $[0, \tau]$ is*

$$W[0,\tau] := \int_0^\tau \mathrm{tr}\left(\rho(t)\, \frac{\partial H_{\mathrm{ctrl}}}{\partial t}\right) dt. \tag{12.7}$$

6. ***Entropy-production functional.*** *The* generalised entropy production *over $[0, \tau]$ is*

$$\Sigma[0,\tau] := \beta\big(W[0,\tau] - \Delta F\big), \tag{12.8}$$

   where $\Delta F = F(\rho(\tau)) - F(\rho(0))$. For uncorrelated (product) initial states, $\Sigma \geq 0$ recovers the standard second-law bound. For initially correlated states, Σ can be negative*, reflecting the consumption of pre-existing correlations (see Remark 12.25).*

Remark 12.5 (Why H_R is fixed)**.** *The bare Hamiltonian H_R defines the system's energy scale and hence the reference state ρ_{eq}. The agent acts on the world through $H_{\mathrm{ctrl}}(t)$, which may be time-dependent. This separation ensures that ρ_{eq} is well-defined and time-independent for the accounting identity. It does* not *make a driven generator satisfy detailed balance with respect to that fixed state. The ceiling result below therefore uses autonomous $H_{\mathrm{ctrl}} = 0$ dynamics.*

Definition 12.6 (Standing Assumptions)**.** *The following minimal assumptions are in force throughout Sections 12.4–12.6 unless stated otherwise. Every main result (Lemma 12.19, Theorem 12.21, Corollary 12.24) relies* only *on items* (A1)–(A5) *below.*

(A1) ***Finite-dimensional bipartite system.*** *$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$, with total Hamiltonian* (12.1) *and global unitary evolution* $U(t) = \mathcal{T}\exp\Big(-i\int_0^t H(s)\,ds\Big)$.

(A2) ***Weak coupling.*** *The system–environment interaction satisfies $\lambda \ll 1$ in* (12.1)*, so that $\Delta\langle H_{\rm int}\rangle = O(\lambda)$ [22]. The first-law bookkeeping is $W = \Delta\langle H_R\rangle + \Delta\langle H_{\rm ctrl}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\rm int}\rangle$. Dropping the last term is the declared weak-coupling approximation $\Delta\langle H_{\rm int}\rangle = O(\lambda)$; it is not an energy-conservation equation with a zero right-hand side.*

(A3) ***Fixed environmental reference.*** *$\rho_E^{\rm th} := e^{-\beta H_E}/Z_E$ is a fixed* bookkeeping *Gibbs state at inverse temperature β. The* actual *initial bath state $\rho_E(0)$ need not coincide with $\rho_E^{\rm th}$; when $\rho_E(0) \neq \rho_E^{\rm th}$, the quantity $D_{\rm KL}(\rho_E(t)\|\rho_E^{\rm th})$ tracks the nonequilibrium free energy stored in the bath relative to this reference. The bath Hamiltonian H_E is time-independent.*

(A4) ***Arbitrary initial state.*** *The total initial state $\rho_{SE}(0)$ is* not *required to be a product state. In particular, initial system–environment correlations $I(S{:}E;\,0) > 0$ and initial bath displacement $D_{\rm KL}(\rho_E(0)\|\rho_E^{\rm th}) > 0$ are both permitted.*

(A5) ***Regularity.*** *All quantum states appearing in the thermodynamic identities are assumed to have full rank (or are restricted to their support), so that all relative entropies $D_{\rm KL}(\rho\|\sigma)$ are finite.*

Remark 12.7 (Bookkeeping conventions)**.** *The heat absorbed by the environment is $Q := \Delta\langle H_E\rangle = \mathrm{Tr}[\rho_E(\tau)H_E] - \mathrm{Tr}[\rho_E(0)H_E]$ (matching Esposito* et al. *[38]). We define $\Sigma := \beta(W - \Delta F)$ as a* generalised entropy-balance functional*; for correlated initial conditions Σ need not be nonnegative (see Remark 12.25).*

12.2.4 The Survival Functional

We now define the central quantity of this paper.

Definition 12.8 (Survival Functional)**.** *For a reduced system R evolving under dynamics Λ over $[0,\tau]$, the* ***survival functional*** *is*

$$\mathcal{S}[\Lambda,\tau] := \Delta F - W[0,\tau] = \big[F(\rho(\tau)) - F(\rho(0))\big] - W[0,\tau]. \tag{12.9}$$

Equivalently, using (12.8)*,*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Sigma[0,\tau]. \tag{12.10}$$

Note on nomenclature. We retain the term "survival functional" to emphasize the biological interpretation of persistence far from equilibrium; mathematically, $\mathcal{S}$ is strictly a *generalized entropy-balance functional* derived from the first and second laws.

Remark 12.9 (Interpretation)**.** *The survival functional has a transparent physical meaning:*

- *$\mathcal{S} > 0$: the system gained more free energy than was invested by the external protocol—a* thermodynamic profit*. The agent has extracted usable work from environmental correlations.*
- *$\mathcal{S} = 0$: the agent breaks even (reversible limit, $\Sigma = 0$).*

- $\mathcal{S} < 0$*: the agent paid more than it gained (the generic irreversible case).*

Under the standard second law ($\Sigma \geq 0$), $\mathcal{S} \leq 0$ always. As Section 12.4 shows, a positive interval requires consumption of a resource present at that interval's initial time, such as system–environment correlation or bath displacement. Whether a memory kernel preserves access to that resource is a model-specific dynamical question.

Remark 12.10 (Connection to Information Thermodynamics)**.** *In the Sagawa–Ueda framework [99, 100], a system under feedback control satisfies the generalized second law*

$$\Sigma \geq -I_{\text{feedback}}, \tag{12.11}$$

where $I_{\text{feedback}} \geq 0$ is the mutual information gained through measurement of the system. This permits $\Sigma < 0$ (and hence $\mathcal{S} > 0$) at the expense of information. Pre-existing correlations can enter the entropy balance in a role analogous to measurement information. A memory kernel is not automatically a measurement or feedback controller; a concrete model must show that it preserves and later exposes those correlations.

12.3 A Restricted Markovian Baseline

The former "Markovian Ceiling" allowed a time-dependent control Hamiltonian while using a fixed bare Gibbs state as the detailed-balance reference. A driven generator need not preserve that state, so the asserted balance law did not follow from Spohn's inequality. That theorem, its open-loop control class, and its temporal-blindness corollary are retracted. The autonomous result below is the narrower statement that survives.

12.3.1 Spohn's Inequality

Throughout this section we assume that the GKSL generator $\mathcal{L}$ is a *thermal Lindbladian*: it is obtained from the weak-coupling (Davies) limit of a system coupled to a single thermal bath at inverse temperature β, and satisfies **quantum detailed balance** (the KMS condition) [112, 22]. Under this assumption, the unique stationary state is the Gibbs state $\rho_{\text{ss}} = \rho_{\text{eq}}$ of (12.4), and the generator is self-adjoint with respect to the KMS inner product. This ensures that the entropy production rate below is well-defined and non-negative.

Definition 12.11 (Autonomous Davies baseline)**.** *The autonomous baseline has $H_{\text{ctrl}} = 0$, fixed system Hamiltonian H_R, and a time-independent Davies/GKSL generator $\mathcal{L}$ satisfying quantum detailed balance with respect to the faithful Gibbs state $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$. No measurement, feedback, or initial system–bath resource is included in this reduced semigroup model.*

Lemma 12.12 (Spohn [112])**.** *For any GKSL dynamical semigroup $\Lambda_t = e^{\mathcal{L}t}$ satisfying quantum detailed balance with unique invariant state ρ_{eq}, the entropy production rate*

$$\sigma(t) := -\operatorname{tr}\big(\mathcal{L}[\rho(t)]\,(\ln \rho(t) - \ln \rho_{\text{eq}})\big) \tag{12.12}$$

satisfies $\sigma(t) \geq 0$, equivalently $dD_{\text{KL}}(\rho(t)\|\rho_{\text{eq}})/dt \leq 0$.

Proof. This follows from the contractivity of CPTP maps under quantum relative entropy [112, 22]: $D_{\text{KL}}(\Lambda_t\rho\|\Lambda_t\rho_{\text{eq}}) \leq D_{\text{KL}}(\rho\|\rho_{\text{eq}})$ for all $t \geq 0$. Differentiating at $t = 0$ yields $\sigma(t) \geq 0$. □

12.3.2 The Restricted Autonomous Ceiling

Theorem 12.13 (Restricted Autonomous Ceiling)**.** *Under Definition 12.11, the survival functional obeys*

$$\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0. \tag{12.13}$$

Equality requires no decrease of relative entropy on the interval (for a primitive semigroup, equilibrium is the stationary equality case).

Proof. Here $W = 0$. By (12.6) and Lemma 12.12, $\beta\Delta F = \Delta D_{\mathrm{KL}} \leq 0$, hence $\mathcal{S} = \Delta F \leq 0$. □

Remark 12.14 (Driven protocols require a different theorem)**.** *For time-dependent $H_S(t)$ one may define the instantaneous Gibbs state of the full driven Hamiltonian and assume a generator satisfying instantaneous detailed balance, together with inclusive work $\int \mathrm{tr}\Big(\rho\,\dot{H}_S\Big)dt$. Those hypotheses describe a different driven-thermodynamics result and are not proved here. Open-loop alone does not repair the missing relation between generator and reference state; feedback protocols require their own information balance.*

Remark 12.15 (Memorylessness and temporal correlations)**.** *Under the Born–Markov approximation, the bath correlation function is replaced by its white-noise limit $C(t,s) \to C_0\,\delta(t-s)$, and the GKSL dissipator depends only on the spectral density $J(\omega)$ evaluated at the system's Bohr frequencies. The reduced semigroup therefore omits explicit inter-epoch bath correlations. This modelling loss does not imply a universal spectral gap formula, a universal maintenance-work bound, or impossibility of continuously driven Markovian persistence.*

Remark 12.16 (Dissipative vs. Self-Nourishing Structures)**.** *The restricted theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It does not say that Markovian systems cannot persist under continuous driving, that non-Markovianity is necessary for persistence, or that a memory kernel alone supplies usable feedback. It is a controlled baseline for comparisons at fixed initial resources.*

12.4 Initial-Correlation Accounting

We now separate a dynamics-agnostic accounting identity from the model-dependent question of memory. Quantum mutual information $I(S{:}E)$ can be a consumable initial resource. A non-Markovian model may preserve such a resource across epochs, but non-Markovianity alone does not fix the sign of $\mathcal{S}$ and is neither necessary nor sufficient for a positive interval.

12.4.1 System–Environment Mutual Information

We work with the total system–environment state $\rho_{SE}(t)$, evolving unitarily under the total Hamiltonian (12.1). The quantum mutual information

$$I(S{:}E;\,t) := S(\rho_S(t)) + S(\rho_E(t)) - S(\rho_{SE}(t)) = D_{\mathrm{KL}}\big(\rho_{SE}(t)\,\big\|\,\rho_S(t)\otimes\rho_E(t)\big) \geq 0 \tag{12.14}$$

quantifies the total correlations (classical and quantum) between the system S and the environment E at time t.

Remark 12.17 (Role of Initial Correlations)**.** *Under the Born approximation, the initial state is taken as a product* $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E^{\mathrm{th}}$*, so* $I(S{:}E;\,0) = 0$*. For a system already interacting with its environment, a restart state* may *be correlated. If a specified memory mechanism preserves those correlations and a later interval consumes them, they contribute to that interval's resource balance. This is a conditional mechanism, not a generic property of every persistent system or memory kernel.*

12.4.2 The Information–Thermodynamic Identity

The following identity is the central technical tool of this section. It holds for **any** initial state—product or correlated—and relies only on unitarity and the definitions of mutual information and relative entropy. It is the Esposito–Lindenberg–Van den Broeck entropy-production decomposition [38], written so that arbitrary initial-correlation terms are retained explicitly.

Remark 12.18 (Relative-entropy chain rule)**.** *We repeatedly use the identity*

$$D_{\mathrm{KL}}\big(\rho_{SE} \,\big\|\, \rho_S \otimes \sigma_E\big) = I(S{:}E)_{\rho_{SE}} + D_{\mathrm{KL}}(\rho_E \| \sigma_E), \tag{12.15}$$

valid for arbitrary *(possibly correlated)* ρ_{SE} *and any full-rank reference state* σ_E*.*[1] *Importantly, this is a* pure algebraic identity *and does not assume product initial conditions.*

Lemma 12.19 (Information–Thermodynamic Identity)**.** *Let* $\rho_{SE}(t)$ *evolve unitarily under the total Hamiltonian. Then, for* ***any*** *initial state* $\rho_{SE}(0)$ *(product or correlated):*

$$\Delta I(S{:}E) + \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) = \Delta S_S + \beta\, \Delta\langle H_E\rangle, \tag{12.16}$$

where $\Delta S_S = S(\rho_S(\tau)) - S(\rho_S(0))$ *is the change in the system's von Neumann entropy and* $\Delta\langle H_E\rangle = \mathrm{Tr}[\rho_E(\tau)\, H_E] - \mathrm{Tr}[\rho_E(0)\, H_E]$ *is the energy absorbed by the environment.*

Proof. Applying the chain rule (12.15) (Remark 12.18) with $\sigma_E = \rho_E^{\mathrm{th}}$:

$$D_{\mathrm{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\mathrm{th}}\big) = I(S{:}E;\,t) + D_{\mathrm{KL}}\big(\rho_E(t) \,\big\|\, \rho_E^{\mathrm{th}}\big). \tag{12.17}$$

Expanding the left side using $\ln \rho_E^{\mathrm{th}} = -\beta H_E - \ln Z_E$:

$$D_{\mathrm{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\mathrm{th}}\big) = -S(\rho_{SE}(t)) + S(\rho_S(t)) + \beta\langle H_E\rangle_t + \ln Z_E. \tag{12.18}$$

Since the total evolution is unitary, $S(\rho_{SE}(t)) = S(\rho_{SE}(0))$ for all t. Taking the difference between times τ and 0 cancels both $S(\rho_{SE})$ and $\ln Z_E$, yielding

$$\Delta\big[I(S{:}E) + D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}})\big] = \Delta S_S + \beta\, \Delta\langle H_E\rangle. \tag{12.19}$$

□

Remark 12.20 (No assumption on the initial state)**.** *The proof of Lemma 12.19 uses* only *unitarity (*$\Delta S(\rho_{SE}) = 0$*) and the algebraic structure of the KL divergence. No assumption is made about the initial state* $\rho_{SE}(0)$*, the coupling strength, or the character (Markovian or non-Markovian) of the reduced dynamics. For a product thermal initial state, autonomous driving, and negligible endpoint interaction energy, the associated entropy balance reduces to the Esposito decomposition [38]:* $\Sigma = I(S{:}E;\,\tau) + D_{\mathrm{KL}}(\rho_E(\tau) \| \rho_E^{\mathrm{th}}) + O(\lambda)$*.*

[1] This follows from the definition of quantum relative entropy and $\ln(\rho_S \otimes \sigma_E) = \ln \rho_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \ln \sigma_E$. See, e.g., M. M. Wilde, *Quantum Information Theory*, 2nd ed., Cambridge University Press (2017), Sec. 11; and M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2000), Ch. 11.

12.4.3 The Survival Identity

We now connect the information–thermodynamic identity (12.16) to the survival functional $\mathcal{S}$ defined in Section 12.2.4.

Theorem 12.21 (Survival Balance: General Form)**.** *Under Assumptions* (A1)–(A5) *of Definition 12.6, let* $\rho_{SE}(t)$ *evolve unitarily from an* arbitrary *(possibly correlated) initial state* $\rho_{SE}(0)$*. The survival functional satisfies*

$$\boxed{\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle,} \tag{12.20}$$

where $\Delta\langle H_{\mathrm{ctrl}}\rangle = \mathrm{Tr}[\rho_S(\tau)\,H_{\mathrm{ctrl}}(\tau)] - \mathrm{Tr}[\rho_S(0)\,H_{\mathrm{ctrl}}(0)]$ *is the change in the control-field energy.*

For ***autonomous evolution*** *with negligible interaction-energy change at the endpoints (*$H_{\mathrm{ctrl}} = 0$ *and* $\Delta\langle H_{\mathrm{int}}\rangle = O(\lambda)$*), the weak-coupling form is*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + O(\lambda). \tag{12.21}$$

Proof. The proof uses three ingredients: the definition of $\mathcal{S}$, the first law, and Lemma 12.19.

Step 1 (First law in weak coupling). Since $H_{\mathrm{ctrl}}(t)$ is the only explicitly time-dependent component of H, inclusive work satisfies the exact first law $W = \Delta\langle H\rangle = \Delta\langle H_R\rangle + \Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle$.

Step 2 (Connecting Σ to the identity). From Definition 12.8 and (12.8), using $\Delta F = \Delta\langle H_R\rangle - \beta^{-1}\Delta S_S$:

$$\begin{aligned}\Sigma &= \beta(W - \Delta F) = \beta\big(W - \Delta\langle H_R\rangle\big) + \Delta S_S \\ &= \beta\big(\Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle\big) + \Delta S_S \\ &= \big(\Delta S_S + \beta\,\Delta\langle H_E\rangle\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle.\end{aligned} \tag{12.22}$$

By Lemma 12.19, the parenthesized term equals $\Delta I(S{:}E) + \Delta D_{\mathrm{KL}}(\rho_E\|\rho_E^{\mathrm{th}})$. Hence

$$\Sigma = \Delta I(S{:}E) + \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle. \tag{12.23}$$

Step 3 (Survival functional). $\beta\,\mathcal{S} = -\Sigma$ by (12.10), yielding (12.20). For $H_{\mathrm{ctrl}} = 0$ and weak endpoint coupling, the last term is $O(\lambda)$, yielding (12.21). □

Remark 12.22 (Nature of the result)**.** *Equation* (12.20) *is an exact* accounting identity*, not an inequality or optimality bound. It establishes that the sign is governed by the declared initial resources and endpoint bookkeeping: changes in correlation, bath displacement, control energy, and interaction energy. The identity itself is dynamics-agnostic and does not make non-Markovianity a resource. A memory kernel matters only when a model shows that it preserves correlations across epochs and later permits their consumption.*

Remark 12.23 (Scope of the theorem)**.** *Theorem 12.21 holds for* any *initial state* $\rho_{SE}(0)$*—product or correlated. The proof requires only Assumptions* (A1)–(A5) *of Definition 12.6 and the definitions of* $\mathcal{S}$*,* $I(S{:}E)$*, and* D_{KL}*. No assumption about the reduced dynamics (Markovian, non-Markovian, or otherwise) is needed. An interacting system may carry correlations at the chosen interval boundary. When present and subsequently consumed, those correlations are one thermodynamic resource; persistence does not require that every model use this resource.*

12.4.4 Three Regimes of Survival

We specialize to the autonomous, endpoint-decoupled weak-coupling convention ($H_{\mathrm{ctrl}} = 0$ and the $O(\lambda)$ interaction term suppressed). This isolates the initial-resource accounting.

Corollary 12.24 (Three Regimes)**.** *Under autonomous evolution, identity* (12.21) *identifies three regimes:*

1. ***Product initial state*** *($I(S{:}E;\,0) = 0$, $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}}) = 0$): Both ΔI and ΔD_{KL} are increases from zero to non-negative final values, so*

$$\beta\,\mathcal{S} = -\big(I(S{:}E;\,\tau) + D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}})\big) \le 0.$$

   This product-state unitary balance is dynamics-agnostic and is compatible with, but logically distinct from, the restricted Davies result (Theorem 12.13).

2. ***Correlated initial state*** *($I(S{:}E;\,0) > 0$): If the dynamics* consumes *pre-existing correlations ($\Delta I < 0$, i.e., $I(S{:}E;\,\tau) < I(S{:}E;\,0)$), the first term contributes* positively *to $\mathcal{S}$. Provided*

$$|\Delta I(S{:}E)| > \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big), \tag{12.24}$$

   the survival functional is strictly positive: $\mathcal{S} > 0$. The agent has converted pre-existing correlations into usable free energy.

3. ***Upper bound:*** *Since $I(S{:}E;\,\tau) \ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) \ge 0$, the maximum survival gain is bounded by*

$$\beta\,\mathcal{S} \le I(S{:}E;\,0) + D_{\mathrm{KL}}\big(\rho_E(0) \,\big\|\, \rho_E^{\mathrm{th}}\big). \tag{12.25}$$

   The thermodynamic profit cannot exceed the total initial "resource budget"—the pre-existing correlations plus the initial displacement of the bath from equilibrium.

12.4.5 The Correlation Battery

The three regimes of Corollary 12.24 raise a natural question: *where do the initial correlations $I(S{:}E;\,0) > 0$ come from?*

Remark 12.25 (The Correlation Battery)**.** *The answer is: from* ***prior interaction epochs****. A persistent agent does not begin its existence in a product state. Over any interaction interval, unitary evolution generically builds system–environment correlations ($\Delta I > 0$), at a thermodynamic cost ($\mathcal{S} < 0$ during this phase by Corollary 12.24(i)). A non-Markovian dilation can preserve correlations across the chosen epoch boundary, but whether they are later consumed with $\mathcal{S} > 0$ must be checked from the full balance.*

The process is analogous to a ***battery****:*

- ***Charging phase*** *(correlation building, $\Delta I > 0$): the agent "pays" free energy to build system–environment correlations. $\mathcal{S} < 0$.*
- ***Discharging phase*** *(correlation consumption, $\Delta I < 0$): the agent extracts free energy from the stored correlations. $\mathcal{S} > 0$.*

The repeated-product Born approximation removes this resource by construction at each reset. A reduced semigroup law by itself does not state that every microscopic dilation is uncorrelated. Conversely, a memory kernel only represents retained history; it does not guarantee that the history is usable as thermodynamic work.

Crucially, global thermodynamics remains respected. For any full cycle starting from an uncorrelated thermal state ($I(S{:}E;\,0)=0$, $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}})=0$), the total survival functional satisfies

$$\beta\,\mathcal{S}[0,t] = -\Sigma[0,t] \;\le\; 0 \qquad \textit{(second law).} \tag{12.26}$$

The local positivity $\mathcal{S}[t^*,t]>0$ during the discharging phase is strictly funded by the free energy dissipated during the earlier charging phase (see Proposition 12.26 for the formal decomposition).

Proposition 12.26 (Full-cycle closure)**.** *Under the conditions of Theorem* 12.21 *with autonomous evolution ($H_{\mathrm{ctrl}}=0$) and negligible interaction-energy changes at the endpoints of each displayed balance, partition $[0,\tau]$ at any intermediate time t^* into a charging phase $[0,t^*]$ and a discharging phase $[t^*,\tau]$.*

(i) ***Charging*** *(product initial state, $I(S{:}E;\,0)=0$). By Corollary* 12.24(i)*,*

$$\beta\,\mathcal{S}[0,t^*] = -I(S{:}E;\,t^*) - D_{\mathrm{KL}}\big(\rho_E(t^*)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{12.27}$$

(ii) ***Discharging*** *(correlated initial state at t^*). Applying* (12.21) *to $[t^*,\tau]$:*

$$\beta\,\mathcal{S}[t^*,\tau] = -\big(I(S{:}E;\,\tau) - I(S{:}E;\,t^*)\big) - \big(D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) - D_{\mathrm{KL}}(\rho_E(t^*)\|\rho_E^{\mathrm{th}})\big), \tag{12.28}$$

which is positive whenever the decrease in correlations dominates the change in bath displacement (Corollary 12.24(ii)*).*

(iii) ***Full cycle.*** *Since $\mathcal{S}$ is additive over concatenated intervals, $\beta\,\mathcal{S}[0,\tau] = \beta\,\mathcal{S}[0,t^*] + \beta\,\mathcal{S}[t^*,\tau]$. Equivalently, applying* (12.21) *directly to $[0,\tau]$ with $I(S{:}E;\,0)=0$:*

$$\beta\,\mathcal{S}[0,\tau] = -I(S{:}E;\,\tau) - D_{\mathrm{KL}}\big(\rho_E(\tau)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{12.29}$$

The net thermodynamic profit over the full cycle is non-positive—the "interest" paid during charging meets or exceeds the "dividend" collected during discharging. But the local *positivity of $\mathcal{S}$ during discharge* (12.28) *is what enables the agent to survive through intervals that would kill a memoryless system.*

Proof. The survival functional is additive over concatenated intervals:

$$\mathcal{S}[0,\tau] = \underbrace{(\Delta F[0,t^*] - W[0,t^*])}_{\mathcal{S}[0,t^*]} + \underbrace{(\Delta F[t^*,\tau] - W[t^*,\tau])}_{\mathcal{S}[t^*,\tau]},$$

since both ΔF and W decompose additively. Items (i) and (iii) then follow from Theorem 12.21 (autonomous case) applied to $[0,t^*]$ and $[0,\tau]$ respectively, each starting from a product state. Item (ii) follows from Theorem 12.21 applied to $[t^*,\tau]$ with correlated initial state $\rho_{SE}(t^*)$. Inequality (12.29) holds because $I(S{:}E;\,\tau)\ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}})\ge 0$. □

12.4.6 Connection to Non-Markovianity Measures

Remark 12.27 (The Born Approximation Destroys the Resource)**.** *Under the Born (product-state) approximation, every infinitesimal time step begins from* $\rho_{SE} \approx \rho_S \otimes \rho_E^{\mathrm{th}}$*, enforcing* $I(S{:}E) = 0$ *at all times. Corollary 12.24(i) then guarantees* $\mathcal{S} \leq 0$ *for* every *finite interval. The Born approximation does not merely simplify the dynamics—it* eliminates the thermodynamic resource *(system–environment correlations) that would otherwise be available.*

Remark 12.28 (Connection to BLP Non-Markovianity)**.** *The Breuer–Laine–Piilo (BLP) measure of non-Markovianity [21] is defined via the temporary increase of trace distance between pairs of initial states:* $\mathcal{N}_{\mathrm{BLP}} := \max_{\rho_{1,2}} \int_{\dot{D}>0} \frac{d}{dt} D(\rho_1(t), \rho_2(t))\, dt$*. BLP backflow and correlation consumption can coincide in particular models, but neither condition implies the other in general [94]. The BLP measure therefore diagnoses one reduced-dynamics notion of non-Markovianity; it is not a witness for the sign of* $\mathcal{S}$ *without the accompanying energy and correlation balance.*

Remark 12.29 (Consistency with the Sagawa–Ueda Framework)**.** *In the Sagawa–Ueda framework [99, 100], measurement-based feedback permits* $\Sigma \geq -I_{\mathrm{feedback}}$*, where* I_{feedback} *is the mutual information gained through measurement. Pre-existing correlations are an analogous accounting resource, but a memory kernel is not itself a measurement record and the two protocols must not be identified. The total system (agent + bath) still satisfies* $\Sigma_{\mathrm{total}} \geq 0$*; the apparent "profit" for the agent is paid for by the correlations consumed from the system–environment entanglement. The bound* (12.25) *is the non-Markovian analogue of the Sagawa–Ueda bound* $\beta\,\mathcal{S} \leq I_{\mathrm{feedback}}$*.*

12.4.7 Mechanism: The Surfer Analogy

The physical mechanism admits an intuitive picture.

- **The Markovian Agent (The Stone):** A stone thrown into the ocean sinks. It interacts with the water only at the instant of contact, dissipates its kinetic energy, and thermalizes ($\mathcal{S} \leq 0$). Each collision builds system–environment correlations that are immediately discarded (Born approximation), so $I(S{:}E) = 0$ at all times. The wave structure is invisible to it.
- **The Non-Markovian Agent (The Surfer):** A surfer carries *memory* of past wave patterns—encoded in the correlations $I(S{:}E;\, t_0) > 0$ built up over previous interactions (the "charging phase" of Remark 12.25). During backflow intervals ($\Delta I < 0$), the surfer *spends* these stored correlations to extract free energy from the wave itself. The surfer remains far from equilibrium not by fighting the environment, but by converting temporal correlations into thermodynamic profit.

Remark 12.30 (Thermodynamic Rectification)**.** *The "surfing" mechanism is* ***thermodynamic rectification****: the memory kernel* $\mathcal{K}(t, s)$ *functions as a temporal filter that enables the system to accumulate correlations during one phase and consume them during another. Formally, the kernel enables access to the resource* $I(S{:}E;\, 0)$ *accumulated during previous interaction epochs—converting the environment's temporal correlations into the system's structural persistence via the* ΔI *term in Theorem 12.21.*

Remark 12.31 (Memory as Implicit Maxwell's Demon)**.** *The memory kernel functions as an* implicit Maxwell's demon*. A Markovian system interacts with each environmental fluctuation exactly once, at the moment of contact; the Born approximation resets $I(S{:}E) = 0$ after each step. A non-Markovian system retains a trace of past fluctuations (via $\mathcal{K}(t, s)$ with $s < t$) and can exploit correlations between past and present environmental states. This is not a violation of the second law but an instance of the Sagawa–Ueda generalization: the demon's cost is paid in the currency of memory maintenance (Landauer erasure), a point we quantify in Section 12.7. The total budget for "demonic profit" is capped by the bound* (12.25)*.*

12.5 Operational Temporal Relations

The accounting identity does not derive a temporal order. This section records two heuristic structures that can be studied after a reduced memory model is specified: an operational influence relation between times and a comparison with pointer-record growth. Neither is a universal partial order, and no HAFF theorem is inherited.

12.5.1 Thresholded Memory-Influence Relation

A non-Markovian memory kernel $\mathcal{K}(t, s)$ defines a causal link between a past state at s and the present dynamics at t. We define a thresholded relation based on the effective response.

Definition 12.32 (Operational memory-influence relation)**.** *For a declared reduced model, let $\mathcal{R}_{t\leftarrow s}$ denote the linear response of the predicted state or read-out at time t to a perturbation applied at time $s < t$. At tolerance $\epsilon > 0$, define*

$$s \rightsquigarrow_K t \quad \Longleftrightarrow \quad \|\mathcal{R}_{t\leftarrow s}\| > \epsilon. \tag{12.30}$$

The response norm, perturbation class, and tolerance are part of the protocol. This relation need not be transitive or antisymmetric, so it is not asserted to be a partial order.

An ideal delta-memory reduced model has no off-diagonal response in this description. A finite-memory model can have $s \rightsquigarrow_K t$ within its response window. This statement is about the chosen reduced model, not about the existence of time or a universal thermodynamic arrow.

12.5.2 A Fisher-Accretion Conjecture

One may ask whether a specified survival optimizer tends to retain parameter information in the forward protocol direction. The following statement is a conjecture; the survival identity alone does not create statistical irreversibility.

Conjecture 12.33 (Fisher Information Accretion)**.** *Let $\mathcal{I}_F(\theta;\, \rho(t))$ denote the Fisher information contained in the system state $\rho(t)$ regarding a parameter θ encoded in the environment at time $s < t$. For an agent whose dynamics maximize the survival functional* (12.9)*, the time-averaged Fisher information satisfies*

$$\overline{\frac{d}{dt}\,\mathcal{I}_F(\theta;\, \rho(t))} \geq 0, \tag{12.31}$$

where the overbar denotes a time average over scales larger than the bath correlation time τ_B.

Remark 12.34 (Status of the conjecture)**.** *The survival balance supplies no optimization domain, controller class, or monotonicity theorem for Fisher information. Mutual information and Fisher information are distinct functionals, and information backflow can make either non-monotone. Establishing the displayed inequality requires a concrete controlled process and objective; it is not used as an input to later theorems.*

12.5.3 The Bridge to HAFF

Revised HAFF Paper F [151] distinguishes an algebra-inclusion preorder from a protocol-relative redundancy diagnostic and does not prove generic monotone record expansion. The following comparison is therefore a heuristic interface, not a theorem.

Conjecture 12.35 (Conditional pointer-record comparison)**.** *Suppose a concrete branching/record model supplies (i) a pointer coarse-graining* Φ*, (ii) a fixed fragmentation, and (iii) a monotone record channel for a specified observation window. One may then test whether an influence* $s \rightsquigarrow_K t$ *is accompanied by nested record algebras,*

$$\mathcal{R}_s \subseteq \mathcal{R}_t. \tag{12.32}$$

The memory-influence relation alone does not imply this inclusion; backflow, record erasure, or a changed fragmentation can violate it.

Remark 12.36 (Scope of the Correspondence)**.** *Conjecture 12.35 states a test that can succeed in a branching record model. No general monotonicity, embedding, or duality between T-DOME and HAFF is established.*

Remark 12.37 (Dynamical and Algebraic Time)**.** *The two papers use distinct, protocol-dependent notions:*

	Paper F (HAFF)	***Paper I (T-DOME)***
Nature	*Algebraic*	*Dynamical*
Mechanism	*Record/algebra diagnostics*	*Reduced-model response*
Formalism	*Preorders/diagnostics*	*Relation* $\rightsquigarrow_K$ *on times*
Asymmetry source	*Declared record protocol*	*Declared kernel/response model*
Domain	*Classical (pointer) sector*	*Full quantum dynamics*

An explicit morphism between them remains open.

Remark 12.38 (The Seed and the Tree)**.** *The seed/tree language is a programme metaphor. It supplies neither a time variable nor a theorem connecting memory dynamics to geometry.*

12.6 Worked Example: The Quantum Predictive Agent

To illustrate free-energy monotonicity and a correlation-funded backflow interval *within one explicit model*, we employ an open quantum system model: the spin-boson model with Lorentz–Drude spectral density, which admits an exact analytic solution for the decoherence dynamics [22].

12.6.1 Model Setup

The total Hamiltonian is $H = H_S + H_B + H_I$. The agent is a two-level system with energy gap ω_0: $H_S = \frac{\omega_0}{2}\sigma_z$. The environment is a bosonic bath: $H_B = \sum_k \omega_k b_k^\dagger b_k$. The interaction is of the pure-dephasing form $H_I = \sigma_z \otimes \sum_k (g_k\, b_k + g_k^*\, b_k^\dagger)$.

The spectral density $J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k)$ characterizes the environment. We choose the Lorentz–Drude form:

$$J(\omega) = \frac{2\lambda\,\gamma\,\omega}{\omega^2 + \gamma^2}, \tag{12.33}$$

where λ is the reorganization energy and γ is the bath memory rate (inverse correlation time $\tau_B = 1/\gamma$). We place the system in the low-temperature regime $\beta\omega_0 \gg 1$ (i.e., $k_B T \ll \omega_0$). The bath correlation function in the $T \to 0$ limit is $C(t) = \lambda\gamma\, e^{-\gamma|t|}$, so the parameter γ directly controls the bath memory depth. For $\beta\omega_0 \geq 10$ the finite-temperature corrections to all quantities below are of order $O(e^{-\beta\omega_0}) \lesssim 5 \times 10^{-5}$ and are neglected throughout.[2]

12.6.2 Exact Decoherence Function

For the pure-dephasing spin-boson model in the $T \to 0$ limit, the off-diagonal element of the reduced density matrix $\rho_{01}(t) = \rho_{01}(0)\, p(t)$ is governed by the **decoherence function** [22]:

$$p(t) = e^{-\gamma t/2}\left[\cos(\Omega t) + \frac{\gamma}{2\Omega}\,\sin(\Omega t)\right], \tag{12.34}$$

where $\Omega := \frac{1}{2}\sqrt{4\lambda\gamma - \gamma^2}$. This solution is exact for the Lorentz–Drude spectral density.

Remark 12.39 (Non-Markovian Regime)**.** *The character of the dynamics is controlled by the discriminant $\Delta := 4\lambda\gamma - \gamma^2 = \gamma(4\lambda - \gamma)$:*

- *$\gamma > 4\lambda$ ($\Delta < 0$): Ω is imaginary, $p(t)$ decays monotonically. The dynamics is Markovian (no backflow).*
- *$\gamma = 4\lambda$ ($\Delta = 0$): Critical damping. $p(t) = (1 + \gamma t/2)\, e^{-\gamma t/2}$.*
- *$\gamma < 4\lambda$ ($\Delta > 0$): Ω is real and positive. $p(t)$ oscillates with envelope $e^{-\gamma t/2}$. The dynamics is* non-Markovian*: intervals where $|p(t)|$ increases correspond to information backflow [21].*

The non-Markovian regime $\gamma < 4\lambda$ is thus the regime of structured, long-memory baths.

12.6.3 Quantitative Evaluation

We now evaluate the survival functional explicitly. For the pure-dephasing model, populations are conserved ($p_0(t) = p_0(0)$, $p_1(t) = p_1(0)$), and the non-equilibrium free energy depends only on the coherence:

$$F(\rho(t)) - F(\rho_{\mathrm{eq}}) = \beta^{-1}\, D_{\mathrm{KL}}(\rho(t) \| \rho_{\mathrm{eq}}). \tag{12.35}$$

[2] All plots and numerical values use the standard $T \to 0$ analytic expression for the decoherence function (12.34) (see Breuer and Petruccione [22], Sec. 12.3, for the Lorentz–Drude pure-dephasing solution), which provides an accurate proxy in the low-temperature regime; β is a well-defined bookkeeping parameter and β^{-1} a finite energy scale.

For a qubit with initial state $\rho(0) = \frac{1}{2}(\mathbb{1} + \vec{r}\cdot\vec{\sigma})$ and $r_z = 0$ (maximal coherence in the x–y plane), the relative entropy reduces to $D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) \approx |p(t)|^2\,|\rho_{01}(0)|^2$ to leading order in the coherence (see, e.g., [22]). Since there is no external driving ($H_{\mathrm{ctrl}} = 0$, $W = 0$), the survival functional is simply

$$\beta\,\mathcal{S}(t) = D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) - D_{\mathrm{KL}}(\rho(0)\|\rho_{\mathrm{eq}}) \propto |p(t)|^2 - 1. \tag{12.36}$$

The proportionality in (12.36) is specific to the **pure-dephasing model** with the chosen maximally coherent initial state ($r_z = 0$) and measurement in the pointer basis (σ_z). Under these conditions, the exact solution [22] ensures that population terms vanish from the free energy ($\Delta\langle H_S\rangle = 0$), leaving only the coherence contribution: $\beta\,\mathcal{S} = -\Delta S_S$ depends only on the coherence trajectory $|p(t)|$. The proxy $|p(t)|^2$ thus rigorously captures the sign and monotone behaviour of $\beta\,\mathcal{S}$; the exact numerical prefactor depends on the initial state and on β, but the qualitative conclusion—$\mathcal{S} > 0$ during backflow intervals—is robust and does not depend on the proxy normalization.

For a Markovian evolution, $|p(t)|$ decreases monotonically, so $|p(t)|^2 - 1 \le 0$ for all t: $\mathcal{S} \le 0$ always in this autonomous model (consistent with Theorem 12.13). For non-Markovian evolution with $\gamma < 4\lambda$, the oscillations in $p(t)$ produce intervals where $|p(t)|$ *increases after a previous decrease*, i.e., the system *re-coheres*.

Concrete parameters. Set $\omega_0 = 1$ (energy units), $\lambda = 1$, $\gamma = 0.5$ (deep non-Markovian regime: $\gamma/4\lambda = 0.125 \ll 1$). Then:

$$\Omega = \tfrac{1}{2}\sqrt{4\cdot 1\cdot 0.5 - 0.25} = \tfrac{1}{2}\sqrt{1.75} \approx 0.661. \tag{12.37}$$

The decoherence function (12.34) first reaches $p(t^*) = 0$ at $t^* = [\pi/2 + \arctan\big(\gamma/(2\Omega)\big)]/\Omega \approx 1.93/\Omega \approx 2.92$ (in units of ω_0^{-1}), where the system has fully decohered. Subsequently, the environment *returns* coherence: $|p(t)|$ increases, reaching a local maximum $|p(t_1)| \approx 0.31$ at $t_1 \approx 4.75/\omega_0$.

Over the backflow interval $[t^*, t_1]$, for the pure-dephasing qubit with the chosen initial state ($r_z = 0$, maximal coherence) and in the autonomous setting ($H_{\mathrm{ctrl}} = 0$, $W = 0$), the survival proxy (12.36) gives

$$\beta\,\mathcal{S}[t^*, t_1] \propto |p(t_1)|^2 - |p(t^*)|^2 \approx 0.093 - 0 = 0.093 > 0. \tag{12.38}$$

Equivalently, $\mathcal{S} \approx 0.093\,\beta^{-1}$ in the bookkeeping units set by β. The agent has gained a dimensionless survival advantage $\beta\,\mathcal{S} \approx +0.093$ *with zero work input* (autonomous evolution, $H_{\mathrm{ctrl}} = 0$), solely by exploiting the non-Markovian backflow. Figure 12.1 illustrates the contrast between Markovian and non-Markovian evolution.

Consistency with Theorem 12.21 and Proposition 12.26. Since this is autonomous evolution ($H_{\mathrm{ctrl}} = 0$), identity (12.21) applies exactly: $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E\|\rho_E^{\mathrm{th}})$. The example realizes the *correlation battery* of Remark 12.25, with the charge–discharge decomposition of Proposition 12.26:

- **Charging** ($[0, t^*]$, eq. (12.27)): the system decoheres, building correlations $I(S{:}E;\,t^*) > 0$ at the cost of $\mathcal{S} < 0$.
- **Discharging** ($[t^*, t_1]$, eq. (12.28)): the correlations are consumed ($\Delta I < 0$ over this interval), returning $\beta\,\mathcal{S} \approx +0.093 > 0$.

The bound (12.25) is satisfied: $\beta\,\mathcal{S}[t^*, t_1] = 0.093 \le I(S{:}E;\,t^*)$. Full-cycle closure (12.29) is confirmed: $\beta\,\mathcal{S}[0, t_1] < 0$.

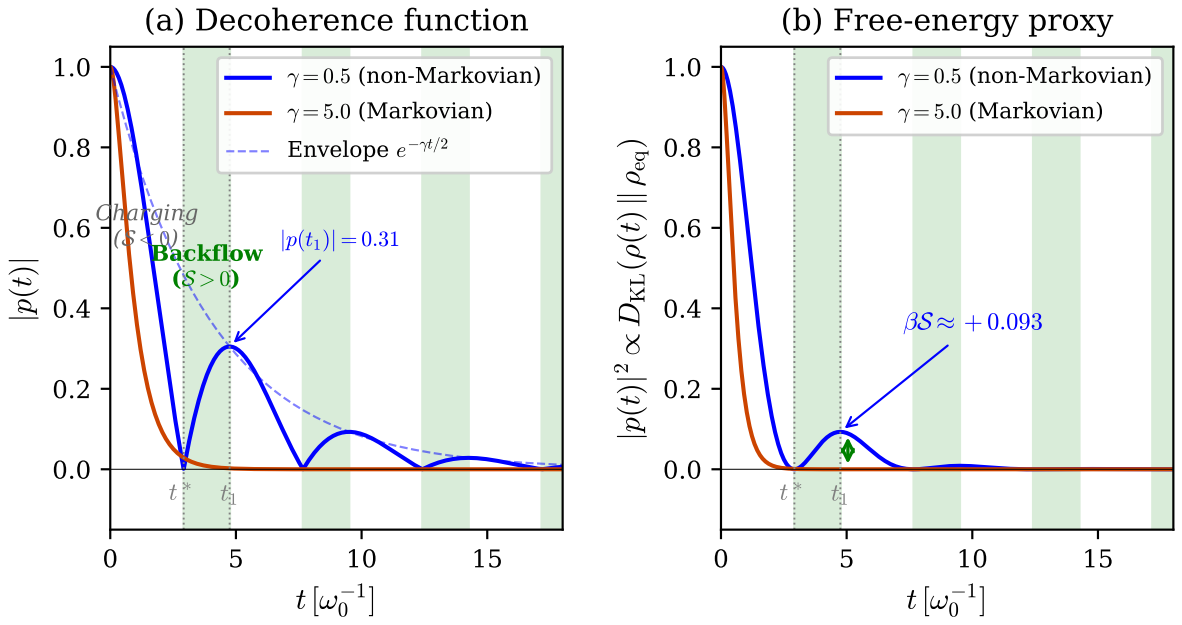


Figure 12.1: Pure-dephasing spin-boson model (Section 12.6) with Lorentz–Drude spectral density (12.33). **Parameters:** $\omega_0 = 1$ (energy unit), $\lambda = 1$ (reorganization energy). **Units:** all times in ω_0^{-1}; energies in $\hbar\omega_0$. **Regime:** low temperature ($\beta\omega_0 \gg 1$); the standard $T \to 0$ analytic expression (12.34) [22] is used as an accurate proxy. **(a)** Decoherence amplitude $|p(t)|$ (eq. (12.34)). Blue: non-Markovian ($\gamma = 0.5$, $\gamma/4\lambda = 0.125$). Orange: Markovian ($\gamma = 5.0$, $\gamma/4\lambda = 1.25$). Dashed: exponential envelope $e^{-\gamma t/2}$. Green bands indicate backflow (Remark 12.39: $d|p|/dt > 0$, $\Gamma(t) < 0$ per (12.39)). **(b)** Survival proxy $|p(t)|^2 \propto \beta\,\mathcal{S}$ (eq. (12.36)). At the first revival ($t_1 = \pi/\Omega \approx 4.75\,\omega_0^{-1}$), the non-Markovian agent achieves $\beta\,\mathcal{S}[t^*, t_1] \approx +0.093$ (eq. (12.38)), consistent with the closed-form prediction (12.40), funded by the consumption of pre-existing correlations (Proposition 12.26). The Markovian comparison decays monotonically: $\mathcal{S} \leq 0$ in this autonomous model (Theorem 12.13).

Instantaneous decoherence rate. The rate of coherence loss is

$$\Gamma(t) := -\frac{d}{dt}\ln|p(t)| = \frac{\gamma}{2} - \frac{\Omega\sin(\Omega t) + \frac{\gamma}{2}\cos(\Omega t)}{\cos(\Omega t) + \frac{\gamma}{2\Omega}\sin(\Omega t)}\,. \tag{12.39}$$

In the Markovian limit $\gamma \gg 4\lambda$, $\Gamma(t) \to \gamma/2 > 0$ for all t (monotone decoherence). In the non-Markovian regime $\gamma < 4\lambda$, $\Gamma(t)$ oscillates and becomes *negative* during the backflow intervals where $|p(t)|$ increases. These are precisely the intervals where $\mathcal{S} > 0$.

Closed-form revival amplitude. The decoherence function (12.34) can be written as $p(t) = R\,e^{-\gamma t/2}\cos(\Omega t - \phi)$, where $R = \sqrt{1 + (\gamma/2\Omega)^2}$ and $\phi = \arctan(\gamma/2\Omega)$, with $R\cos\phi = 1$. The extrema of $|p(t)|$ occur at $t_n = n\pi/\Omega$ ($n = 0, 1, 2, \ldots$), and the first revival peak after the first zero is at $t_1 = \pi/\Omega$. Its amplitude is *exactly*

$$|p(t_1)| = e^{-\gamma\pi/(2\Omega)}, \qquad \beta\,\mathcal{S}[t^*, t_1] \approx |p(t_1)|^2 = e^{-\gamma\pi/\Omega}. \tag{12.40}$$

This is the paper's central computable prediction: the survival gain at first backflow is determined by a single dimensionless ratio γ/Ω.

Remark 12.40 (Parameter Survey)**.** *Table 12.1 demonstrates the transition from the Markovian regime ($\mathcal{S} \leq 0$) to the non-Markovian regime ($\mathcal{S} > 0$) as the bath memory rate γ decreases below the critical value 4λ. All entries use $\omega_0 = 1$, $\lambda = 1$, $W = 0$ (autonomous evolution), with revival amplitudes computed from* (12.40)*.*

γ	$\gamma/4\lambda$	*Regime*	$\lvert p(t_1)\rvert$	$\beta\,\mathcal{S}(t_1)$	$\Gamma_{\min}$
20.0	5.0	*Markov*	—	≤ 0	> 0
4.0	1.0	*Critical*	—	≤ 0	$= 0$
2.0	0.50	*Non-Markov*	0.043	$+0.002$	< 0
1.0	0.25	*Non-Markov*	0.163	$+0.027$	< 0
0.5	0.125	*Deep NM*	0.305	$+0.093$	< 0
0.1	0.025	*Deep NM*	0.605	$+0.37$	< 0

Table 12.1: Survival functional at first backflow revival as a function of the bath memory rate γ, for the pure-dephasing spin-boson model with Lorentz–Drude spectral density. $|p(t_1)|$ is computed from (12.40); $\Gamma_{\min}$ is the sign of the minimum of the instantaneous decoherence rate (12.39). The transition $\mathcal{S} \leq 0 \to \mathcal{S} > 0$ occurs precisely at the non-Markovian threshold $\gamma = 4\lambda$. For $\gamma = 0.1$ (deep non-Markovian), the agent achieves $\beta\,\mathcal{S} \approx +0.37$ per backflow cycle in the autonomous setting ($H_{\text{ctrl}} = 0$).

Remark 12.41 (The Two Regimes: Summary)**.**

	Markovian *($\gamma = 20$)*	***Non-Markovian*** *($\gamma = 0.5$)*
$\gamma/4\lambda$	5.0 *(overdamped)*	0.125 *(underdamped)*
τ_B	$0.05\,\omega_0^{-1}$	$2.0\,\omega_0^{-1}$
$p(t)$	*Monotone decay*	*Oscillatory with envelope*
$\lvert p(t_1)\rvert$ *at first revival*	0 *(no revival)*	≈ 0.31
$\beta\,\mathcal{S}$ *at revival*	≤ 0	$\approx +0.093$
$\Gamma(t)$	> 0 *always*	*Oscillates,* < 0 *during backflow*
Interpretation	*Stone (sinks)*	*Surfer (rides backflow)*

The non-Markovian agent achieves $\beta\,\mathcal{S} \approx +0.093$ per backflow cycle (autonomous, $H_{\text{ctrl}} = 0$), while the Markovian agent can only lose free energy. As the coupling deepens ($\gamma/4\lambda \to 0$), the revival amplitude grows and $\mathcal{S}$ increases (Table 12.1), bounded above by $\beta\,\mathcal{S} \leq I(S{:}E;\, t^*)$ (Corollary 12.24(iii)).

12.7 The Cost of Memory

The preceding identity allows a correlated initial resource to fund a later interval. It does not assign a universal energetic price to storing or maintaining that resource. This section separates Landauer's erasure bound from a conditional archive-overload model used only to motivate Paper II.

12.7.1 The Landauer Debt

Landauer's principle prices a logically irreversible reset, not passive storage, sensing, correlation maintenance, or reversible computation.

Remark 12.42 (Conditional erasure cost)**.** *If a declared memory implementation irreversibly erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys*

$$W_{\text{reset}} \geq k_B T \ln 2 \, N_{\text{erase}}. \tag{12.41}$$

No lower bound proportional to total stored mutual information follows without an implementation model. Reversible storage and reversible measurement can in principle avoid this reset cost until a logically irreversible operation is performed [68, 14]. The former universal "Landauer Cost of Memory" proposition is withdrawn.

12.7.2 Raw Archive Overload Scenario

The crisis arises from the scaling of $\mathcal{I}_{\text{stored}}$ with time. To quantify this, we borrow two quantities from computational mechanics [32, 105]:

Definition 12.43 (Entropy Rate and Predictive Information)**.** *Let $\{X_t\}$ be the stochastic process describing the environment's influence on the agent (e.g., the sequence of bath correlation values).*

1. *The **entropy rate** of the environment is*

$$h_\mu := \lim_{n\to\infty} H(X_n \mid X_{n-1}, \ldots, X_1), \tag{12.42}$$

   measuring the intrinsic unpredictability per time step.

2. *The **predictive information** (excess entropy) is*

$$I_{\text{pred}} := I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big[H(X_k) - h_\mu\big], \tag{12.43}$$

   where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the past and future half-chains. This is the total amount of information about the future that is encoded in the past—the useful *memory.*

Finite predictive information does not by itself prove that a finite physical memory captures every useful feature, and positive entropy rate does not lower-bound predictive information by $h_\mu \tau_{\text{mem}}$. In particular, a stationary mixing process commonly has finite excess entropy.

Remark 12.44 (Conditional archive-overload scenario)**.** *Consider a deliberately specified implementation: an uncompressed raw archive receives records at rate r_{raw} bits per unit time, has capacity M bits, and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives*

$$\dot{W}_{\text{reset}} \geq k_B T \ln 2 \, r_{\text{erase}}. \tag{12.44}$$

For a long-memory or non-stationary source with divergent I_{pred}, the amount of predictively relevant history may also grow, but that is an additional source assumption. This archive can exceed a finite storage, throughput, or reset budget. The result is an engineering overload scenario, not a universal thermodynamic catastrophe and not a theorem that every memory implementation must compress.

12.7.3 Conditional Motivation for Compression

An agent using the raw-archive implementation may respond by increasing capacity, lowering sample rate, reversible uncomputation, sufficient statistics, lossy compression, or accepting stale data. Paper II studies one coordinate-selection model among these options.

- **Compression requires a criterion.** To decide what to keep and what to erase, the agent needs a *relevance function*—a mapping from stored correlations to survival value. This is a reference frame that ranks information by its contribution to $\mathcal{S}$.
- **Coordinate selection uses a reference frame.** If the admissible compressor is restricted to retaining a subset of supplied coordinates, choosing that subset privileges a basis. More general invariant or learned sufficient-statistic compressors need not have this form.

Remark 12.45 (The Origin of Paper II)**.** *Remark 12.44 supplies Paper II with a conditional motivation: a bounded agent attempting to evaluate an uncompressed archive needs some resource-allocation policy. It does not prove that compression, Clifford coordinates, spontaneous symmetry breaking, or a "self" is thermodynamically necessary. Those are modelling choices to be assessed in Paper II.*

12.8 Numerical Demonstration

The preceding sections establish an autonomous thermodynamic baseline and an accounting identity. The following HMM is a separate classical performance illustration: it compares finite observation windows with a Bayes filter in an aliased environment. Reward here is not the thermodynamic survival functional, so the demonstration does not test either theorem. Full code and parameters are provided for reproducibility.

12.8.1 Model

Environment. A two-hidden-state HMM with aliased observations. The hidden state $s_t \in \{0, 1\}$ evolves as a persistent Markov chain with $\Pr(s_{t+1} = s_t) = 1-\varepsilon$; the parameter $\varepsilon \in [10^{-3}, 10^{-1}]$ controls the correlation length $\ell \sim 1/\varepsilon$. Observations $o_t \in \{A, B\}$ are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5+\delta$, $\Pr(o_t = A \mid s_t = 1) = 0.5-\delta$, with $\delta = 0.05$ (mutual information $I(O; S) \approx 0.007$ bits). Reward: $r_t = 1$ if $a_t = s_t$, 0 otherwise.

Agents. All agents use the true model parameters and compute exact Bayesian posteriors; the only difference is how many observations each agent retains.

- **Markov-k** ($k \in \{1, 2, 4, 8\}$): runs an exact Bayes filter over the most recent k observations (sliding window, uniform prior at each window start); acts by MAP.
- **Memory (Bayes filter)**: maintains the full belief state $b_t = \Pr(s_t = 1 \mid o_{1:t})$ via the exact predict–update cycle over all past observations; acts by MAP.

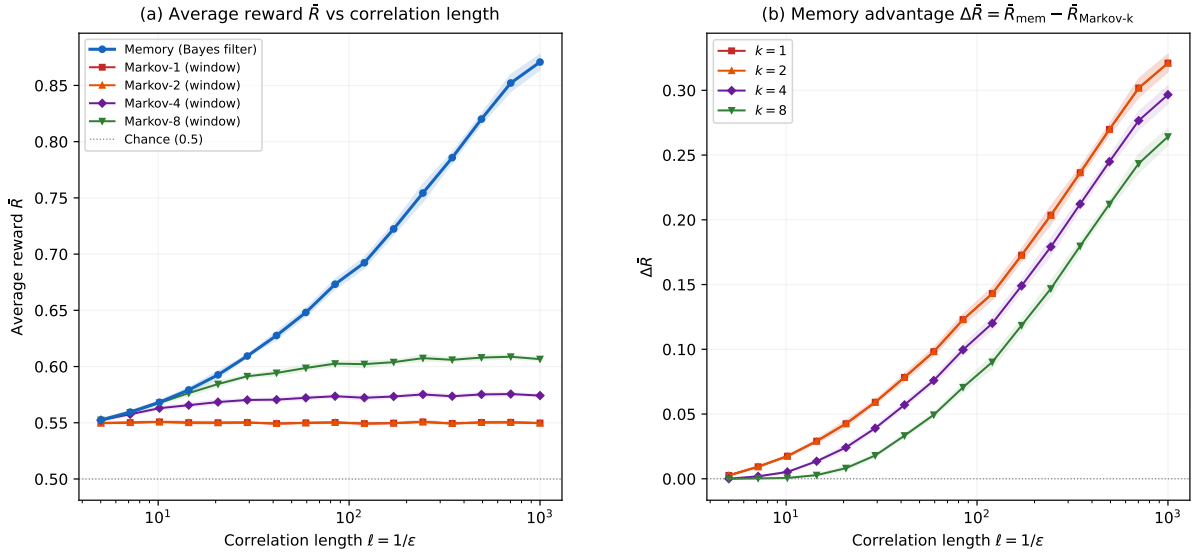


Figure 12.2: **Finite-window saturation and Bayes-filter advantage.** $T = 100{,}000$, 10 seeds, 95% CI bands. **(a)** Average reward $\bar{R}$ vs correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent ceilings. **(b)** Performance gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases with ℓ; smaller k yields a larger gap.

Parameters.

Quantity	Value	Role
T	100,000	horizon per trial
Seeds	10	independent replications
δ	0.05	observation asymmetry
k	$\{1, 2, 4, 8\}$	Markov window sizes
ε	logspace(10^{-3}, 10^{-1}, 15)	transition noise grid
Burn-in	5,000	discarded steps

12.8.2 Results

Figure 12.2 shows the two key signatures.

Result 1: finite-window performance saturation (Figure 12.2a). The average reward $\bar{R}$ of the Bayes filter (memory agent) increases monotonically with correlation length $\ell = 1/\varepsilon$, while each Markov-k agent saturates at a distinct ceiling. The ceilings are ordered: $k = 1$ (lowest) through $k = 8$ (highest), and all fall below the memory agent for $\ell \gtrsim 20$. This is a property of the specified aliased HMM and agent classes, not a consequence of the thermodynamic ceiling.

Result 2: Memory advantage (Figure 12.2b). The gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases monotonically with ℓ, and is larger for smaller k. Shaded bands show 95% confidence intervals across 10 seeds. The Markov-1 and Markov-2 curves nearly overlap at small ℓ, reflecting the fact that short observation windows provide negligible additional information in this aliasing regime—a consistency check, not a deficiency.

12.8.3 Scope of This Demonstration

These simulations illustrate finite-window saturation under the stated model class; they do not constitute a thermodynamic test.

This demonstration **does** show:

1. A reproducible regime in which finite-order Markov agents exhibit a performance ceiling while a memory-carrying (Bayes filter) agent improves.
2. The performance gap widens with correlation length and tightens with window size in this HMM. This is conceptually related to retained history but is not the quantity in Theorem 12.21.

This demonstration does **not** show:

1. Universality across environments, observation models, or agent architectures. The model uses a two-state HMM with binary aliased observations.
2. Tight constants or the functional form of the ceiling boundary $\ell_c(k)$.
3. That the Bayes filter is optimal among all possible memory-carrying agents.

Reproducibility. The complete simulation is a self-contained Python script (`paper1_markov_ceiling_demo.py`, $\sim$560 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper1_markov_ceiling_demo.py` — simulation script
- `fig_paper1_markov_ceiling.pdf` — Figure 12.2
- `markov_ceiling_data.csv` — raw sweep data
- `markov_ceiling_boundary.csv` — extracted ceiling boundaries $\ell_c(k)$

12.9 Discussion

12.9.1 Summary of Results

Result	**Statement**	**Sec.**
Restricted autonomous ceiling	$\mathcal{S} \leq 0$ for a fixed-Hamiltonian Davies semigroup	12.3
Initial-resource balance	$\beta\mathcal{S} = -\Delta I - \Delta D_{\mathrm{KL}} - \beta\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\Delta\langle H_{\mathrm{int}}\rangle$	12.4
Quantitative demo	Spin-boson: $\beta\mathcal{S} \approx +0.093 > 0$ at first backflow revival (Fig. 12.1, Table 12.1)	12.6
Memory/record interface	operational influence relation plus a conditional record test	12.5
Archive overload	conditional on raw storage, finite capacity, and irreversible reset	12.7
Numerical demo	finite-window saturation in an aliased HMM (Fig. 12.2)	12.8

12.9.2 What This Paper Does and Does Not Show

This paper shows:

1. Under an autonomous fixed-Hamiltonian Davies semigroup, the survival functional $\mathcal{S} \leq 0$ (Theorem 12.13).

2. For any initial state (product or correlated), the survival functional satisfies the exact identity $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle$ (Theorem 12.21). Under autonomous evolution, when pre-existing system–environment correlations are consumed ($\Delta I < 0$), $\mathcal{S} > 0$ is achievable, bounded by the initial correlation budget (Corollary 12.24).

3. A quantitative spin-boson example illustrates: $\beta\,\mathcal{S} \approx +0.093 > 0$ at the first non-Markovian revival (Section 12.6).

4. The memory-influence relation and HAFF record ordering define a conditional model test, not a proved embedding (Conjecture 12.35).

5. Landauer supplies a conditional reset cost; a raw finite archive can overload under the implementation assumptions in Remark 12.44.

6. A computational demonstration exhibits finite-window saturation in a two-state HMM with aliased observations (Section 12.8, Figure 12.2).

This paper does not show:

1. That memory or non-Markovianity is necessary or sufficient for persistence.

2. That non-Markovian systems optimize a larger survival supremum than all Markovian systems.

3. That driven or measurement-feedback Markovian agents obey the restricted autonomous ceiling (Remark 12.14).

4. That the specific form of the optimal memory kernel can be derived from first principles without specifying the environment.

5. That memory implies or requires consciousness.

6. That storage or maintenance has a universal Landauer cost, or that compression and symmetry breaking are thermodynamic necessities.

Chapter 13

Spontaneous Symmetry Breaking of Reference Frames as a Computational Cost Minimization Strategy

Paper II — "The Ego"

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Abstract

We investigate a conditional implementation model for bounded agents that process memory-like state under finite computational resources. Revised Paper I [163] supplies a conditional correlation-resource account; it does not establish universal memory necessity, a universal Markovian ceiling, or a Landauer charge for merely storing records.

The present paper therefore studies a declared raw D-channel archive processed under a finite throughput budget. Within that model, symmetric processing of all declared channels can exceed the available budget at a finite time, while a supplied coordinate selector can reduce the declared load. This is a conditional overload result for the stated archive and algorithm, not a theorem that every persistent agent must compress memory or break a symmetry.

Reference-frame selection is treated as a modelling strategy: a privileged basis or coordinate projection packages a lower-dimensional representation of the archive. The rate-distortion calculation gives a budget-dependent target dimension under the stated source, distortion, and budget assumptions. It does not prove that the "self" is a universal survival-optimal structure.

The projected representation is accompanied by four diagnostic labels: basis selection, frame drag, objective centering, and model incompleteness. Only basis selection follows from projection alone; the other diagnostics require additional controller, update, objective, or task assumptions. A simple drift model gives a visibility crossover for a fixed frame, motivating Paper III's calibration problem without establishing an undetectable universal Delusion Trap.

13.1 Introduction

13.1.1 Context: The Problem of Overload

Revised Paper I [163] gives an ELV-type accounting identity in which initial correlations can fund a later interval and proves a ceiling only for a fixed-Hamiltonian autonomous Davies baseline. It retracts both universal memory necessity and a Landauer charge for merely storing or maintaining records.

The present paper therefore begins with an independent implementation model: a raw, uncompressed D-channel archive processed under a finite throughput budget. Its overload is conditional on that archive and algorithm. The question is when a supplied coordinate selector can reduce this declared load, not whether every persistent agent must compress or break symmetry.

13.1.2 Position within the Series

This paper is the second of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	**Question**		**Result**	**Status**
HAFF [148, 149]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [160, 161]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I [163]	What can inter-epoch correlations fund?	Seed	Restricted baseline; accounting identity	Conditional
T-DOME II (this work)	When can a model select a frame?	Ego	Conditional coordinate selection	**This paper**
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence rather than an irreversible theorem chain:

1. **Paper I (The Seed):** An ELV-type balance identifies initial correlations as one possible interval resource. Raw-archive overload is an implementation scenario, not a thermodynamic necessity.

2. **Paper II (The Ego, this work):** Unbounded memory under finite computational resources causes processing overload in the coordinate-processing model. This paper studies conditional feature selection; it does not prove universal SSB.

3. **Paper III (The Loop):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop (monitoring one's own prediction error) resolves the bias but requires the system to "observe its own observation"—closing the self-calibration loop.

13.1.3 Relation to Q-RAIF

Revised Q-RAIF Papers A and B [160, 161] use a Clifford algebra $Cl(V, q)$ as a supplied representation choice; persistence does not select it. Paper C [162] proves only that the image of a separately supplied homomorphism lies in the external algebra. This paper adopts both the Clifford coordinates and the representation map as model inputs.

The Clifford algebra $Cl(V, q)$, however, admits a non-trivial *automorphism group* $G = \mathrm{Aut}(Cl(V, q))$. In the absence of external constraints, all elements of G yield physically equivalent representations—the choice of basis within the algebra is a *gauge freedom*. This gauge freedom is the mathematical substrate of the symmetry considered in the conditional breaking analysis below.

The "ego" names a selected computational frame in that supplied representation. Whether a concrete optimum is a genuine gauge fixing is a model condition, not an inherited theorem.

13.1.4 Relation to HAFF Paper G

Revised HAFF Paper G [152] offers philosophy-of-science motivation and no fixed-point no-go theorem. It does not imply basis selection here. Any connection between its meta-level concern and this finite-budget model is motivation only.

13.1.5 Scope and Disclaimers

To prevent interpretational overreach, we state at the outset what this paper does *not* claim:

1. We do not claim that symmetry breaking is *sufficient* for persistence. Paper III addresses the additional requirements.

2. We do not claim that basis selection or spontaneous symmetry breaking is necessary for every bounded agent. The results concern a declared coordinate-retention model.

3. The term "ego" or "self" is used in the control-theoretic sense: a fixed reference frame within the agent's internal algebra. It carries no implication of consciousness or subjective experience.

4. A broader structural analogy with classical philosophical concepts of selfhood exists but is outside the scope of this paper.

Related work. The idea that bounded agents must compress their representations has roots in Simon's bounded rationality [108], Shannon's rate-distortion theory [106, 30], and Sims's rational inattention [109], which models finite-capacity decision-makers as solving a rate-distortion problem—precisely the economic counterpart of our $\mathcal{C}_{\mathrm{budget}}$ formalism. The information bottleneck [120] formalises relevance-weighted compression and has been applied to neural coding and deep learning. The role of decoherence in selecting preferred bases (pointer states) is well established via quantum Darwinism [143]; here it motivates one model-specific coordinate-selection rule. Measures of non-Markovianity and their thermodynamic consequences are reviewed in [94, 22]; the connection to survival was established in Paper I.

Summary of contributions. This paper separates three levels of result:

1. **Raw-channel load scaling law** a conditional accounting identity for a raw D-channel archive.
2. **Survival-weighted rate-distortion bound** a compactness-based existence result; the budget supplies only a capacity $k_{\max}$, not a universal optimal retained dimension.
3. **Delusion dynamics** a linear-drift visibility crossover plus an explicit two-mode example showing when gradient anti-correlation can occur.

Result-status ledger. The original Computational Ceiling and SSB-necessity chain is retracted ('TD-II-T3.5'). Raw-channel load is an FD/I accounting proposition; compression existence is conditional on a compact feasible model; SSB is a model interpretation. The Delusion Trap is downgraded ('TD-II-T6.7'): visibility crossover is exact in the declared two-mode geometry, while gradient sign and self-detection require extra dynamics.

13.2 Mathematical Preliminaries

13.2.1 Inherited Framework from Paper I

We briefly recall the key objects from Paper I [163] that the present work builds upon. The reader is referred to Paper I for full definitions and proofs.

Survival functional. For an open quantum system S coupled to an environment E at inverse temperature β, with dynamics Λ and external control protocol $H_{\text{ctrl}}(t)$, the survival functional is

$$\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau], \tag{13.1}$$

where $\Delta F = F(\rho(\tau)) - F(\rho(0))$ is the change in non-equilibrium free energy and $W = \int_0^\tau \operatorname{tr}\Big(\rho(t)\, \dot{H}_{\text{ctrl}}(t)\Big)\, dt$ is the work performed by the external protocol.

Restricted autonomous baseline. For a fixed-Hamiltonian Davies semigroup with no control, Paper I proves

$$\mathcal{S}[\Lambda^{\text{M}}, \tau] \leq 0 \qquad \text{for all } \tau \geq 0. \tag{13.2}$$

It does not extend this result to driven Markovian agents.

Non-Markovian advantage identity. For arbitrary initial states:

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\text{KL}}(\rho_E \| \rho_E^{\text{th}}) - \beta\, \Delta\langle H_{\text{ctrl}} \rangle - \beta\, \Delta\langle H_{\text{int}} \rangle. \tag{13.3}$$

Conditional raw-archive model. Paper I retracts a universal Landauer maintenance bound. This paper independently posits an uncompressed record stream. Let h_μ be its *per-component* entropy rate [32], defined by

$$h_\mu := \lim_{T\to\infty} \frac{1}{T} H(X_{0:T}), \tag{13.4}$$

measuring the asymptotic information (in bits per unit time) generated by one declared record channel (we work in units where the sampling interval equals the environmental correlation time τ_E)[1] The predictive information I_{pred} is the *excess entropy* (excess entropy) [15, 105], defined as the mutual information between past and future of the environmental process:

$$I_{\text{pred}} := I\Big(\overleftarrow{X};\, \overrightarrow{X}\Big) = H(\overrightarrow{X}) - H(\overrightarrow{X} \mid \overleftarrow{X}), \tag{13.5}$$

where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the semi-infinite past and future, respectively. For a stationary process, I_{pred} relates to h_μ via the entropy-rate decomposition $H(X_{1:T}) = I_{\text{pred}} + h_\mu T + o(1)$ as $T \to \infty$ when the relevant limits exist [32]. No lower bound $I_{\text{stored}} \geq \min(I_{\text{pred}}, h_\mu \tau)$ is inherited.

13.2.2 The Agent's Internal Algebra

As a modelling choice motivated by revised Q-RAIF [161, 162], the agent's internal control algebra is taken to be a Clifford algebra $\mathcal{O}_{\text{int}} = Cl(V, q)$ for a real vector space V equipped with a non-degenerate quadratic form q. The algebra satisfies the fundamental relation $v^2 = q(v)\,\mathbf{1}$ for all $v \in V$.

The *automorphism group*

$$G := \text{Aut}(Cl(V, q)) \tag{13.6}$$

is the group of algebra automorphisms that preserve the grading and quadratic form.[2] For $Cl(1, 3)$, G contains the spin group $\text{Spin}(1, 3) \cong SL(2, \mathbb{C})$ as a subgroup—a six-real-dimensional Lie group.

In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions of the agent's internal state. The choice of basis within $Cl(V, q)$ is a *gauge freedom*—the symmetry examined in the conditional coordinate-selection model below.

A representation map $\phi : Cl(V, q) \to \mathcal{A}_{\text{ext}}$ is supplied as an interface. Q-RAIF Paper C proves only the image-level algebraic statement once this map is given; it does not select a preferred $Cl(1, 3)$ target or preserve quadratic signature.

Dimensional convention. Two distinct notions of dimension appear throughout:

Symbol	Meaning	Scaling
$n := \dim V$	number of generators (degrees of freedom)	—
$D := \dim Cl(V, q) = 2^n$	full multivector space (algebra basis size)	exponential in n

[1] For a continuous-valued process sampled at resolution b bits, h_μ includes the quantisation cost: $h_\mu = h_\mu^{(\text{diff})} + b\, f_s$, where $h_\mu^{(\text{diff})}$ is the differential entropy rate and f_s the sampling frequency. All raw-channel accounting identities in this paper use h_μ so defined.

[2] We use G as an effective symmetry group acting transitively on admissible frames. The detailed Lie-algebraic structure of G is not required for our results; only the existence of a non-trivial symmetry that may be broken under the additional encoder conditions in Remark 13.18. In concrete models, one may replace G by its image under the adjoint representation—typically $O(V, q)$ or a pin/spin subgroup.

The raw-channel load model (Section 13.3) scales with D, not n; the distinction matters whenever one compares generator-level and algebra-level quantities.

13.2.3 Rate-Distortion Theory

We require the classical rate-distortion framework of Shannon [106].

Definition 13.1 (Rate-distortion function)**.** *Let X be a random source with distribution $p(x)$, $\hat{X}$ a reconstruction, and $d : \mathcal{X} \times \hat{\mathcal{X}} \to [0, \infty)$ a distortion measure. The* rate-distortion function *is*

$$R(D) := \min_{\substack{p(\hat{x}|x): \\ \mathbb{E}[d(X,\hat{X})] \leq D}} I(X; \hat{X}), \tag{13.7}$$

the minimum mutual information between source and reconstruction that achieves average distortion at most D.

$R(D)$ is a convex, non-increasing function of D with $R(0) = H(X)$ (lossless) and $R(D_{\max}) = 0$ (maximum distortion). It provides the fundamental limit on lossy compression [30]. The *information bottleneck* method of Tishby et al. [120] generalises this framework to the case where the relevant variable is not the source itself but a downstream prediction target—precisely the situation in our survival-weighted compression problem (Section 13.4.2). For source X, representation Z, and declared survival-relevant target Y, the bottleneck objective is

$$\min_{p(z|x)} I(X; Z) - \beta_{\mathrm{IB}} I(Z; Y). \tag{13.8}$$

The coordinate-retention model below does not solve this unrestricted optimization. Rate-distortion and information bottleneck therefore provide comparison frameworks, not authority for the component count.

13.2.4 Bounded Rationality

Following Simon [108], we model computational limitations as a hard constraint on the agent's information processing rate.

Definition 13.2 (Computational budget)**.** *The agent's* computational budget $\mathcal{C}_{\mathrm{budget}}$ *(measured in bits per unit time) is the maximum rate at which the agent can evaluate functions of its stored correlations. We assume $\mathcal{C}_{\mathrm{budget}} < \infty$.*

Physically, finiteness of $\mathcal{C}_{\mathrm{budget}}$ reflects the finite number of degrees of freedom in the agent's physical substrate: finite Hilbert space dimension, finite memory register size, and finite energy available for computation (Landauer's principle [68, 14]).

13.2.5 Fiber Bundle Formalism

The geometric setting for reference-frame selection is a principal fiber bundle.

Definition 13.3 (Gauge bundle)**.** *The* gauge bundle *is the principal G-bundle*

$$\pi : P \to M, \qquad G = \mathrm{Aut}(Cl(V, q)), \tag{13.9}$$

where:

- *M is the base space of* effective memory kernels*—equivalently, the space of induced sufficient-statistic processes accessible to the agent (a finite-dimensional manifold that admits local parametrisation by the environmental spectral-density couplings);*
- *G is the structure group acting transitively on admissible frames (see footnote 2 for the effective subgroup);*
- *the fiber $\pi^{-1}(\kappa)$ over a kernel $\kappa \in M$ is the G-orbit of equivalent algebraic representations (frames) for describing κ in $Cl(V, q)$;*
- *a* section *$\sigma : M \to P$ constitutes a global gauge-fixing policy—a systematic choice of reference frame for every kernel configuration.*

A *connection* on P specifies how the reference frame is parallel-transported as the agent's state evolves. The curvature of this connection measures the extent to which the reference frame "twists" along different paths through state space.

13.2.6 Standing Assumptions

Definition 13.4 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(B1) ***Inherited framework.*** *All assumptions (A1)–(A5) of Paper I [163] remain in force (open quantum system coupled to a thermal bath, well-defined free energy, non-equilibrium initial state, finite-dimensional system Hilbert space, and weak-coupling or controlled-coupling regime). Additionally, the agent possesses an internal control algebra $\mathcal{O}_{\text{int}} = Cl(V, q)$ with a supplied representation map $\phi : Cl(V, q) \to \mathcal{A}_{\text{ext}}$ (Q-RAIF [162]). Neither the Clifford choice nor the map is derived from persistence.*

(B2) ***Finite computational budget.*** *The agent's information processing rate satisfies $\mathcal{C}_{\text{budget}} < \infty$ (Definition 13.2).*

(B3) ***Raw-channel load model.*** *The supplied record process has D declared component channels, per-channel innovation rate $h_\mu > 0$, memory depth $\tau_{\text{mem}} > 0$, and no shared-code or group-theoretic compression before the accounting in Section 13.3. This worst-case assumption, not environmental non-triviality alone, carries the load calculation.*

(B4) ***Survival imperative.*** *The agent's dynamics must maintain $\mathcal{S} \geq \mathcal{S}_{\min}$ over survival horizons $T \gg \tau_{\text{mem}}$. This is a persistence constraint, not an optimization objective.*

(B5) ***Gauge symmetry of bare algebra.*** *The automorphism group $G = \text{Aut}(Cl(V, q))$ is non-trivial ($G \neq \{e\}$). In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions.*

13.3 Raw-Channel Load Model

We now record the throughput of the raw-channel model in (B3). This is a bookkeeping identity for an explicitly compression-free representation, not a lower bound over all symmetric algorithms and not a consequence of Paper I.

13.3.1 The Information Processing Inequality for Bounded Agents

Accounting convention. To ensure dimensional consistency throughout, we distinguish two quantities:

- $\mathcal{C}_{\text{budget}}$: the agent's processing *rate* (bits per unit time).
- $\mathcal{I}_{\text{proc}}(\tau)$: the total information (bits) that must be processed per evaluation cycle when the memory archive has depth τ.

The agent must complete one evaluation cycle per environmental correlation time τ_E. The *processing rate* required for a memory depth τ is

$$\mathcal{R}_{\text{proc}}(\tau) := \frac{\mathcal{I}_{\text{proc}}(\tau)}{\tau_E}. \tag{13.10}$$

Paralysis occurs when $\mathcal{R}_{\text{proc}}(\tau_{\text{mem}}) > \mathcal{C}_{\text{budget}}$. Hereafter we measure time in units of τ_E (i.e., set $\tau_E = 1$), so that rates and per-cycle information quantities are numerically equal.

Definition 13.5 (Symmetric processing)**.** *An agent processes its memory* symmetrically *if both its cost functional $\mathcal{C}[\cdot]$ and its distortion measure $D(\cdot)$ are G-invariant: $\mathcal{C}[g \cdot \mathcal{K}] = \mathcal{C}[\mathcal{K}]$ and $D(g \cdot \mathcal{F}) = D(\mathcal{F})$ for every $g \in G = \mathrm{Aut}(Cl(V, q))$. In operational terms: for every stored correlation c_i in the memory kernel $\mathcal{K}(t, s)$ and every $g \in G$, the cost of evaluating c_i equals the cost of evaluating $g \cdot c_i$, and no basis direction is* a priori *preferred for survival evaluation.*

Remark 13.6 (Operational meaning of processing rate)**.** *We define the processing rate $\mathcal{R}_{\text{proc}}$ as an* information-throughput *measure: the number of algebraic components that must be updated per unit time, multiplied by the innovation rate h_μ per component. It captures the* bandwidth *cost of maintaining an internal representation, not the algorithmic gate complexity of individual operations.*

Proposition 13.7 (Raw-channel load accounting)**.** *Under (B3), let $n = \dim V$ and $D = \dim Cl(V, q) = 2^n$. A raw archive of depth τ_{mem} contains, by the model definition,*

$$\mathcal{I}_{\text{archive}} = D\, h_\mu\, \tau_{\text{mem}} \qquad \textit{bits.} \tag{13.11}$$

Its incremental input rate is Dh_μ bits per unit time. If the declared algorithm rescans the entire archive once per environmental correlation time τ_E, its required throughput is

$$\mathcal{R}_{\text{sweep}} = \frac{Dh_\mu \tau_{\text{mem}}}{\tau_E}. \tag{13.12}$$

For a processing budget $\mathcal{C}_{\text{budget}}$, this particular full-sweep algorithm fits only when

$$\tau_{\text{mem}} \le \tau_{\max} := \frac{\mathcal{C}_{\text{budget}} \tau_E}{Dh_\mu}. \tag{13.13}$$

Proof. All three statements follow by multiplying the declared number of channels by their per-channel innovation rate and then converting stored bits per sweep to bits per unit time. No lower bound over compressed, streaming, sufficient-statistic, or group-equivariant algorithms is claimed. □

Remark 13.8 (Conditional component-count ratio)**.** *Within the same raw-coordinate implementation, comparing a D-channel sweep with a k-channel sweep gives the component-count ratio D/k. This is not a tax imposed by gauge invariance on all algorithms.*

Remark 13.9 (Effective vs. full dimension)**.** *The ceiling uses $D = \dim Cl(V,q) = 2^n$, the full multivector dimension. In practice, the environment may couple to only a subset of grades (e.g., grade-1 generators), yielding an effective dimension $D_{\text{eff}} \leq D$. For a* structured *environment where the agent knows which grades are active, the ceiling can be tightened to $\mathcal{R}_{\text{proc}} \gtrsim h_\mu \cdot D_{\text{eff}}$. The raw-channel condition in (B3) represents this worst case. A structured environment or algorithm changes the model rather than falling under a universal ceiling.*

13.3.2 Processing Collapse

Remark 13.10 (Raw full-sweep overload)**.** *If $\tau_{\text{mem}} > \tau_{\max}$, the full-sweep algorithm in Proposition 13.7 misses its declared cycle deadline. This does not imply that every record is useless or that another streaming/compressed algorithm fails.*

Remark 13.11 (Comparison with revised Paper I)**.** *Paper I now treats storage/reset cost as implementation-dependent. The present overload is likewise conditional on a raw full-sweep algorithm. Compression is one response; neither paper proves it must take the form of symmetry breaking.*

13.4 Conditional Frame-Selection Model

13.4.1 Reference Frame as Gauge Fixing

Definition 13.12 (Reference frame)**.** *A* reference frame *$\mathcal{F}$ is a section $\sigma : M \to P$ of the gauge bundle (Definition 13.3). Choosing σ is equivalent to selecting a preferred orthonormal basis $\{e_1, \ldots, e_n\}$ of the generating vector space V at each point in state space M, thereby fixing the gauge freedom of $Cl(V,q)$.*

Definition 13.13 (Projected memory kernel)**.** *Given a reference frame $\mathcal{F}$, let $V_{\text{fg}}(\mathcal{F}) \subset Cl(V,q)$ be a k-dimensional* foreground subspace *selected by the declared coordinate model, with $0 \leq k \leq D$. Let $\Pi_{\mathcal{F}}$ denote the orthogonal projection onto $V_{\text{fg}}(\mathcal{F})$ with respect to the trace inner product $\langle A, B\rangle := \operatorname{tr}\big(A^\dagger B\big)$. The* projected memory kernel *is*

$$\mathcal{K}_{\mathcal{F}}(t,s) := \Pi_{\mathcal{F}}\, \mathcal{K}(t,s)\, \Pi_{\mathcal{F}}. \tag{13.14}$$

The complementary projection $\Pi_{\mathcal{F}}^{\perp} = \mathbf{1} - \Pi_{\mathcal{F}}$ defines the background subspace *$V_{\text{bg}}(\mathcal{F})$. The decomposition $Cl(V,q) = V_{\text{fg}} \oplus V_{\text{bg}}$ is determined by $\mathcal{F}$, not by any* a priori *ordering of basis vectors.*

13.4.2 A Coordinate-Compression Model

We now define a restricted coordinate-compression problem and compare it with rate-distortion theory.

Processing rate of a frame. If the agent retains k algebraic components (the foreground subspace V_{fg}), each generating h_μ bits per unit time, the processing rate of frame $\mathcal{F}$ is

$$R_{\mathcal{F}}(k) \;=\; k \cdot h_\mu \qquad \text{(bits per unit time)}. \tag{13.15}$$

The budget constraint $R_{\mathcal{F}} \le \mathcal{C}_{\text{budget}}$ thus bounds the number of maintainable components. Define the capacity

$$k_{\max} := \min\!\left(D,\, \left\lfloor \frac{\mathcal{C}_{\text{budget}}}{h_\mu} \right\rfloor\right). \tag{13.16}$$

This is a maximum feasible component count, not yet an optimizer.

Definition 13.14 (Survival distortion)**.** *The* survival distortion *of a reference frame $\mathcal{F}$ is*

$$D(\mathcal{F}) := \mathbb{E}_\xi\big[\ell\big(\mathcal{S}_{\text{full}}(\xi) - \mathcal{S}_{\mathcal{F}}(\xi)\big)\big], \tag{13.17}$$

where ξ denotes environmental realizations, $\ell : \mathbb{R} \to [0,\infty)$ is a convex, non-decreasing loss function (we use squared error $\ell(x) = x^2$ throughout), $\mathcal{S}_{\text{full}}(\xi)$ is the survival functional evaluated using the full memory kernel $\mathcal{K}(t,s)$, and $\mathcal{S}_{\mathcal{F}}(\xi)$ is evaluated using the projected kernel $\mathcal{K}_{\mathcal{F}}(t,s)$.

Remark 13.15 (Information-theoretic objects)**.** *Strictly speaking, rate-distortion theory and mutual information apply to stochastic processes, not to superoperator kernels directly. Throughout Sections 13.4–13.7, $I(\mathcal{K}_{\mathcal{F}};\mathcal{K})$ is shorthand for $I(\hat{X};X)$, where $X = \{c_i(t)\}_{i=1}^{D}$ is the sufficient-statistic record process induced by the full kernel $\mathcal{K}$ acting on the agent's internal coordinates, and $\hat{X} = \{c_i(t)\}_{i\in V_{\text{fg}}}$ is the projected record induced by $\mathcal{K}_{\mathcal{F}}$. The distortion measure* (13.17) *acts on the survival functional $\mathcal{S}$ evaluated on these records.*

Proposition 13.16 (Existence in the coordinate model)**.** *For each $0 \le k \le k_{\max}$, let $\mathfrak{F}_k$ be a compact set of admissible k-component frames. Assume the feasible union $\mathfrak{F}_{\le k_{\max}}$ is nonempty and the distortion $D(\mathcal{F})$ is lower semicontinuous. Then*

$$\mathfrak{F}^* := \operatorname*{arg\,min}_{\mathcal{F}\in\mathfrak{F}_{\le k_{\max}}} D(\mathcal{F}) \tag{13.18}$$

is nonempty. Every optimizer has some retained dimension $k_{\text{opt}} \le k_{\max}$. It follows only by counting that its discarded component fraction is

$$1 - \frac{k_{\text{opt}}}{D}. \tag{13.19}$$

Budget saturation $k_{\text{opt}} = k_{\max}$ requires the additional strict-improvement condition that every feasible added component lowers the best attainable distortion. Without it, unused capacity is possible.

Proof. The feasible union is compact (a finite union of compact sets), so a lower-semicontinuous function attains its minimum. The dimension and discard fraction statements are definitions. Mere non-increase of distortion does not imply strict improvement or budget saturation. □

Remark 13.17 (Relation to Shannon and information bottleneck)**.** *Proposition 13.16 is not a Shannon converse or an information-bottleneck solution. Those theories optimize over stochastic encoders; this proposition optimizes over a supplied compact family of coordinate projections. The original theorem claiming $k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ as the universal optimum is withdrawn.*

For the remainder of the paper, k^* denotes the retained dimension of one chosen optimizer from $\mathfrak{F}^*$, when such a choice is made. Thus $k^* = k_{\text{opt}} \leq k_{\max}$; it is not identified with $\lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ unless the stated strict-improvement condition is separately verified.

13.4.3 Conditional Symmetry-Breaking Interpretation

Remark 13.18 (Conditional symmetry-breaking interpretation)**.** *Suppose, in addition to Proposition 13.16, that (i) every admissible compressor is a proper coordinate projection, (ii) the effective group G acts on the candidate frames, and (iii) no G-invariant feasible encoder attains the required distortion. Then choosing any feasible optimizer privileges a foreground subspace of the D-dimensional algebra representation and breaks that model symmetry operationally. This conclusion is conditional on the encoder class; it is not a universal necessity theorem.*

If $D(\mathcal{F}) \neq D(\mathcal{F}')$ for distinct frames, that condition supports uniqueness of a global minimizer, not multiple local minima. Multiple "ego attractors" require a separate non-convex landscape with several demonstrated minima. The original SSB theorem and its generic non-uniqueness claim are withdrawn.

13.4.4 The Four Bias Terms

Definition 13.19 (Four frame-dependent diagnostics)**.** *For a declared coordinate projection $\Pi_{\mathcal{F}}$ and a specified controller, we use the following four labels as diagnostics. They organize possible departures from a symmetric representation; they are not asserted to be universal consequences of gauge fixing.*

(i) ***Basis selection bias*** *($\mathcal{B}_{\text{select}}$): The choice of $\{e_1, \ldots, e_{k^*}\}$ privileges certain algebraic components over others. Information aligned with the chosen basis is processed efficiently; misaligned information is discarded or distorted.* Observable consequence: *systematic blindness to off-basis environmental perturbations (orthogonal masking).*

(ii) ***Frame drag*** *($\mathcal{B}_{\text{frame}}$): If the update rule transports the current frame with memory or a restoring penalty, it may prefer states near that choice.* Possible observable consequence: *hysteresis in belief updating and lag behind environmental shifts. Holonomy alone does not supply a restoring force.*

(iii) ***Objective centering*** *($\mathcal{B}_{\text{center}}$): A projected surrogate objective depends only on the retained coordinates. Unless it is sufficient for the full objective, its stationary points need not be global stationary points.* Possible observable consequence: *failure to detect optima whose relevant directions lie in the background subspace.*

(iv) ***Model incompleteness*** *($\mathcal{B}_{\text{inc}}$): The compression from $Cl(V, q)$ to V_{fg} is lossy. The discarded components V_{bg} may contain correlations absent from the retained representation.* Possible observable consequence: *underestimation of total uncertainty when those correlations matter to the task.*

Only the first item follows from the projection alone: components in $\ker \Pi_{\mathcal{F}}$ are removed. Frame drag requires a stateful update law, objective centering requires a specified

surrogate loss, and overconfidence requires task-relevant information in the discarded coordinates. Proposition 13.16 supplies no generic information fraction for these diagnostics.

Bias	**Origin**	**Observable consequence**	**Determines**
$\mathcal{B}_{\text{select}}$ (selection)	projection $\Pi_{\mathcal{F}}$	Systematic blindness to off-basis perturbations (orthogonal masking)	*what* is seen
$\mathcal{B}_{\text{frame}}$ (frame drag)	bundle connection / holonomy	Hysteresis in belief updating; model lags behind rapid drift	*duration*
$\mathcal{B}_{\text{center}}$ (centering)	$\nabla\mathcal{S} \in V_{\text{fg}}$	Local frame-relative optima; global background optima invisible	*target*
$\mathcal{B}_{\text{inc}}$ (incompleteness)	lossy compression $k^* \ll D$	Underestimation of thermodynamic uncertainty (structural overconfidence)	*blind spot*

Table 13.1: The four bias terms of the broken phase. Only the first follows from projection alone; the others require the additional update, objective, or task assumptions stated in the text.

Remark 13.20 (Nature of the bias terms)**.** *The four bias terms (Table 13.1) are not pathologies. They are diagnostic names for distinct ways in which a projected controller may depart from a symmetric representation. Projection alone gives basis selection; the remaining terms become live only when the corresponding update, objective, or task structure is specified.*

13.5 Emergent Structure: The Architecture of Ego

We consolidate the gauge-fixed compressed representation into a single mathematical object. Throughout this section, "ego" is used purely as shorthand for a gauge-fixed compressed representation; no claims about phenomenal consciousness, subjective experience, or qualia are intended or implied.

Definition 13.21 (Ego)**.** *The* ego *of an agent satisfying* (B1)–(B5) *is the pair*

$$\mathfrak{E} := \big(\mathcal{F}^*,\ V_{\text{fg}}^*\big), \tag{13.20}$$

where $\mathcal{F}^* \in \mathfrak{F}^*$ (Proposition 13.16) is a chosen frame (providing the coordinate system) and $V_{\text{fg}}^* := V_{\text{fg}}(\mathcal{F}^*)$ is the k^*-dimensional foreground subspace selected within the coordinate model (providing the compression). The projected memory kernel $\mathcal{K}_{\mathfrak{E}} := \Pi_{V_{\text{fg}}^*}\,\mathcal{K}\,\Pi_{V_{\text{fg}}^*}$ is induced by this pair. All bias terms, distortion bounds, and delusion dynamics are functions of $\mathfrak{E}$.

13.5.1 The Ego as a Fiber Bundle Section

The reference frame $\mathcal{F}$, understood as a section $\sigma : M \to P$, is the mathematical object we call the *ego*. It has three key properties:

Smoothness. The section σ varies continuously with the agent's state $\rho \in M$. Small changes in ρ produce small changes in the preferred basis—the ego is not a discrete switch but a smooth deformation of perspective.

Holonomy. If the agent's state traces a closed loop $\gamma : [0, 1] \to M$ with $\gamma(0) = \gamma(1) = \rho_0$, the parallel-transported frame need not return to its initial value:

$$\sigma(\gamma(1)) = \mathrm{Hol}(\gamma) \cdot \sigma(\gamma(0)), \tag{13.21}$$

where $\mathrm{Hol}(\gamma) \in G$ is the holonomy of the connection around γ. Non-trivial holonomy means the agent can "learn"—its reference frame shifts after a complete cycle of experience.

Topological obstruction. In general, a *global* section $\sigma : M \to P$ may not exist. The obstruction is measured by the characteristic classes of the bundle P. When a global section does not exist, the ego must have "singularities"—states where the preferred basis is undefined or discontinuous. This connects to the crisis of Paper III: the delusion trap can be understood as the agent approaching a topological obstruction of its own reference frame.

13.5.2 The Effective Survival Functional

Proposition 13.22 (Survival decomposition)**.** *In the broken phase, the survival functional decomposes as*

$$\mathcal{S} = \mathcal{S}_{\mathrm{vis}}(\mathcal{F}) + \mathcal{S}_{\mathrm{hid}}(\mathcal{F}), \tag{13.22}$$

where:

- $\mathcal{S}_{\mathrm{vis}}(\mathcal{F})$ *is the contribution from the foreground subspace* V_{fg}*, computable within the agent's reference frame;*
- $\mathcal{S}_{\mathrm{hid}}(\mathcal{F})$ *is the contribution from the background subspace* V_{bg}*, invisible to the agent.*

The agent maximizes $\mathcal{S}_{\mathrm{vis}}$ *while being structurally blind to* $\mathcal{S}_{\mathrm{hid}}$*.*

Proof. The survival functional $\mathcal{S} = \Delta F - W$ depends on $\rho(t)$, which is a function of the full memory kernel $\mathcal{K}(t, s)$. Decomposing $\mathcal{K} = \Pi_{\mathcal{F}} \mathcal{K} \Pi_{\mathcal{F}} + \Pi_{\mathcal{F}}^{\perp} \mathcal{K} \Pi_{\mathcal{F}}^{\perp}$ + cross terms, the leading contributions are $\mathcal{S}_{\mathrm{vis}} := \mathcal{S}[\Pi_{\mathcal{F}} \mathcal{K} \Pi_{\mathcal{F}}]$ and $\mathcal{S}_{\mathrm{hid}} := \mathcal{S} - \mathcal{S}_{\mathrm{vis}}$ (collecting background and cross terms). The agent computes only $\mathcal{S}_{\mathrm{vis}}$, as the projected kernel $\mathcal{K}_{\mathcal{F}}$ discards all background components. □

13.5.3 The Computational Speedup

Proposition 13.23 (Ego dividend)**.** *After symmetry breaking, the computational cost of processing memory drops from* $\mathcal{C}_{\text{proc}} \sim h_\mu \cdot \tau_{\text{mem}} \cdot D$ *(symmetric case,* $D = \dim Cl(V, q)$*) to*

$$\mathcal{C}_{\text{proc}}^{(\mathcal{F})} \sim h_\mu \cdot \tau_{\text{mem}} \cdot k^*. \tag{13.23}$$

The speedup factor is

$$\frac{D}{k^*} = \frac{2^n}{k^*}. \tag{13.24}$$

This is the computational advantage of reference-frame selection. For $Cl(1, 3)$ ($D = 16$) with $k^* = 2$, the speedup is $8\times$. For higher-dimensional algebras, the speedup grows exponentially in n.

13.5.4 Uniform-Channel Information Trade-off

Theorem 13.24 (Uniform-channel predictive-loss relation)**.** *Let* $X = \{c_i(t)\}_{i=1}^{D}$ *denote the full stochastic record process induced by the memory kernel* $\mathcal{K}$ *on the agent's internal coordinates, and let* $\hat{X} = \{c_i(t)\}_{i \in V_{\text{fg}}}$ *denote the projected record retained by the ego. The mutual information between compressed and full records satisfies*

$$I(\hat{X};\, X) \;\le\; H(\hat{X}) \;\le\; k^* \cdot h_\mu \cdot \tau_{\text{mem}}. \tag{13.25}$$

Under the additional assumption that I_{pred} (13.5) *is approximately uniformly distributed across the* D *algebraic components in the symmetric phase*[3]*, define the predictive information lost by projection as*

$$I_{\text{lost}}^{\text{pred}} := I(X_{\text{past}}; X_{\text{future}}) - I(\hat{X}_{\text{past}}; X_{\text{future}}) \approx \left(1 - \frac{k^*}{D}\right) I_{\text{pred}}, \tag{13.26}$$

up to the error in the uniform-allocation approximation. This is a model relation, not a distribution-free lower bound.

Proof. By the data processing inequality, $I(\hat{X}; X) \le H(\hat{X})$. The projected record $\hat{X}$ has k^* components, each carrying at most h_μ bits per unit time over a window of τ_{mem}, giving $H(\hat{X}) \le k^* \cdot h_\mu \cdot \tau_{\text{mem}}$ [30]. This yields (13.25). Data processing also gives $I(\hat{X}_{\text{past}}; X_{\text{future}}) \le I(X_{\text{past}}; X_{\text{future}}) = I_{\text{pred}}$. Under the uniformity assumption, each of the D components carries $\sim I_{\text{pred}}/D$, so the k^* retained components account for $\sim (k^*/D)\, I_{\text{pred}}$. Equation (13.26) follows by subtraction under that allocation assumption. □

Remark 13.25 (The price of selfhood)**.** *Equation* (13.26) *quantifies predictive information loss for the uniform raw-channel model. It does not say that every* k^**-coordinate encoder loses this fraction, or that the selected encoder is globally optimal over all stochastic encoders.*

[3]This "uniformity assumption" is the information-theoretic counterpart of the raw-channel condition (B3). When some components carry disproportionately more predictive information, the bound tightens or loosens depending on the alignment between V_{fg} and the high-information subspace.

13.6 Worked Example: Qubit in a Two-Channel Bath

13.6.1 Model Setup

We extend Paper I's spin-boson model to demonstrate symmetry breaking explicitly. Consider a qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ (the quaternions, $\dim = 4$).

Symbol mapping. The general framework of Sections 13.3–13.5 specialises as follows:

General	This example	Value
$Cl(V,q)$	$Cl(0,2) \cong \mathbb{H}$	$D = 4$
$G = \mathrm{Aut}(Cl(V,q))$	$SO(3)$	acting on $\{\mathbf{i},\mathbf{j},\mathbf{k}\}$
$\mathcal{C}_{\mathrm{budget}}$	$2\,h_\mu$	bits/time
$k_{\max}$ (capacity)	$\lfloor 2h_\mu/h_\mu \rfloor = 2$	components
V_{fg}	$\mathrm{span}\{1,\mathbf{k}\}$	dephasing subspace
V_{bg}	$\mathrm{span}\{\mathbf{i},\mathbf{j}\}$	dissipative subspace
$\tau_{\max}$ (Prop. 13.7)	$2h_\mu/(4h_\mu) = 0.5$	ω_0^{-1}

The qubit is coupled to a bosonic environment through *two* independent channels:

- A *dephasing channel* via σ_z, with spectral density

$$J_z(\omega) = \frac{2\lambda_z\,\gamma_z\,\omega}{\omega^2 + \gamma_z^2} \qquad \text{(Lorentz–Drude)}, \tag{13.27}$$

  producing a memory kernel $\mathcal{K}_z(t,s)$ with non-Markovian backflow.

- A *dissipative channel* via σ_x, with spectral density

$$J_x(\omega) = \frac{2\lambda_x\,\gamma_x\,\omega}{\omega^2 + \gamma_x^2} \qquad \text{(Lorentz–Drude)}, \tag{13.28}$$

  producing a memory kernel $\mathcal{K}_x(t,s)$.

The full memory kernel is $\mathcal{K}(t,s) = \mathcal{K}_z(t,s) \oplus \mathcal{K}_x(t,s)$, and the quaternionic algebra $\mathbb{H} = \mathrm{span}\{1,\mathbf{i},\mathbf{j},\mathbf{k}\}$ has automorphism group $G = \mathrm{Aut}(\mathbb{H}) \cong SO(3)$ (rotations of the pure quaternion subspace).

Parameters. We set $\omega_0 = 1$ (energy unit), $\lambda_z = 1$, $\gamma_z = 0.5$ (underdamped, strong non-Markovian effects in the dephasing channel), $\lambda_x = 0.3$, $\gamma_x = 5.0$ (overdamped, approximately Markovian in the dissipative channel), and the low-temperature regime $\beta\omega_0 \gg 1$.

Computational budget. The agent has $\mathcal{C}_{\mathrm{budget}} = 2\,h_\mu$ bits per unit time—sufficient to track two components of $\mathbb{H}$ but not all four.

Parameter-to-model mapping. Table 13.2 collects the example parameters and checks the raw full-sweep budget.

Quantity	Symbol	Value	Theorem check
Full dimension	D	4	model input
Entropy rate	h_μ	1.0 (normalised)	per-component rate
Budget	$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Def. 13.2
Raw update check	$h_\mu D$ vs $\mathcal{C}_{\text{budget}}$	$4 > 2$	full raw update does not fit
Capacity	$k_{\max}$	$\lfloor 2/1 \rfloor = 2$	budget identity
Discard fraction	$1 - k^*/D$	$1/2 = 50\%$	Thm. 13.24
Full-sweep depth	$\tau_{\max}$	$2/(4) = 0.5$	Eq. (13.13)

Table 13.2: Parameter mapping for the two-channel qubit example. This example chooses a two-component frame and verifies that it fits the budget; it does not prove symmetry-breaking necessity or budget saturation for every optimum.

13.6.2 Raw Full-Sweep Budget Limit

In the declared raw full-sweep implementation, the agent tracks all four quaternionic components $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ simultaneously. The computational cost is

$$\mathcal{C}_{\text{proc}} = h_\mu \cdot \tau_{\text{mem}} \cdot D = 4\,h_\mu \cdot \tau_{\text{mem}}, \qquad D := \dim Cl(0,2) = 4. \tag{13.29}$$

The maximum raw full-sweep archive depth is

$$\tau_{\text{par}} = \frac{\mathcal{C}_{\text{budget}}}{h_\mu \cdot D} = \frac{2\,h_\mu}{4\,h_\mu} = 0.5 \qquad \text{(in units of } \omega_0^{-1}\text{)}. \tag{13.30}$$

Beyond $\tau_{\text{mem}} = 0.5\,\omega_0^{-1}$, this uncompressed implementation does not fit the stated budget. This does not exclude other symmetric compressed algorithms.

13.6.3 Frame Selection: Choosing σ_z

For comparison, we choose a frame that selects σ_z as the privileged basis direction, retaining the $\{1, \mathbf{k}\}$ subspace (the dephasing channel) as foreground and discarding $\{\mathbf{i}, \mathbf{j}\}$ (the dissipative channel) as background:

$$\mathbb{H} = \underbrace{\text{span}\{1, \mathbf{k}\}}_{V_{\text{fg}}\ (k^*=2)} \oplus \underbrace{\text{span}\{\mathbf{i}, \mathbf{j}\}}_{V_{\text{bg}}}. \tag{13.31}$$

Why σ_z? The dephasing channel ($\lambda_z = 1$, $\gamma_z = 0.5$) is strongly non-Markovian and carries the dominant survival-relevant information (the backflow revivals that enable $\mathcal{S} > 0$, as demonstrated in Paper I). The dissipative channel ($\lambda_x = 0.3$, $\gamma_x = 5.0$) is approximately Markovian and contributes primarily to decoherence—its survival value is negative.

This choice coincides with the *pointer basis* selected by environmental decoherence (quantum Darwinism [143]): the σ_z eigenstates are the states that survive decoherence and become redundantly encoded in the environment. The ego "accepts the suggestion" of decoherence, aligning its computational resources with the environmentally stable basis.

13.6.4 The Broken Phase: Effective Processing

In the broken phase, the projected memory kernel $\mathcal{K}_{\mathcal{F}} = \mathcal{K}_z$ retains only the dephasing-channel dynamics. The computational cost drops to

$$\mathcal{C}_{\text{proc}}^{(\mathcal{F})} = h_\mu \cdot \tau_{\text{mem}} \cdot k^* = 2\, h_\mu \cdot \tau_{\text{mem}}, \tag{13.32}$$

exactly half the symmetric cost (13.29). This implementation can process a raw archive up to depth $\tau_{\text{mem}} = 1\,\omega_0^{-1}$ before reaching its budget—twice the full-sweep depth of the four-channel baseline.

The survival functional in the broken phase is

$$\mathcal{S}_{\text{vis}}(\mathcal{F}) = \mathcal{S}[\mathcal{K}_z], \tag{13.33}$$

which gives the plotted positive backflow proxy at the first revival. This proxy comparison is not a proof of thermodynamic optimality.

The hidden component $\mathcal{S}_{\text{hid}} = \mathcal{S}[\mathcal{K}_x]$ is the survival contribution from the dissipative channel, which the agent can no longer evaluate. For the chosen parameters, $|\mathcal{S}_{\text{hid}}| \ll |\mathcal{S}_{\text{vis}}|$ (the dissipative channel contributes primarily negative survival value), so the distortion is small.

13.6.5 Quantitative Evaluation

We now evaluate the declared implementation comparison explicitly. Each channel's decoherence function follows from the exact $T \to 0$ solution of the Lorentz–Drude pure-dephasing model [22, 163]:

$$p_\alpha(t) = e^{-\gamma_\alpha t/2}\left[\cos(\Omega_\alpha t) + \frac{\gamma_\alpha}{2\Omega_\alpha}\,\sin(\Omega_\alpha t)\right], \quad \Omega_\alpha := \tfrac{1}{2}\sqrt{4\lambda_\alpha\gamma_\alpha - \gamma_\alpha^2}, \tag{13.34}$$

for $\alpha \in \{z, x\}$. When $4\lambda_\alpha\gamma_\alpha < \gamma_\alpha^2$ (the overdamped regime), Ω_α becomes imaginary and the trigonometric functions are replaced by hyperbolic functions (monotonic decay, no backflow).

For our parameters:

- **z-channel** ($\lambda_z = 1$, $\gamma_z = 0.5$): $\Omega_z = \frac{1}{2}\sqrt{1.75} \approx 0.661$. Underdamped; $|p_z(t)|$ exhibits oscillatory backflow.
- **x-channel** ($\lambda_x = 0.3$, $\gamma_x = 5.0$): Discriminant $4\lambda_x\gamma_x - \gamma_x^2 = 6 - 25 = -19 < 0$. Overdamped; $|p_x(t)|$ decays monotonically with no backflow.

For this model we plot the proxy used in Paper I, $\beta\,\mathcal{S} \propto |p(t)|^2 - 1$ (for the pure-dephasing model with maximally coherent initial state and pointer-basis measurement), evaluated for each channel independently. Backflow intervals are those where $d|p_\alpha|/dt > 0$. The proxy increases there; interpreting that increase as thermodynamic survival requires the initial-state and interaction-energy conditions of the revised Paper I accounting identity.

Model comparison. For the z-channel with $\gamma_z = 0.5$, the first backflow interval begins at $t^* \approx 2.9\,\omega_0^{-1}$—well after the raw full-sweep depth $\tau_{\max} = 0.5\,\omega_0^{-1}$. The stipulated raw full-sweep implementation stops at its budget limit before the plotted backflow interval. The selected z-channel implementation remains within its declared component budget and records the three plotted revivals.

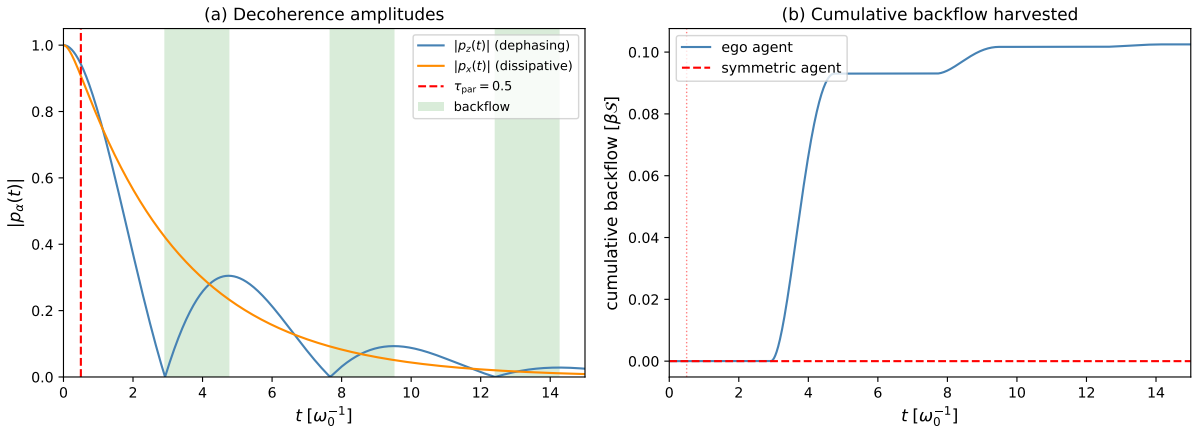


Figure 13.1: Two-channel qubit model (Section 13.6) with Lorentz–Drude spectral density. **Parameters:** $\omega_0 = 1$ (energy unit); $\lambda_z = 1$, $\gamma_z = 0.5$ (dephasing, non-Markovian); $\lambda_x = 0.3$, $\gamma_x = 5.0$ (dissipative, ∼Markovian); $\mathcal{C}_{\text{budget}} = 2\,h_\mu$. **Units:** time in ω_0^{-1}. **Regime:** low temperature ($\beta\omega_0 \gg 1$); using the standard $T \to 0$ analytic expression (13.34) [22]. **(a)** Decoherence amplitudes $|p_z(t)|$ (blue, non-Markovian, with backflow in green bands) and $|p_x(t)|$ (orange, monotonic decay). Red dashed: raw full-sweep depth $\tau_{\max} = 0.5$. **(b)** Cumulative backflow harvested. Blue: ego agent (broken $\to \sigma_z$) exploits all three revival intervals. Red dashed: the stipulated raw full-sweep agent stops at $\tau_{\max}$; all plotted backflow occurs later. The growing gap is the *ego dividend*.

The cumulative backflow harvested by the ego agent (Figure 13.1(b)) totals approximately 0.10 (in dimensionless $\beta\,\mathcal{S}$ units) over $t \in [0, 15\,\omega_0^{-1}]$. The raw baseline records zero after its stipulated stopping point. The result compares these two implementations only; it is not an infinite performance ratio against all symmetric algorithms and does not show that backflow is accessible only after symmetry breaking.

Accounting check. The raw update load is $h_\mu D = 4h_\mu$ while the budget is $2h_\mu$, so that declared representation does not fit. The example then *chooses* the two-component dephasing frame ($k^* = 2$); for that choice, $R_{\mathcal{F}} = 2\,h_\mu = \mathcal{C}_{\text{budget}}$. Saturation is an example property, not a general consequence of monotonic distortion.

13.6.6 The Pointer-State Connection

In this example the chosen basis coincides with an einselection (environment-induced superselection) basis of the dephasing model [143]. The coordinate optimization can select pointer-aligned components when the declared distortion assigns them the highest survival value per bit. This alignment is verified only within a concrete model and is not generic rate-distortion theorem.

Remark 13.26 (Decoherence as symmetry-breaking catalyst)**.** *The environment does not* force *a specific gauge fixing. A pointer-aligned frame is selected only if the declared distortion and encoder class rank it above the alternatives. The present example chooses that ranking; it does not derive it from bounded computation.*

13.7 The Cost of Ego

Coordinate selection reduces the declared raw load but can introduce a blind subspace. A fixed reference frame is a *static* gauge choice in a *dynamic* environment. If the environment changes, the ego becomes progressively maladaptive.

Drift layer. Environmental change can occur at multiple levels: parameter drift ($\lambda_\alpha(t)$, $\gamma_\alpha(t)$), spectral-density deformation ($J(\omega, t)$), or full process-distribution shift ($P_t(X)$). For analytical tractability, we model drift at the *spectral-density parameter level* throughout this section. No monotone extension to arbitrary distribution shift is claimed.

13.7.1 The Rigidity Trap

Proposition 13.27 (Frame Rigidity under Drift)**.** *Let the environment undergo slow drift: the spectral density parameters change as $\lambda_\alpha(t) = \lambda_\alpha^{(0)} + \varepsilon\, f_\alpha(t)$ for $\alpha \in \{z, x\}$, with drift rate $\varepsilon > 0$. Assume the supplied distortion has a locally unique C^1 minimizer $\mathcal{F}^*(\lambda) \in G$ and $\|D_\lambda \mathcal{F}^*\| \le K$ along the path.*

If the agent's reference frame $\mathcal{F}$ is held fixed (no recalibration), the mismatch between $\mathcal{F}$ and $\mathcal{F}^*(t)$ then obeys

$$\delta(t) := d_G\big(\mathcal{F},\, \mathcal{F}^*(t)\big) \le K\varepsilon \int_0^t \|\dot{f}(s)\|\, ds, \tag{13.35}$$

where d_G is the geodesic distance in the gauge group.

Proof. The chain rule and the stated sensitivity bound give $\|\dot{\mathcal{F}}^*(t)\| \le K\varepsilon \|\dot{f}(t)\|$. Integrating path length bounds the geodesic distance and yields (13.35). □

13.7.2 Stylized Drift Model

To quantify the loss of visibility of a fixed frame, we use the linear drift supported by Proposition 13.27. No unstable environmental dynamics or Lyapunov exponent is assumed.

Definition 13.28 (Rotating optimal frame)**.** *Let the mismatch angle $\theta(t)$ between the agent's fixed frame $\mathcal{F}$ and the instantaneous optimal frame $\mathcal{F}^*(t)$ evolve as*

$$\theta(t) = \theta_0 + \varepsilon t, \tag{13.36}$$

where $\theta_0 \in (0, \pi/4)$ is the initial misalignment (so that the crossover time below is positive) and $\varepsilon > 0$ is the angular drift rate. This is a local model up to the first visibility crossover, not a claim about long-time motion on the compact gauge group.

The visible and hidden survival components decompose geometrically:

$$\mathcal{S}_{\text{vis}}(t) = \mathcal{S}_{\text{tot}}\, \cos^2\theta(t), \qquad \mathcal{S}_{\text{hid}}(t) = \mathcal{S}_{\text{tot}}\, \sin^2\theta(t), \tag{13.37}$$

where $\mathcal{S}_{\text{tot}}$ is the full survival functional (invariant under frame rotation).

13.7.3 The Prediction Error Divergence

Proposition 13.29 (Growth of the hidden component)**.** *Under the drift model* (13.36)–(13.37)*, the hidden survival component grows as*

$$\big|\mathcal{S}_{\text{hid}}(t)\big| = |\mathcal{S}_{\text{tot}}|\,\sin^2\!\big(\theta_0 + \varepsilon t\big). \tag{13.38}$$

For small angles ($\theta_0 + \varepsilon t \ll 1$)*:* $|\mathcal{S}_{\text{hid}}| \approx |\mathcal{S}_{\text{tot}}|(\theta_0 + \varepsilon t)^2$*.*

Proof. Direct substitution of (13.36) into (13.37). The small-angle expansion $\sin^2\theta \approx \theta^2$ gives the quadratic form. □

13.7.4 Visibility Crossover and Conditional Gradient Reversal

Proposition 13.30 (Visibility crossover and a conditional gradient reversal)**.** *Under* (B1)–(B5) *with the drift model* (13.36) *and initial misalignment* $\theta_0 \in (0, \pi/4)$*, an agent with a fixed reference frame* $\mathcal{F}$ *reaches the visibility crossover*

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}, \tag{13.39}$$

at which the hidden and visible magnitudes are equal. For $t > t_{\text{vis}}$*, the hidden magnitude is larger while* $\theta(t) < \pi/2$*. This geometric crossover alone implies neither gradient anti-correlation nor failure of every in-frame detector.*

There nevertheless exist specified controllers for which gradients do reverse. In the two-mode model

$$\mathcal{S}_{\text{vis}}(u) = -\tfrac{1}{2}(u-1)^2, \qquad \mathcal{S}_{\text{full}}(u,\theta) = -\tfrac{1}{2}\|ue_1 - y^*(\theta)\|^2, \quad y^*(\theta) = \cos\theta\, e_1 + \sin\theta\, e_2, \tag{13.40}$$

the two scalar update directions are anti-correlated exactly when

$$\cos\theta(t) < u < 1.$$

Thus the reversal time is controller- and state-dependent; it need not coincide with t_{vis}*.*

Proof. The crossover $|\mathcal{S}_{\text{hid}}| = |\mathcal{S}_{\text{vis}}|$ occurs when $\sin^2\theta = \cos^2\theta$, i.e., $\theta(t_{\text{vis}}) = \pi/4$. Substituting (13.36): $\theta_0 + \varepsilon t_{\text{vis}} = \pi/4$, which gives (13.39). For (13.40), $\partial_u \mathcal{S}_{\text{vis}} = 1 - u$ and $\partial_u \mathcal{S}_{\text{full}} = \cos\theta - u$. Their product is negative exactly when u lies strictly between $\cos\theta$ and 1. Outside that interval the two gradients agree in sign or one vanishes. The calculation exhibits the claimed effect without promoting it to a universal consequence of visibility loss. □

Remark 13.31 (Model scope)**.** *For* $\theta_0 = 0$*, the crossover occurs at* $t_{\text{vis}} = \pi/(4\varepsilon)$*. With* $\varepsilon = 0.01\,\omega_0$*,* $t_{\text{vis}} \approx 79\,\omega_0^{-1}$*—long enough for the agent to accumulate a false sense of security, yet short on environmental timescales. Neither this number nor the two-mode sign condition proves that an agent cannot detect drift from residual statistics; Paper III studies one such detector.*

Remark 13.32 (Why dithering does not help)**.** *One might ask whether the agent could escape the delusion trap by randomly "probing" the background subspace* V_{bg}*—temporarily rotating its frame to sample hidden components. In the raw-coordinate model, a full probe competes with foreground processing at the component-count rate in Remark 13.8. This*

does not prove dithering fails: randomized exploration can reveal hidden directions, with a model-dependent cost–benefit trade-off. A deterministic in-frame controller may lack a signal indicating when or where to probe. As long as $|\mathcal{S}_{\text{hid}}| < |\mathcal{S}_{\text{vis}}|$ (pre-delusion), the in-frame performance metric $\mathcal{S}_{\text{vis}}$ shows no anomaly. The linear-drift visibility model below motivates monitoring residual statistics, the subject of Paper III, but does not rule out other exploration policies.

Remark 13.33 (The ego as medicine and poison)**.** *In this model, coordinate selection reduces raw load but creates a blind subspace. The visibility crossover below motivates Paper III. This is a conditional engineering trade-off, not an irreversible logic chain.*

13.7.5 The Origin of Paper III

To respond to the visibility loss in this model, the agent may use a mechanism to monitor the quality of its own reference frame—to "observe its own observation." This requires a *second-order control loop*: a meta-controller that adjusts the gauge fixing σ in response to accumulated prediction errors.

The distribution of in-frame prediction residuals can itself change as the frame becomes stale. Paper III studies whether a specified likelihood model makes that change statistically identifiable. This paper does not prove that Fisher information is the unique detector or that every controller must add such a loop.

This is the subject of Paper III: the Fisher information geometry of self-referential calibration, and the thermodynamic cost of the loop that closes the chain *Chaos* $\rightarrow$ *Time* $\rightarrow$ *Self* $\rightarrow$ *Calibration*.

13.8 Numerical Demonstration

The preceding sections establish analytic bounds and a worked example with a qubit in a two-channel bath. We now compare budgeted coordinate-selection rules in a minimal multi-dimensional system. The observed attention concentration and selection advantage are properties of this supplied algorithm and environment. Full code and parameters are provided for reproducibility.

13.8.1 Model

Environment. A D-dimensional linear prediction task with sparse rotating support: $y(t) = \mathbf{w}^*(t)^\top \mathbf{x}(t) + \xi(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, $\xi \sim \mathcal{N}(0, \sigma^2)$. Only $m \ll D$ dimensions carry nonzero weight at any time; the active support rotates every τ_{switch} steps, modelling environmental drift.

Hard budget constraint. Per step, the agent may update only k coordinates of its weight vector (a hard processing budget), mirroring the bounded computation assumption (B2).

Agents.

- **Budgeted selector**: selects the top-k dimensions by importance score—an exponential moving average of the signed per-coordinate gradient. Signed accumulation

ensures that noise dimensions (zero expected signal) cancel over time while signal dimensions persist, enabling reliable discrimination without access to the true support.

- **Random-k baseline**: selects k dimensions uniformly at random each step. This provides a budget-fair comparison with no persistent coordinate preference.

The choice of *signed* gradient EMA (rather than squared-gradient magnitude) is structurally motivated: for noise dimensions $\mathbb{E}[r\,x_i] = 0$, so the signed accumulation cancels over time; for signal dimensions $\mathbb{E}[r\,x_i] \neq 0$, so a consistent directional bias persists. The signed EMA thus acts as a *directional coherence filter* that discriminates signal from noise without access to the true support—a minimal model of a persistent coordinate preference.

Parameters.

Quantity	Value	Role
D	64	ambient dimension
m	8	signal dimensions (sparse support)
T	10,000	horizon per trial
Seeds	10	independent replications
σ	0.3	observation noise std
η	0.02	SGD learning rate
λ	0.995	weight decay per step
k	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	budget grid
τ_{switch}	$\{500, 1000, 2000\}$	support rotation period

Attention entropy. Let n_i be the number of updates coordinate i receives in a measurement window of the last 1,000 steps. The normalised update frequency $p_i = n_i / \sum_j n_j$ defines the attention entropy:

$$H_{\text{attn}} = -\sum_{i=1}^{D} p_i \ln p_i. \tag{13.41}$$

Under uniform coordinate selection, $p_i = 1/D$ and $H_{\text{attn}} = \ln D$. Under budget-constrained selection, H_{attn} collapses away from $\ln D$, measuring concentration of the update rule. It is not by itself a thermodynamic order parameter.

Oracle metric. Neither agent has access to $\mathbf{w}^*(t)$. Performance is evaluated externally using the weight-space mean-squared error $\text{MSE} = \|\hat{\mathbf{w}} - \mathbf{w}^*\|^2$, averaged over post-burn-in steps.

13.8.2 Results

Figure 13.2 shows the two key signatures.

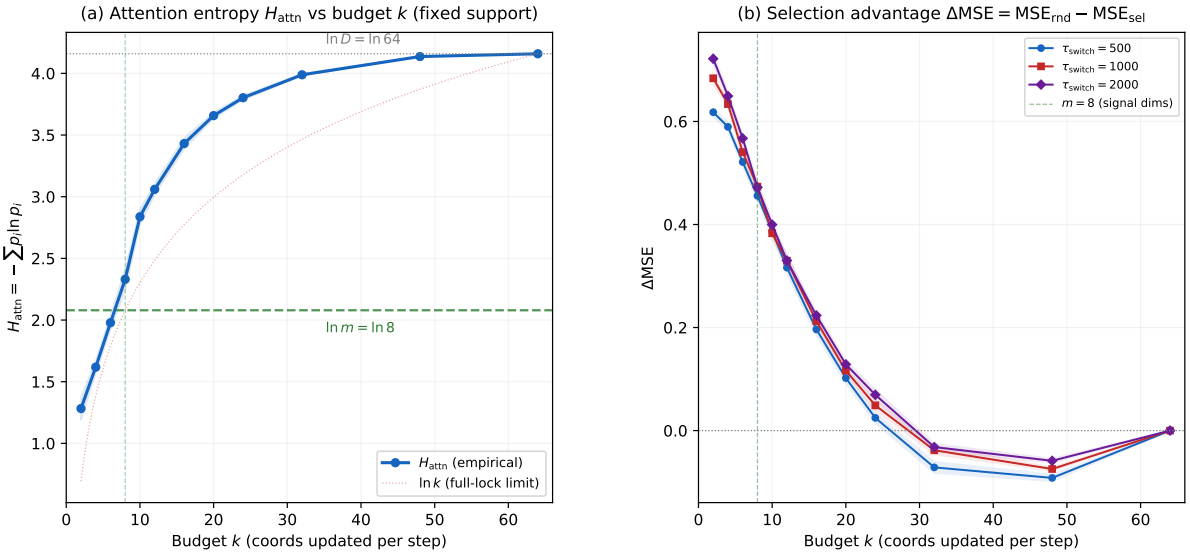


Figure 13.2: **Budgeted coordinate selection in the declared linear model.** $D = 64$, $m = 8$, $T = 10{,}000$, 10 seeds, 95% CI bands. **(a)** Attention entropy H_{attn} vs budget k (fixed support). The empirical curve (blue) collapses from $\ln D \approx 4.16$ toward an $O(\ln m)$ floor as budget tightens. For $k \leq m$, H_{attn} remains below $\ln m \approx 2.08$ (green dashed), consistent with confinement to the signal subspace. **(b)** Selection advantage $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ vs budget k under rotating support. The gap is positive for $k \lesssim 3m$ (selection helps), turns slightly negative at large k (commitment cost exceeds diversification), and returns to zero at $k = D$. Slower drift ($\tau = 2000$) yields a larger peak advantage.

Result 1: Attention entropy collapse (Figure 13.2a). Under fixed support (no rotation), the attention entropy H_{attn} exhibits a sharp collapse away from $\ln D = \ln 64 \approx 4.16$ and increases monotonically with k, consistent with a budget-induced concentration of update mass onto signal-carrying dimensions. For budgets near and below the signal scale ($k \leq m$), H_{attn} remains $O(\ln m)$, consistent with confinement to the signal subspace. We use the collapse of H_{attn} away from $\ln D$ as the order parameter of symmetry breaking; a strict plateau at $\ln m$ is not expected under the present re-selection dynamics and finite-window estimator.

Result 2: Selection advantage (Figure 13.2b). Under rotating support, the mean-squared error gap $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ is positive for $k \lesssim 3m$ and peaks at tight budgets ($k = 2$) where the selection advantage is strongest. For $k \gg m$ the gap turns slightly negative (the selector's commitment to stale dimensions costs more than the random baseline's diversification), before returning to zero at $k = D$. The three τ_{switch} curves are ordered: slower drift (larger τ) yields a larger peak gap, with the ordering most visible at small k.

13.8.3 Scope of This Demonstration

These simulations compare two supplied coordinate-selection rules in the stated sparse linear model. They do not establish a universal symmetry-breaking transition or identify the capacity $k_{\max}$ with an information-bottleneck optimum.

This demonstration **does** show:

1. Under hard budget constraints, attention entropy collapses sharply away from $\ln D$ and remains $O(\ln m)$ for $k \leq m$—the agent confines its updates to the signal subspace. This measures concentration of the chosen update rule.

2. A budgeted selector that exploits importance-weighted selection systematically outperforms a budget-fair random baseline, consistent with a survival advantage for the parameter range shown.

3. The advantage scales with both budget tightness (smaller k) and environmental stability (larger τ_{switch}).

In summary, this demonstration establishes a model-specific selection advantage and a measurable attention-concentration statistic; it does not derive selection from the budget or claim universality across architectures or environment classes.

This demonstration does **not** show:

1. That H_{attn} reaches a strict plateau at $\ln m$ for all $k \leq m$. Under the adaptive re-selection dynamics used here, the selector cycles within the signal subspace, producing H_{attn} values near but not locked to $\ln m$. The relevant signature is the collapse *away from* $\ln D$, not convergence to a specific lower bound.

2. That the specific form of the importance score (signed gradient EMA) is optimal. It is one realisation of the selection mechanism.

3. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and sparse rotating support.

4. That the delusion–correction cycle is addressed. This is the subject of Paper III.

5. That a sharp thermodynamic phase transition occurs, or that $k = k_{\max}$ is the distortion-minimizing retained dimension.

Reproducibility. The complete simulation is a self-contained Python script (`paper2_kstar_scaling_demo.py`, $\sim$540 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper2_kstar_scaling_demo.py` — simulation script
- `fig_paper2_kstar_scaling.pdf` — Figure 13.2
- `kstar_scaling_data.csv` — raw performance gap data

13.9 Discussion

13.9.1 Summary of Results

Result	Statement	Sec.
Raw-channel load	A declared D-channel archive costs Dh_μ per incremental update; this is accounting, not a lower bound	13.3
Compression existence	A minimizer exists in the supplied compact encoder class and has $k_{\text{opt}} \leq k_{\max}$	13.4.2
Conditional frame selection	A chosen proper projection breaks the declared model symmetry when no invariant feasible encoder meets the target	13.4.3
Four diagnostics	Frame-dependent models can be audited using $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$	13.4.4
Survival Decomposition	$\mathcal{S} = \mathcal{S}_{\text{vis}} + \mathcal{S}_{\text{hid}}$	13.5.2
Uniform-channel trade-off	Predictive loss is approximately $1 - k^*/D$ only under the uniform raw-channel assumption	13.5.4
Visibility crossover	Linear drift gives $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$; gradient reversal requires a separate controller condition	13.7.4
Numerical demo	Attention concentration and model-specific selection advantage (Fig. 13.2)	13.8

13.9.2 Testable Predictions

The conditional model claims can be tested without treating them as universal laws. First, vary k, D, and the support-switching time in the declared sparse linear environment and compare the supplied selector with random and alternative compressed baselines. A sharp transition is not predicted without a separately defined thermodynamic limit.

Second, estimate $I(X_{\text{past}}; X_{\text{future}})$ and $I(\hat{X}_{\text{past}}; X_{\text{future}})$ while varying k/D. Linear $1 - k/D$ scaling tests the uniform-allocation assumption in Theorem 13.24; systematic curvature falsifies that assumption rather than information theory.

Third, in a controlled rotating-feature task, measure both the visibility ratio and the inner product of surrogate and oracle gradients. The linear model predicts the visibility crossover at $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Proposition 13.30 predicts gradient reversal only for controllers entering the region $\cos\theta(t) < u(t) < 1$; coincidence of the two events is not required.

13.9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. A declared raw D-channel archive has the stated component-count load, and a minimizer exists in the supplied compact encoder class.

2. A proper coordinate projection privileges a frame; under additional feasibility assumptions this can be interpreted as operational symmetry breaking (Remark 13.18).

3. Definition 13.19 supplies four model diagnostics, with the extra controller and task assumptions needed by each made explicit.

4. Linear frame drift produces a visibility crossover, while the two-mode model exhibits a separate, conditional gradient reversal (Proposition 13.30).

5. A minimal computational demonstration measures attention concentration and a selection advantage over a budget-fair random baseline (Section 13.8, Figure 13.2).

This paper does **not** show:

1. That bounded computation uniquely determines a privileged basis, forces symmetry breaking, or makes the chosen frame survival-optimal.

2. That symmetry breaking is necessary or sufficient for persistence.

3. That the "ego" implies or requires consciousness, subjective experience, or phenomenal awareness. The term is used strictly in the control-theoretic sense.

4. That this framework constitutes a theory of consciousness. It is a theory of computational optimality under thermodynamic constraints.

5. That the four diagnostics exhaust self-reference or follow from gauge fixing alone.

6. That the coordinate capacity is a Shannon rate–distortion bound or is achievable by a specific physical implementation.

7. That visibility loss forces gradient reversal or is undetectable. Paper III analyzes one conditional calibration model.

8. That the framework constitutes or implies a philosophical or metaphysical claim about the nature of selfhood.

9. That this framework applies to all possible physical systems. It applies to systems satisfying the declared channel, encoder, controller, and drift assumptions.

10. That the Clifford algebra is the only possible algebraic setting. It is supplied here as a model input; Q-RAIF does not uniquely derive it.

Chapter 14

Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration

Paper III — "The Loop"

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Abstract

Papers I and II of the T-DOME series [163, 164] supply conditional resource and reference-frame models for persistent agents. They do not prove that all persistent agents require non-Markovian memory or spontaneous Clifford gauge-symmetry breaking. Paper II's linear-drift visibility crossover is likewise a declared coordinate model, not a universal logarithmic "Delusion Trap" timescale.

This paper studies **self-referential calibration** under explicit residual-likelihood, prior, drift, convexity, and hardware assumptions. The agent uses the score statistics of a supplied prediction-residual model to estimate frame mismatch. The principal conditional results are:

1. **Gaussian Fisher Signal** (Proposition 14.14): in the declared even Gaussian residual model, per-sample local Fisher information grows quadratically and accumulated information grows cubically. Detection precedes the visibility crossover if and only if the threshold condition in Corollary 14.16 holds.

2. **Self-Referential Cramér–Rao Bound** (Theorem 14.18): the van Trees lower bound applies under the stated regular likelihood, conditional-independence, compact-support, and prior assumptions, with data and prior Fisher information entering separately.

3. **Conditional Implementation Cost** (Proposition 14.28): a separable implementation with declared irreversible sensor and computational resets and linear-response actuation obeys $\dot{W}_{\mathrm{loop}} \geq k_B T \ln 2\,(R_s + R_c) + \mathcal{L}^2/\tau^2_{\mathrm{recalib}}$, a bound for that implementation rather than a universal cost of sensing, Fisher evaluation, or self-reference.

The calibration loop has a local tracking bound only under geodesic strong convexity and bounded drift (Theorem 14.23). The four supplied layers in Definition 14.27 define the running model architecture; they are not proved necessary, sufficient, minimal, or complete for persistent agency. "Reflexivity" here denotes one implementation of second-order control, with no claim of architectural uniqueness or phenomenal consciousness.

14.1 Introduction

14.1.1 Context: The Delusion Trap

Revised Paper II [164] studies a supplied coordinate projection under a raw-channel budget. It does not prove that persistence forces Clifford gauge symmetry breaking. Its drift model uses a linearly changing mismatch $\theta(t) = \theta_0 + \varepsilon t$ and identifies the geometric visibility crossover

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}. \tag{14.1}$$

That crossover does not by itself imply gradient reversal or undetectability. The present paper asks a narrower question: for a specified residual likelihood, can score statistics detect mismatch, and under what conditions can a natural-gradient controller track a moving optimum?

14.1.2 Position within Papers I–III

This paper is the third and final of the T-DOME framework, closing the three-paper sequence.

Framework	**Question**	**Result**	**Status**
HAFF [148, 149]	How does geometry emerge?	Conditional algebra/dictionary models	Revised
Q-RAIF [160, 161]	What algebra must an observer have?	Clifford representations as model inputs	Revised
T-DOME I [163]	What can inter-epoch correlations fund?	Restricted baseline; correlation accounting	Conditional
T-DOME II [164]	When can a model select a frame?	Conditional coordinate selection	Conditional
T-DOME III (this work)	How does self-calibration arise?	Fisher self-referential bound; thermodynamic cost of reflexivity	**This paper**

The three T-DOME papers form a conditional modelling sequence:

1. **Paper I:** Initial correlations can enter an ELV-type thermodynamic balance; a restricted autonomous baseline is proved.

2. **Paper II:** A raw-coordinate archive motivates a supplied compression model and a linear visibility-crossover example.

3. **Paper III (this work):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop—monitoring the Fisher information of one's own prediction stream—resolves mismatch under explicit likelihood and convexity assumptions. A conditional hardware model assigns reset and actuation costs.

The links are motivation and model interfaces, not an irreversible theorem chain. None of the three papers proves that this architecture is necessary for persistence.

14.1.3 The Information-Geometric Insight

The key observation that resolves the Delusion Trap is subtle: *while the agent cannot observe V_{bg} directly, it can observe the statistical properties of its own prediction residuals in V_{fg}.*

The prediction residual $e(t) := \mathcal{S}_{\text{vis}}(t) - \mathcal{S}_{\text{vis}}^{(\text{pred})}(t)$ lies in V_{fg} by construction. In the declared even Gaussian model, a record cannot identify the sign of mismatch. Its *distribution*—the probability law $p(e \,|\, \sigma)$, parametrised by the gauge-fixing parameter σ—does depend on σ, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured and which are discarded.

When the frame σ drifts away from the optimal σ^*, the residual distribution shifts. The *Fisher information metric*

$$g_{ij}(\sigma) = \mathbb{E}_\sigma\left[\frac{\partial \log p(e \,|\, \sigma)}{\partial \sigma^i}\,\frac{\partial \log p(e \,|\, \sigma)}{\partial \sigma^j}\right] \tag{14.2}$$

measures the sensitivity of this distribution to changes in σ. Whether g_{ij} changes with staleness is likelihood-dependent. The Gaussian model below provides one explicit case.

This is the mathematical realisation of the "second-order operation" demanded by Paper II, Section 7.5: the agent does not need to see the truth (the full $Cl(V, q)$), but only the *rate of change of its own prediction error* as a function of its frame parameters. Fisher information is one local sensitivity measure, not the unique possible detector.

14.1.4 Relation to Architectural Incompleteness

Revised HAFF Paper G [152] supplies philosophy-of-science motivation rather than a fixed-point no-go theorem. In that motivational sense, the self-referential calibration loop cannot *eliminate* architectural incompleteness, but it can *track* the consequences of incompleteness in real time. The Lyapunov function $V(\sigma)$ monitors the distance between the agent's frame and the optimal frame without requiring access to the "complete" description—it operates entirely within the agent's own predictive statistics.

14.1.5 Scope and Disclaimers

1. *Reflexivity* refers throughout to second-order control: the ability of a system to monitor and adjust its own monitoring process. It carries *no* implication of phenomenal consciousness, subjective experience, or qualia.

2. The self-referential calibration loop does not *eliminate* the ego's bias; it tracks and compensates for drift in the bias. The four bias terms of Paper II persist in the calibrated phase.

3. The thermodynamic cost statement is conditional on irreversible reset, separable hardware, and a linear-response actuation model.

4. The framework applies to systems satisfying (C1)–(C5) (Section 14.2.6). It is not a universal theory of agency.

Related work. The Fisher information metric on statistical manifolds was introduced by Rao [93] and shown to be unique by Čencov [26]. The natural gradient and information geometry were developed by Amari [5, 6]. Thermodynamic length and optimal finite-time transformations were established by Crooks [31] and Sivak–Crooks [110]. The connection between Fisher information and entropy production was formalised by Ito [59] and Barato–Seifert [11]. Second-order cybernetics originates with Ashby [9] and von Foerster [146]. Adaptive control and self-tuning regulators are treated in [10]. The Bayesian Cramér–Rao bound (van Trees inequality) is from [145].

Summary of contributions. This paper separates three levels of result:

1. **Drift Detectability** (Proposition 14.14): an explicit Gaussian residual likelihood has quadratic per-sample and cubic accumulated local Fisher signals; early detection is a threshold condition.

2. **Self-Referential Cramér–Rao Bound** (Theorem 14.18): drift-estimation precision is bounded by the sum of data Fisher information and ego rigidity.

3. **Thermodynamic Cost** (Proposition 14.28): a separable implementation with irreversible resets and finite-time actuation obeys a conditional additive bound.

Result-status ledger. The original drift-detectability theorem `TD-III-T3.3` is retracted; Proposition 14.14 is an FD/I model calculation and Corollary 14.16 is conditional on a declared threshold. Ego rigidity `TD-III-ego` remains a supplied prior model. The tracking theorem is conditional on local geodesic strong convexity. The universal additive-cost claim `TD-III-T6.2` is downgraded to Proposition 14.28, and the inherited persistence budget is withdrawn.

14.2 Mathematical Preliminaries

14.2.1 Inherited Framework from Papers I and II

We briefly recall the key objects; the reader is referred to Papers I and II for full definitions and proofs.

From Paper I [163].

- **Survival functional.** $\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau]$ (Paper I, Eq. (9)).

- **Restricted autonomous baseline.** $\mathcal{S}[\Lambda^{\mathrm{D}}, \tau] \leq 0$ for a fixed-Hamiltonian, no-control Davies semigroup under its stated initial conditions. It is not a ceiling for all Markovian agents.

- **Memory kernel.** $\mathcal{K}(t,s)$: the non-Markovian superoperator encoding system–environment correlations.
- **Entropy rate.** $h_\mu := \lim_{T\to\infty} T^{-1} H(X_{0:T})$ (bits per unit time per algebraic component).
- **Predictive information.** $I_{\text{pred}} := I(\overleftarrow{X};\, \overrightarrow{X})$.

From Paper II [164].

- **Internal algebra.** $\mathcal{O}_{\text{int}} = Cl(V,q)$, $D = \dim Cl(V,q) = 2^n$, gauge group $G = \text{Aut}(Cl(V,q))$.
- **Gauge bundle.** $\pi : P \to M$, structure group G; a section $\sigma : M \to P$ is a reference frame.
- **Ego.** $\mathfrak{E} := (\mathcal{F}^*, V_{\text{fg}}^*)$ denotes a chosen optimizer in a supplied encoder class, with $k^* = k_{\text{opt}} \le k_{\max}$. The budget capacity is not a universal optimizer formula.
- **Projected kernel.** $\mathcal{K}_{\mathcal{F}}(t,s) = \Pi_{\mathcal{F}}\, \mathcal{K}(t,s)\, \Pi_{\mathcal{F}}$.
- **Survival decomposition.** $\mathcal{S} = \mathcal{S}_{\text{vis}}(\mathcal{F}) + \mathcal{S}_{\text{hid}}(\mathcal{F})$.
- **Four bias terms.** $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$ (revised Paper II, four-diagnostics definition). These are model diagnostics, not generic consequences of gauge fixing.
- **Visibility crossover.** $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$ in the supplied linear drift geometry. Gradient reversal requires an additional controller condition.
- **Information objects.** Mutual information is evaluated on explicitly induced record processes; algebra symbols are not themselves random variables.

14.2.2 Fisher Information Metric

Definition 14.1 (Fisher information matrix)**.** *Let* $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ *be a parametric family of probability densities satisfying standard regularity conditions (interchange of differentiation and integration). The* Fisher information matrix *is*

$$g_{ij}(\theta) := \mathbb{E}_\theta\left[\frac{\partial \log p(x \mid \theta)}{\partial \theta^i}\, \frac{\partial \log p(x \mid \theta)}{\partial \theta^j}\right] = -\mathbb{E}_\theta\left[\frac{\partial^2 \log p(x \mid \theta)}{\partial \theta^i\, \partial \theta^j}\right]. \tag{14.3}$$

The pair (Θ, g) *is a Riemannian manifold called the* statistical manifold.

Remark 14.2 (Uniqueness)**.** *By Čencov's theorem [26], the Fisher–Rao metric* g^{FR} *is, up to a positive scalar multiple, the unique Riemannian metric on the space of probability distributions that is invariant under all Markov morphisms (sufficient-statistic embeddings). Within the hypotheses and category used by Čencov's theorem, this makes the Fisher metric a canonical local distinguishability metric. It does not make a Fisher-based detector physically mandatory, nor does it choose a likelihood or control objective for the agent.*

Proposition 14.3 (Cramér–Rao bound)**.** *For any unbiased estimator $\hat{\theta}$ of θ based on n independent observations:*

$$\mathrm{Cov}(\hat{\theta}) \succeq \frac{1}{n}\big[g(\theta)\big]^{-1} \tag{14.4}$$

in the Löwner order. The scalar case reads $\mathrm{Var}(\hat{\theta}) \geq 1/\big(n\, g(\theta)\big)$.

Remark 14.4 (Effective independence)**.** *Throughout this paper, references to "independent observations" in the context of continuous-time residual streams should be read as* effective independence *after thinning by the environmental decorrelation time τ_E, yielding an effective sample size $n_{\mathrm{eff}} \approx T/\tau_E$. In particular, the sample count n in* (14.4) *becomes n_{eff} in the self-referential setting of Section 14.4.2.*

Remark 14.5 (Fisher metric and KL divergence)**.** *The Fisher metric arises as the Hessian of the Kullback–Leibler divergence [30]:*

$$D_{\mathrm{KL}}\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O\big(|d\theta|^3\big)\,. \tag{14.5}$$

This identifies the Fisher metric as the infinitesimal measure of statistical distinguishability.

14.2.3 Information Geometry

Following Amari [4, 6], the statistical manifold $(\Theta,\, g)$ carries additional geometric structure beyond the Riemannian metric.

α-connections. For each $\alpha \in [-1,\, 1]$, Amari defines an affine connection $\nabla^{(\alpha)}$ on Θ. The cases $\alpha = 1$ (exponential connection, $\nabla^{(e)}$) and $\alpha = -1$ (mixture connection, $\nabla^{(m)}$) are *dual* with respect to g: $\partial_k\, g(X, Y) = g(\nabla^{(e)}_k X,\, Y) + g(X,\, \nabla^{(m)}_k Y)$. For exponential families, $\nabla^{(e)}$ is flat in natural parameters and $\nabla^{(m)}$ is flat in expectation parameters—the *dually flat structure*. The case $\alpha = 0$ recovers the Levi-Civita connection of the Fisher metric.

Natural gradient. Standard gradient descent in parameter space ignores the curvature of the statistical manifold. The *natural gradient* [5]

$$\dot{\theta} = -\eta\, g^{-1}(\theta)\, \nabla_\theta L(\theta), \tag{14.6}$$

where $\eta > 0$ is the learning rate and $L(\theta)$ is a loss function, provides the steepest descent direction in the Fisher metric. It is reparametrisation-invariant. Statistical efficiency requires additional estimator and asymptotic assumptions and is not implied by the update formula alone.

Pythagorean theorem. In a dually flat space, the KL divergence satisfies a generalised Pythagorean relation: $D_{\mathrm{KL}}(p \,\|\, r) = D_{\mathrm{KL}}(p \,\|\, q) + D_{\mathrm{KL}}(q \,\|\, r)$ when q is the m-projection of p onto a submanifold containing r. This decomposition will be applied to separate the foreground-recoverable and background-irrecoverable components of drift.

14.2.4 Thermodynamic Length

Definition 14.6 (Thermodynamic length)**.** *Let $\lambda(t)$ for $t \in [0, \tau]$ be a path through control parameter space, and let $\zeta_{ij}(\lambda)$ be the* friction tensor *(the time-integrated equilibrium force–force correlation function at λ). The* thermodynamic length *of the path [31] is*

$$\mathcal{L} := \int_0^\tau \sqrt{\zeta_{ij}(\lambda)\, \dot{\lambda}^i\, \dot{\lambda}^j}\; dt. \tag{14.7}$$

Proposition 14.7 (Sivak–Crooks bound)**.** *The excess (dissipated) work during a finite-time transformation of duration τ satisfies [110]*

$$W_{\mathrm{ex}} \;\geq\; \frac{\mathcal{L}^2}{\tau}. \tag{14.8}$$

The minimum is achieved by the geodesic of the friction tensor ζ. In the linear-response regime, the friction tensor is related to the Fisher metric of the equilibrium distribution at λ by $\zeta_{ij}(\lambda) \sim \tau_{\mathrm{relax}}\, g^{\mathrm{Fisher}}_{ij}(\lambda)$, where τ_{relax} is the relaxation time.

14.2.5 Second-Order Cybernetics

Von Foerster [146] distinguished two levels of control:

- **First-order cybernetics**: feedback control of observed systems. The controller adjusts its actions based on the output of a sensor. Paper II's ego is a first-order structure: it processes environmental data within a fixed frame.
- **Second-order cybernetics**: feedback control of the *observing* system itself. The controller adjusts the *sensor*—or equivalently, the reference frame within which the sensor operates. This is what Paper III provides.

Ashby's Law of Requisite Variety [9] provides a lower bound on the complexity of the meta-controller:

$$\dim(\text{meta-controller state space}) \;\geq\; \dim(\text{environmental drift subspace})\,. \tag{14.9}$$

The meta-observer must have at least as many adjustable parameters as there are independent modes of environmental drift.

In adaptive control theory [10], the analogous result is the *persistent excitation* condition: parameter estimates converge if and only if the input signal is "rich enough" to excite all modes of the system. In our framework, persistent excitation corresponds to $h_\mu > 0$—the environment must continue to generate novelty for the self-calibration loop to function.

Remark 14.8 (Operational content)**.** *The second-order cybernetic structure in this paper is* not *a philosophical metaphor. It has concrete operational content: the natural gradient update* (14.6) *is a specific algorithm that takes as input the Fisher information of the residual stream and produces as output an update to the frame parameter σ. An implementation must supply a differentiable residual likelihood, evaluate or approximate its score, and realize the update law. Merely accumulating an unlabelled residual second moment is not generally a Fisher computation.*

14.2.6 Standing Assumptions

Definition 14.9 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(C1) ***Supplied representation model.*** *A finite-dimensional agent model, a foreground projection, and a control algebra are specified. A Clifford representation and any external map ϕ are model inputs, not conclusions inherited from revised Q-RAIF or T-DOME II.*

(C2) ***Environmental drift.*** *The instantaneous optimal frame $\mathcal{F}^*(t)$ rotates continuously in G at bounded speed. In the scalar worked model the mismatch is $\theta(t) = \theta_0 + \varepsilon t$ up to the first visibility crossover, as in revised Paper II.*

(C3) ***Finite meta-observer budget.*** *The self-calibration loop has a computational budget $\mathcal{C}_{\text{meta}} < \infty$ (bits per unit time), distinct from the ego's processing budget $\mathcal{C}_{\text{budget}}$.*

(C4) ***Regularity.*** *The agent's predictive family $\{p(e \mid \sigma) : \sigma \in G/H\}$ satisfies standard Fisher information regularity: finite Fisher matrix and interchange of differentiation and integration. Any scalar sensitivity or uniform lower bound used later must be assumed separately.*

(C5) ***Persistent excitation.*** *The time-averaged Fisher matrix obeys the explicit lower bound in Definition 14.24. A positive entropy rate alone is not sufficient for this condition.*

14.3 The Drift Detection Problem

14.3.1 Identifiability in an Even Residual Model

Proposition 14.10 (Sign non-identifiability in an even residual model)**.** *Suppose the scalar residual model is $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$ with $a \neq 0$ and $\sigma_e > 0$. Then $p(e \mid \theta) = p(e \mid -\theta)$ for every sample and every record length. Residual data alone therefore cannot identify the sign of the frame mismatch without an orientation-breaking input.*

Proof. The likelihood depends on θ only through θ^2; hence the joint likelihood of any record is invariant under $\theta \mapsto -\theta$. No statistic of that record can distinguish the two parameter values. □

This proposition is a property of the declared even likelihood. It does not prove that every instantaneous or "first-order" controller fails: directional probes, asymmetric residual models, and exogenous orientation data can remove the ambiguity.

14.3.2 The Agent's Statistical Manifold

The agent's prediction-residual stream $\{e(t)\}_{t \geq 0}$ defines a stochastic process whose distribution depends on the gauge-fixing parameter σ. We model this dependence as a parametric family.

Definition 14.11 (Predictive family)**.** *The* predictive family *of the agent is the set*

$$\mathcal{P} := \{p(e \,|\, \sigma) : \sigma \in \mathcal{M}_G\}, \tag{14.10}$$

where $\mathcal{M}_G := G/H$ is the space of gauge-fixing orbits (H is the stabiliser of the foreground subspace), e denotes the prediction-residual time series over a window of length T, and $p(e \,|\, \sigma)$ is the likelihood of the observed residuals given the gauge parameter σ.

The key insight is that $p(e \,|\, \sigma)$ depends on σ even though $e(t) \in V_{\text{fg}}$, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured. When σ drifts from the optimal σ^*:

- The variance of the residuals increases (the discarded background components contribute unmodelled noise).
- The temporal correlations of the residuals change (the projected kernel $\mathcal{K}_{\mathcal{F}}$ no longer captures the dominant environmental modes).
- Higher-order statistics (kurtosis, spectral shape) shift systematically.

These distributional changes are invisible to the raw error $e(t)$ but detectable by the Fisher metric of $\mathcal{P}$.

14.3.3 Self-Referential Fisher Information

Definition 14.12 (Self-referential Fisher information)**.** *For one residual with likelihood $p(e \mid \sigma)$, define the* per-sample Fisher information

$$\mathcal{I}_1(\sigma) := \mathbb{E}_\sigma\left[(\partial_\sigma \log p(e \mid \sigma))^2\right]. \tag{14.11}$$

For conditionally independent residuals observed at times t_m, the accumulated local information *is $J_T := \sum_{t_m \le T} \mathcal{I}_1(\sigma(t_m))$. The Fisher metric g, the local quadratic form $g_{ij}\delta\sigma^i\delta\sigma^j$, and J_T are distinct objects. Fisher information is measured in inverse-squared parameter units, not bits.*

Remark 14.13 (What the agent "measures")**.** *Computing $\mathcal{I}_1(\sigma)$ does not require access to V_{bg} or to the "true" environment. It requires only: (i) the agent's own prediction residuals $\{e(t)\}$ (which lie in V_{fg}), and (ii) the ability to evaluate the score function $\partial_\sigma \log p(e \,|\, \sigma)$— the sensitivity of its own predictive model to frame perturbations. This is a computation entirely within the agent's internal algebra, using only quantities already available from the ego's processing pipeline.*

Proposition 14.14 (Fisher signal in a Gaussian residual model)**.** *Let residuals sampled at rate r be conditionally independent with*

$$e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \qquad \theta(t) = \theta_0 + \varepsilon t, \tag{14.12}$$

where $a \neq 0$, $\sigma_e > 0$, and $\varepsilon > 0$. The per-sample Fisher information with respect to the current mismatch is

$$\mathcal{I}_1(\theta) = \frac{4a^2\theta^2}{\sigma_e^2}. \tag{14.13}$$

Consequently the expected accumulated local information over a window of length T is

$$J_T \simeq \frac{4a^2 r}{3\sigma_e^2 \varepsilon} \left[(\theta_0 + \varepsilon T)^3 - \theta_0^3 \right]. \tag{14.14}$$

For a freshly calibrated frame ($\theta_0 = 0$), the per-sample signal is quadratic in time and $J_T = 4a^2 r \varepsilon^2 T^3/(3\sigma_e^2)$.

Proof. For a normal family with fixed variance and mean $\mu(\theta) = a\theta^2$, the score is $(e - \mu)\mu'(\theta)/\sigma_e^2$. Taking its squared expectation gives $(\mu')^2/\sigma_e^2$, which is (14.13). Summing over samples and using the sampling rate gives $J_T \simeq r \int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t)dt$, which evaluates to (14.14). □

Remark 14.15 (Scope of the sensitivity claim)**.** *Full-rank Fisher information for σ does not imply that a scalar mismatch map $\theta(\sigma)$ has a Jacobian bounded away from zero. Proposition 14.14 therefore works directly in the identified scalar mismatch coordinate. A multi-parameter extension must supply or derive the relevant sensitivity map.*

Corollary 14.16 (Conditional detection before visibility crossover)**.** *In the Gaussian residual model, let $J_{\min} > 0$ be a declared detection threshold. For a freshly calibrated frame,*

$$T_{\det} = \left(\frac{3\sigma_e^2 J_{\min}}{4a^2 r \varepsilon^2} \right)^{1/3}. \tag{14.15}$$

Detection precedes Paper II's visibility crossover if and only if $T_{\det} < t_{\text{vis}}$, where for the same linear drift $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Equivalently, for general θ_0 the condition is $J_{t_{\text{vis}}} > J_{\min}$. No unconditional ordering follows from regularity alone.

Proof. Set the fresh-calibration expression for J_T in Proposition 14.14 equal to $J_{\min}$ and solve for T. Comparing that value with the independently defined visibility time gives the stated condition. □

14.4 The Self-Referential Bound

14.4.1 The Bayesian Framework

The agent's ego structure (Paper II) provides a *prior belief* about the correct gauge parameter: the current frame σ_0 is the ego's "preferred" value. We encode this as a prior distribution $\pi_{\text{ego}}(\sigma)$, concentrated around σ_0.

Definition 14.17 (Ego rigidity)**.** *The* ego rigidity *is the prior Fisher information*

$$\mathcal{I}_{\text{ego}} := \int_{\mathcal{M}_G} \left(\frac{\partial \log \pi_{\text{ego}}(\sigma)}{\partial \sigma} \right)^2 \pi_{\text{ego}}(\sigma)\, d\sigma. \tag{14.16}$$

High $\mathcal{I}_{\text{ego}}$ corresponds to a sharply peaked prior (rigid ego); low $\mathcal{I}_{\text{ego}}$ to a diffuse prior (flexible ego). This prior and its Fisher information are supplied model data. The diagnostics of Paper II do not determine a prior without an additional statistical construction.

14.4.2 The Self-Referential Cramér–Rao Bound

Theorem 14.18 (Self-Referential Cramér–Rao Bound)**.** *Let the scalar drift parameter $\delta\sigma$ lie in a compact interval $[a, b]$. Assume that the residual likelihood is regular, that the effective samples are conditionally independent, and that the prior π_{ego} is C^1, has finite prior Fisher information, and vanishes at a and b. Let $\delta\hat{\sigma}$ be any estimator based only on a residual record of duration T. Define the* effective sample size *$n_{\text{eff}} := T/\tau_E$, where τ_E is the decorrelation time of the residual process $\{e(t)\}$ (the time beyond which consecutive residuals carry approximately independent information about σ). If the effective samples share per-sample information $\mathcal{I}_1(\delta\sigma)$, the van Trees inequality [145] gives*

$$\mathbb{E}\Big[\big|\delta\hat{\sigma} - \delta\sigma\big|^2\Big] \;\geq\; \frac{1}{n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1(\delta\sigma)] + \mathcal{I}_{\text{ego}}}. \tag{14.17}$$

Proof. This is a direct application of the van Trees (Bayesian Cramér–Rao) inequality [145]. The total information about the drift parameter $\delta\sigma$ consists of two contributions:

- $n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1]$: the data Fisher information. In continuous time, the residual process is correlated with decorrelation time τ_E set by the bath memory kernel. Over a window of duration T, the process yields $n_{\text{eff}} \approx T/\tau_E$ effectively independent samples, each carrying Fisher information about $\delta\sigma$.
- $\mathcal{I}_{\text{ego}}$: the prior Fisher information from the ego's preference for σ_0 (Definition 14.17).

The boundary and regularity assumptions remove the integration-by-parts boundary term in the standard van Trees proof. Fisher information is not measured in bits; the denominator has inverse-squared parameter units. □

Remark 14.19 (The ego as help and hindrance)**.** *The ego rigidity $\mathcal{I}_{\text{ego}}$ acts as both help and hindrance:*

- ***Help**: when the ego is well-aligned ($\sigma_0 \approx \sigma^*$), the prior tightens the bound, reducing estimation variance.*
- ***Hindrance**: when the ego is misaligned ($|\sigma_0 - \sigma^*|$ large), the prior pulls the estimate toward the wrong value, creating a* confirmation bias *that resists recalibration.*

For a conjugate scalar Gaussian location model only, the posterior mean has the precision-weighted form below, where $\bar{\mathcal{I}}_1$ is its constant per-sample data precision:

$$\hat{\sigma}_{\text{opt}} = \frac{n_{\text{eff}}\,\bar{\mathcal{I}}_1\,\hat{\sigma}_{\text{MLE}} + \mathcal{I}_{\text{ego}}\,\sigma_0}{n_{\text{eff}}\,\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{ego}}}, \tag{14.18}$$

a weighted average of the maximum-likelihood estimate $\hat{\sigma}_{\text{MLE}}$ and the ego's prior belief σ_0, with weights proportional to their respective Fisher informations. This formula is not implied by van Trees for a general likelihood or prior. In the conjugate Gaussian case, as $n_{\text{eff}}\,\bar{\mathcal{I}}_1 \gg \mathcal{I}_{\text{ego}}$ (enough data to overwhelm the ego), the estimator converges to the MLE.

14.4.3 The Rigidity-Sensitivity Trade-off

Proposition 14.20 (Optimal ego rigidity)**.** *Let the total expected loss be* $\mathcal{L}_{\text{total}}(\mathcal{I}_{\text{ego}}) = \mathcal{L}_{\text{estimation}} + \lambda\,\mathcal{L}_{\text{calibration}}$*, where* $\mathcal{L}_{\text{estimation}}$ *is the mean-squared drift-estimation error (bounded by* (14.17)*) and* $\mathcal{L}_{\text{calibration}}$ *is the cost of adjusting the frame (proportional to the frame rotation distance, hence larger when the ego is rigid and must be overcome). If the admissible rigidity interval is nonempty and compact and the two supplied loss terms are lower semicontinuous, an optimal rigidity* $\mathcal{I}^*_{\text{ego}}$ *exists. Uniqueness or an interior optimum additionally requires a model-specific strict-convexity or monotonicity condition.*

Too rigid *(*$\mathcal{I}_{\text{ego}} \gg n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the ego overwhelms the data; the agent is blind to drift.* Too soft *(*$\mathcal{I}_{\text{ego}} \ll n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the agent overreacts to noise; calibration cost is high. The optimum balances sensitivity against stability.*

Proof. A lower-semicontinuous real function attains its minimum on a nonempty compact set. The existence conclusion uses only that fact; the qualitative "too rigid/too soft" interpretation is not a convexity proof. □

14.5 The Calibration Loop

14.5.1 The Natural Gradient Update Law

The meta-observer updates the gauge parameter σ following the natural gradient on the statistical manifold $(\mathcal{M}_G,\, g)$:

$$\dot{\sigma} = -\eta\, g^{-1}(\sigma)\, \nabla_\sigma L_{\text{frame}}(\sigma), \tag{14.19}$$

where $\eta > 0$ is the adaptation rate and the *frame loss* is

$$L_{\text{frame}}(\sigma) := \mathbb{E}_\sigma[-\mathcal{S}_{\text{vis}}(\sigma)]\,. \tag{14.20}$$

The frame loss is minimised at the optimal gauge σ^* that maximises visible survival. The Fisher metric enters through the inverse g^{-1} in the natural gradient, not as a penalty term: it defines the *geometry* of the update, ensuring reparametrisation invariance.

Remark 14.21 (Reparametrisation invariance)**.** *The natural gradient* (14.19) *is invariant under reparametrisation of the gauge manifold* $\mathcal{M}_G$*: the update direction does not depend on the choice of coordinates for* σ*. This is essential because the gauge manifold inherits its geometry from the Clifford algebra, and no canonical coordinate system is preferred.*

14.5.2 Lyapunov Stability of the Loop

Drift velocity. Let $\sigma^*(t)$ denote the instantaneous optimal gauge parameter, defined as a minimiser of L_{frame} at time t in Eq. (14.20). Define the *drift velocity* $\dot{\sigma}^* := d\sigma^*/dt$, measured with respect to the Fisher metric g; its norm $\|\dot{\sigma}^*\|_g := \sqrt{g_{ij}\,\dot{\sigma}^{*i}\,\dot{\sigma}^{*j}}$ is the instantaneous rate at which the environment's optimal frame rotates on the gauge manifold.

Definition 14.22 (Lyapunov monitoring function)**.** *The* Lyapunov monitoring function *is the squared geodesic distance on the statistical manifold from the current frame to the instantaneous optimal frame:*

$$V(\sigma) := d_{\text{geo}}(\sigma,\, \sigma^*(t))^2\,, \tag{14.21}$$

where d_{geo} *is the geodesic distance in the Fisher metric* g*.*

Theorem 14.23 (Loop Tracking Bound)**.** *Assume the natural-gradient trajectory remains in a geodesically convex normal neighbourhood of $\sigma^*(t)$ and that the frame loss is geodesically m_L-strongly convex there, with* $\operatorname{Hess} L_{\text{frame}} \succeq m_L g$ *for some $m_L > 0$. Then the update* (14.19) *satisfies*

$$\frac{dV}{dt} \;\le\; -2\eta\, m_L\, V \;+\; 2\sqrt{V}\,\big\|\dot\sigma^*\big\|_g, \tag{14.22}$$

where $\|\dot\sigma^*\|_g$ is the instantaneous drift speed of the optimal frame. Consequently:

(a) ***Tracking.*** *Whenever $\sqrt{V} > \|\dot\sigma^*\|_g/(\eta\, m_L)$, we have $dV/dt < 0$: the loop actively reduces the mismatch.*

(b) ***Tracking neighbourhood.*** *The mismatch converges to a neighbourhood of the set of stationary points of L_{frame}. This neighbourhood has size*

$$V_\infty \;:=\; \frac{\big\|\dot\sigma^*\big\|_g^2}{(\eta\, m_L)^2}. \tag{14.23}$$

For bounded drift ($\|\dot\sigma^*\|_g \le \Lambda_{\max}$), the mismatch is bounded: $\limsup_{t\to\infty} V(t) \le \Lambda_{\max}^2/(\eta m_L)^2$.

(c) ***Static limit.*** *When $\sigma^* = \text{const}$ ($\dot\sigma^* = 0$), the bound reduces to $dV/dt \le -2\eta m_L\, V$, giving exponential convergence $V(t) \le V(0)\, e^{-2\eta m_L\, t}$.*

Proof. Since $V(\sigma) = d_{\text{geo}}(\sigma,\, \sigma^*(t))^2$ and $\sigma^*(t)$ is time-varying, the total derivative has two contributions:

$$\frac{dV}{dt} = \underbrace{\frac{\partial V}{\partial \sigma}\cdot \dot\sigma}_{\text{control}} \;+\; \underbrace{\frac{\partial V}{\partial \sigma^*}\cdot \dot\sigma^*}_{\text{drift}}.$$

Control term. In normal coordinates centred at σ^*, let $\delta\sigma := \sigma - \sigma^*$. The control contribution is $2\, g(\delta\sigma,\, \dot\sigma) = 2\, g(\delta\sigma,\, -\eta\, g^{-1}\nabla L_{\text{frame}}) = -2\eta\, \langle \delta\sigma,\, \nabla L_{\text{frame}}\rangle$. By the explicit strong-convexity hypothesis, $\langle \delta\sigma,\, \nabla L_{\text{frame}}\rangle \ge m_L\, |\delta\sigma|_g^2 = m_L\, V$, giving a control contribution $\le -2\eta\, m_L\, V$.

Drift term. The drift contribution is $-2\, g(\delta\sigma,\, \dot\sigma^*)$. By Cauchy–Schwarz: $|g(\delta\sigma,\, \dot\sigma^*)| \le |\delta\sigma|_g\, \|\dot\sigma^*\|_g = \sqrt{V}\, \|\dot\sigma^*\|_g$. Hence the drift contribution is bounded by $+2\sqrt{V}\, \|\dot\sigma^*\|_g$.

Combined. Adding both contributions gives (14.22). Part (a) follows by setting $dV/dt < 0$; part (b) by solving $dV/dt = 0$ for the fixed point $\sqrt{V_\infty} = \|\dot\sigma^*\|_g/(\eta m_L)$; part (c) by setting $\dot\sigma^* = 0$. □

14.5.3 Convergence Rate under Persistent Excitation

Definition 14.24 (Persistent excitation constant)**.** *The* persistent excitation constant $\alpha_{\text{PE}} > 0$ *is the minimum eigenvalue of the time-averaged Fisher information matrix:*

$$\bar g(t) := \frac{1}{T}\int_t^{t+T} g(\sigma(s))\, ds \;\succeq\; \alpha_{\text{PE}}\, I \qquad \textit{for all } t, \tag{14.24}$$

when the residual likelihood is persistently exciting. Pointwise full-rank regularity and a positive environmental entropy rate do not by themselves provide a uniform lower bound.

Remark 14.25 (Tracking vs convergence)**.** *In the static case ($\sigma^* = \text{const}$), Theorem 14.23(c) gives pure exponential convergence: $V(t) \leq V(0)\, e^{-2\eta m_L t}$. Under environmental drift, convergence to zero is* not *possible—instead the loop maintains the mismatch within the tracking neighbourhood* (14.23)*. The tracking error V_∞ grows with drift speed $\|\dot{\sigma}^*\|_g$ and decreases with loop parameters η and m_L. The distinct constant α_{PE} controls statistical conditioning of the Fisher metric; no equality $m_L = \alpha_{\mathrm{PE}}$ is assumed. A physical implementation must separately relate adaptation rate to its energy budget.*

Remark 14.26 (The necessity of novelty)**.** *A frozen or uninformative residual likelihood can violate persistent excitation, but $h_\mu \to 0$ alone does not determine α_{PE} or m_L. Loss of excitation therefore has to be checked in the supplied observation model rather than inferred from entropy rate alone.*

14.5.4 Four-Part Model Template

The following is an architecture template for the declared model, not a necessity or sufficiency theorem for persistent agents.

Definition 14.27 (Four-part model architecture)**.** *The running T-DOME model uses four supplied structural layers:*

(I) ***External observable geometry.*** *The environmental observable algebra supports a metric structure; $Cl(1,3)$ serves as the running example throughout the programme, but the argument applies to any algebra satisfying (C1). This is a modelling premise; revised HAFF/Q-RAIF do not derive a general metric dictionary.*

(II) ***Internal control algebra.*** *The agent carries a supplied internal algebra $Cl(V,q)$ and a supplied representation map ϕ into an external algebra. Revised Q-RAIF neither uniquely selects the Clifford algebra nor proves a signature-preserving embedding.*

(III) ***Self-monitoring function.*** *The agent maintains a Lyapunov function $V(\sigma)$* (14.21) *satisfying the tracking bound* (14.22)*, via a second-order control loop on the agent's Fisher information, keeping the mismatch within the tracking neighbourhood* (14.23)*.* Conditional source: *Theorem 14.23, under its explicit local strong-convexity assumptions.*

(IV) ***Biased, non-Markovian memory.*** *The agent carries path-dependent state (non-Markovian memory kernel $\mathcal{K}(t,s)$) compressed through a gauge-fixed reference frame (the ego $\mathfrak{E}$). This is a model choice motivated by Papers I and II, not a necessity result inherited from them.*

The template is sufficient only for performance separately proved by a fully specified implementation. Omitting a layer may be benign in a different architecture.

14.6 Thermodynamic Cost

14.6.1 The Three Cost Components

One implementation may use sensing, computing, and actuation. Their thermodynamic costs depend on how they are physically realized.

1. Sensing cost. The meta-observer must read the prediction residuals from the ego's processing pipeline. Measurement or copying can in principle be reversible. If sensor registers irreversibly erase Shannon information at rate R_s bits per unit time, Landauer's principle gives

$$\dot{W}_{\text{sense}} \;\geq\; k_B T \,\ln 2 \cdot R_s. \tag{14.25}$$

No positive sensing bound follows merely from reading a stream.

2. Computing cost. Reversible logic likewise evades a gate-count Landauer charge. If computational work registers irreversibly erase Shannon information at rate R_c bits per unit time, then

$$\dot{W}_{\text{compute}} \;\geq\; k_B T \,\ln 2 \cdot R_c. \tag{14.26}$$

3. Actuating cost. Rotating the gauge parameter from the current frame σ to the estimated optimal frame $\hat{\sigma}^*$ is a finite-time thermodynamic transformation on the gauge manifold. By the Sivak–Crooks bound (Proposition 14.7):

$$\dot{W}_{\text{actuate}} \;\geq\; \frac{\mathcal{L}^2(\sigma,\, \hat{\sigma}^*)}{\tau_{\text{recalib}}^2}, \tag{14.27}$$

where $\mathcal{L}(\sigma,\, \hat{\sigma}^*)$ is the thermodynamic length (14.7) of the geodesic from σ to $\hat{\sigma}^*$, and τ_{recalib} is the recalibration time.

14.6.2 Conditional Implementation Bound

Proposition 14.28 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, and actuation use separable physical degrees of freedom with no shared erasure accounting; (ii) sensor and computational registers erase Shannon information at rates R_s and R_c; and (iii) frame actuation follows a linear-response protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}$. Then*

$$\dot{W}_{\text{loop}} \;\geq\; k_B T \,\ln 2 \,[R_s + R_c] + \frac{\mathcal{L}^2(\sigma,\, \sigma^*)}{\tau_{\text{recalib}}^2}. \tag{14.28}$$

This is a bound for the declared implementation, not a universal cost of sensing, Fisher-information evaluation, or self-reference.

Proof. Landauer's bound applies to the two declared irreversible reset streams, giving the first two terms. The Sivak–Crooks work bound divided by the protocol duration gives the actuation rate. The separability hypothesis permits addition without double counting. Dropping that hypothesis, or implementing reversible sensing/computation with no reset in the window, invalidates the corresponding additive terms. □

14.6.3 Withdrawal of the Universal Persistence Budget

Remark 14.29 (Withdrawal of the universal persistence budget)**.** *The former additive persistence corollary is withdrawn. Revised Papers I and II provide no universal memory-maintenance or ego-processing Landauer terms that can be inherited here. Proposition 14.28 may be combined with independently established costs for a concrete hardware implementation, provided separability and reset accounting are verified afresh.*

14.7 Worked Example: Qubit in a Drifting Two-Channel Bath

14.7.1 Model Setup

We extend the two-channel qubit model from Paper II (Section 6) by introducing environmental drift.

Inherited setup. A qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ ($D = 4$), coupled to two bosonic channels:

- Dephasing channel (σ_z): $J_z(\omega) = 2\lambda_z\gamma_z\omega/(\omega^2 + \gamma_z^2)$.
- Dissipative channel (σ_x): $J_x(\omega) = 2\lambda_x\gamma_x\omega/(\omega^2 + \gamma_x^2)$.

Paper II's ego selects $V_{\text{fg}} = \text{span}\{1, \mathbf{k}\}$ (the dephasing subspace), discarding $V_{\text{bg}} = \text{span}\{\mathbf{i}, \mathbf{j}\}$.

Environmental drift and residual channel. We use Paper II's revised linear mismatch model and the Gaussian residual channel of Proposition 14.14:

$$\theta(t) = \varepsilon t, \qquad e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \quad a = 1.2, \quad \sigma_e = 0.35, \quad \varepsilon = 0.04\,\omega_0. \tag{14.29}$$

At sampling rate $r = 50\,\omega_0$, the visibility crossover is $t_{\text{vis}} = \pi/(4\varepsilon) \approx 19.63\,\omega_0^{-1}$. The Gaussian likelihood is a declared observation model, not derived from the bosonic spectral density.

Parameter mapping.

Quantity	Value	Source
$D = \dim Cl(0,2)$	4	Paper II
k^*	2	chosen Paper II model frame
$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Paper II
θ_0 (initial misalignment)	0	fresh calibration
ε (angular drift rate)	$0.04\,\omega_0$	Eq. (14.29)
t_{vis}	$19.63\,\omega_0^{-1}$	Paper II visibility model
η (adaptation rate)	0.5	meta-observer

14.7.2 Fisher Information under Drift

For the declared residual channel, the per-sample and accumulated local Fisher signals are

$$\mathcal{I}_1(t) = \frac{4a^2\varepsilon^2 t^2}{\sigma_e^2}, \qquad J_T = \frac{4a^2 r\varepsilon^2 T^3}{3\sigma_e^2}. \tag{14.30}$$

The first quantity grows quadratically in time; the second grows cubically because it also counts samples. For $J_{\min} = 20$, Equation (14.15) gives $T_{\text{det}} \approx 2.52\,\omega_0^{-1} < t_{\text{vis}}$. This ordering is a parameter check, not a general theorem.

14.7.3 Loop Dynamics: Self-Calibration in Action

Under the natural gradient update (14.19), the frame angle $\phi(t)$ tracks the drifting optimal frame $\phi^*(t)$. The Lyapunov function $V(t) = (\phi(t) - \phi^*(t))^2$ is governed by the tracking bound (14.22): the loop drives V toward the tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|_g^2/(\eta m_L)^2$, with the approach rate set by the loss strong-convexity modulus m_L and adaptation rate η.

Comparison.

- **Without loop** (Paper II agent): the mismatch grows as $\theta(t) = \varepsilon t$, reaching the visibility crossover at t_{vis}.
- **With loop** (Paper III agent): the mismatch oscillates around zero, bounded by the estimation neighbourhood supplied by Theorem 14.23 when its local strong-convexity and bounded-drift hypotheses hold.

A multi-dimensional numerical evaluation extending this qubit illustration to continuous drift is presented in Section 14.8.

14.7.4 Thermodynamic Cost Evaluation

For illustration, suppose a particular implementation irreversibly resets sensor and work registers at rates $R_s = 2$ and $R_c = 1$ bits per unit time and uses separable actuation hardware. Then

$$\dot{W}_{\text{sense}} \geq k_B T \ln 2 \cdot R_s = 2\, k_B T \ln 2, \tag{14.31}$$

$$\dot{W}_{\text{compute}} \geq k_B T \ln 2 \cdot R_c = k_B T \ln 2, \tag{14.32}$$

$$\dot{W}_{\text{actuate}} \geq \frac{\mathcal{L}^2}{\tau_{\text{recalib}}^2} \approx \frac{(\Delta\theta)^2\, \tau_{\text{relax}}}{\tau_{\text{recalib}}^2}\, k_B T. \tag{14.33}$$

The two Landauer terms are present because this example stipulates irreversible reset. A reversible implementation may lower them, and no percentage overhead relative to Papers I or II follows.

14.8 Numerical Demonstration

We report two separate numerical checks. The first directly samples the Gaussian likelihood and computes its score-based Fisher signal. The second retains the earlier multi-dimensional calibration controller, whose top-k gradient-overlap sentinel is a heuristic proxy rather than Fisher information.

14.8.1 Direct Fisher-Signal Test

We sample Equation (14.12) at rate $r = 50$ with $a = 1.2$, $\sigma_e = 0.35$, $\varepsilon = 0.04$, and 4,000 Monte Carlo residuals per plotted time. For each residual we evaluate the score

$$s_\theta(e) = \frac{(e - a\theta^2)(2a\theta)}{\sigma_e^2}$$

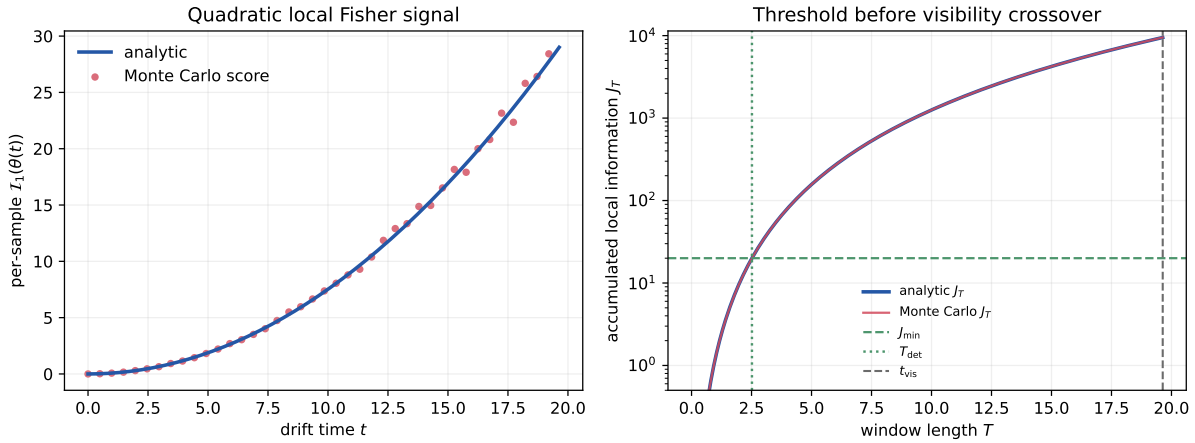


Figure 14.1: **Direct score-based Fisher test. Left:** analytic per-sample Fisher information and Monte Carlo score-squared estimate, both quadratic in drift time. **Right:** analytic and empirical accumulated local information; the $J_{\min} = 20$ crossing precedes the independently defined visibility crossover for this parameter set.

and estimate $\mathcal{I}_1(\theta) = \mathbb{E}[s_\theta^2]$ directly. Figure 14.1 shows agreement with the analytic quadratic curve and with the cubic accumulated curve. For $J_{\min} = 20$, the analytic and empirical crossings are respectively $T_{\det} = 2.517$ and 2.524, both before $t_{\text{vis}} = 19.635$. This verifies Proposition 14.14 and Corollary 14.16 for the declared parameter set; it does not establish the ordering for other likelihoods or thresholds.

14.8.2 Heuristic Calibration-Control Model

Environment. A d-dimensional linear prediction task: $y(t) = \mathbf{w}(t)^\top \mathbf{x}(t) + \sigma\,\epsilon(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $\epsilon \sim \mathcal{N}(0,1)$. The weight vector $\mathbf{w}(t) \in \mathbb{S}^{d-1}$ drifts by receiving random perturbations on *background* dimensions only (indices $k, \ldots, d-1$), then renormalising. Signal therefore migrates progressively from the ego's foreground to its blind sector.

Agents.

- **Fixed ego** (Paper II analogue): learns a linear model on a *fixed* foreground subspace of dimension k via stochastic gradient descent (SGD, rate η, decay λ). Embodies the "frozen gauge" of Paper II.
- **Calibrated loop** (Paper III analogue): identical ego plus a staleness sentinel and recalibration mechanism. The sentinel tracks $g_i = \text{EMA}(|e\,x_i|)$ for each dimension i (an absolute-gradient proxy), computes the fraction of top-k gradient dimensions *not* in the current foreground as a frame-staleness index $m \in [0,1]$, and triggers recalibration when $\text{EMA}(m) > \theta$. After recalibration, a settling period of τ steps elapses before the sentinel resumes monitoring.

Parameters.

Quantity	Value	Role
d	20	full ambient dimension
k	5	ego foreground dimension ($k/d = 0.25$)
σ	0.1	observation noise std
η	0.01	SGD learning rate
λ	0.998	SGD weight decay
θ	0.25	staleness threshold
Λ_{demo}	variable	simulator drift amplitude per step
τ	variable	settling period (cooldown)

Oracle metrics. Neither agent has access to $\mathbf{w}(t)$. We evaluate performance externally using the *oracle full-space error*:

$$\mathcal{E}_{\text{full}} = \left\|\mathbf{w}_{\text{ego}} - \mathbf{w}^*_{\text{fg}}\right\|^2 + \left\|\mathbf{w}^*_{\text{bg}}\right\|^2, \tag{14.34}$$

where $\mathbf{w}^*_{\text{fg}}$ and $\mathbf{w}^*_{\text{bg}}$ denote the true weight vector restricted to foreground and background coordinates respectively, and $\mathbf{w}_{\text{ego}}$ is the ego's foreground-supported estimator lifted to the full space. The first term captures foreground tracking error (accessible to the ego); the second captures hidden-sector signal (invisible).

14.8.3 Results

Result 1: Delusion-correction separation (Figure 14.2). At simulator drift amplitude $\Lambda_{\text{demo}} = 0.02$, settling period $\tau = 200$, and $T = 5\,000$ steps, three phenomena are visible:

(a) *Delusion trap.* The ego's foreground tracking error converges to near zero, while the true full-space error rises toward ~ 1 and stabilises. The growing gap between the two is the hidden sector in this supplied linear model. It does not prove a no-go theorem for all first-order controllers.

(b) *Detectability.* The staleness sentinel produces a clean sawtooth: rising from zero after each recalibration, crossing the threshold $\theta = 0.25$, and triggering frame reset (25 events over $T = 5\,000$). This demonstrates that the chosen overlap proxy can trigger a useful reset. The sentinel is not Fisher information; the direct Fisher test is reported in Section 14.8.1.

(c) *Net benefit.* The calibrated loop achieves $\mathcal{E}_{\text{full}} \approx 0.74$ versus the fixed ego's ≈ 1.02: a 27% reduction in true prediction error.

Result 2: Phase structure and optimal calibration budget (Figures 14.3–14.4). We scan 16 simulator drift amplitudes $\Lambda_{\text{demo}} \in [0.005, 0.08]$ and 16 settling periods $\tau \in [15, 800]$ (logarithmically spaced), running both agents for $T = 4\,000$ steps across 6 random seeds per grid point.

Figure 14.3(a) shows the performance gain $\Delta = \mathcal{E}_{\text{ego}} - \mathcal{E}_{\text{loop}}$: the loop improves over the ego (green) across most of the parameter space, with a boundary at $\Delta = 0$ (dashed) below which recalibration is counterproductive (very low drift, where the overhead of re-learning exceeds the benefit of tracking). The solid curve traces the *optimal settling period*

$\tau_{\text{opt}}(\Lambda_{\text{demo}})$—the recalibration period minimising $\mathcal{E}_{\text{loop}}$—which decreases monotonically from ~ 370 steps at $\Lambda_{\text{demo}} = 0.005$ to ~ 100 at $\Lambda_{\text{demo}} = 0.08$. Figure 14.3(b) shows that calibration frequency increases smoothly with drift and with shorter settling period, exhibiting the cost–performance trade-off of this supplied controller and cost proxy. It does not test Proposition 14.28.

Extracting $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ yields the *optimal calibration frequency* $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ (Figure 14.4). The curve is smooth and monotonically increasing: faster drift demands tighter calibration. It saturates at high Λ_{demo} near $\alpha_{\text{opt}} \approx 0.01$ per step ($\tau_{\text{opt}} \approx 100$), of the same order as the learner's settling time. This is consistent with the intuition that drift estimation requires a minimum observation window; the self-referential Cramér–Rao bound (Theorem 14.18) provides the analytic counterpart of this computational floor.

14.8.4 Scope of This Demonstration

This demonstration **does** show:

1. The explicit Gaussian residual model has a directly measured score-squared Fisher signal with the quadratic and cubic scalings in Proposition 14.14.

2. For the declared Gaussian parameters, the Fisher threshold is crossed before the linear visibility crossover.

3. A separate, non-Fisher frame-staleness proxy has clean threshold dynamics and triggers effective recalibration in the supplied linear controller.

4. An optimal calibration frequency $\alpha_{\text{opt}}(\Lambda_{\text{demo}})$ exists, increases monotonically with drift rate, and saturates at the learner's settling timescale in the scanned grid.

In summary, the direct experiment validates the declared Fisher likelihood, while the controller experiment establishes a model-specific proxy trade-off. Neither is universal across architectures or environment classes.

This demonstration does **not** show:

1. That the specific functional form of $\alpha_{\text{opt}}(\Lambda)$ matches the analytic Cramér–Rao prediction in the large-d limit. The demonstration confirms the monotonic trend and saturation; deriving the exact scaling exponent from Theorem 14.18 remains open.

2. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and isotropic background drift; extensions to non-linear, non-Gaussian, or structured-drift settings require further investigation.

3. That the calibration loop is optimal among all possible adaptive strategies. It implements one specific realisation of the calibration-loop architecture.

4. That the gradient-overlap sentinel estimates Fisher information, or that recalibration-event frequency is a thermodynamic dissipation measurement.

Reproducibility. The controller simulation is in `tdome_demo.py`. The direct Fisher experiment is in `fisher_residual_demo.py`, with raw output in `fisher_residual_data.csv`. Both require only NumPy and Matplotlib and use fixed random seeds.

14.9 Discussion

14.9.1 Summary of Results

Result	**Statement**	**Sec.**
Even-likelihood non-identifiability	The Gaussian residual record cannot identify mismatch sign	14.3.1
Drift Detectability	Per-sample FI is quadratic and accumulated local FI is cubic in the Gaussian linear-drift model	14.3.3
Self-Referential CR Bound	Drift estimation bounded by $1/(n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1] + \mathcal{I}_{\text{ego}})$ under regular prior/likelihood data	14.4.2
Loop Tracking Bound	Lyapunov V with tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|^2/(\eta m_L)^2$ under local geodesic strong convexity	14.5.2
Four-Part Structure	Four supplied layers define the running model architecture	14.5.4
Loop Cost	$\dot{W}_{\text{loop}} \geq k_B T \ln 2\,(R_s + R_c) + \mathcal{L}^2/\tau^2_{\text{recalib}}$ for a separable reset-based implementation	14.6.2
Persistence Budget	Universal inherited budget withdrawn	14.6.3
Numerical Demonstration	Direct Fisher test plus a separate heuristic-controller scan	14.8

14.9.2 Conditional Interface Sequence

Papers I–III provide a conditional interface sequence:

Paper	**Crisis**	**Resolution**	**What is born**
Paper I	Restricted autonomous baseline	Correlation-aware accounting	**Conditional resource model**
Paper II	Raw archive exceeds a declared budget	Supplied coordinate projection	**Frame model**
Paper III	Linear visibility loss in a fixed frame	Likelihood-based detection; conditional tracking	**Calibration model**

The arrows are motivation or model interfaces. The tracking theorem is local and conditional, the universal persistence budget is withdrawn, and no termination or completeness result follows.

14.9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. The even Gaussian residual likelihood cannot identify mismatch sign, while its score has the explicit Fisher scalings in Proposition 14.14.

2. Detection precedes visibility loss exactly when the threshold condition in Corollary 14.16 holds.

3. Drift estimation precision is bounded by the Self-Referential Cramér–Rao bound (Theorem 14.18).

4. The calibration loop tracks the optimal frame within a bounded neighbourhood under a Lyapunov tracking bound (Theorem 14.23).

5. A separable reset-based hardware model obeys the conditional cost bound in Proposition 14.28.

This paper does **not** show:

1. That self-referential calibration implies or requires phenomenal consciousness, subjective experience, or qualia. "Reflexivity" as used here denotes second-order control, nothing more.

2. That the Lyapunov function V is a measure of "awareness." It is a control-theoretic stability condition, not a consciousness metric.

3. That the four-part template is necessary, sufficient, or a complete characterisation of agency.

4. That Fisher information requires the agent to "know" it is computing Fisher information. The computation can be implemented implicitly by any physical system whose dynamics approximate the natural gradient.

5. That the calibration loop eliminates the ego's bias. It tracks and compensates for drift in the bias; the four bias terms of Paper II persist.

6. That sensing or computing is intrinsically dissipative. The Landauer terms require irreversible register reset.

7. That this framework applies to all possible systems. It applies to systems satisfying (C1)–(C5).

8. That the structural parallel with philosophical concepts of self-awareness constitutes a philosophical or metaphysical claim.

9. That the Clifford algebra is the only possible algebraic setting. Other control algebras may yield analogous results with different quantitative bounds.

We have established a checkable Gaussian Fisher signal, a regular Bayesian estimation bound, and a local tracking result. Their use in a physical agent remains conditional on the declared observation, optimization, and hardware models.

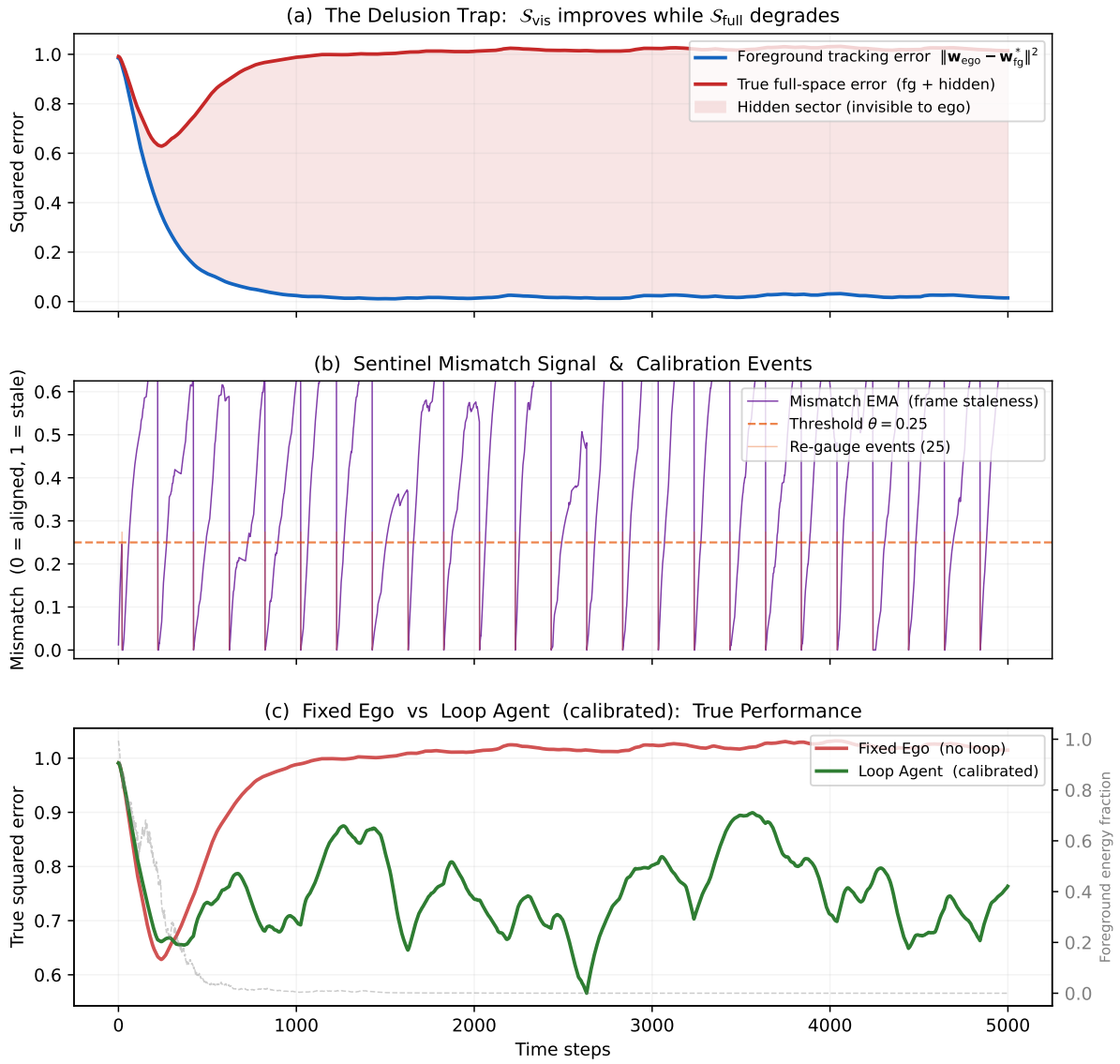


Figure 14.2: **Delusion trap and calibration loop.** $d = 20$, $k = 5$, $\Lambda_{\text{demo}} = 0.02$, $\tau = 200$, $T = 5\,000$. **(a)** Foreground tracking error (blue) decreases toward zero while true full-space error (red) increases; the shaded region is the hidden sector, invisible to the ego. **(b)** Frame-staleness sentinel (purple) rises monotonically between recalibration events (orange), producing a sawtooth with 25 threshold crossings. **(c)** The calibrated loop (green) maintains lower true error than the fixed ego (red); the grey dashed line shows the foreground energy fraction decaying as signal migrates to the background.

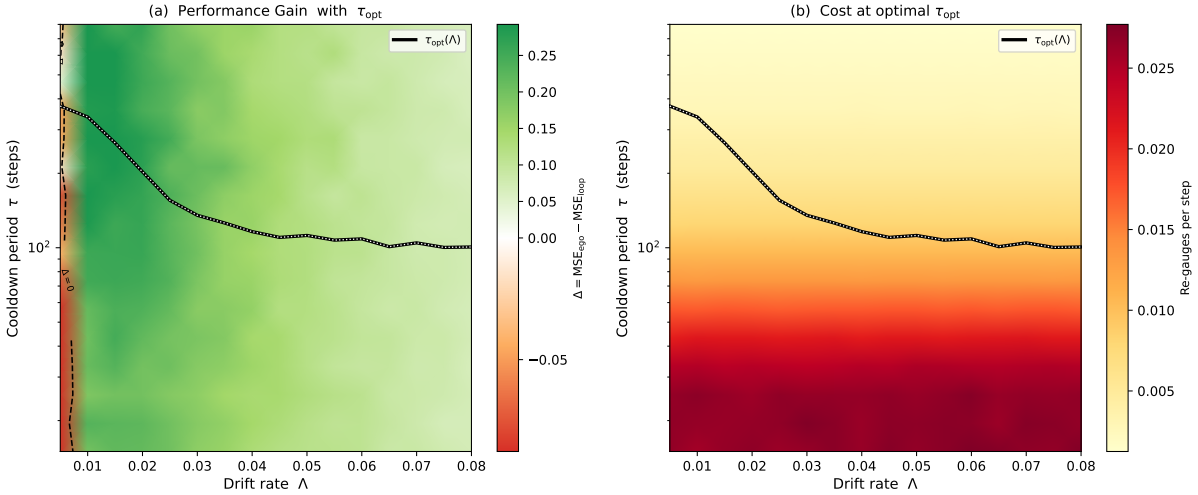


Figure 14.3: **Phase structure and calibration cost.** 16×16 grid, $T = 4\,000$, 6 seeds per point. **(a)** Performance gain Δ; green = loop improves on ego, red = counterproductive. Solid curve: $\tau_{\text{opt}}(\Lambda_{\text{demo}})$. Dashed: $\Delta = 0$ boundary. **(b)** Calibration frequency (thermodynamic cost proxy: recalibration events per step, proportional to energy expenditure under a fixed per-recalibration cost model); τ_{opt} overlaid.

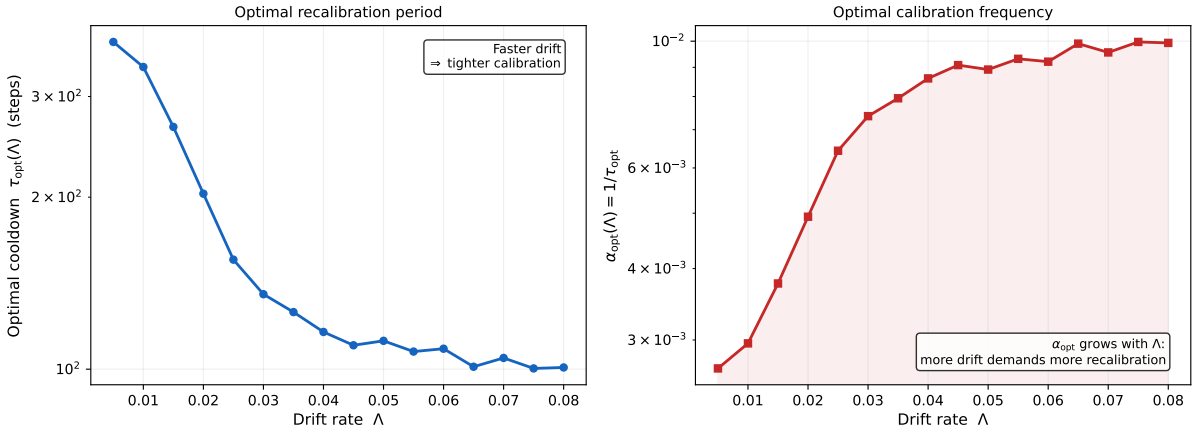


Figure 14.4: **Optimal calibration frequency. Left:** $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ decreases monotonically with drift rate. **Right:** $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ increases with drift rate and saturates at the learner's settling timescale (~ 100 steps), consistent with an observation-window floor.

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